Atlas of lower bounds for multicolour Ramsey numbers $R(k_1,...,k_r)$ - version 22

Julien Courties, November 2026. Each row gives the best lower bound we know for the multicolour Ramsey number, its origin (published result, our bricks, tabu search, ...) and the construction that produces it. Section 1: 2 to 7 colours, limits 3 to 10 (6,426 rows). Section 2: three colours, limits 11 to 15 (335 rows). All constructions rest on explicitly verified colourings (atoms and templates, library Atome_Modele.txt); machine-readable versions (xlsx, md) are published alongside this PDF.

Section 1 - Atlas, 2 to 7 colours, limits 3 to 10

Parameters	Bound (R >=)	Origin	Construction
R(3,3)	6	Published	published value
R(3,4)	9	Published	published value
R(3,5)	14	Published	published value
R(3,6)	18	Published	published value
R(3,7)	23	Published	published value
R(3,8)	28	Published	published value
R(3,9)	36	Published	published value
R(3,10)	40	Published	published value
R(4,4)	18	Published	published value
R(4,5)	25	Published	published value
R(4,6)	36	Published	published value
R(4,7)	49	Published	published value
R(4,8)	59	Published	published value
R(4,9)	73	Published	published value
R(4,10)	92	Published	published value
R(5,5)	43	Published	published value
R(5,6)	59	Published	published value
R(5,7)	80	Published	published value
R(5,8)	101	Published	published value
R(5,9)	133	Published	published value
R(5,10)	149	Published	published value
R(6,6)	102	Published	published value
R(6,7)	115	Published	published value
R(6,8)	135	Published	published value
R(6,9)	183	Published	published value
R(6,10)	204	Published	published value
R(7,7)	205	Published	published value
R(7,8)	219	Published	published value
R(7,9)	252	Published	published value
R(7,10)	292	Published	published value
R(8,8)	282	Published	published value
R(8,9)	329	Published	published value
R(8,10)	345	Published	published value
R(9,9)	565	Published	published value
R(9,10)	581	Published	published value
R(10,10)	798	Published	published value
R(3,3,3)	17	Published	published value
R(3,3,4)	30	Published	published value

Parameters	Bound (R >=)	Origin	Construction
R(3,3,5)	45	Published	published value
R(3,3,6)	61	Published	published value
R(3,3,7)	85	Published	published value
R(3,3,8)	103	Published	published value
R(3,3,9)	133	Our bricks	atom (3,3,9 ; 132) (ours) -> 132 vertices
R(3,3,10)	150	Published	published value
R(3,4,4)	55	Published	published value
R(3,4,5)	89	Published	published value
R(3,4,6)	117	Published	published value
R(3,4,7)	162	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices
R(3,4,8)	193	Published	published value
R(3,4,9)	242	Published	published value
R(3,4,10)	289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 73 – 3 = 289; with R(4,9) >= 73
R(3,5,5)	147	Our bricks	atom (3,5,5 ; 146) (ours) -> 146 vertices
R(3,5,6)	192	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [K2] -> 191 vertices
R(3,5,7)	267	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices
R(3,5,8)	337	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 85 – 3 = 337; with R(3,3,7) >= 85
R(3,5,9)	446	Our bricks	[template (5,9,3 ; 314) (ours)] x [K2] -> 445 vertices
R(3,5,10)	529	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 133 – 3 = 529; with R(5,9) >= 133
R(3,6,6)	338	Published	published value
R(3,6,7)	405	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 102 – 3 = 405; with R(6,6) >= 102
R(3,6,8)	457	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 115 – 3 = 457; with R(6,7) >= 115
R(3,6,9)	584	Our bricks	[template (6,9,3 ; 402) (ours)] x [K2] -> 583 vertices
R(3,6,10)	729	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 183 – 3 = 729; with R(6,9) >= 183
R(3,7,7)	674	Published	published value
R(3,7,8)	817	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 205 – 3 = 817; with R(7,7) >= 205
R(3,7,9)	873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 219 – 3 = 873; with R(7,8) >= 219
R(3,7,10)	1 005	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 252 – 3 = 1,005; with R(7,9) >= 252
R(3,8,8)	941	Published	published value
R(3,8,9)	1 125	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 282 – 3 = 1,125; with R(8,8) >= 282
R(3,8,10)	1 313	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 329 – 3 = 1,313; with R(8,9) >= 329
R(3,9,9)	1 844	Published	published value
R(3,9,10)	2 257	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 565 – 3 = 2,257; with R(9,9) >= 565
R(3,10,10)	2 841	Published	published value
R(4,4,4)	128	Published	published value
R(4,4,5)	154	Our bricks	non-linear colouring with 153 vertices found by SAT search over the Gaussian-integer ring Z[i]/(12+3i), invariant under multiplication by i (matrix R445_matrice_N153_Zi.txt)
R(4,4,6)	195	Tabu	non-linear colouring with 194 vertices found by tabu search, starting from the Atelier's linear colouring (matrix R446_matrice_N194_tabu.txt)
R(4,4,7)	277	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K3] -> 276 vertices
R(4,4,8)	312	Published bricks + theorems	[template (8,3 ; 20)] x [Paley-17 (4,4;17)] -> 311 vertices
R(4,4,9)	409	Published bricks + theorems	[template (9,3 ; 26)] x [Paley-17 (4,4;17)] -> 408 vertices
R(4,4,10)	474	Published bricks + theorems	[template (10,3 ; 30)] x [Paley-17 (4,4;17)] -> 473 vertices
R(4,5,5)	241	Tabu	non-linear colouring with 240 vertices found by tabu search, starting from the Atelier's linear colouring (matrix R455_matrice_N240_tabu.txt)
R(4,5,6)	353	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 89 – 3 = 353; with R(3,4,5) >= 89
R(4,5,7)	509	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 128 – 3 = 509; with R(4,4,4) >= 128
R(4,5,8)	645	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 1 quadrupling -> 644 vertices
R(4,5,9)	769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 193 – 3 = 769; with R(3,4,8) >= 193
R(4,5,10)	965	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 242 – 3 = 965; with R(3,4,9) >= 242

Parameters	Bound (R >=)	Origin	Construction
R(4,6,6)	570	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices
R(4,6,7)	593	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 1 quadrupling -> 592 vertices
R(4,6,8)	645	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 1 quadrupling -> 644 vertices (by monotonicity, from (4,5,8))
R(4,6,9)	985	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [K3] -> 984 vertices
R(4,6,10)	1 014	Published bricks + theorems	[template (4,3;6)] x [atom (6,10 ; 203)] -> 1,013 vertices
R(4,7,7)	1 145	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices
R(4,7,8)	1 145	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices (by monotonicity, from (4,7,7))
R(4,7,9)	1 409	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 353 – 3 = 1,409; with R(3,4,5) >= 89
R(4,7,10)	1 633	Published bricks + theorems	[template (9,3 ; 26)] x [Paley-17 (4,4;17)] -> 408 vertices; then 1 quadrupling -> 1,632 vertices
R(4,8,8)	1 580	Published bricks + theorems	[template (8,8,3 ; 650)] x [K3] -> 1,579 vertices
R(4,8,9)	2 033	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 509 – 3 = 2,033; with R(4,4,4) >= 128
R(4,8,10)	2 033	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 2 quadruplings -> 2,032 vertices (by monotonicity, from (4,8,9))
R(4,9,9)	3 125	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K3] -> 3,124 vertices
R(4,9,10)	3 125	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K3] -> 3,124 vertices (by monotonicity, from (4,9,9))
R(4,10,10)	3 984	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (10,10 ; 797)]
R(5,5,5)	454	Published	published value
R(5,5,6)	557	Our bricks	[template (5,5,3 ; 99) (ours)] x [K2] -> 139 vertices; then 1 quadrupling -> 556 vertices
R(5,5,7)	765	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [K2] -> 191 vertices; then 1 quadrupling -> 764 vertices
R(5,5,8)	1 065	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices; then 1 quadrupling -> 1,064 vertices
R(5,5,9)	1 345	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 337 – 3 = 1,345; with R(3,3,7) >= 85
R(5,5,10)	1 781	Our bricks	[template (5,9,3 ; 314) (ours)] x [K2] -> 445 vertices; then 1 quadrupling -> 1,780 vertices
R(5,6,6)	905	Published bricks + theorems	[template (5,3;10)] x [Paley-101 (6,6;101)] -> 904 vertices
R(5,6,7)	1 349	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 338 – 3 = 1,349; with R(3,6,6) >= 338
R(5,6,8)	1 617	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 405 – 3 = 1,617; with R(6,6) >= 102
R(5,6,9)	1 825	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 457 – 3 = 1,825; with R(6,7) >= 115
R(5,6,10)	2 333	Our bricks	[template (6,9,3 ; 402) (ours)] x [K2] -> 583 vertices; then 1 quadrupling -> 2,332 vertices
R(5,7,7)	1 814	Published bricks + theorems	[template (5,3;10)] x [Mathon-doubling (7,7;202)] -> 1,813 vertices
R(5,7,8)	2 693	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 674 – 3 = 2,693; with R(3,7,7) >= 674
R(5,7,9)	3 265	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 817 – 3 = 3,265; with R(7,7) >= 205
R(5,7,10)	3 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 873 – 3 = 3,489; with R(7,8) >= 219
R(5,8,8)	3 265	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 817 – 3 = 3,265; with R(7,7) >= 205
R(5,8,9)	3 761	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 941 – 3 = 3,761; with R(3,8,8) >= 941
R(5,8,10)	4 497	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,125 – 3 = 4,497; with R(8,8) >= 282
R(5,9,9)	5 054	Published bricks + theorems	[template (5,3;10)] x [atom (9,9 ; 562)] -> 5,053 vertices
R(5,9,10)	7 373	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,844 – 3 = 7,373; with R(3,9,9) >= 1,844
R(5,10,10)	9 025	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,257 – 3 = 9,025; with R(9,9) >= 565
R(6,6,6)	1 106	Published	published value
R(6,6,7)	1 507	Published bricks + theorems	[template (7,3;16)] x [Paley-101 (6,6;101)] -> 1,506 vertices
R(6,6,8)	1 908	Published bricks + theorems	[template (8,3 ; 20)] x [Paley-101 (6,6;101)] -> 1,907 vertices
R(6,6,9)	2 509	Published bricks + theorems	[template (9,3 ; 26)] x [Paley-101 (6,6;101)] -> 2,508 vertices
R(6,6,10)	2 910	Published bricks + theorems	[template (10,3 ; 30)] x [Paley-101 (6,6;101)] -> 2,909 vertices
R(6,7,7)	2 277	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 1 quadrupling -> 2,276 vertices
R(6,7,8)	2 693	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 1 quadrupling -> 2,692 vertices (by monotonicity, from (5,7,8))
R(6,7,9)	3 617	Published bricks + theorems	[template (5,3;10)] x [Paley-101 (6,6;101)] -> 904 vertices; then 1 quadrupling -> 3,616 vertices
R(6,7,10)	3 937	Our bricks	[template (6,9,3 ; 402) (ours)] x [K3] -> 984 vertices; then 1 quadrupling -> 3,936 vertices
R(6,8,8)	3 233	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 2 quadruplings -> 3,232 vertices (by monotonicity, from (5,8,8))
R(6,8,9)	5 393	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,349 – 3 = 5,393; with R(3,6,6) >= 338

Parameters	Bound (R >=)	Origin	Construction
R(6,8,10)	5 393	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 2 quadruplings -> 5,392 vertices (by monotonicity, from (6,8,9))
R(6,9,9)	6 465	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,617 – 3 = 6,465; with R(6,6) >= 102
R(6,9,10)	7 373	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K2] -> 1,843 vertices; then 1 quadrupling -> 7,372 vertices (by monotonicity, from (5,9,10))
R(6,10,10)	8 993	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 2 quadruplings -> 8,992 vertices (by monotonicity, from (5,10,10))
R(7,7,7)	3 214	Published	published value
R(7,7,8)	4 577	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 1 quadrupling -> 4,576 vertices
R(7,7,9)	5 393	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,349 – 3 = 5,393; with R(3,6,6) >= 338
R(7,7,10)	5 839	Published bricks + theorems	[template (10,3 ; 30)] x [Mathon-doubling (7,7;202)] -> 5,838 vertices
R(7,8,8)	4 577	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 1 quadrupling -> 4,576 vertices (by monotonicity, from (7,7,8))
R(7,8,9)	7 253	Published bricks + theorems	[template (5,3;10)] x [Mathon-doubling (7,7;202)] -> 1,813 vertices; then 1 quadrupling -> 7,252 vertices
R(7,8,10)	8 129	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,033 – 3 = 8,129; with R(4,4,4) >= 128
R(7,9,9)	10 769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,693 – 3 = 10,769; with R(3,7,7) >= 674
R(7,9,10)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(7,10,10)	12 929	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices (by monotonicity, from (7,9,10))
R(8,8,8)	7 174	Published	published value
R(8,8,9)	10 769	Published	published value
R(8,8,10)	10 769	Published (deduced)	R(8,8,10) >= 10,769: published value R(8,8,9) >= 10,769 (Radziszowski rev. 18 (Rowley)); by monotonicity, R(8,8,10) >= R(8,8,9) because each parameter of R(8,8,9) is ≤ the corresponding parameter of R(8,8,1
R(8,9,9)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(8,9,10)	15 041	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,761 – 3 = 15,041; with R(3,8,8) >= 941
R(8,10,10)	15 132	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [atome (10,10 ; 797)]
R(9,9,9)	15 041	Published	published value
R(9,9,10)	20 213	Published bricks + theorems	[template (5,3;10)] x [atom (9,9 ; 562)] -> 5,053 vertices; then 1 quadrupling -> 20,212 vertices
R(9,10,10)	29 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 7,373 – 3 = 29,489; with R(3,9,9) >= 1,844
R(10,10,10)	29 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K2] -> 1,843 vertices; then 2 quadruplings -> 29,488 vertices (by monotonicity, from (9,10,10))
R(3,3,3,3)	51	Published	published value
R(3,3,3,4)	97	Published	published value
R(3,3,3,5)	162	Published	published value
R(3,3,3,6)	186	Published (deduced)	Robertson formula: 3 x 61 + 6 – 3 = 186; with R(3,3,6) >= 61
R(3,3,3,7)	259	Published (deduced)	Robertson formula: 3 x 85 + 7 – 3 = 259; with R(3,3,7) >= 85
R(3,3,3,8)	337	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 85 – 3 = 337; with R(3,3,7) >= 85
R(3,3,3,9)	409	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 103 – 3 = 409; with R(3,3,8) >= 103
R(3,3,3,10)	529	Our bricks	atom (3,3,9 ; 132) (ours) -> 132 vertices; then 1 quadrupling -> 528 vertices
R(3,3,4,4)	174	Published	published value
R(3,3,4,5)	267	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [K3] -> 266 vertices
R(3,3,4,6)	353	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 89 – 3 = 353; with R(3,4,5) >= 89
R(3,3,4,7)	507	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [(3,3;5)] -> 506 vertices
R(3,3,4,8)	645	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 1 quadrupling -> 644 vertices
R(3,3,4,9)	769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 193 – 3 = 769; with R(3,4,8) >= 193
R(3,3,4,10)	965	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 242 – 3 = 965; with R(3,4,9) >= 242
R(3,3,5,5)	438	Our bricks + published	[generic template 2m-1 of K2] x [atom (3,5,5 ; 146) (ours)] -> 437 vertices
R(3,3,5,6)	645	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 162 – 3 = 645; with R(3,3,3,5) >= 162
R(3,3,5,7)	828	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [(3,3;5)] -> 827 vertices
R(3,3,5,8)	1 065	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices; then 1 quadrupling -> 1,064 vertices
R(3,3,5,9)	1 385	Our bricks	[template (5,9,3 ; 314) (ours)] x [(3,3;5)] -> 1,384 vertices
R(3,3,5,10)	1 781	Our bricks	[template (5,9,3 ; 314) (ours)] x [K2] -> 445 vertices; then 1 quadrupling -> 1,780 vertices
R(3,3,6,6)	1 044	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,6,7)	1 349	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 338 – 3 = 1,349; with R(3,6,6) >= 338
R(3,3,6,8)	1 617	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 405 – 3 = 1,617; with R(6,6) >= 102
R(3,3,6,9)	1 825	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 457 – 3 = 1,825; with R(6,7) >= 115
R(3,3,6,10)	2 333	Our bricks	[template (6,9,3 ; 402) (ours)] x [K2] -> 583 vertices; then 1 quadrupling -> 2,332 vertices
R(3,3,7,7)	2 087	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices
R(3,3,7,8)	2 693	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 674 – 3 = 2,693; with R(3,7,7) >= 674
R(3,3,7,9)	3 265	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 817 – 3 = 3,265; with R(7,7) >= 205
R(3,3,7,10)	3 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 873 – 3 = 3,489; with R(7,8) >= 219
R(3,3,8,8)	3 265	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 817 – 3 = 3,265; with R(7,7) >= 205
R(3,3,8,9)	3 761	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 941 – 3 = 3,761; with R(3,8,8) >= 941
R(3,3,8,10)	4 497	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,125 – 3 = 4,497; with R(8,8) >= 282
R(3,3,9,9)	5 687	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices
R(3,3,9,10)	7 373	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,844 – 3 = 7,373; with R(3,9,9) >= 1,844
R(3,3,10,10)	9 025	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,257 – 3 = 9,025; with R(9,9) >= 565
R(3,4,4,4)	381	Published	published value
R(3,4,4,5)	509	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 128 – 3 = 509; with R(4,4,4) >= 128
R(3,4,4,6)	641	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [Paley-17 (4,4;17)] -> 640 vertices
R(3,4,4,7)	934	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [Paley-17 (4,4;17)] -> 933 vertices
R(3,4,4,8)	1 179	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [Paley-17 (4,4;17)] -> 1,178 vertices
R(3,4,4,9)	1 334	Published bricks + theorems	[template (9,3 ; 26)] x [atom (3,4,4 ; 54)] -> 1,333 vertices
R(3,4,4,10)	1 714	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (4,10 ; 91) (ours)] -> 1,713 vertices
R(3,4,5,5)	770	Our bricks	[template (5,3;10)] x [atom (3,4,5 ; 86) (ours)] -> 769 vertices
R(3,4,5,6)	1 065	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [K3] -> 266 vertices; then 1 quadrupling -> 1,064 vertices
R(3,4,5,7)	1 521	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 381 – 3 = 1,521; with R(3,4,4,4) >= 381
R(3,4,5,8)	2 033	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 509 – 3 = 2,033; with R(4,4,4) >= 128
R(3,4,5,9)	2 577	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 2 quadruplings -> 2,576 vertices
R(3,4,5,10)	3 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 769 – 3 = 3,073; with R(3,4,8) >= 193
R(3,4,6,6)	1 904	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices
R(3,4,6,7)	2 277	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 1 quadrupling -> 2,276 vertices
R(3,4,6,8)	2 577	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 2 quadruplings -> 2,576 vertices
R(3,4,6,9)	3 443	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (6,9 ; 182)] -> 3,442 vertices
R(3,4,6,10)	3 937	Our bricks	[template (6,9,3 ; 402) (ours)] x [K3] -> 984 vertices; then 1 quadrupling -> 3,936 vertices
R(3,4,7,7)	3 823	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices
R(3,4,7,8)	4 577	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 1 quadrupling -> 4,576 vertices
R(3,4,7,9)	4 754	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (7,9 ; 251)] -> 4,753 vertices
R(3,4,7,10)	5 633	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,409 – 3 = 5,633; with R(3,4,5) >= 89
R(3,4,8,8)	5 324	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (8,8 ; 281)] -> 5,323 vertices
R(3,4,8,9)	6 317	Published bricks + theorems	[template (8,8,3 ; 650)] x [K3] -> 1,579 vertices; then 1 quadrupling -> 6,316 vertices
R(3,4,8,10)	8 129	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,033 – 3 = 8,129; with R(4,4,4) >= 128
R(3,4,9,9)	10 663	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices
R(3,4,9,10)	12 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K3] -> 3,124 vertices; then 1 quadrupling -> 12,496 vertices
R(3,4,10,10)	15 128	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [atome (10,10 ; 797)]
R(3,5,5,5)	1 430	Published	published value
R(3,5,5,6)	1 813	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 454 – 3 = 1,813; with R(5,5,5) >= 454
R(3,5,5,7)	2 577	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 645 – 3 = 2,577; with R(3,3,3,5) >= 162
R(3,5,5,8)	3 309	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [(3,3;5)] -> 827 vertices; then 1 quadrupling -> 3,308 vertices
R(3,5,5,9)	4 257	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices; then 2 quadruplings -> 4,256 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,5,10)	5 537	Our bricks	[template (5,9,3 ; 314) (ours)] x [(3,3;5)] -> 1,384 vertices; then 1 quadrupling -> 5,536 vertices
R(3,5,6,6)	3 029	Published bricks + theorems	[template (5,3;10)] x [atom (3,6,6 ; 337)] -> 3,028 vertices
R(3,5,6,7)	4 173	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 1 quadrupling -> 4,172 vertices
R(3,5,6,8)	5 393	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,349 – 3 = 5,393; with R(3,6,6) >= 338
R(3,5,6,9)	6 465	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,617 – 3 = 6,465; with R(6,6) >= 102
R(3,5,6,10)	7 297	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,825 – 3 = 7,297; with R(6,7) >= 115
R(3,5,7,7)	6 053	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [K2]] -> 6,052 vertices
R(3,5,7,8)	8 345	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 1 quadrupling -> 8,344 vertices
R(3,5,7,9)	10 769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,693 – 3 = 10,769; with R(3,7,7) >= 674
R(3,5,7,10)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(3,5,8,8)	10 769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,693 – 3 = 10,769; with R(3,7,7) >= 674
R(3,5,8,9)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(3,5,8,10)	15 041	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,761 – 3 = 15,041; with R(3,8,8) >= 941
R(3,5,9,9)	16 844	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (9,9 ; 562)] -> 16,843 vertices
R(3,5,9,10)	22 745	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 1 quadrupling -> 22,744 vertices
R(3,5,10,10)	29 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 7,373 – 3 = 29,489; with R(3,9,9) >= 1,844
R(3,6,6,6)	3 917	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [Paley-101 (6,6;101)] -> 3,916 vertices
R(3,6,6,7)	5 722	Our bricks	[template (3,7,3 ; 58) (ours)] x [Paley-101 (6,6;101)] -> 5,721 vertices
R(3,6,6,8)	7 227	Our bricks	[template (3,8,3 ; 73) (ours)] x [Paley-101 (6,6;101)] -> 7,226 vertices
R(3,6,6,9)	8 409	Published bricks + theorems	[template (9,3 ; 26)] x [atom (3,6,6 ; 337)] -> 8,408 vertices
R(3,6,6,10)	10 033	Published bricks + theorems	[template (9,3 ; 26)] x [Paley-101 (6,6;101)] -> 2,508 vertices; then 1 quadrupling -> 10,032 vertices
R(3,6,7,7)	7 856	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [Mathon-doubling (7,7;202)] -> 7,855 vertices
R(3,6,7,8)	10 769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,693 – 3 = 10,769; with R(3,7,7) >= 674
R(3,6,7,9)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(3,6,7,10)	14 465	Published bricks + theorems	[template (5,3;10)] x [Paley-101 (6,6;101)] -> 904 vertices; then 2 quadruplings -> 14,464 vertices
R(3,6,8,8)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(3,6,8,9)	16 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 2 quadruplings -> 16,688 vertices
R(3,6,8,10)	21 569	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 5,393 – 3 = 21,569; with R(3,6,6) >= 338
R(3,6,9,9)	21 896	Our bricks	[template (3,6,3 ; 40) (ours)] x [atom (9,9 ; 562)] -> 21,895 vertices
R(3,6,9,10)	29 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 7,373 – 3 = 29,489; with R(3,9,9) >= 1,844
R(3,6,10,10)	36 097	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 9,025 – 3 = 36,097; with R(9,9) >= 565
R(3,7,7,7)	11 479	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [Mathon-doubling (7,7;202)] -> 11,478 vertices
R(3,7,7,8)	15 289	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 1 quadrupling -> 15,288 vertices
R(3,7,7,9)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(3,7,7,10)	21 569	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 5,393 – 3 = 21,569; with R(3,6,6) >= 338
R(3,7,8,8)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(3,7,8,9)	24 209	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [K2]] -> 6,052 vertices; then 1 quadrupling -> 24,208 vertices
R(3,7,8,10)	29 009	Published bricks + theorems	[template (5,3;10)] x [Mathon-doubling (7,7;202)] -> 1,813 vertices; then 2 quadruplings -> 29,008 vertices
R(3,7,9,9)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(3,7,9,10)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(3,7,10,10)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(3,8,8,8)	20 187	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (8,8 ; 281)] -> 20,186 vertices
R(3,8,8,9)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(3,8,8,10)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(8,8,9) >= 10,769
R(3,8,9,9)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(3,8,9,10)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(3,8,10,10)	60 161	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 15,041 – 3 = 60,161; with R(3,8,8) >= 941
R(3,9,9,9)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205

Parameters	Bound (R >=)	Origin	Construction
R(3,9,9,10)	67 373	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (9,9 ; 562)] -> 16,843 vertices; then 1 quadrupling -> 67,372 vertices
R(3,9,10,10)	90 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 2 quadruplings -> 90,976 vertices
R(3,10,10,10)	117 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 29,489 – 3 = 117,953; with R(3,9,9) >= 1,844
R(4,4,4,4)	634	Published	published value
R(4,4,4,5)	1 139	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices
R(4,4,4,6)	1 392	Published bricks + theorems	[template (6,3;12)] x [atom (4,4,4 ; 127)] -> 1,391 vertices
R(4,4,4,7)	1 897	Published bricks + theorems	[template (7,3;16)] x [atom (4,4,4 ; 127)] -> 1,896 vertices
R(4,4,4,8)	2 402	Published bricks + theorems	[template (8,3 ; 20)] x [atom (4,4,4 ; 127)] -> 2,401 vertices
R(4,4,4,9)	3 159	Published bricks + theorems	[template (9,3 ; 26)] x [atom (4,4,4 ; 127)] -> 3,158 vertices
R(4,4,4,10)	3 664	Published bricks + theorems	[template (10,3 ; 30)] x [atom (4,4,4 ; 127)] -> 3,663 vertices
R(4,4,5,5)	1 610	Our bricks	[template (5,5,3 ; 99) (ours)] x [Paley-17 (4,4;17)] -> 1,609 vertices
R(4,4,5,6)	2 217	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Paley-17 (4,4;17)] -> 2,216 vertices
R(4,4,5,7)	3 072	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [Paley-17 (4,4;17)] -> 3,071 vertices
R(4,4,5,8)	3 861	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [Paley-17 (4,4;17)] -> 3,860 vertices
R(4,4,5,9)	5 141	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [Paley-17 (4,4;17)] -> 5,140 vertices
R(4,4,5,10)	5 423	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,10 ; 91) (ours)] -> 5,422 vertices
R(4,4,6,6)	3 846	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices
R(4,4,6,7)	4 553	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 1 quadrupling -> 4,552 vertices
R(4,4,6,8)	4 553	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 1 quadrupling -> 4,552 vertices (by monotonicity, from (4,4,6,7))
R(4,4,6,9)	6 599	Our bricks	[template (6,9,3 ; 402) (ours)] x [Paley-17 (4,4;17)] -> 6,598 vertices
R(4,4,6,10)	7 144	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (4,10 ; 91) (ours)] -> 7,143 vertices
R(4,4,7,7)	7 739	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices
R(4,4,7,8)	7 739	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices (by monotonicity, from (4,4,7,7))
R(4,4,7,9)	9 605	Published bricks + theorems	[template (8,3 ; 20)] x [atom (4,4,4 ; 127)] -> 2,401 vertices; then 1 quadrupling -> 9,604 vertices
R(4,4,7,10)	12 633	Published bricks + theorems	[template (9,3 ; 26)] x [atom (4,4,4 ; 127)] -> 3,158 vertices; then 1 quadrupling -> 12,632 vertices
R(4,4,8,8)	10 666	Published bricks + theorems	[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)] -> 10,665 vertices
R(4,4,8,9)	12 285	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [Paley-17 (4,4;17)] -> 3,071 vertices; then 1 quadrupling -> 12,284 vertices
R(4,4,8,10)	12 633	Published bricks + theorems	[template (9,3 ; 26)] x [atom (4,4,4 ; 127)] -> 3,158 vertices; then 1 quadrupling -> 12,632 vertices (by monotonicity, from (4,4,7,10))
R(4,4,9,9)	21 059	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices
R(4,4,9,10)	21 059	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices (by monotonicity, from (4,4,9,9))
R(4,4,10,10)	26 286	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,4,3 ; 34)] x [atome (10,10 ; 797)]
R(4,5,5,5)	2 423	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [(5,5;41)] -> 2,422 vertices
R(4,5,5,6)	3 323	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (5,6 ; 56)] -> 3,322 vertices
R(4,5,5,7)	4 703	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (5,7 ; 79)] -> 4,702 vertices
R(4,5,5,8)	6 081	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,521 – 3 = 6,081; with R(3,4,4,4) >= 381
R(4,5,5,9)	8 129	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,033 – 3 = 8,129; with R(4,4,4) >= 128
R(4,5,5,10)	10 305	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 3 quadruplings -> 10,304 vertices
R(4,5,6,6)	6 023	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Paley-101 (6,6;101)] -> 6,022 vertices
R(4,5,6,7)	7 613	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 1 quadrupling -> 7,612 vertices
R(4,5,6,8)	9 105	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 2 quadruplings -> 9,104 vertices
R(4,5,6,9)	10 883	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (6,9 ; 182)] -> 10,882 vertices
R(4,5,6,10)	13 769	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (6,9 ; 182)] -> 3,442 vertices; then 1 quadrupling -> 13,768 vertices
R(4,5,7,7)	12 083	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Mathon-doubling (7,7;202)] -> 12,082 vertices
R(4,5,7,8)	15 289	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 1 quadrupling -> 15,288 vertices
R(4,5,7,9)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(4,5,7,10)	20 561	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [Paley-17 (4,4;17)] -> 5,140 vertices; then 1 quadrupling -> 20,560 vertices
R(4,5,8,8)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,8,9)	21 293	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (8,8 ; 281)] -> 5,323 vertices; then 1 quadrupling -> 21,292 vertices
R(4,5,8,10)	25 265	Published bricks + theorems	[template (8,8,3 ; 650)] x [K3] -> 1,579 vertices; then 2 quadruplings -> 25,264 vertices
R(4,5,9,9)	33 683	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (9,9 ; 562)] -> 33,682 vertices
R(4,5,9,10)	42 649	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 1 quadrupling -> 42,648 vertices
R(4,5,10,10)	49 985	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K3] -> 3,124 vertices; then 2 quadruplings -> 49,984 vertices
R(4,6,6,6)	7 934	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Paley-101 (6,6;101)] -> 7,933 vertices
R(4,6,6,7)	11 547	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)] -> 11,546 vertices
R(4,6,6,8)	12 152	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [Paley-101 (6,6;101)] -> 12,151 vertices
R(4,6,6,9)	15 780	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,9 ; 68)] -> 15,779 vertices
R(4,6,6,10)	21 162	Our bricks + published	[template (6,6,3;235)] x [atom (4,10 ; 91) (ours)] -> 21,161 vertices
R(4,6,7,7)	15 913	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices
R(4,6,7,8)	18 209	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 2 quadruplings -> 18,208 vertices
R(4,6,7,9)	24 089	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Paley-101 (6,6;101)] -> 6,022 vertices; then 1 quadrupling -> 24,088 vertices
R(4,6,7,10)	26 393	Our bricks	[template (6,9,3 ; 402) (ours)] x [Paley-17 (4,4;17)] -> 6,598 vertices; then 1 quadrupling -> 26,392 vertices
R(4,6,8,8)	22 154	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (8,8 ; 281)] -> 22,153 vertices
R(4,6,8,9)	30 449	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 2 quadruplings -> 30,448 vertices
R(4,6,8,10)	30 449	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 2 quadruplings -> 30,448 vertices (by monotonicity, from (4,6,8,9))
R(4,6,9,9)	44 353	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (9,9 ; 562)] -> 44,352 vertices
R(4,6,9,10)	44 353	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (9,9 ; 562)] -> 44,352 vertices (by monotonicity, from (4,6,9,9))
R(4,6,10,10)	62 918	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,6,3 ; 80)] x [atome (10,10 ; 797)]
R(4,7,7,7)	23 162	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Mathon-doubling (7,7;202)] -> 23,161 vertices
R(4,7,7,8)	30 953	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 1 quadrupling -> 30,952 vertices
R(4,7,7,9)	31 760	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,9 ; 68)] -> 31,759 vertices
R(4,7,7,10)	42 593	Our bricks + published	[template (7,7,3;472)] x [atom (4,10 ; 91) (ours)] -> 42,592 vertices
R(4,7,8,8)	32 247	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (8,8 ; 281)] -> 32,246 vertices
R(4,7,8,9)	48 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Mathon-doubling (7,7;202)] -> 12,082 vertices; then 1 quadrupling -> 48,328 vertices
R(4,7,8,10)	50 529	Published bricks + theorems	[template (9,3 ; 26)] x [atom (4,4,4 ; 127)] -> 3,158 vertices; then 2 quadruplings -> 50,528 vertices
R(4,7,9,9)	64 562	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)] -> 64,561 vertices
R(4,7,9,10)	84 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 1 quadrupling -> 84,232 vertices
R(4,7,10,10)	91 587	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,7,3 ; 116)] x [atome (10,10 ; 797)]
R(4,8,8,8)	33 932	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [atom (8,8 ; 281)] -> 33,931 vertices
R(4,8,8,9)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices
R(4,8,8,10)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices (by monotonicity, from (4,8,8,9))
R(4,8,9,9)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(4,8,9,10)	85 169	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (8,8 ; 281)] -> 5,323 vertices; then 2 quadruplings -> 85,168 vertices
R(4,8,10,10)	96 368	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,8,3 ; 122)] x [atome (10,10 ; 797)]
R(4,9,9,9)	87 585	Our bricks + published	[modele (4,9,3 ; 157)] x [atome (9,9 ; 562)] [new template (4,9;157), v21]
R(4,9,9,10)	134 729	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (9,9 ; 562)] -> 33,682 vertices; then 1 quadrupling -> 134,728 vertices
R(4,9,10,10)	170 593	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 2 quadruplings -> 170,592 vertices
R(4,10,10,10)	170 593	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 2 quadruplings -> 170,592 vertices (by monotonicity, from (4,9,10,10))
R(5,5,5,5)	4 073	Published	published value
R(5,5,5,6)	5 717	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,430 – 3 = 5,717; with R(3,5,5,5) >= 1,430
R(5,5,5,7)	7 686	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [atom (5,7 ; 79)] -> 7,685 vertices
R(5,5,5,8)	10 305	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,577 – 3 = 10,305; with R(3,3,3,5) >= 162
R(5,5,5,9)	13 233	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [(3,3;5)] -> 827 vertices; then 2 quadruplings -> 13,232 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,5,10)	17 025	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices; then 3 quadruplings -> 17,024 vertices
R(5,5,6,6)	9 842	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)] -> 9,841 vertices
R(5,5,6,7)	12 113	Published bricks + theorems	[template (5,3;10)] x [atom (3,6,6 ; 337)] -> 3,028 vertices; then 1 quadrupling -> 12,112 vertices
R(5,5,6,8)	16 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 2 quadruplings -> 16,688 vertices
R(5,5,6,9)	21 569	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 5,393 – 3 = 21,569; with R(3,6,6) >= 338
R(5,5,6,10)	25 857	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,465 – 3 = 25,857; with R(6,6) >= 102
R(5,5,7,7)	19 740	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)] -> 19,739 vertices
R(5,5,7,8)	24 209	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [K2]] -> 6,052 vertices; then 1 quadrupling -> 24,208 vertices
R(5,5,7,9)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(5,5,7,10)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(5,5,8,8)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(5,5,8,9)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(5,5,8,10)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(5,5,9,9)	55 020	Our bricks	[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)] -> 55,019 vertices
R(5,5,9,10)	67 373	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (9,9 ; 562)] -> 16,843 vertices; then 1 quadrupling -> 67,372 vertices
R(5,5,10,10)	90 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 2 quadruplings -> 90,976 vertices
R(5,6,6,6)	13 557	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Paley-101 (6,6;101)] -> 13,556 vertices
R(5,6,6,7)	18 780	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [Paley-101 (6,6;101)] -> 18,779 vertices
R(5,6,6,8)	23 601	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [Paley-101 (6,6;101)] -> 23,600 vertices
R(5,6,6,9)	31 433	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [Paley-101 (6,6;101)] -> 31,432 vertices
R(5,6,6,10)	33 633	Published bricks + theorems	[template (9,3 ; 26)] x [atom (3,6,6 ; 337)] -> 8,408 vertices; then 1 quadrupling -> 33,632 vertices
R(5,6,7,7)	27 192	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices
R(5,6,7,8)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(5,6,7,9)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(5,6,7,10)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(5,6,8,8)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(5,6,8,9)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(5,6,8,10)	66 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 3 quadruplings -> 66,752 vertices
R(5,6,9,9)	75 792	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices
R(5,6,9,10)	90 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 2 quadruplings -> 90,976 vertices
R(5,6,10,10)	117 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 29,489 – 3 = 117,953; with R(3,9,9) >= 1,844
R(5,7,7,7)	37 667	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)] -> 37,666 vertices
R(5,7,7,8)	48 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Mathon-doubling (7,7;202)] -> 12,082 vertices; then 1 quadrupling -> 48,328 vertices
R(5,7,7,9)	63 046	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)] -> 63,045 vertices
R(5,7,7,10)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(5,7,8,8)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices
R(5,7,8,9)	78 957	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)] -> 19,739 vertices; then 1 quadrupling -> 78,956 vertices
R(5,7,8,10)	96 833	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [K2]] -> 6,052 vertices; then 2 quadruplings -> 96,832 vertices
R(5,7,9,9)	104 987	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)] -> 104,986 vertices
R(5,7,9,10)	134 729	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (9,9 ; 562)] -> 33,682 vertices; then 1 quadrupling -> 134,728 vertices
R(5,7,10,10)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(5,8,8,8)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(5,8,8,9)	96 833	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [K2]] -> 6,052 vertices; then 2 quadruplings -> 96,832 vertices
R(5,8,8,10)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(5,8,9,9)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(5,8,9,10)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(5,8,10,10)	208 897	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 52,225 – 3 = 208,897; with R(7,7) >= 205
R(5,9,9,9)	175 726	Our bricks	[template (5,9,3 ; 314) (ours)] x [atom (9,9 ; 562)] -> 175,725 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,9,9,10)	220 077	Our bricks	[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)] -> 55,019 vertices; then 1 quadrupling -> 220,076 vertices
R(5,9,10,10)	269 489	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (9,9 ; 562)] -> 16,843 vertices; then 2 quadruplings -> 269,488 vertices
R(5,10,10,10)	363 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 3 quadruplings -> 363,904 vertices
R(6,6,6,6)	23 502	Published	published value
R(6,6,6,7)	24 204	Our bricks	[modele (6,6,3;235)] x [atome (6,7 ; 104)] [new atom (6,7;104), v19]
R(6,6,6,8)	24 089	Our bricks	[template (4,5,3 ; 61) (ours)] x [Paley-101 (6,6;101)] -> 6,022 vertices; then 1 quadrupling -> 24,088 vertices (by monotonicity, from (6,6,6,7))
R(6,6,6,9)	42 456	Published bricks + theorems	[template (6,6,3;235)] x [atom (6,9 ; 182)] -> 42,455 vertices
R(6,6,6,10)	47 370	Published bricks + theorems	[template (6,6,3;235)] x [atom (6,10 ; 203)] -> 47,369 vertices
R(6,6,7,7)	47 303	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices
R(6,6,7,8)	47 303	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices (by monotonicity, from (6,6,7,7))
R(6,6,7,9)	58 602	Published bricks + theorems	[template (6,6,3;235)] x [atom (7,9 ; 251)] -> 58,601 vertices
R(6,6,7,10)	67 962	Published bricks + theorems	[template (6,6,3;235)] x [atom (7,10 ; 291)] -> 67,961 vertices
R(6,6,8,8)	65 622	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,8 ; 281)] -> 65,621 vertices
R(6,6,8,9)	75 117	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [Paley-101 (6,6;101)] -> 18,779 vertices; then 1 quadrupling -> 75,116 vertices
R(6,6,8,10)	79 896	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,10 ; 342)] -> 79,895 vertices
R(6,6,9,9)	131 376	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices
R(6,6,9,10)	131 376	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices (by monotonicity, from (6,6,9,9))
R(6,6,10,10)	186 366	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [atome (10,10 ; 797)]
R(6,7,7,7)	48 716	Our bricks	[modele (7,7,3;472)] x [atome (6,7 ; 104)] [new atom (6,7;104), v19]
R(6,7,7,8)	63 649	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 1 quadrupling -> 63,648 vertices
R(6,7,7,9)	85 454	Published bricks + theorems	[template (7,7,3;472)] x [atom (6,9 ; 182)] -> 85,453 vertices
R(6,7,7,10)	96 353	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Paley-101 (6,6;101)] -> 6,022 vertices; then 2 quadruplings -> 96,352 vertices
R(6,7,8,8)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(6,7,8,9)	108 765	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 1 quadrupling -> 108,764 vertices
R(6,7,8,10)	121 793	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 3 quadruplings -> 121,792 vertices
R(6,7,9,9)	134 729	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (9,9 ; 562)] -> 33,682 vertices; then 1 quadrupling -> 134,728 vertices
R(6,7,9,10)	177 409	Our bricks	[template (4,6,3 ; 80) (ours)] x [atom (9,9 ; 562)] -> 44,352 vertices; then 1 quadrupling -> 177,408 vertices
R(6,7,10,10)	199 937	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K3] -> 3,124 vertices; then 3 quadruplings -> 199,936 vertices
R(6,8,8,8)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices (by monotonicity, from (5,8,8,8))
R(6,8,8,9)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(6,8,8,10)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices (by monotonicity, from (5,8,8,10))
R(6,8,9,9)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(6,8,9,10)	208 897	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 52,225 – 3 = 208,897; with R(7,7) >= 205
R(6,8,10,10)	206 849	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 5 quadruplings -> 206,848 vertices (by monotonicity, from (5,8,10,10))
R(6,9,9,9)	232 424	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)] -> 232,423 vertices
R(6,9,9,10)	303 165	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices; then 1 quadrupling -> 303,164 vertices
R(6,9,10,10)	363 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 3 quadruplings -> 363,904 vertices
R(6,10,10,10)	363 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 3 quadruplings -> 363,904 vertices (by monotonicity, from (5,10,10,10))
R(7,7,7,7)	94 874	Published	published value
R(7,7,7,8)	94 874	Published (deduced)	R(7,7,7,8) >= 94,874: published value R(7,7,7,7) >= 94,874 (Radziszowski rev. 18 Table XIa); by monotonicity, R(7,7,7,8) >= R(7,7,7,7) because each parameter of R(7,7,7,7) is ≤ the corresponding parameter
R(7,7,7,9)	123 809	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 2 quadruplings -> 123,808 vertices
R(7,7,7,10)	136 793	Published bricks + theorems	[template (7,7,3;472)] x [atom (7,10 ; 291)] -> 136,792 vertices
R(7,7,8,8)	132 083	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices
R(7,7,8,9)	150 665	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)] -> 37,666 vertices; then 1 quadrupling -> 150,664 vertices
R(7,7,8,10)	193 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Mathon-doubling (7,7;202)] -> 12,082 vertices; then 2 quadruplings -> 193,312 vertices
R(7,7,9,9)	264 434	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices

Parameters	Bound (R >=)	Origin	Construction
R(7,7,9,10)	264 434	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices (by monotonicity, from (7,7,9,9))
R(7,7,10,10)	375 119	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(7,8,8,8)	132 083	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices (by monotonicity, from (7,7,8,8))
R(7,8,8,9)	193 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Mathon-doubling (7,7;202)] -> 12,082 vertices; then 2 quadruplings -> 193,312 vertices
R(7,8,8,10)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(7,8,9,9)	264 434	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices (by monotonicity, from (7,7,9,9))
R(7,8,9,10)	336 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 2 quadruplings -> 336,928 vertices
R(7,8,10,10)	375 119	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(7,9,9,9)	320 813	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (7,9 ; 251)] -> 320,812 vertices
R(7,9,9,10)	419 945	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)] -> 104,986 vertices; then 1 quadrupling -> 419,944 vertices
R(7,9,10,10)	538 913	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (9,9 ; 562)] -> 33,682 vertices; then 2 quadruplings -> 538,912 vertices
R(7,10,10,10)	682 369	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 3 quadruplings -> 682,368 vertices
R(8,8,8,8)	182 002	Published	published value
R(8,8,8,9)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(8,8,8,10)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices (by monotonicity, from (7,8,8,10))
R(8,8,9,9)	364 371	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (9,9 ; 562)] -> 364,370 vertices
R(8,8,9,10)	387 329	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [K2]] -> 6,052 vertices; then 3 quadruplings -> 387,328 vertices
R(8,8,10,10)	516 886	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,8,3 ; 650)] x [atome (10,10 ; 797)]
R(8,9,9,9)	419 945	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)] -> 104,986 vertices; then 1 quadrupling -> 419,944 vertices
R(8,9,9,10)	538 913	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (9,9 ; 562)] -> 33,682 vertices; then 2 quadruplings -> 538,912 vertices
R(8,9,10,10)	689 153	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 172,289 – 3 = 689,153; with R(3,7,7) >= 674
R(8,10,10,10)	689 153	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 5 quadruplings -> 689,152 vertices (by monotonicity, from (8,9,10,10))
R(9,9,9,9)	719 204	Published	published value
R(9,9,9,10)	719 204	Published (deduced)	R(9,9,9,10) >= 719,204: published value R(9,9,9,9) >= 719,204 (Radziszowski rev. 18 Table XIa); by monotonicity, R(9,9,9,10) >= R(9,9,9,9) because each parameter of R(9,9,9,9) is ≤ the corresponding parameter
R(9,9,10,10)	1 020 239	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]
R(9,10,10,10)	1 077 953	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (9,9 ; 562)] -> 16,843 vertices; then 3 quadruplings -> 1,077,952 vertices
R(10,10,10,10)	1 268 826	Published	published value
R(3,3,3,3,3)	162	Published	published value
R(3,3,3,3,4)	297	Published (deduced)	Robertson formula: 3 x 97 + 9 – 3 = 297; with R(3,3,3,4) >= 97, R(3,4) >= 9
R(3,3,3,3,5)	497	Published (deduced)	Robertson formula: 3 x 162 + 14 – 3 = 497; with R(3,3,3,5) >= 162, R(3,5) >= 14
R(3,3,3,3,6)	645	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 162 – 3 = 645; with R(3,3,3,5) >= 162
R(3,3,3,3,7)	804	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,3,7 ; 81) (ours)] -> 803 vertices
R(3,3,3,3,8)	1 036	Published (deduced)	Robertson formula: 3 x 337 + 28 – 3 = 1,036; with R(3,3,7) >= 85, R(3,8) >= 28
R(3,3,3,3,9)	1 345	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 337 – 3 = 1,345; with R(3,3,7) >= 85
R(3,3,3,3,10)	1 633	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 409 – 3 = 1,633; with R(3,3,8) >= 103
R(3,3,3,4,4)	537	Published (deduced)	Robertson formula: 3 x 174 + 18 – 3 = 537; with R(3,3,4,4) >= 174, R(4,4) >= 18
R(3,3,3,4,5)	854	Published bricks + theorems	[template (3,5,3 ; 31)] x [(3,3,4;29)] -> 853 vertices
R(3,3,3,4,6)	1 124	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,4,6 ; 113) (ours)] -> 1,123 vertices
R(3,3,3,4,7)	1 618	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [(3,3,4;29)] -> 1,617 vertices
R(3,3,3,4,8)	2 043	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [(3,3,4;29)] -> 2,042 vertices
R(3,3,3,4,9)	2 577	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 2 quadruplings -> 2,576 vertices
R(3,3,3,4,10)	3 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 769 – 3 = 3,073; with R(3,4,8) >= 193
R(3,3,3,5,5)	1 454	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 1,453 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,5,6)	1 991	Published (deduced)	Robertson formula: 3 x 645 + 59 – 3 = 1,991; with R(3,3,3,5) >= 162, R(5,6) >= 59
R(3,3,3,5,7)	2 654	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [K2]] -> 2,653 vertices
R(3,3,3,5,8)	3 344	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,8,3 ; 236) (ours)] x [K2]] -> 3,343 vertices
R(3,3,3,5,9)	4 444	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 4,443 vertices
R(3,3,3,5,10)	5 537	Our bricks	[template (5,9,3 ; 314) (ours)] x [(3,3;5)] -> 1,384 vertices; then 1 quadrupling -> 5,536 vertices
R(3,3,3,6,6)	3 364	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices
R(3,3,3,6,7)	4 173	Our bricks	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 1 quadrupling -> 4,172 vertices
R(3,3,3,6,8)	5 393	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,349 – 3 = 5,393; with R(3,6,6) >= 338
R(3,3,3,6,9)	6 465	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,617 – 3 = 6,465; with R(6,6) >= 102
R(3,3,3,6,10)	7 297	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,825 – 3 = 7,297; with R(6,7) >= 115
R(3,3,3,7,7)	6 724	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices
R(3,3,3,7,8)	8 345	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 1 quadrupling -> 8,344 vertices
R(3,3,3,7,9)	10 769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,693 – 3 = 10,769; with R(3,7,7) >= 674
R(3,3,3,7,10)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(3,3,3,8,8)	10 769	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,693 – 3 = 10,769; with R(3,7,7) >= 674
R(3,3,3,8,9)	13 057	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,265 – 3 = 13,057; with R(7,7) >= 205
R(3,3,3,8,10)	15 041	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 3,761 – 3 = 15,041; with R(3,8,8) >= 941
R(3,3,3,9,9)	18 521	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices
R(3,3,3,9,10)	22 745	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 1 quadrupling -> 22,744 vertices
R(3,3,3,10,10)	29 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 7,373 – 3 = 29,489; with R(3,9,9) >= 1,844
R(3,3,4,4,4)	1 264	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (4,4,4 ; 127)] -> 1,263 vertices
R(3,3,4,4,5)	1 821	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-17 (4,4;17)] -> 1,820 vertices
R(3,3,4,4,6)	2 246	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [(3,3,4;29)] -> 2,245 vertices
R(3,3,4,4,7)	3 267	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [(3,3,4;29)] -> 3,266 vertices
R(3,3,4,4,8)	3 843	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (3,4,4 ; 54)] -> 3,842 vertices
R(3,3,4,4,9)	4 713	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [Paley-17 (4,4;17)] -> 1,178 vertices; then 1 quadrupling -> 4,712 vertices
R(3,3,4,4,10)	5 880	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (4,10 ; 91) (ours)] -> 5,879 vertices
R(3,3,4,5,5)	2 786	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [(3,3,4;29)] -> 2,785 vertices
R(3,3,4,5,6)	3 837	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [(3,3,4;29)] -> 3,836 vertices
R(3,3,4,5,7)	5 316	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [(3,3,4;29)] -> 5,315 vertices
R(3,3,4,5,8)	6 681	Our bricks	[template (5,8,3 ; 236) (ours)] x [(3,3,4;29)] -> 6,680 vertices
R(3,3,4,5,9)	8 897	Our bricks	[template (5,9,3 ; 314) (ours)] x [(3,3,4;29)] -> 8,896 vertices
R(3,3,4,5,10)	10 305	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 3 quadruplings -> 10,304 vertices
R(3,3,4,6,6)	6 654	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices
R(3,3,4,6,7)	7 613	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 1 quadrupling -> 7,612 vertices
R(3,3,4,6,8)	9 105	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 2 quadruplings -> 9,104 vertices
R(3,3,4,6,9)	11 795	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (6,9 ; 182)] -> 11,794 vertices
R(3,3,4,6,10)	13 769	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (6,9 ; 182)] -> 3,442 vertices; then 1 quadrupling -> 13,768 vertices
R(3,3,4,7,7)	13 391	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices
R(3,3,4,7,8)	15 289	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 1 quadrupling -> 15,288 vertices
R(3,3,4,7,9)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(3,3,4,7,10)	19 013	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (7,9 ; 251)] -> 4,753 vertices; then 1 quadrupling -> 19,012 vertices
R(3,3,4,8,8)	18 454	Published bricks + theorems	[template (8,8,3 ; 650)] x [(3,3,4;29)] -> 18,453 vertices
R(3,3,4,8,9)	21 293	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (8,8 ; 281)] -> 5,323 vertices; then 1 quadrupling -> 21,292 vertices
R(3,3,4,8,10)	25 265	Published bricks + theorems	[template (8,8,3 ; 650)] x [K3] -> 1,579 vertices; then 2 quadruplings -> 25,264 vertices
R(3,3,4,9,9)	36 495	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices
R(3,3,4,9,10)	42 649	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 1 quadrupling -> 42,648 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,4,10,10)	51 770	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [atome (10,10 ; 797)]
R(3,3,5,5,5)	4 524	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (5,5,5 ; 453)] -> 4,523 vertices
R(3,3,5,5,6)	6 150	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [atom (5,6 ; 56)] -> 6,149 vertices
R(3,3,5,5,7)	8 703	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,7 ; 79)] -> 8,702 vertices
R(3,3,5,5,8)	11 034	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,8 ; 100)] -> 11,033 vertices
R(3,3,5,5,9)	14 586	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,9 ; 132)] -> 14,585 vertices
R(3,3,5,5,10)	17 773	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 4,443 vertices; then 1 quadrupling -> 17,772 vertices
R(3,3,5,6,6)	11 145	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices
R(3,3,5,6,7)	13 453	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 1 quadrupling -> 13,452 vertices
R(3,3,5,6,8)	16 689	Our bricks	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 2 quadruplings -> 16,688 vertices
R(3,3,5,6,9)	21 569	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 5,393 – 3 = 21,569; with R(3,6,6) >= 338
R(3,3,5,6,10)	25 857	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,465 – 3 = 25,857; with R(6,6) >= 102
R(3,3,5,7,7)	22 356	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices
R(3,3,5,7,8)	26 893	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 1 quadrupling -> 26,892 vertices
R(3,3,5,7,9)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(3,3,5,7,10)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(3,3,5,8,8)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(3,3,5,8,9)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(3,3,5,8,10)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(3,3,5,9,9)	62 316	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices
R(3,3,5,9,10)	74 081	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices; then 1 quadrupling -> 74,080 vertices
R(3,3,5,10,10)	90 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 2 quadruplings -> 90,976 vertices
R(3,3,6,6,6)	13 674	Our bricks + published	[template (6,6,3;235)] x [atom (3,3,6 ; 59) (ours)] -> 13,673 vertices
R(3,3,6,6,7)	19 174	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [atom (3,6,6 ; 337)] -> 19,173 vertices
R(3,3,6,6,8)	24 219	Our bricks	[template (3,8,3 ; 73) (ours)] x [atom (3,6,6 ; 337)] -> 24,218 vertices
R(3,3,6,6,9)	30 756	Our bricks + published	[template (6,6,3;235)] x [atom (3,3,9 ; 132) (ours)] -> 30,755 vertices
R(3,3,6,6,10)	34 734	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,3,10 ; 149)] -> 34,733 vertices
R(3,3,6,7,7)	27 521	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,6 ; 59) (ours)] -> 27,520 vertices
R(3,3,6,7,8)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(3,3,6,7,9)	44 577	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 1 quadrupling -> 44,576 vertices
R(3,3,6,7,10)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(3,3,6,8,8)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(3,3,6,8,9)	53 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 2 quadruplings -> 53,808 vertices
R(3,3,6,8,10)	66 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 3 quadruplings -> 66,752 vertices
R(3,3,6,9,9)	74 861	Our bricks + published	[template (9,9,3 ; 1282)] x [atom (3,3,6 ; 59) (ours)] -> 74,860 vertices
R(3,3,6,9,10)	90 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 2 quadruplings -> 90,976 vertices
R(3,3,6,10,10)	117 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 29,489 – 3 = 117,953; with R(3,9,9) >= 1,844
R(3,3,7,7,7)	38 326	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [K2]] -> 38,325 vertices
R(3,3,7,7,8)	53 561	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 1 quadrupling -> 53,560 vertices
R(3,3,7,7,9)	61 904	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,9 ; 132) (ours)] -> 61,903 vertices
R(3,3,7,7,10)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(3,3,7,8,8)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices
R(3,3,7,8,9)	89 421	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 1 quadrupling -> 89,420 vertices
R(3,3,7,8,10)	96 833	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [K2]] -> 6,052 vertices; then 2 quadruplings -> 96,832 vertices
R(3,3,7,9,9)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,3,7,9,10)	145 977	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 1 quadrupling -> 145,976 vertices
R(3,3,7,10,10)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674

Parameters	Bound (R >=)	Origin	Construction
R(3,3,8,8,8)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(3,3,8,8,9)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,3,8,8,10)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,3,8,9,9)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,3,8,9,10)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(3,3,8,10,10)	208 897	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 52,225 – 3 = 208,897; with R(7,7) >= 205
R(3,3,9,9,9)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(3,3,9,9,10)	249 261	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 1 quadrupling -> 249,260 vertices
R(3,3,9,10,10)	296 321	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices; then 2 quadruplings -> 296,320 vertices
R(3,3,10,10,10)	363 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 3 quadruplings -> 363,904 vertices
R(3,4,4,4,4)	2 398	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (4,4,4 ; 127)] -> 2,397 vertices
R(3,4,4,4,5)	3 794	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (4,4,4 ; 127)] -> 3,793 vertices
R(3,4,4,4,6)	4 931	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [atom (4,4,4 ; 127)] -> 4,930 vertices
R(3,4,4,4,7)	7 204	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [atom (4,4,4 ; 127)] -> 7,203 vertices
R(3,4,4,4,8)	9 099	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices
R(3,4,4,4,9)	9 605	Published bricks + theorems	[template (8,3 ; 20)] x [atom (4,4,4 ; 127)] -> 2,401 vertices; then 1 quadrupling -> 9,604 vertices
R(3,4,4,4,10)	12 633	Published bricks + theorems	[template (9,3 ; 26)] x [atom (4,4,4 ; 127)] -> 3,158 vertices; then 1 quadrupling -> 12,632 vertices
R(3,4,4,5,5)	5 236	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [atom (3,4,4 ; 54)] -> 5,235 vertices
R(3,4,4,5,6)	7 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-17 (4,4;17)] -> 1,820 vertices; then 1 quadrupling -> 7,280 vertices
R(3,4,4,5,7)	9 991	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,4,4 ; 54)] -> 9,990 vertices
R(3,4,4,5,8)	13 065	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [(3,3,4;29)] -> 3,266 vertices; then 1 quadrupling -> 13,064 vertices
R(3,4,4,5,9)	16 722	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,4,4 ; 54)] -> 16,721 vertices
R(3,4,4,5,10)	20 561	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [Paley-17 (4,4;17)] -> 5,140 vertices; then 1 quadrupling -> 20,560 vertices
R(3,4,4,6,6)	12 504	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,4,4 ; 54)] -> 12,503 vertices
R(3,4,4,6,7)	15 381	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices; then 1 quadrupling -> 15,380 vertices
R(3,4,4,6,8)	18 209	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 2 quadruplings -> 18,208 vertices
R(3,4,4,6,9)	21 436	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (3,4,4 ; 54)] -> 21,435 vertices
R(3,4,4,6,10)	26 393	Our bricks	[template (6,9,3 ; 402) (ours)] x [Paley-17 (4,4;17)] -> 6,598 vertices; then 1 quadrupling -> 26,392 vertices
R(3,4,4,7,7)	25 166	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices
R(3,4,4,7,8)	30 953	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 1 quadrupling -> 30,952 vertices
R(3,4,4,7,9)	36 393	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 1 quadrupling -> 36,392 vertices
R(3,4,4,7,10)	38 417	Published bricks + theorems	[template (8,3 ; 20)] x [atom (4,4,4 ; 127)] -> 2,401 vertices; then 2 quadruplings -> 38,416 vertices
R(3,4,4,8,8)	34 679	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (3,4,4 ; 54)] -> 34,678 vertices
R(3,4,4,8,9)	42 661	Published bricks + theorems	[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)] -> 10,665 vertices; then 1 quadrupling -> 42,660 vertices
R(3,4,4,8,10)	50 529	Published bricks + theorems	[template (9,3 ; 26)] x [atom (4,4,4 ; 127)] -> 3,158 vertices; then 2 quadruplings -> 50,528 vertices
R(3,4,4,9,9)	68 456	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices
R(3,4,4,9,10)	84 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 1 quadrupling -> 84,232 vertices
R(3,4,4,10,10)	91 595	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,4,4,3 ; 116)] x [atome (10,10 ; 797)]
R(3,4,5,5,5)	8 723	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 8,722 vertices
R(3,4,5,5,6)	11 532	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (3,4,5 ; 86) (ours)] -> 11,531 vertices
R(3,4,5,5,7)	16 722	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 16,721 vertices
R(3,4,5,5,8)	21 261	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [(3,3,4;29)] -> 5,315 vertices; then 1 quadrupling -> 21,260 vertices
R(3,4,5,5,9)	26 738	Our bricks	[template (5,9,3 ; 314) (ours)] x [atom (3,4,5 ; 86) (ours)] -> 26,737 vertices
R(3,4,5,5,10)	35 585	Our bricks	[template (5,9,3 ; 314) (ours)] x [(3,3,4;29)] -> 8,896 vertices; then 1 quadrupling -> 35,584 vertices
R(3,4,5,6,6)	20 183	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices
R(3,4,5,6,7)	26 613	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 1 quadrupling -> 26,612 vertices
R(3,4,5,6,8)	30 449	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 2 quadruplings -> 30,448 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,5,6,9)	36 417	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 3 quadruplings -> 36,416 vertices
R(3,4,5,6,10)	47 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (6,9 ; 182)] -> 11,794 vertices; then 1 quadrupling -> 47,176 vertices
R(3,4,5,7,7)	40 343	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices
R(3,4,5,7,8)	53 561	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 1 quadrupling -> 53,560 vertices
R(3,4,5,7,9)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices
R(3,4,5,7,10)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(3,4,5,8,8)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices
R(3,4,5,8,9)	73 813	Published bricks + theorems	[template (8,8,3 ; 650)] x [(3,3,4;29)] -> 18,453 vertices; then 1 quadrupling -> 73,812 vertices
R(3,4,5,8,10)	85 169	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (8,8 ; 281)] -> 5,323 vertices; then 2 quadruplings -> 85,168 vertices
R(3,4,5,9,9)	110 543	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices
R(3,4,5,9,10)	145 977	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 1 quadrupling -> 145,976 vertices
R(3,4,5,10,10)	170 593	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 2 quadruplings -> 170,592 vertices
R(3,4,6,6,6)	26 578	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (3,6,6 ; 337)] -> 26,577 vertices
R(3,4,6,6,7)	38 687	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices
R(3,4,6,6,8)	46 185	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)] -> 11,546 vertices; then 1 quadrupling -> 46,184 vertices
R(3,4,6,6,9)	56 262	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,4,9 ; 241)] -> 56,261 vertices
R(3,4,6,6,10)	63 878	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,10 ; 91) (ours)] -> 63,877 vertices
R(3,4,6,7,7)	53 122	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [K2]] -> 53,121 vertices
R(3,4,6,7,8)	63 649	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 1 quadrupling -> 63,648 vertices
R(3,4,6,7,9)	80 729	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 1 quadrupling -> 80,728 vertices
R(3,4,6,7,10)	96 353	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Paley-101 (6,6;101)] -> 6,022 vertices; then 2 quadruplings -> 96,352 vertices
R(3,4,6,8,8)	74 215	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (3,8,8 ; 940)] -> 74,214 vertices
R(3,4,6,8,9)	106 449	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 2 quadruplings -> 106,448 vertices
R(3,4,6,8,10)	121 793	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 3 quadruplings -> 121,792 vertices
R(3,4,6,9,9)	145 552	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 145,551 vertices
R(3,4,6,9,10)	177 409	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (9,9 ; 562)] -> 44,352 vertices; then 1 quadrupling -> 177,408 vertices
R(3,4,6,10,10)	199 937	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K3] -> 3,124 vertices; then 3 quadruplings -> 199,936 vertices
R(3,4,7,7,7)	77 327	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [K2]] -> 77,326 vertices
R(3,4,7,7,8)	100 661	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 1 quadrupling -> 100,660 vertices
R(3,4,7,7,9)	123 809	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 2 quadruplings -> 123,808 vertices
R(3,4,7,7,10)	145 569	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 2 quadruplings -> 145,568 vertices
R(3,4,7,8,8)	123 809	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 2 quadruplings -> 123,808 vertices
R(3,4,7,8,9)	161 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 1 quadrupling -> 161,368 vertices
R(3,4,7,8,10)	193 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [Mathon-doubling (7,7;202)] -> 12,082 vertices; then 2 quadruplings -> 193,312 vertices
R(3,4,7,9,9)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,4,7,9,10)	273 821	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 1 quadrupling -> 273,820 vertices
R(3,4,7,10,10)	336 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 2 quadruplings -> 336,928 vertices
R(3,4,8,8,8)	128 985	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (8,8 ; 281)] -> 32,246 vertices; then 1 quadrupling -> 128,984 vertices
R(3,4,8,8,9)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,4,8,8,10)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(3,4,8,9,9)	258 245	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)] -> 64,561 vertices; then 1 quadrupling -> 258,244 vertices
R(3,4,8,9,10)	336 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 2 quadruplings -> 336,928 vertices
R(3,4,8,10,10)	366 345	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,7,3 ; 116)] x [atome (10,10 ; 797)]
R(3,4,9,9,9)	308 003	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,9 ; 241)] -> 308,002 vertices
R(3,4,9,9,10)	442 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 1 quadrupling -> 442,168 vertices
R(3,4,9,10,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,4,10,10,10)	682 369	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 3 quadruplings -> 682,368 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,5,5,5)	14 252	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 14,251 vertices
R(3,5,5,5,6)	19 632	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 19,631 vertices
R(3,5,5,5,7)	27 195	Our bricks	[template (5,7,3 ; 188) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 27,194 vertices
R(3,5,5,5,8)	34 809	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,7 ; 79)] -> 8,702 vertices; then 1 quadrupling -> 34,808 vertices
R(3,5,5,5,9)	45 518	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 45,517 vertices
R(3,5,5,5,10)	58 341	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,9 ; 132)] -> 14,585 vertices; then 1 quadrupling -> 58,340 vertices
R(3,5,5,6,6)	34 032	Our bricks + published	[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)] -> 34,031 vertices
R(3,5,5,6,7)	44 577	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 1 quadrupling -> 44,576 vertices
R(3,5,5,6,8)	53 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 2 quadruplings -> 53,808 vertices
R(3,5,5,6,9)	66 753	Our bricks	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 3 quadruplings -> 66,752 vertices
R(3,5,5,6,10)	86 273	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 21,569 – 3 = 86,273; with R(3,6,6) >= 338
R(3,5,5,7,7)	68 498	Our bricks + published	[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)] -> 68,497 vertices
R(3,5,5,7,8)	89 421	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 1 quadrupling -> 89,420 vertices
R(3,5,5,7,9)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,5,5,7,10)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,5,5,8,8)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,5,5,8,9)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,5,5,8,10)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(3,5,5,9,9)	186 308	Our bricks	[template (9,9,3 ; 1282)] x [atom (3,5,5 ; 146) (ours)] -> 186,307 vertices
R(3,5,5,9,10)	249 261	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 1 quadrupling -> 249,260 vertices
R(3,5,5,10,10)	296 321	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices; then 2 quadruplings -> 296,320 vertices
R(3,5,6,6,6)	45 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (3,6,6 ; 337)] -> 45,416 vertices
R(3,5,6,6,7)	62 912	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,6,6 ; 337)] -> 62,911 vertices
R(3,5,6,6,8)	79 061	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [atom (3,6,6 ; 337)] -> 79,060 vertices
R(3,5,6,6,9)	105 301	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,6,6 ; 337)] -> 105,300 vertices
R(3,5,6,6,10)	125 729	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [Paley-101 (6,6;101)] -> 31,432 vertices; then 1 quadrupling -> 125,728 vertices
R(3,5,6,7,7)	90 777	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices
R(3,5,6,7,8)	108 765	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 1 quadrupling -> 108,764 vertices
R(3,5,6,7,9)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,5,6,7,10)	178 305	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 2 quadruplings -> 178,304 vertices
R(3,5,6,8,8)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,5,6,8,9)	178 305	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 2 quadruplings -> 178,304 vertices
R(3,5,6,8,10)	215 233	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 3 quadruplings -> 215,232 vertices
R(3,5,6,9,9)	249 261	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 1 quadrupling -> 249,260 vertices
R(3,5,6,9,10)	303 165	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices; then 1 quadrupling -> 303,164 vertices
R(3,5,6,10,10)	363 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 3 quadruplings -> 363,904 vertices
R(3,5,7,7,7)	125 744	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [K2]] -> 125,743 vertices
R(3,5,7,7,8)	161 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 1 quadrupling -> 161,368 vertices
R(3,5,7,7,9)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,5,7,7,10)	252 181	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)] -> 63,045 vertices; then 1 quadrupling -> 252,180 vertices
R(3,5,7,8,8)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,5,7,8,9)	263 589	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]] -> 65,897 vertices; then 1 quadrupling -> 263,588 vertices
R(3,5,7,8,10)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,5,7,9,9)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,5,7,9,10)	442 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 1 quadrupling -> 442,168 vertices
R(3,5,7,10,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,5,8,8,8)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(3,5,8,8,9)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,8,8,10)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(3,5,8,9,9)	432 971	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 432,970 vertices
R(3,5,8,9,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,5,8,10,10)	689 153	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 172,289 – 3 = 689,153; with R(3,7,7) >= 674
R(3,5,9,9,9)	576 679	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 576,678 vertices
R(3,5,9,9,10)	722 229	Our bricks	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 180,557 vertices; then 1 quadrupling -> 722,228 vertices
R(3,5,9,10,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(3,5,10,10,10)	1 185 281	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices; then 3 quadruplings -> 1,185,280 vertices
R(3,6,6,6,6)	78 726	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices
R(3,6,6,6,7)	94 005	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 23,502 – 3 = 94,005; with R(6,6,6,6) >= 23,502
R(3,6,6,6,8)	96 353	Our bricks	[template (4,5,3 ; 61) (ours)] x [Paley-101 (6,6;101)] -> 6,022 vertices; then 2 quadruplings -> 96,352 vertices
R(3,6,6,6,9)	136 290	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 136,289 vertices
R(3,6,6,6,10)	169 821	Published bricks + theorems	[template (6,6,3;235)] x [atom (6,9 ; 182)] -> 42,455 vertices; then 1 quadrupling -> 169,820 vertices
R(3,6,6,7,7)	158 459	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices
R(3,6,6,7,8)	189 209	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 1 quadrupling -> 189,208 vertices
R(3,6,6,7,9)	189 209	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 1 quadrupling -> 189,208 vertices (by monotonicity, from (3,6,6,7,8))
R(3,6,6,7,10)	234 405	Published bricks + theorems	[template (6,6,3;235)] x [atom (7,9 ; 251)] -> 58,601 vertices; then 1 quadrupling -> 234,404 vertices
R(3,6,6,8,8)	219 828	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices
R(3,6,6,8,9)	262 485	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,8 ; 281)] -> 65,621 vertices; then 1 quadrupling -> 262,484 vertices
R(3,6,6,8,10)	300 465	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [Paley-101 (6,6;101)] -> 18,779 vertices; then 2 quadruplings -> 300,464 vertices
R(3,6,6,9,9)	431 130	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices
R(3,6,6,9,10)	525 501	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 1 quadrupling -> 525,500 vertices
R(3,6,6,10,10)	562 314	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]
R(3,6,7,7,7)	189 209	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 1 quadrupling -> 189,208 vertices
R(3,6,7,7,8)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,6,7,7,9)	274 325	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 274,324 vertices
R(3,6,7,7,10)	341 813	Published bricks + theorems	[template (7,7,3;472)] x [atom (6,9 ; 182)] -> 85,453 vertices; then 1 quadrupling -> 341,812 vertices
R(3,6,7,8,8)	262 485	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,8 ; 281)] -> 65,621 vertices; then 1 quadrupling -> 262,484 vertices
R(3,6,7,8,9)	363 105	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 1 quadrupling -> 363,104 vertices
R(3,6,7,8,10)	435 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 2 quadruplings -> 435,056 vertices
R(3,6,7,9,9)	525 501	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 1 quadrupling -> 525,500 vertices
R(3,6,7,9,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,6,7,10,10)	745 461	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [atome (10,10 ; 797)]
R(3,6,8,8,8)	292 865	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 4 quadruplings -> 292,864 vertices
R(3,6,8,8,9)	435 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 2 quadruplings -> 435,056 vertices
R(3,6,8,8,10)	534 017	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 4 quadruplings -> 534,016 vertices
R(3,6,8,9,9)	538 913	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (9,9 ; 562)] -> 33,682 vertices; then 2 quadruplings -> 538,912 vertices
R(3,6,8,9,10)	713 217	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 3 quadruplings -> 713,216 vertices
R(3,6,8,10,10)	835 585	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 208,897 – 3 = 835,585; with R(7,7) >= 205
R(3,6,9,9,9)	746 105	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 746,104 vertices
R(3,6,9,9,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(3,6,9,10,10)	1 212 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices; then 2 quadruplings -> 1,212,656 vertices
R(3,6,10,10,10)	1 455 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 4 quadruplings -> 1,455,616 vertices
R(3,7,7,7,7)	316 715	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices
R(3,7,7,7,8)	379 493	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 94,874 – 3 = 379,493; with R(7,7,7,7) >= 94,874
R(3,7,7,7,9)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,7,7,7,10)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,7,7,8,8)	442 472	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices
R(3,7,7,8,9)	528 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 1 quadrupling -> 528,328 vertices
R(3,7,7,8,10)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices
R(3,7,7,9,9)	867 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices
R(3,7,7,9,10)	1 057 733	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 1 quadrupling -> 1,057,732 vertices
R(3,7,7,10,10)	1 125 422	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(3,7,8,8,8)	528 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 1 quadrupling -> 528,328 vertices
R(3,7,8,8,9)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices
R(3,7,8,8,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,7,8,9,9)	1 057 733	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 1 quadrupling -> 1,057,732 vertices
R(3,7,8,9,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(3,7,8,10,10)	1 500 473	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,7,9,9,9)	1 057 733	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 1 quadrupling -> 1,057,732 vertices (by monotonicity, from (3,7,8,9,9))
R(3,7,9,9,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,7,9,10,10)	1 768 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 2 quadruplings -> 1,768,672 vertices
R(3,7,10,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,8,8,8,8)	609 693	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (3,8,8 ; 940)] -> 609,692 vertices
R(3,8,8,8,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,8,8,8,10)	978 433	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 4 quadruplings -> 978,432 vertices
R(3,8,8,9,9)	1 203 422	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices
R(3,8,8,9,10)	1 457 481	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (9,9 ; 562)] -> 364,370 vertices; then 1 quadrupling -> 1,457,480 vertices
R(3,8,8,10,10)	1 550 743	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,8,3 ; 650)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(3,8,9,9,9)	1 457 481	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (9,9 ; 562)] -> 364,370 vertices; then 1 quadrupling -> 1,457,480 vertices
R(3,8,9,9,10)	1 768 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 2 quadruplings -> 1,768,672 vertices
R(3,8,9,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,8,10,10,10)	2 756 609	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 689,153 – 3 = 2,756,609; with R(3,7,7) >= 674
R(3,9,9,9,9)	2 360 165	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices
R(3,9,9,9,10)	2 876 813	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 719,204 – 3 = 2,876,813; with R(9,9,9,9) >= 719,204
R(3,9,9,10,10)	3 060 872	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(3,9,10,10,10)	4 080 953	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]
R(3,10,10,10,10)	4 311 809	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (9,9 ; 562)] -> 16,843 vertices; then 4 quadruplings -> 4,311,808 vertices
R(4,4,4,4,4)	4 176	Published	published value
R(4,4,4,4,5)	7 583	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)] -> 7,582 vertices
R(4,4,4,4,6)	9 988	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (4,4,4 ; 127)] -> 9,987 vertices
R(4,4,4,4,7)	14 537	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)] -> 14,536 vertices
R(4,4,4,4,8)	15 298	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [atom (4,4,4 ; 127)] -> 15,297 vertices
R(4,4,4,4,9)	19 725	Our bricks + published	[modele (4,9,3 ; 157)] x [atome (4,4,4 ; 127)] [new template (4,9;157), v21]
R(4,4,4,4,10)	22 898	Our bricks + published	[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,10 ; 91) (ours)] -> 22,897 vertices
R(4,4,4,5,5)	12 390	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [atom (4,4,4 ; 127)] -> 12,389 vertices
R(4,4,4,5,6)	17 067	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices
R(4,4,4,5,7)	23 642	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)] -> 23,641 vertices
R(4,4,4,5,8)	29 711	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)] -> 29,710 vertices
R(4,4,4,5,9)	39 571	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,5,10)	39 571	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices (by monotonicity, from (4,4,4,5,9))
R(4,4,4,6,6)	29 586	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices
R(4,4,4,6,7)	30 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)] -> 7,582 vertices; then 1 quadrupling -> 30,328 vertices
R(4,4,4,6,8)	30 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)] -> 7,582 vertices; then 1 quadrupling -> 30,328 vertices (by monotonicity, from (4,4,4,6,7))
R(4,4,4,6,9)	50 709	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)] -> 50,708 vertices
R(4,4,4,6,10)	51 234	Published bricks + theorems	[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (6,10 ; 203)] -> 51,233 vertices
R(4,4,4,7,7)	59 549	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices
R(4,4,4,7,8)	59 549	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices (by monotonicity, from (4,4,4,7,7))
R(4,4,4,7,9)	68 265	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices; then 1 quadrupling -> 68,264 vertices
R(4,4,4,7,10)	75 369	Our bricks + published	[template (4,9,3 ; 150) (ours)] x [atom (4,4,4 ; 127)] -> 18,842 vertices; then 1 quadrupling -> 75,368 vertices
R(4,4,4,8,8)	82 056	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices
R(4,4,4,8,9)	94 565	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)] -> 23,641 vertices; then 1 quadrupling -> 94,564 vertices
R(4,4,4,8,10)	94 565	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)] -> 23,641 vertices; then 1 quadrupling -> 94,564 vertices (by monotonicity, from (4,4,4,8,9))
R(4,4,4,9,9)	161 969	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices
R(4,4,4,9,10)	161 969	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices (by monotonicity, from (4,4,4,9,9))
R(4,4,4,10,10)	201 516	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]
R(4,4,5,5,5)	16 070	Published bricks + theorems	[template (5,5,5,3 ; 977)] x [Paley-17 (4,4;17)] -> 16,069 vertices
R(4,4,5,5,6)	20 941	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [atom (3,4,4 ; 54)] -> 5,235 vertices; then 1 quadrupling -> 20,940 vertices
R(4,4,5,5,7)	30 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)] -> 7,582 vertices; then 1 quadrupling -> 30,328 vertices
R(4,4,5,5,8)	39 961	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,4,4 ; 54)] -> 9,990 vertices; then 1 quadrupling -> 39,960 vertices
R(4,4,5,5,9)	52 257	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [(3,3,4;29)] -> 3,266 vertices; then 2 quadruplings -> 52,256 vertices
R(4,4,5,5,10)	66 885	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,4,4 ; 54)] -> 16,721 vertices; then 1 quadrupling -> 66,884 vertices
R(4,4,5,6,6)	34 601	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 34,600 vertices
R(4,4,5,6,7)	50 013	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,4,4 ; 54)] -> 12,503 vertices; then 1 quadrupling -> 50,012 vertices
R(4,4,5,6,8)	61 521	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices; then 2 quadruplings -> 61,520 vertices
R(4,4,5,6,9)	72 833	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 3 quadruplings -> 72,832 vertices
R(4,4,5,6,10)	85 741	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (3,4,4 ; 54)] -> 21,435 vertices; then 1 quadrupling -> 85,740 vertices
R(4,4,5,7,7)	69 638	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 69,637 vertices
R(4,4,5,7,8)	100 661	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 1 quadrupling -> 100,660 vertices
R(4,4,5,7,9)	123 809	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 2 quadruplings -> 123,808 vertices
R(4,4,5,7,10)	158 281	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 1 quadrupling -> 158,280 vertices
R(4,4,5,8,8)	123 809	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 2 quadruplings -> 123,808 vertices
R(4,4,5,8,9)	145 569	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 2 quadruplings -> 145,568 vertices
R(4,4,5,8,10)	170 641	Published bricks + theorems	[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)] -> 10,665 vertices; then 2 quadruplings -> 170,640 vertices
R(4,4,5,9,9)	189 518	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 189,517 vertices
R(4,4,5,9,10)	273 821	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 1 quadrupling -> 273,820 vertices
R(4,4,5,10,10)	336 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 2 quadruplings -> 336,928 vertices
R(4,4,6,6,6)	44 906	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [K3]] -> 44,905 vertices
R(4,4,6,6,7)	68 265	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices; then 1 quadrupling -> 68,264 vertices
R(4,4,6,6,8)	73 044	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 73,043 vertices
R(4,4,6,6,9)	96 109	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 96,108 vertices
R(4,4,6,6,10)	111 486	Published bricks + theorems	[template (10,3 ; 30)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 111,485 vertices
R(4,4,6,7,7)	118 341	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 1 quadrupling -> 118,340 vertices
R(4,4,6,7,8)	121 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)] -> 7,582 vertices; then 2 quadruplings -> 121,312 vertices
R(4,4,6,7,9)	158 281	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 1 quadrupling -> 158,280 vertices
R(4,4,6,7,10)	202 833	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)] -> 50,708 vertices; then 1 quadrupling -> 202,832 vertices
R(4,4,6,8,8)	124 696	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (8,8,3 ; 650)] x [K3]] -> 124,695 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,6,8,9)	200 049	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,4,4 ; 54)] -> 12,503 vertices; then 2 quadruplings -> 200,048 vertices
R(4,4,6,8,10)	202 833	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)] -> 50,708 vertices; then 1 quadrupling -> 202,832 vertices (by monotonicity, from (4,4,6,7,10))
R(4,4,6,9,9)	246 751	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 246,750 vertices
R(4,4,6,9,10)	291 329	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 4 quadruplings -> 291,328 vertices
R(4,4,6,10,10)	336 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 2 quadruplings -> 336,928 vertices (by monotonicity, from (4,4,5,10,10))
R(4,4,7,7,7)	131 492	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [K3]] -> 131,491 vertices
R(4,4,7,7,8)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices
R(4,4,7,7,9)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices (by monotonicity, from (4,4,7,7,8))
R(4,4,7,7,10)	273 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices; then 2 quadruplings -> 273,056 vertices
R(4,4,7,8,8)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices (by monotonicity, from (4,4,7,7,8))
R(4,4,7,8,9)	328 221	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 1 quadrupling -> 328,220 vertices
R(4,4,7,8,10)	378 257	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)] -> 23,641 vertices; then 2 quadruplings -> 378,256 vertices
R(4,4,7,9,9)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(4,4,7,9,10)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices
R(4,4,7,10,10)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices (by monotonicity, from (4,4,7,9,10))
R(4,4,8,8,8)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices (by monotonicity, from (4,4,7,7,8))
R(4,4,8,8,9)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(4,4,8,8,10)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices (by monotonicity, from (4,4,8,8,9))
R(4,4,8,9,9)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(4,4,8,9,10)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices (by monotonicity, from (4,4,7,9,10))
R(4,4,8,10,10)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices (by monotonicity, from (4,4,7,9,10))
R(4,4,9,9,9)	582 273	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 3 quadruplings -> 582,272 vertices
R(4,4,9,9,10)	758 069	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 189,517 vertices; then 1 quadrupling -> 758,068 vertices
R(4,4,9,10,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(4,4,10,10,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices (by monotonicity, from (4,4,9,10,10))
R(4,5,5,5,5)	27 143	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (5,5,5 ; 453)] -> 27,142 vertices
R(4,5,5,5,6)	35 742	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (5,5,5 ; 453)] -> 35,741 vertices
R(4,5,5,5,7)	52 027	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (5,5,5 ; 453)] -> 52,026 vertices
R(4,5,5,5,8)	63 689	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,7,3 ; 188) (ours)] x [K2]] -> 15,922 vertices; then 1 quadrupling -> 63,688 vertices
R(4,5,5,5,9)	85 041	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [(3,3,4;29)] -> 5,315 vertices; then 2 quadruplings -> 85,040 vertices
R(4,5,5,5,10)	106 881	Our bricks	[template (5,8,3 ; 236) (ours)] x [(3,3,4;29)] -> 6,680 vertices; then 2 quadruplings -> 106,880 vertices
R(4,5,5,6,6)	55 706	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [K3]] -> 55,705 vertices
R(4,5,5,6,7)	80 729	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 1 quadrupling -> 80,728 vertices
R(4,5,5,6,8)	106 449	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 2 quadruplings -> 106,448 vertices
R(4,5,5,6,9)	121 793	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 3 quadruplings -> 121,792 vertices
R(4,5,5,6,10)	145 665	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 4 quadruplings -> 145,664 vertices
R(4,5,5,7,7)	112 056	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K3]] -> 112,055 vertices
R(4,5,5,7,8)	161 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 1 quadrupling -> 161,368 vertices
R(4,5,5,7,9)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(4,5,5,7,10)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(4,5,5,8,8)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(4,5,5,8,9)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(4,5,5,8,10)	295 249	Published bricks + theorems	[template (8,8,3 ; 650)] x [(3,3,4;29)] -> 18,453 vertices; then 2 quadruplings -> 295,248 vertices
R(4,5,5,9,9)	306 096	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 306,095 vertices
R(4,5,5,9,10)	442 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 1 quadrupling -> 442,168 vertices
R(4,5,5,10,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(4,5,6,6,6)	80 729	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 1 quadrupling -> 80,728 vertices
R(4,5,6,6,7)	118 341	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 1 quadrupling -> 118,340 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,6,6,8)	154 745	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 1 quadrupling -> 154,744 vertices
R(4,5,6,6,9)	184 737	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)] -> 11,546 vertices; then 2 quadruplings -> 184,736 vertices
R(4,5,6,6,10)	202 581	Our bricks + published	[template (6,6,3;235)] x [[template (4,9,3 ; 150) (ours)] x [K2]] -> 50,645 vertices; then 1 quadrupling -> 202,580 vertices
R(4,5,6,7,7)	161 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 1 quadrupling -> 161,368 vertices
R(4,5,6,7,8)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(4,5,6,7,9)	254 593	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 2 quadruplings -> 254,592 vertices
R(4,5,6,7,10)	322 913	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 2 quadruplings -> 322,912 vertices
R(4,5,6,8,8)	254 593	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 2 quadruplings -> 254,592 vertices
R(4,5,6,8,9)	322 913	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 2 quadruplings -> 322,912 vertices
R(4,5,6,8,10)	425 793	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 3 quadruplings -> 425,792 vertices
R(4,5,6,9,9)	442 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 1 quadrupling -> 442,168 vertices
R(4,5,6,9,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(4,5,6,10,10)	709 633	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (9,9 ; 562)] -> 44,352 vertices; then 2 quadruplings -> 709,632 vertices
R(4,5,7,7,7)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices
R(4,5,7,7,8)	309 305	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [K2]] -> 77,326 vertices; then 1 quadrupling -> 309,304 vertices
R(4,5,7,7,9)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(4,5,7,7,10)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(4,5,7,8,8)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(4,5,7,8,9)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(4,5,7,8,10)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices
R(4,5,7,9,9)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices
R(4,5,7,9,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(4,5,7,10,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(4,5,8,8,8)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(4,5,8,8,9)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices
R(4,5,8,8,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(4,5,8,9,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(4,5,8,9,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(4,5,8,10,10)	1 347 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 3 quadruplings -> 1,347,712 vertices
R(4,5,9,9,9)	1 032 977	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)] -> 64,561 vertices; then 2 quadruplings -> 1,032,976 vertices
R(4,5,9,9,10)	1 347 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 3 quadruplings -> 1,347,712 vertices
R(4,5,9,10,10)	1 768 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 2 quadruplings -> 1,768,672 vertices
R(4,5,10,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(4,6,6,6,6)	133 014	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]] -> 133,013 vertices
R(4,6,6,6,7)	138 401	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 34,600 vertices; then 1 quadrupling -> 138,400 vertices
R(4,6,6,6,8)	154 745	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 1 quadrupling -> 154,744 vertices (by monotonicity, from (4,5,6,6,8))
R(4,6,6,6,9)	230 124	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K3]] -> 230,123 vertices
R(4,6,6,6,10)	236 910	Published bricks + theorems	[template (6,6,3;235)] x [[template (4,3;6)] x [atom (6,10 ; 203)]] -> 236,909 vertices
R(4,6,6,7,7)	267 731	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices
R(4,6,6,7,8)	273 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices; then 2 quadruplings -> 273,056 vertices
R(4,6,6,7,9)	322 913	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 2 quadruplings -> 322,912 vertices
R(4,6,6,7,10)	384 433	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 96,108 vertices; then 1 quadrupling -> 384,432 vertices
R(4,6,6,8,8)	369 354	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [K3]] -> 369,353 vertices
R(4,6,6,8,9)	473 361	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 2 quadruplings -> 473,360 vertices
R(4,6,6,8,10)	473 361	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 2 quadruplings -> 473,360 vertices (by monotonicity, from (4,6,6,8,9))
R(4,6,6,9,9)	730 884	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices
R(4,6,6,9,10)	738 945	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)] -> 11,546 vertices; then 3 quadruplings -> 738,944 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,6,6,10,10)	931 890	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (4,3;6)] x [atome (10,10 ; 797)]]
R(4,6,7,7,7)	278 549	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 69,637 vertices; then 1 quadrupling -> 278,548 vertices
R(4,6,7,7,8)	473 361	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 2 quadruplings -> 473,360 vertices
R(4,6,7,7,9)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(4,6,7,7,10)	633 121	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 2 quadruplings -> 633,120 vertices
R(4,6,7,8,8)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(4,6,7,8,9)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices
R(4,6,7,8,10)	811 329	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)] -> 50,708 vertices; then 2 quadruplings -> 811,328 vertices
R(4,6,7,9,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(4,6,7,9,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(4,6,7,10,10)	1 347 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 3 quadruplings -> 1,347,712 vertices
R(4,6,8,8,8)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices (by monotonicity, from (4,5,8,8,8))
R(4,6,8,8,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(4,6,8,8,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices (by monotonicity, from (4,5,8,8,10))
R(4,6,8,9,9)	1 018 369	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 3 quadruplings -> 1,018,368 vertices
R(4,6,8,9,10)	1 291 649	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 3 quadruplings -> 1,291,648 vertices
R(4,6,8,10,10)	1 347 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 3 quadruplings -> 1,347,712 vertices (by monotonicity, from (4,5,8,10,10))
R(4,6,9,9,9)	1 291 649	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 3 quadruplings -> 1,291,648 vertices
R(4,6,9,9,10)	1 768 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 2 quadruplings -> 1,768,672 vertices
R(4,6,9,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(4,6,10,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices (by monotonicity, from (4,5,10,10,10))
R(4,7,7,7,7)	538 556	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices
R(4,7,7,7,8)	538 556	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices (by monotonicity, from (4,7,7,7,7))
R(4,7,7,7,9)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(4,7,7,7,10)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices (by monotonicity, from (4,7,7,7,9))
R(4,7,7,8,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(4,7,7,8,9)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(4,7,7,8,10)	1 312 881	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 2 quadruplings -> 1,312,880 vertices
R(4,7,7,9,9)	1 471 136	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices
R(4,7,7,9,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(4,7,7,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(4,7,8,8,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices (by monotonicity, from (4,7,7,8,8))
R(4,7,8,8,9)	1 312 881	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 2 quadruplings -> 1,312,880 vertices
R(4,7,8,8,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(4,7,8,9,9)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(4,7,8,9,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(4,7,8,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices (by monotonicity, from (4,7,7,10,10))
R(4,7,9,9,9)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(4,7,9,9,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(4,7,9,10,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(4,7,10,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(4,8,8,8,8)	1 024 404	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [K3]] -> 1,024,403 vertices
R(4,8,8,8,9)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(4,8,8,8,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices (by monotonicity, from (4,7,8,8,10))
R(4,8,8,9,9)	2 027 109	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [K3]] -> 2,027,108 vertices
R(4,8,8,9,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices (by monotonicity, from (4,7,8,9,10))
R(4,8,8,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices (by monotonicity, from (4,7,7,10,10))

Parameters	Bound (R >=)	Origin	Construction
R(4,8,9,9,9)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(4,8,9,9,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(4,8,9,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(4,8,10,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices (by monotonicity, from (4,7,10,10,10))
R(4,9,9,9,9)	4 001 126	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]] -> 4,001,125 vertices
R(4,9,9,9,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(4,9,9,10,10)	5 390 849	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 4 quadruplings -> 5,390,848 vertices
R(4,9,10,10,10)	7 074 689	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 3 quadruplings -> 7,074,688 vertices
R(4,10,10,10,10)	7 074 689	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 3 quadruplings -> 7,074,688 vertices (by monotonicity, from (4,9,10,10,10))
R(5,5,5,5,5)	44 338	Our bricks	[template (5,5,3 ; 99) (ours)] x [atom (5,5,5 ; 453)] -> 44,337 vertices
R(5,5,5,5,6)	61 077	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (5,5,5 ; 453)] -> 61,076 vertices
R(5,5,5,5,7)	84 604	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (5,5,5 ; 453)] -> 84,603 vertices
R(5,5,5,5,8)	106 321	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [atom (5,5,5 ; 453)] -> 106,320 vertices
R(5,5,5,5,9)	141 609	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (5,5,5 ; 453)] -> 141,608 vertices
R(5,5,5,5,10)	176 529	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,8 ; 100)] -> 11,033 vertices; then 2 quadruplings -> 176,528 vertices
R(5,5,5,6,6)	105 870	Published bricks + theorems	[template (6,6,3;235)] x [atom (5,5,5 ; 453)] -> 105,869 vertices
R(5,5,5,6,7)	131 877	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [atom (3,6,6 ; 337)] -> 32,969 vertices; then 1 quadrupling -> 131,876 vertices
R(5,5,5,6,8)	178 305	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 2 quadruplings -> 178,304 vertices
R(5,5,5,6,9)	215 233	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 3 quadruplings -> 215,232 vertices
R(5,5,5,6,10)	267 009	Our bricks	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 4 quadruplings -> 267,008 vertices
R(5,5,5,7,7)	213 095	Published bricks + theorems	[template (7,7,3;472)] x [atom (5,5,5 ; 453)] -> 213,094 vertices
R(5,5,5,7,8)	263 589	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]] -> 65,897 vertices; then 1 quadrupling -> 263,588 vertices
R(5,5,5,7,9)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(5,5,5,7,10)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(5,5,5,8,8)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(5,5,5,8,9)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(5,5,5,8,10)	534 017	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 4 quadruplings -> 534,016 vertices
R(5,5,5,9,9)	579 575	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)] -> 579,574 vertices
R(5,5,5,9,10)	722 229	Our bricks	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 180,557 vertices; then 1 quadrupling -> 722,228 vertices
R(5,5,5,10,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(5,5,6,6,6)	131 877	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [atom (3,6,6 ; 337)] -> 32,969 vertices; then 1 quadrupling -> 131,876 vertices
R(5,5,6,6,7)	181 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (3,6,6 ; 337)] -> 45,416 vertices; then 1 quadrupling -> 181,664 vertices
R(5,5,6,6,8)	251 645	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,6,6 ; 337)] -> 62,911 vertices; then 1 quadrupling -> 251,644 vertices
R(5,5,6,6,9)	316 241	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [atom (3,6,6 ; 337)] -> 79,060 vertices; then 1 quadrupling -> 316,240 vertices
R(5,5,6,6,10)	421 201	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,6,6 ; 337)] -> 105,300 vertices; then 1 quadrupling -> 421,200 vertices
R(5,5,6,7,7)	263 589	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]] -> 65,897 vertices; then 1 quadrupling -> 263,588 vertices
R(5,5,6,7,8)	363 105	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 1 quadrupling -> 363,104 vertices
R(5,5,6,7,9)	435 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 2 quadruplings -> 435,056 vertices
R(5,5,6,7,10)	534 017	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 4 quadruplings -> 534,016 vertices
R(5,5,6,8,8)	435 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 2 quadruplings -> 435,056 vertices
R(5,5,6,8,9)	534 017	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 4 quadruplings -> 534,016 vertices
R(5,5,6,8,10)	713 217	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 3 quadruplings -> 713,216 vertices
R(5,5,6,9,9)	722 229	Our bricks	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 180,557 vertices; then 1 quadrupling -> 722,228 vertices
R(5,5,6,9,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(5,5,6,10,10)	1 212 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices; then 2 quadruplings -> 1,212,656 vertices
R(5,5,7,7,7)	363 105	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 1 quadrupling -> 363,104 vertices
R(5,5,7,7,8)	502 973	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [K2]] -> 125,743 vertices; then 1 quadrupling -> 502,972 vertices
R(5,5,7,7,9)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,7,7,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(5,5,7,8,8)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices
R(5,5,7,8,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(5,5,7,8,10)	1 054 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]] -> 65,897 vertices; then 2 quadruplings -> 1,054,352 vertices
R(5,5,7,9,9)	1 054 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]] -> 65,897 vertices; then 2 quadruplings -> 1,054,352 vertices
R(5,5,7,9,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(5,5,7,10,10)	1 768 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 2 quadruplings -> 1,768,672 vertices
R(5,5,8,8,8)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(5,5,8,8,9)	1 054 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]] -> 65,897 vertices; then 2 quadruplings -> 1,054,352 vertices
R(5,5,8,8,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(5,5,8,9,9)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(5,5,8,9,10)	1 768 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 2 quadruplings -> 1,768,672 vertices
R(5,5,8,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(5,5,9,9,9)	1 731 881	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 432,970 vertices; then 1 quadrupling -> 1,731,880 vertices
R(5,5,9,9,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(5,5,9,10,10)	2 888 913	Our bricks	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 180,557 vertices; then 2 quadruplings -> 2,888,912 vertices
R(5,5,10,10,10)	3 988 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 3 quadruplings -> 3,988,160 vertices
R(5,6,6,6,6)	211 505	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 211,504 vertices
R(5,6,6,6,7)	314 901	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 1 quadrupling -> 314,900 vertices
R(5,6,6,6,8)	376 017	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 94,005 – 3 = 376,017; with R(6,6,6,6) >= 23,502
R(5,6,6,6,9)	423 477	Published bricks + theorems	[template (6,6,3;235)] x [atom (5,5,5 ; 453)] -> 105,869 vertices; then 1 quadrupling -> 423,476 vertices
R(5,6,6,6,10)	545 157	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 136,289 vertices; then 1 quadrupling -> 545,156 vertices
R(5,6,6,7,7)	425 714	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 425,713 vertices
R(5,6,6,7,8)	633 833	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 1 quadrupling -> 633,832 vertices
R(5,6,6,7,9)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices
R(5,6,6,7,10)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices (by monotonicity, from (5,6,6,7,9))
R(5,6,6,8,8)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices
R(5,6,6,8,9)	879 309	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 1 quadrupling -> 879,308 vertices
R(5,6,6,8,10)	1 049 937	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,8 ; 281)] -> 65,621 vertices; then 2 quadruplings -> 1,049,936 vertices
R(5,6,6,9,9)	1 182 371	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 1,182,370 vertices
R(5,6,6,9,10)	1 724 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 1 quadrupling -> 1,724,516 vertices
R(5,6,6,10,10)	2 102 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 2 quadruplings -> 2,102,000 vertices
R(5,6,7,7,7)	633 833	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 1 quadrupling -> 633,832 vertices
R(5,6,7,7,8)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices
R(5,6,7,7,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(5,6,7,7,10)	1 097 297	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 274,324 vertices; then 1 quadrupling -> 1,097,296 vertices
R(5,6,7,8,8)	879 309	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 1 quadrupling -> 879,308 vertices
R(5,6,7,8,9)	1 097 297	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 274,324 vertices; then 1 quadrupling -> 1,097,296 vertices
R(5,6,7,8,10)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(5,6,7,9,9)	1 724 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 1 quadrupling -> 1,724,516 vertices
R(5,6,7,9,10)	2 102 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 2 quadruplings -> 2,102,000 vertices
R(5,6,7,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(5,6,8,8,8)	1 049 937	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,8 ; 281)] -> 65,621 vertices; then 2 quadruplings -> 1,049,936 vertices
R(5,6,8,8,9)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(5,6,8,8,10)	1 740 225	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 3 quadruplings -> 1,740,224 vertices
R(5,6,8,9,9)	2 102 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 2 quadruplings -> 2,102,000 vertices
R(5,6,8,9,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,6,8,10,10)	2 981 841	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [atome (10,10 ; 797)]
R(5,6,9,9,9)	2 318 297	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)] -> 579,574 vertices; then 1 quadrupling -> 2,318,296 vertices
R(5,6,9,9,10)	2 984 417	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 746,104 vertices; then 1 quadrupling -> 2,984,416 vertices
R(5,6,9,10,10)	3 988 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 3 quadruplings -> 3,988,160 vertices
R(5,6,10,10,10)	4 850 625	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices; then 3 quadruplings -> 4,850,624 vertices
R(5,7,7,7,7)	853 853	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 853,852 vertices
R(5,7,7,7,8)	1 266 857	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 1 quadrupling -> 1,266,856 vertices
R(5,7,7,7,9)	1 517 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 379,493 – 3 = 1,517,969; with R(7,7,7,7) >= 94,874
R(5,7,7,7,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(5,7,7,8,8)	1 517 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 379,493 – 3 = 1,517,969; with R(7,7,7,7) >= 94,874
R(5,7,7,8,9)	1 769 885	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 1 quadrupling -> 1,769,884 vertices
R(5,7,7,8,10)	2 113 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 2 quadruplings -> 2,113,312 vertices
R(5,7,7,9,9)	2 379 893	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 2,379,892 vertices
R(5,7,7,9,10)	3 471 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 1 quadrupling -> 3,471,136 vertices
R(5,7,7,10,10)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(5,7,8,8,8)	1 769 885	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 1 quadrupling -> 1,769,884 vertices
R(5,7,8,8,9)	2 113 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 2 quadruplings -> 2,113,312 vertices
R(5,7,8,8,10)	2 581 889	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 3 quadruplings -> 2,581,888 vertices
R(5,7,8,9,9)	3 471 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 1 quadrupling -> 3,471,136 vertices
R(5,7,8,9,10)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(5,7,8,10,10)	4 501 685	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(5,7,9,9,9)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(5,7,9,9,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(5,7,9,10,10)	6 001 889	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(5,7,10,10,10)	7 074 689	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 3 quadruplings -> 7,074,688 vertices
R(5,8,8,8,8)	2 113 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 2 quadruplings -> 2,113,312 vertices
R(5,8,8,8,9)	2 581 889	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 3 quadruplings -> 2,581,888 vertices
R(5,8,8,8,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(5,8,8,9,9)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(5,8,8,9,10)	4 813 685	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 1 quadrupling -> 4,813,684 vertices
R(5,8,8,10,10)	6 001 889	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(5,8,9,9,9)	4 813 685	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 1 quadrupling -> 4,813,684 vertices
R(5,8,9,9,10)	5 829 921	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (9,9 ; 562)] -> 364,370 vertices; then 2 quadruplings -> 5,829,920 vertices
R(5,8,9,10,10)	7 074 689	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 3 quadruplings -> 7,074,688 vertices
R(5,8,10,10,10)	9 342 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 4 quadruplings -> 9,342,464 vertices
R(5,9,9,9,9)	6 472 823	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 6,472,822 vertices
R(5,9,9,9,10)	9 440 657	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 1 quadrupling -> 9,440,656 vertices
R(5,9,9,10,10)	11 507 249	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,876,813 – 3 = 11,507,249; with R(9,9,9,9) >= 719,204
R(5,9,10,10,10)	12 243 485	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(5,10,10,10,10)	16 323 809	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]
R(6,6,6,6,6)	258 506	Published	published value
R(6,6,6,6,7)	352 507	Published bricks + theorems	[template (7,3;16)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 352,506 vertices
R(6,6,6,6,8)	446 508	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,507 vertices

Parameters	Bound (R >=)	Origin	Construction
R(6,6,6,6,9)	587 509	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 587,508 vertices
R(6,6,6,6,10)	681 510	Published bricks + theorems	[template (10,3 ; 30)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 681,509 vertices
R(6,6,6,7,7)	532 053	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]] -> 133,013 vertices; then 1 quadrupling -> 532,052 vertices
R(6,6,6,7,8)	633 833	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 1 quadrupling -> 633,832 vertices (by monotonicity, from (5,6,6,7,8))
R(6,6,6,7,9)	846 017	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 211,504 vertices; then 1 quadrupling -> 846,016 vertices
R(6,6,6,7,10)	920 493	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K3]] -> 230,123 vertices; then 1 quadrupling -> 920,492 vertices
R(6,6,6,8,8)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices (by monotonicity, from (5,6,6,8,8))
R(6,6,6,8,9)	1 259 601	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 2 quadruplings -> 1,259,600 vertices
R(6,6,6,8,10)	1 259 601	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 2 quadruplings -> 1,259,600 vertices (by monotonicity, from (6,6,6,8,9))
R(6,6,6,9,9)	1 504 065	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 376,017 – 3 = 1,504,065; with R(6,6,6,6) >= 23,502
R(6,6,6,9,10)	1 724 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 1 quadrupling -> 1,724,516 vertices (by monotonicity, from (5,6,6,9,10))
R(6,6,6,10,10)	2 102 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 2 quadruplings -> 2,102,000 vertices (by monotonicity, from (5,6,6,10,10))
R(6,6,7,7,7)	709 522	Published bricks + theorems	[template (7,3;16)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 709,521 vertices
R(6,6,7,7,8)	1 070 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 1 quadrupling -> 1,070,920 vertices
R(6,6,7,7,9)	1 259 601	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 2 quadruplings -> 1,259,600 vertices
R(6,6,7,7,10)	1 371 739	Published bricks + theorems	[template (10,3 ; 30)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 1,371,738 vertices
R(6,6,7,8,8)	1 092 225	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices; then 3 quadruplings -> 1,092,224 vertices
R(6,6,7,8,9)	1 702 853	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 425,713 vertices; then 1 quadrupling -> 1,702,852 vertices
R(6,6,7,8,10)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(6,6,7,9,9)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(6,6,7,9,10)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(6,6,7,10,10)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices (by monotonicity, from (6,6,7,9,10))
R(6,6,8,8,8)	1 246 788	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 1,246,787 vertices
R(6,6,8,8,9)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(6,6,8,8,10)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices (by monotonicity, from (6,6,8,8,9))
R(6,6,8,9,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(6,6,8,9,10)	3 517 233	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 2 quadruplings -> 3,517,232 vertices
R(6,6,8,10,10)	3 540 924	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,6,9,9,9)	3 517 233	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 2 quadruplings -> 3,517,232 vertices
R(6,6,9,9,10)	4 729 481	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 1,182,370 vertices; then 1 quadrupling -> 4,729,480 vertices
R(6,6,9,10,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(6,6,10,10,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices (by monotonicity, from (6,6,9,10,10))
R(6,7,7,7,7)	1 070 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 1 quadrupling -> 1,070,920 vertices
R(6,7,7,7,8)	1 266 857	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 1 quadrupling -> 1,266,856 vertices (by monotonicity, from (5,7,7,7,8))
R(6,7,7,7,9)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(6,7,7,7,10)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(6,7,7,8,8)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(6,7,7,8,9)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(6,7,7,8,10)	2 581 889	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 3 quadruplings -> 2,581,888 vertices
R(6,7,7,9,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(6,7,7,9,10)	3 471 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 1 quadrupling -> 3,471,136 vertices (by monotonicity, from (5,7,7,9,10))
R(6,7,7,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(6,7,8,8,8)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(6,7,8,8,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(6,7,8,8,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(6,7,8,9,9)	3 517 233	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 2 quadruplings -> 3,517,232 vertices
R(6,7,8,9,10)	4 389 185	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 274,324 vertices; then 2 quadruplings -> 4,389,184 vertices

Parameters	Bound (R >=)	Origin	Construction
R(6,7,8,10,10)	5 390 849	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 4 quadruplings -> 5,390,848 vertices
R(6,7,9,9,9)	4 729 481	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 1,182,370 vertices; then 1 quadrupling -> 4,729,480 vertices
R(6,7,9,9,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(6,7,9,10,10)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(6,7,10,10,10)	9 342 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 4 quadruplings -> 9,342,464 vertices
R(6,8,8,8,8)	2 113 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 2 quadruplings -> 2,113,312 vertices (by monotonicity, from (5,8,8,8,8))
R(6,8,8,8,9)	3 517 233	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 2 quadruplings -> 3,517,232 vertices
R(6,8,8,8,10)	3 517 233	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 2 quadruplings -> 3,517,232 vertices (by monotonicity, from (6,8,8,8,9))
R(6,8,8,9,9)	4 389 185	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 274,324 vertices; then 2 quadruplings -> 4,389,184 vertices
R(6,8,8,9,10)	5 809 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 3 quadruplings -> 5,809,664 vertices
R(6,8,8,10,10)	6 001 889	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(6,8,9,9,9)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(6,8,9,9,10)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(6,8,9,10,10)	9 342 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 4 quadruplings -> 9,342,464 vertices
R(6,8,10,10,10)	9 342 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 4 quadruplings -> 9,342,464 vertices (by monotonicity, from (5,8,10,10,10))
R(6,9,9,9,9)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(6,9,9,9,10)	9 440 657	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 1 quadrupling -> 9,440,656 vertices (by monotonicity, from (5,9,9,9,10))
R(6,9,9,10,10)	11 937 665	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 746,104 vertices; then 2 quadruplings -> 11,937,664 vertices
R(6,9,10,10,10)	15 952 641	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 4 quadruplings -> 15,952,640 vertices
R(6,10,10,10,10)	16 323 809	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]
R(7,7,7,7,7)	1 423 087	Published	published value
R(7,7,7,7,8)	2 154 221	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 1 quadrupling -> 2,154,220 vertices
R(7,7,7,7,9)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(7,7,7,7,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(7,7,7,8,8)	2 154 221	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 1 quadrupling -> 2,154,220 vertices (by monotonicity, from (7,7,7,7,8))
R(7,7,7,8,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(7,7,7,8,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(7,7,7,9,9)	5 067 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 2 quadruplings -> 5,067,424 vertices
R(7,7,7,9,10)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(7,7,7,10,10)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(7,7,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(7,7,8,8,9)	5 067 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 2 quadruplings -> 5,067,424 vertices
R(7,7,8,8,10)	5 251 521	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 3 quadruplings -> 5,251,520 vertices
R(7,7,8,9,9)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(7,7,8,9,10)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(7,7,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(7,7,9,9,9)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(7,7,9,9,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(7,7,9,10,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(7,7,10,10,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices (by monotonicity, from (7,7,9,10,10))
R(7,8,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices (by monotonicity, from (7,7,8,8,8))
R(7,8,8,8,9)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(7,8,8,8,10)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(7,8,8,9,9)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(7,8,8,9,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(7,8,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices (by monotonicity, from (7,7,8,10,10))

Parameters	Bound (R >=)	Origin	Construction
R(7,8,9,9,9)	9 519 569	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 2,379,892 vertices; then 1 quadrupling -> 9,519,568 vertices
R(7,8,9,9,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(7,8,9,10,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(7,8,10,10,10)	17 524 481	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 4 quadruplings -> 17,524,480 vertices
R(7,9,9,9,9)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(7,9,9,9,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(7,9,9,10,10)	18 006 737	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(7,9,10,10,10)	24 007 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(7,10,10,10,10)	28 298 753	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 4 quadruplings -> 28,298,752 vertices
R(8,8,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices (by monotonicity, from (7,7,8,8,8))
R(8,8,8,8,9)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(8,8,8,8,10)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices (by monotonicity, from (8,8,8,8,9))
R(8,8,8,9,9)	8 453 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 3 quadruplings -> 8,453,248 vertices
R(8,8,8,9,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices (by monotonicity, from (7,8,8,9,10))
R(8,8,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices (by monotonicity, from (7,7,8,10,10))
R(8,8,9,9,9)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(8,8,9,9,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(8,8,9,10,10)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices
R(8,8,10,10,10)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices (by monotonicity, from (8,8,9,10,10))
R(8,9,9,9,9)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(8,9,9,9,10)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices
R(8,9,9,10,10)	24 007 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(8,9,10,10,10)	28 298 753	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 4 quadruplings -> 28,298,752 vertices
R(8,10,10,10,10)	28 298 753	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 4 quadruplings -> 28,298,752 vertices (by monotonicity, from (7,10,10,10,10))
R(9,9,9,9,9)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices
R(9,9,9,9,10)	25 891 289	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 6,472,822 vertices; then 1 quadrupling -> 25,891,288 vertices
R(9,9,9,10,10)	37 762 625	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 2 quadruplings -> 37,762,624 vertices
R(9,9,10,10,10)	46 028 993	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 11,507,249 – 3 = 46,028,993; with R(9,9,9,9) >= 719,204
R(9,10,10,10,10)	48 973 937	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(10,10,10,10,10)	48 973 937	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(3,3,3,3,3,3)	539	Published	published value (explicit sum-free partition of {1..537} into six sets, S(6) >= 537); atom (3,3,3,3,3,3 ; 538)
R(3,3,3,3,3,4)	934	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 933 vertices
R(3,3,3,3,3,5)	1 544	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 1,543 vertices
R(3,3,3,3,3,6)	1 993	Published (deduced)	Robertson formula: 3 x 645 + 61 – 3 = 1,993; with R(3,3,3,5) >= 162, R(3,3,6) >= 61
R(3,3,3,3,3,7)	2 648	Our bricks	[template (3,3,3,3 ; 34)] x [atom (3,3,7 ; 81) (ours)] -> 2,647 vertices
R(3,3,3,3,3,8)	3 275	Our bricks	[template (3,3,3,3 ; 34)] x [atom (3,3,8 ; 100) (ours)] -> 3,274 vertices
R(3,3,3,3,3,9)	4 331	Our bricks	[template (3,3,3,3 ; 34)] x [atom (3,3,9 ; 132) (ours)] -> 4,330 vertices
R(3,3,3,3,3,10)	5 377	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,345 – 3 = 5,377; with R(3,3,7) >= 85
R(3,3,3,3,4,4)	1 850	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [(3,3,4;29)] -> 1,849 vertices
R(3,3,3,3,4,5)	3 153	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [(3,3,4;29)] -> 3,152 vertices
R(3,3,3,3,4,6)	3 800	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,3,6 ; 59) (ours)] -> 3,799 vertices
R(3,3,3,3,4,7)	5 323	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 5,322 vertices
R(3,3,3,3,4,8)	6 723	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 6,722 vertices
R(3,3,3,3,4,9)	8 545	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,3,9 ; 132) (ours)] -> 8,544 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,3,4,10)	10 305	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 3 quadruplings -> 10,304 vertices
R(3,3,3,3,5,5)	4 818	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,3,5 ; 44)] -> 4,817 vertices
R(3,3,3,3,5,6)	6 483	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [atom (3,3,6 ; 59) (ours)] -> 6,482 vertices
R(3,3,3,3,5,7)	8 925	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,3,7 ; 81) (ours)] -> 8,924 vertices
R(3,3,3,3,5,8)	11 115	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 11,114 vertices
R(3,3,3,3,5,9)	14 660	Our bricks + published	[template (3,3,3,3 ; 34)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 14,659 vertices
R(3,3,3,3,5,10)	17 773	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 4,443 vertices; then 1 quadrupling -> 17,772 vertices
R(3,3,3,3,6,6)	11 145	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [Paley-101 (6,6;101)] -> 11,144 vertices
R(3,3,3,3,6,7)	13 453	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 1 quadrupling -> 13,452 vertices
R(3,3,3,3,6,8)	16 689	Our bricks	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 2 quadruplings -> 16,688 vertices
R(3,3,3,3,6,9)	21 569	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 5,393 – 3 = 21,569; with R(3,6,6) >= 338
R(3,3,3,3,6,10)	25 857	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,465 – 3 = 25,857; with R(6,6) >= 102
R(3,3,3,3,7,7)	22 356	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices
R(3,3,3,3,7,8)	26 893	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 1 quadrupling -> 26,892 vertices
R(3,3,3,3,7,9)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(3,3,3,3,7,10)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(3,3,3,3,8,8)	33 377	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 2 quadruplings -> 33,376 vertices
R(3,3,3,3,8,9)	43 073	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 10,769 – 3 = 43,073; with R(3,7,7) >= 674
R(3,3,3,3,8,10)	52 225	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 13,057 – 3 = 52,225; with R(7,7) >= 205
R(3,3,3,3,9,9)	62 316	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [atom (9,9 ; 562)] -> 62,315 vertices
R(3,3,3,3,9,10)	74 081	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices; then 1 quadrupling -> 74,080 vertices
R(3,3,3,3,10,10)	91 305	Published (deduced)	Robertson formula: 3 x 29,489 + 2,841 – 3 = 91,305; with R(3,9,9) >= 1,844, R(3,10,10) >= 2,841
R(3,3,3,4,4,4)	4 166	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (4,4,4 ; 127)] -> 4,165 vertices
R(3,3,3,4,4,5)	5 928	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,4 ; 54)] -> 5,927 vertices
R(3,3,3,4,4,6)	7 381	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 7,380 vertices
R(3,3,3,4,4,7)	10 742	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 10,741 vertices
R(3,3,3,4,4,8)	13 065	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [(3,3,4;29)] -> 3,266 vertices; then 1 quadrupling -> 13,064 vertices
R(3,3,3,4,4,9)	15 630	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,4,9 ; 241)] -> 15,629 vertices
R(3,3,3,4,4,10)	18 849	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [Paley-17 (4,4;17)] -> 1,178 vertices; then 2 quadruplings -> 18,848 vertices
R(3,3,3,4,5,5)	9 480	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,5 ; 86) (ours)] -> 9,479 vertices
R(3,3,3,4,5,6)	12 612	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 12,611 vertices
R(3,3,3,4,5,7)	17 805	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [K2]] -> 17,804 vertices
R(3,3,3,4,5,8)	21 956	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 21,955 vertices
R(3,3,3,4,5,9)	29 242	Our bricks	[template (5,9,3 ; 314) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 29,241 vertices
R(3,3,3,4,5,10)	35 585	Our bricks	[template (5,9,3 ; 314) (ours)] x [(3,3,4;29)] -> 8,896 vertices; then 1 quadrupling -> 35,584 vertices
R(3,3,3,4,6,6)	21 870	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices
R(3,3,3,4,6,7)	26 613	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 1 quadrupling -> 26,612 vertices
R(3,3,3,4,6,8)	30 449	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 2 quadruplings -> 30,448 vertices
R(3,3,3,4,6,9)	37 860	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 37,859 vertices
R(3,3,3,4,6,10)	47 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (6,9 ; 182)] -> 11,794 vertices; then 1 quadrupling -> 47,176 vertices
R(3,3,3,4,7,7)	44 006	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices
R(3,3,3,4,7,8)	53 561	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 1 quadrupling -> 53,560 vertices
R(3,3,3,4,7,9)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices
R(3,3,3,4,7,10)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(3,3,3,4,8,8)	61 153	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 2 quadruplings -> 61,152 vertices
R(3,3,3,4,8,9)	73 813	Published bricks + theorems	[template (8,8,3 ; 650)] x [(3,3,4;29)] -> 18,453 vertices; then 1 quadrupling -> 73,812 vertices
R(3,3,3,4,8,10)	85 169	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (8,8 ; 281)] -> 5,323 vertices; then 2 quadruplings -> 85,168 vertices
R(3,3,3,4,9,9)	119 760	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,4,9,10)	145 977	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 1 quadrupling -> 145,976 vertices
R(3,3,3,4,10,10)	170 593	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (9,9 ; 562)] -> 10,662 vertices; then 2 quadruplings -> 170,592 vertices
R(3,3,3,5,5,5)	16 140	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 16,139 vertices
R(3,3,3,5,5,6)	21 135	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [K2]] -> 21,134 vertices
R(3,3,3,5,5,7)	29 460	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [K2]] -> 29,459 vertices
R(3,3,3,5,5,8)	37 119	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [K2]] -> 37,118 vertices
R(3,3,3,5,5,9)	49 329	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 49,328 vertices
R(3,3,3,5,5,10)	58 637	Our bricks + published	[template (3,3,3,3 ; 34)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 14,659 vertices; then 1 quadrupling -> 58,636 vertices
R(3,3,3,5,6,6)	37 341	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices
R(3,3,3,5,6,7)	44 577	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 1 quadrupling -> 44,576 vertices
R(3,3,3,5,6,8)	53 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 2 quadruplings -> 53,808 vertices
R(3,3,3,5,6,9)	66 753	Our bricks	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 3 quadruplings -> 66,752 vertices
R(3,3,3,5,6,10)	86 273	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 21,569 – 3 = 86,273; with R(3,6,6) >= 338
R(3,3,3,5,7,7)	74 637	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices
R(3,3,3,5,7,8)	89 421	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 1 quadrupling -> 89,420 vertices
R(3,3,3,5,7,9)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,3,3,5,7,10)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,3,3,5,8,8)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,3,3,5,8,9)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,3,3,5,8,10)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(3,3,3,5,9,9)	204 507	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices
R(3,3,3,5,9,10)	249 261	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 1 quadrupling -> 249,260 vertices
R(3,3,3,5,10,10)	296 321	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices; then 2 quadruplings -> 296,320 vertices
R(3,3,3,6,6,6)	44 577	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 1 quadrupling -> 44,576 vertices
R(3,3,3,6,6,7)	59 416	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 59,415 vertices
R(3,3,3,6,6,8)	76 693	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [atom (3,6,6 ; 337)] -> 19,173 vertices; then 1 quadrupling -> 76,692 vertices
R(3,3,3,6,6,9)	96 873	Our bricks	[template (3,8,3 ; 73) (ours)] x [atom (3,6,6 ; 337)] -> 24,218 vertices; then 1 quadrupling -> 96,872 vertices
R(3,3,3,6,6,10)	123 021	Our bricks + published	[template (6,6,3;235)] x [atom (3,3,9 ; 132) (ours)] -> 30,755 vertices; then 1 quadrupling -> 123,020 vertices
R(3,3,3,6,7,7)	89 421	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 1 quadrupling -> 89,420 vertices
R(3,3,3,6,7,8)	110 081	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,6 ; 59) (ours)] -> 27,520 vertices; then 1 quadrupling -> 110,080 vertices
R(3,3,3,6,7,9)	149 361	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 1 quadrupling -> 149,360 vertices
R(3,3,3,6,7,10)	178 305	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 2 quadruplings -> 178,304 vertices
R(3,3,3,6,8,8)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,3,3,6,8,9)	178 305	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 2 quadruplings -> 178,304 vertices
R(3,3,3,6,8,10)	215 233	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 3 quadruplings -> 215,232 vertices
R(3,3,3,6,9,9)	249 261	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 1 quadrupling -> 249,260 vertices
R(3,3,3,6,9,10)	299 441	Our bricks + published	[template (9,9,3 ; 1282)] x [atom (3,3,6 ; 59) (ours)] -> 74,860 vertices; then 1 quadrupling -> 299,440 vertices
R(3,3,3,6,10,10)	363 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [(3,3;5)] -> 5,686 vertices; then 3 quadruplings -> 363,904 vertices
R(3,3,3,7,7,7)	118 867	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 118,866 vertices
R(3,3,3,7,7,8)	176 021	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 1 quadrupling -> 176,020 vertices
R(3,3,3,7,7,9)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,3,3,7,7,10)	247 613	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,9 ; 132) (ours)] -> 61,903 vertices; then 1 quadrupling -> 247,612 vertices
R(3,3,3,7,8,8)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,3,3,7,8,9)	298 545	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 1 quadrupling -> 298,544 vertices
R(3,3,3,7,8,10)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,3,7,9,9)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,3,7,9,10)	479 037	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 1 quadrupling -> 479,036 vertices
R(3,3,3,7,10,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,8,8,8)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(3,3,3,8,8,9)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,3,8,8,10)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(3,3,3,8,9,9)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(3,3,3,8,9,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,3,3,8,10,10)	689 153	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 172,289 – 3 = 689,153; with R(3,7,7) >= 674
R(3,3,3,9,9,9)	534 017	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 4 quadruplings -> 534,016 vertices
R(3,3,3,9,9,10)	818 025	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 1 quadrupling -> 818,024 vertices
R(3,3,3,9,10,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(3,3,3,10,10,10)	1 185 281	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [atom (9,9 ; 562)] -> 18,520 vertices; then 3 quadruplings -> 1,185,280 vertices
R(3,3,4,4,4,4)	8 220	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (4,4,4 ; 127)] -> 8,219 vertices
R(3,3,4,4,4,5)	14 031	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices
R(3,3,4,4,4,6)	15 173	Published bricks + theorems	[template (3,5,3 ; 31)] x [atom (4,4,4 ; 127)] -> 3,793 vertices; then 1 quadrupling -> 15,172 vertices
R(3,3,4,4,4,7)	21 625	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 21,624 vertices
R(3,3,4,4,4,8)	28 813	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [atom (4,4,4 ; 127)] -> 7,203 vertices; then 1 quadrupling -> 28,812 vertices
R(3,3,4,4,4,9)	36 393	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 1 quadrupling -> 36,392 vertices
R(3,3,4,4,4,10)	38 417	Published bricks + theorems	[template (8,3 ; 20)] x [atom (4,4,4 ; 127)] -> 2,401 vertices; then 2 quadruplings -> 38,416 vertices
R(3,3,4,4,5,5)	16 730	Our bricks + published	[template (3,4,4,3 ; 116) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 16,729 vertices
R(3,3,4,4,5,6)	23 709	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,4 ; 54)] -> 5,927 vertices; then 1 quadrupling -> 23,708 vertices
R(3,3,4,4,5,7)	32 877	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (4,4,4 ; 127)] -> 8,219 vertices; then 1 quadrupling -> 32,876 vertices
R(3,3,4,4,5,8)	42 965	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 10,741 vertices; then 1 quadrupling -> 42,964 vertices
R(3,3,4,4,5,9)	56 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,3,4,4,5,10)	66 885	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,4,4 ; 54)] -> 16,721 vertices; then 1 quadrupling -> 66,884 vertices
R(3,3,4,4,6,6)	39 414	Our bricks + published	[template (6,6,3;235)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 39,413 vertices
R(3,3,4,4,6,7)	56 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,3,4,4,6,8)	61 521	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices; then 2 quadruplings -> 61,520 vertices
R(3,3,4,4,6,9)	72 833	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 3 quadruplings -> 72,832 vertices
R(3,3,4,4,6,10)	85 741	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (3,4,4 ; 54)] -> 21,435 vertices; then 1 quadrupling -> 85,740 vertices
R(3,3,4,4,7,7)	79 331	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices
R(3,3,4,4,7,8)	100 661	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 1 quadrupling -> 100,660 vertices
R(3,3,4,4,7,9)	123 809	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 2 quadruplings -> 123,808 vertices
R(3,3,4,4,7,10)	145 569	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 2 quadruplings -> 145,568 vertices
R(3,3,4,4,8,8)	123 809	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 2 quadruplings -> 123,808 vertices
R(3,3,4,4,8,9)	145 569	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 2 quadruplings -> 145,568 vertices
R(3,3,4,4,8,10)	170 641	Published bricks + theorems	[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)] -> 10,665 vertices; then 2 quadruplings -> 170,640 vertices
R(3,3,4,4,9,9)	215 771	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices
R(3,3,4,4,9,10)	273 821	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 1 quadrupling -> 273,820 vertices
R(3,3,4,4,10,10)	336 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 2 quadruplings -> 336,928 vertices
R(3,3,4,5,5,5)	29 410	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (5,5,5 ; 453)] -> 29,409 vertices
R(3,3,4,5,5,6)	37 917	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,5 ; 86) (ours)] -> 9,479 vertices; then 1 quadrupling -> 37,916 vertices
R(3,3,4,5,5,7)	56 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,3,4,5,5,8)	71 217	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [K2]] -> 17,804 vertices; then 1 quadrupling -> 71,216 vertices
R(3,3,4,5,5,9)	87 821	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 21,955 vertices; then 1 quadrupling -> 87,820 vertices
R(3,3,4,5,5,10)	116 965	Our bricks	[template (5,9,3 ; 314) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 29,241 vertices; then 1 quadrupling -> 116,964 vertices
R(3,3,4,5,6,6)	63 093	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [K3]] -> 63,092 vertices
R(3,3,4,5,6,7)	87 477	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 1 quadrupling -> 87,476 vertices
R(3,3,4,5,6,8)	106 449	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 2 quadruplings -> 106,448 vertices
R(3,3,4,5,6,9)	121 793	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 3 quadruplings -> 121,792 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,4,5,6,10)	151 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,5 ; 86) (ours)] -> 9,479 vertices; then 2 quadruplings -> 151,664 vertices
R(3,3,4,5,7,7)	126 918	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices
R(3,3,4,5,7,8)	176 021	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 1 quadrupling -> 176,020 vertices
R(3,3,4,5,7,9)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,3,4,5,7,10)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(3,3,4,5,8,8)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,3,4,5,8,9)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(3,3,4,5,8,10)	295 249	Published bricks + theorems	[template (8,8,3 ; 650)] x [(3,3,4;29)] -> 18,453 vertices; then 2 quadruplings -> 295,248 vertices
R(3,3,4,5,9,9)	346 698	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices
R(3,3,4,5,9,10)	479 037	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 1 quadrupling -> 479,036 vertices
R(3,3,4,5,10,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,3,4,6,6,6)	82 352	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 82,351 vertices
R(3,3,4,6,6,7)	119 877	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 119,876 vertices
R(3,3,4,6,6,8)	154 745	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 1 quadrupling -> 154,744 vertices
R(3,3,4,6,6,9)	184 737	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)] -> 11,546 vertices; then 2 quadruplings -> 184,736 vertices
R(3,3,4,6,6,10)	225 045	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,4,9 ; 241)] -> 56,261 vertices; then 1 quadrupling -> 225,044 vertices
R(3,3,4,6,7,7)	164 749	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 164,748 vertices
R(3,3,4,6,7,8)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,3,4,6,7,9)	254 593	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 2 quadruplings -> 254,592 vertices
R(3,3,4,6,7,10)	322 913	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 2 quadruplings -> 322,912 vertices
R(3,3,4,6,8,8)	254 593	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 2 quadruplings -> 254,592 vertices
R(3,3,4,6,8,9)	349 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 2 quadruplings -> 349,904 vertices
R(3,3,4,6,8,10)	425 793	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 3 quadruplings -> 425,792 vertices
R(3,3,4,6,9,9)	449 149	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 449,148 vertices
R(3,3,4,6,9,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,3,4,6,10,10)	709 633	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (9,9 ; 562)] -> 44,352 vertices; then 2 quadruplings -> 709,632 vertices
R(3,3,4,7,7,7)	239 822	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 239,821 vertices
R(3,3,4,7,7,8)	317 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 1 quadrupling -> 317,320 vertices
R(3,3,4,7,7,9)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(3,3,4,7,7,10)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,3,4,7,8,8)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(3,3,4,7,8,9)	507 669	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 1 quadrupling -> 507,668 vertices
R(3,3,4,7,8,10)	645 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 2 quadruplings -> 645,472 vertices
R(3,3,4,7,9,9)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,4,7,9,10)	863 081	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 1 quadrupling -> 863,080 vertices
R(3,3,4,7,10,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(3,3,4,8,8,8)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,3,4,8,8,9)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,4,8,8,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,4,8,9,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,4,8,9,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(3,3,4,8,10,10)	1 347 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 3 quadruplings -> 1,347,712 vertices
R(3,3,4,9,9,9)	1 032 977	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)] -> 64,561 vertices; then 2 quadruplings -> 1,032,976 vertices
R(3,3,4,9,9,10)	1 386 789	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 1 quadrupling -> 1,386,788 vertices
R(3,3,4,9,10,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,3,4,10,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,3,5,5,5,5)	50 217	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,5,5 ; 453)] -> 50,216 vertices
R(3,3,5,5,5,6)	64 557	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 16,139 vertices; then 1 quadrupling -> 64,556 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,5,5,5,7)	84 537	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [K2]] -> 21,134 vertices; then 1 quadrupling -> 84,536 vertices
R(3,3,5,5,5,8)	117 837	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [K2]] -> 29,459 vertices; then 1 quadrupling -> 117,836 vertices
R(3,3,5,5,5,9)	148 473	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [K2]] -> 37,118 vertices; then 1 quadrupling -> 148,472 vertices
R(3,3,5,5,5,10)	197 313	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 49,328 vertices; then 1 quadrupling -> 197,312 vertices
R(3,3,5,5,6,6)	102 708	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (3,5,5 ; 146) (ours)] -> 102,707 vertices
R(3,3,5,5,6,7)	149 361	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 1 quadrupling -> 149,360 vertices
R(3,3,5,5,6,8)	178 305	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 2 quadruplings -> 178,304 vertices
R(3,3,5,5,6,9)	215 233	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 3 quadruplings -> 215,232 vertices
R(3,3,5,5,6,10)	267 009	Our bricks	[template (3,6,6,3 ; 707) (ours)] x [K2] -> 1,043 vertices; then 4 quadruplings -> 267,008 vertices
R(3,3,5,5,7,7)	205 559	Our bricks + published	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (3,5,5 ; 146) (ours)]] -> 205,558 vertices
R(3,3,5,5,7,8)	298 545	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 1 quadrupling -> 298,544 vertices
R(3,3,5,5,7,9)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,5,5,7,10)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(3,3,5,5,8,8)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,5,5,8,9)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(3,3,5,5,8,10)	534 017	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 4 quadruplings -> 534,016 vertices
R(3,3,5,5,9,9)	560 831	Our bricks + published	[template (5,3;10)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 560,830 vertices
R(3,3,5,5,9,10)	818 025	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 1 quadrupling -> 818,024 vertices
R(3,3,5,5,10,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(3,3,5,6,6,6)	149 361	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 1 quadrupling -> 149,360 vertices
R(3,3,5,6,6,7)	194 934	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 194,933 vertices
R(3,3,5,6,6,8)	251 645	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,6,6 ; 337)] -> 62,911 vertices; then 1 quadrupling -> 251,644 vertices
R(3,3,5,6,6,9)	326 279	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 326,278 vertices
R(3,3,5,6,6,10)	421 201	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,6,6 ; 337)] -> 105,300 vertices; then 1 quadrupling -> 421,200 vertices
R(3,3,5,6,7,7)	298 545	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 1 quadrupling -> 298,544 vertices
R(3,3,5,6,7,8)	363 105	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 1 quadrupling -> 363,104 vertices
R(3,3,5,6,7,9)	440 321	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,6 ; 59) (ours)] -> 27,520 vertices; then 2 quadruplings -> 440,320 vertices
R(3,3,5,6,7,10)	597 441	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 2 quadruplings -> 597,440 vertices
R(3,3,5,6,8,8)	440 321	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,6 ; 59) (ours)] -> 27,520 vertices; then 2 quadruplings -> 440,320 vertices
R(3,3,5,6,8,9)	597 441	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 2 quadruplings -> 597,440 vertices
R(3,3,5,6,8,10)	713 217	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 3 quadruplings -> 713,216 vertices
R(3,3,5,6,9,9)	818 025	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 1 quadrupling -> 818,024 vertices
R(3,3,5,6,9,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(3,3,5,6,10,10)	1 212 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices; then 2 quadruplings -> 1,212,656 vertices
R(3,3,5,7,7,7)	389 975	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 389,974 vertices
R(3,3,5,7,7,8)	507 669	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 1 quadrupling -> 507,668 vertices
R(3,3,5,7,7,9)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,5,7,7,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,5,7,8,8)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,5,7,8,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,5,7,8,10)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,3,5,7,9,9)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,3,5,7,9,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,3,5,7,10,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,3,5,8,8,8)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,5,8,8,9)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,3,5,8,8,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,3,5,8,9,9)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,5,8,9,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,3,5,8,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,3,5,9,9,9)	1 779 538	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 1,779,537 vertices
R(3,3,5,9,9,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,3,5,9,10,10)	3 272 097	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 2 quadruplings -> 3,272,096 vertices
R(3,3,5,10,10,10)	3 988 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 3 quadruplings -> 3,988,160 vertices
R(3,3,6,6,6,6)	243 930	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices
R(3,3,6,6,6,7)	314 901	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 1 quadrupling -> 314,900 vertices
R(3,3,6,6,6,8)	376 017	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 94,005 – 3 = 376,017; with R(6,6,6,6) >= 23,502
R(3,3,6,6,6,9)	424 544	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 424,543 vertices
R(3,3,6,6,6,10)	545 157	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 136,289 vertices; then 1 quadrupling -> 545,156 vertices
R(3,3,6,6,7,7)	490 985	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices
R(3,3,6,6,7,8)	633 833	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 1 quadrupling -> 633,832 vertices
R(3,3,6,6,7,9)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices
R(3,3,6,6,7,10)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices (by monotonicity, from (3,3,6,6,7,9))
R(3,3,6,6,8,8)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices
R(3,3,6,6,8,9)	879 309	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 1 quadrupling -> 879,308 vertices
R(3,3,6,6,8,10)	1 049 937	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,8 ; 281)] -> 65,621 vertices; then 2 quadruplings -> 1,049,936 vertices
R(3,3,6,6,9,9)	1 335 365	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices
R(3,3,6,6,9,10)	1 724 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 1 quadrupling -> 1,724,516 vertices
R(3,3,6,6,10,10)	2 102 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 2 quadruplings -> 2,102,000 vertices
R(3,3,6,7,7,7)	633 833	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 1 quadrupling -> 633,832 vertices
R(3,3,6,7,7,8)	756 833	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 2 quadruplings -> 756,832 vertices
R(3,3,6,7,7,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,3,6,7,7,10)	1 097 297	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]] -> 274,324 vertices; then 1 quadrupling -> 1,097,296 vertices
R(3,3,6,7,8,8)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,3,6,7,8,9)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,3,6,7,8,10)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(3,3,6,7,9,9)	1 724 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 1 quadrupling -> 1,724,516 vertices
R(3,3,6,7,9,10)	2 102 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 2 quadruplings -> 2,102,000 vertices
R(3,3,6,7,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,3,6,8,8,8)	1 049 937	Published bricks + theorems	[template (6,6,3;235)] x [atom (8,8 ; 281)] -> 65,621 vertices; then 2 quadruplings -> 1,049,936 vertices
R(3,3,6,8,8,9)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(3,3,6,8,8,10)	1 761 281	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,6 ; 59) (ours)] -> 27,520 vertices; then 3 quadruplings -> 1,761,280 vertices
R(3,3,6,8,9,9)	2 102 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 2 quadruplings -> 2,102,000 vertices
R(3,3,6,8,9,10)	2 389 761	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 3 quadruplings -> 2,389,760 vertices
R(3,3,6,8,10,10)	2 981 841	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [atome (10,10 ; 797)]
R(3,3,6,9,9,9)	2 389 761	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 3 quadruplings -> 2,389,760 vertices
R(3,3,6,9,9,10)	3 272 097	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 2 quadruplings -> 3,272,096 vertices
R(3,3,6,9,10,10)	3 988 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 3 quadruplings -> 3,988,160 vertices
R(3,3,6,10,10,10)	4 850 625	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)] -> 75,791 vertices; then 3 quadruplings -> 4,850,624 vertices
R(3,3,7,7,7,7)	982 238	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices
R(3,3,7,7,7,8)	1 266 857	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 1 quadrupling -> 1,266,856 vertices
R(3,3,7,7,7,9)	1 517 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 379,493 – 3 = 1,517,969; with R(7,7,7,7) >= 94,874
R(3,3,7,7,7,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,3,7,7,8,8)	1 517 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 379,493 – 3 = 1,517,969; with R(7,7,7,7) >= 94,874
R(3,3,7,7,8,9)	1 769 885	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 1 quadrupling -> 1,769,884 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,7,7,8,10)	2 113 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 2 quadruplings -> 2,113,312 vertices
R(3,3,7,7,9,9)	2 677 838	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices
R(3,3,7,7,9,10)	3 471 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 1 quadrupling -> 3,471,136 vertices
R(3,3,7,7,10,10)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(3,3,7,8,8,8)	1 769 885	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 1 quadrupling -> 1,769,884 vertices
R(3,3,7,8,8,9)	2 113 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 2 quadruplings -> 2,113,312 vertices
R(3,3,7,8,8,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,7,8,9,9)	3 471 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 1 quadrupling -> 3,471,136 vertices
R(3,3,7,8,9,10)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(3,3,7,8,10,10)	4 501 685	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(3,3,7,9,9,9)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(3,3,7,9,9,10)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(3,3,7,9,10,10)	6 001 889	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,3,7,10,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,3,8,8,8,8)	2 113 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 2 quadruplings -> 2,113,312 vertices
R(3,3,8,8,8,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,8,8,8,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(3,3,8,8,9,9)	4 230 929	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 2 quadruplings -> 4,230,928 vertices
R(3,3,8,8,9,10)	4 813 685	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 1 quadrupling -> 4,813,684 vertices
R(3,3,8,8,10,10)	6 001 889	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,3,8,9,9,9)	4 813 685	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 1 quadrupling -> 4,813,684 vertices
R(3,3,8,9,9,10)	5 829 921	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (9,9 ; 562)] -> 364,370 vertices; then 2 quadruplings -> 5,829,920 vertices
R(3,3,8,9,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,3,8,10,10,10)	9 342 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 4 quadruplings -> 9,342,464 vertices
R(3,3,9,9,9,9)	7 283 048	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices
R(3,3,9,9,9,10)	9 440 657	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 1 quadrupling -> 9,440,656 vertices
R(3,3,9,9,10,10)	11 507 249	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 2,876,813 – 3 = 11,507,249; with R(9,9,9,9) >= 719,204
R(3,3,9,10,10,10)	13 088 385	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 3 quadruplings -> 13,088,384 vertices
R(3,3,10,10,10,10)	16 323 809	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]
R(3,4,4,4,4,4)	14 545	Our bricks + published	[template (3,4,4,3 ; 116) (ours)] x [atom (4,4,4 ; 127)] -> 14,544 vertices
R(3,4,4,4,4,5)	22 763	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 22,762 vertices
R(3,4,4,4,4,6)	30 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)] -> 7,582 vertices; then 1 quadrupling -> 30,328 vertices
R(3,4,4,4,4,7)	43 632	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 43,631 vertices
R(3,4,4,4,4,8)	58 145	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)] -> 14,536 vertices; then 1 quadrupling -> 58,144 vertices
R(3,4,4,4,4,9)	61 189	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [atom (4,4,4 ; 127)] -> 15,297 vertices; then 1 quadrupling -> 61,188 vertices
R(3,4,4,4,4,10)	76 377	Our bricks + published	[modele (4,9,3 ; 152)] x [atome (4,4,4 ; 127)] [new template (4,9;151), v20]
R(3,4,4,4,5,5)	37 184	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 37,183 vertices
R(3,4,4,4,5,6)	56 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,4,4,4,5,7)	70 953	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 70,952 vertices
R(3,4,4,4,5,8)	94 565	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)] -> 23,641 vertices; then 1 quadrupling -> 94,564 vertices
R(3,4,4,4,5,9)	118 841	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)] -> 29,710 vertices; then 1 quadrupling -> 118,840 vertices
R(3,4,4,4,5,10)	158 281	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 1 quadrupling -> 158,280 vertices
R(3,4,4,4,6,6)	89 294	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices
R(3,4,4,4,6,7)	118 341	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 1 quadrupling -> 118,340 vertices
R(3,4,4,4,6,8)	121 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)] -> 7,582 vertices; then 2 quadruplings -> 121,312 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,4,4,6,9)	158 281	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 1 quadrupling -> 158,280 vertices
R(3,4,4,4,6,10)	202 833	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)] -> 50,708 vertices; then 1 quadrupling -> 202,832 vertices
R(3,4,4,4,7,7)	178 712	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices
R(3,4,4,4,7,8)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices
R(3,4,4,4,7,9)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices (by monotonicity, from (3,4,4,4,7,8))
R(3,4,4,4,7,10)	273 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices; then 2 quadruplings -> 273,056 vertices
R(3,4,4,4,8,8)	246 253	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices
R(3,4,4,4,8,9)	328 221	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 1 quadrupling -> 328,220 vertices
R(3,4,4,4,8,10)	378 257	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)] -> 23,641 vertices; then 2 quadruplings -> 378,256 vertices
R(3,4,4,4,9,9)	486 062	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices
R(3,4,4,4,9,10)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices
R(3,4,4,4,10,10)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices (by monotonicity, from (3,4,4,4,9,10))
R(3,4,4,5,5,5)	56 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,4,4,5,5,6)	68 265	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)] -> 17,066 vertices; then 1 quadrupling -> 68,264 vertices
R(3,4,4,5,5,7)	94 833	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,4 ; 54)] -> 5,927 vertices; then 2 quadruplings -> 94,832 vertices
R(3,4,4,5,5,8)	131 505	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (4,4,4 ; 127)] -> 8,219 vertices; then 2 quadruplings -> 131,504 vertices
R(3,4,4,5,5,9)	171 857	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 10,741 vertices; then 2 quadruplings -> 171,856 vertices
R(3,4,4,5,5,10)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,4,4,5,6,6)	118 341	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 1 quadrupling -> 118,340 vertices
R(3,4,4,5,6,7)	157 653	Our bricks + published	[template (6,6,3;235)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 39,413 vertices; then 1 quadrupling -> 157,652 vertices
R(3,4,4,5,6,8)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,4,4,5,6,9)	246 081	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices; then 3 quadruplings -> 246,080 vertices
R(3,4,4,5,6,10)	291 329	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 4 quadruplings -> 291,328 vertices
R(3,4,4,5,7,7)	238 193	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 1 quadrupling -> 238,192 vertices
R(3,4,4,5,7,8)	317 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 1 quadrupling -> 317,320 vertices
R(3,4,4,5,7,9)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(3,4,4,5,7,10)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,4,4,5,8,8)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(3,4,4,5,8,9)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,4,4,5,8,10)	633 121	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 2 quadruplings -> 633,120 vertices
R(3,4,4,5,9,9)	647 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 1 quadrupling -> 647,872 vertices
R(3,4,4,5,9,10)	863 081	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 1 quadrupling -> 863,080 vertices
R(3,4,4,5,10,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(3,4,4,6,6,6)	150 712	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (4,6,3 ; 80) (ours)] x [Paley-101 (6,6;101)]] -> 150,711 vertices
R(3,4,4,6,6,7)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,4,4,6,6,8)	276 795	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 276,794 vertices
R(3,4,4,6,6,9)	312 559	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 312,558 vertices
R(3,4,4,6,6,10)	402 044	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,10 ; 91) (ours)]] -> 402,043 vertices
R(3,4,4,6,7,7)	357 173	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 1 quadrupling -> 357,172 vertices
R(3,4,4,6,7,8)	473 361	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 2 quadruplings -> 473,360 vertices
R(3,4,4,6,7,9)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,4,4,6,7,10)	633 121	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 2 quadruplings -> 633,120 vertices
R(3,4,4,6,8,8)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,4,4,6,8,9)	633 121	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 2 quadruplings -> 633,120 vertices
R(3,4,4,6,8,10)	811 329	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)] -> 50,708 vertices; then 2 quadruplings -> 811,328 vertices
R(3,4,4,6,9,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,4,4,6,9,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(3,4,4,6,10,10)	1 347 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 3 quadruplings -> 1,347,712 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,4,7,7,7)	473 361	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 2 quadruplings -> 473,360 vertices
R(3,4,4,7,7,8)	714 845	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 1 quadrupling -> 714,844 vertices
R(3,4,4,7,7,9)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(3,4,4,7,7,10)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices (by monotonicity, from (3,4,4,7,7,9))
R(3,4,4,7,8,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(3,4,4,7,8,9)	985 009	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 1 quadrupling -> 985,008 vertices
R(3,4,4,7,8,10)	1 312 881	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 2 quadruplings -> 1,312,880 vertices
R(3,4,4,7,9,9)	1 312 881	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 2 quadruplings -> 1,312,880 vertices
R(3,4,4,7,9,10)	1 944 245	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 1 quadrupling -> 1,944,244 vertices
R(3,4,4,7,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(3,4,4,8,8,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices (by monotonicity, from (3,4,4,7,8,8))
R(3,4,4,8,8,9)	1 312 881	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 2 quadruplings -> 1,312,880 vertices
R(3,4,4,8,8,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,4,4,8,9,9)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,4,4,8,9,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(3,4,4,8,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices (by monotonicity, from (3,4,4,7,10,10))
R(3,4,4,9,9,9)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(3,4,4,9,9,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(3,4,4,9,10,10)	3 452 321	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 2 quadruplings -> 3,452,320 vertices
R(3,4,4,10,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(3,4,5,5,5,5)	85 703	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 85,702 vertices
R(3,4,5,5,5,6)	117 637	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (5,5,5 ; 453)] -> 29,409 vertices; then 1 quadrupling -> 117,636 vertices
R(3,4,5,5,5,7)	164 267	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 164,266 vertices
R(3,4,5,5,5,8)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,4,5,5,5,9)	284 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [K2]] -> 17,804 vertices; then 2 quadruplings -> 284,864 vertices
R(3,4,5,5,5,10)	351 281	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 21,955 vertices; then 2 quadruplings -> 351,280 vertices
R(3,4,5,5,6,6)	186 964	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 186,963 vertices
R(3,4,5,5,6,7)	252 369	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [K3]] -> 63,092 vertices; then 1 quadrupling -> 252,368 vertices
R(3,4,5,5,6,8)	349 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 2 quadruplings -> 349,904 vertices
R(3,4,5,5,6,9)	425 793	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 3 quadruplings -> 425,792 vertices
R(3,4,5,5,6,10)	487 169	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 4 quadruplings -> 487,168 vertices
R(3,4,5,5,7,7)	375 026	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 375,025 vertices
R(3,4,5,5,7,8)	507 669	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 1 quadrupling -> 507,668 vertices
R(3,4,5,5,7,9)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,4,5,5,7,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,4,5,5,8,8)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,4,5,5,8,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,4,5,5,8,10)	978 433	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 4 quadruplings -> 978,432 vertices
R(3,4,5,5,9,9)	1 045 346	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 1,045,345 vertices
R(3,4,5,5,9,10)	1 386 789	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 1 quadrupling -> 1,386,788 vertices
R(3,4,5,5,10,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,4,5,6,6,6)	257 549	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Paley-101 (6,6;101)]] -> 257,548 vertices
R(3,4,5,6,6,7)	357 173	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 1 quadrupling -> 357,172 vertices
R(3,4,5,6,6,8)	479 505	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 119,876 vertices; then 1 quadrupling -> 479,504 vertices
R(3,4,5,6,6,9)	618 977	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 2 quadruplings -> 618,976 vertices
R(3,4,5,6,6,10)	738 945	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)] -> 11,546 vertices; then 3 quadruplings -> 738,944 vertices
R(3,4,5,6,7,7)	516 614	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 516,613 vertices
R(3,4,5,6,7,8)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,5,6,7,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,4,5,6,7,10)	1 018 369	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 3 quadruplings -> 1,018,368 vertices
R(3,4,5,6,8,8)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,4,5,6,8,9)	1 018 369	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 3 quadruplings -> 1,018,368 vertices
R(3,4,5,6,8,10)	1 399 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 3 quadruplings -> 1,399,616 vertices
R(3,4,5,6,9,9)	1 440 014	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 1,440,013 vertices
R(3,4,5,6,9,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,4,5,6,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,4,5,7,7,7)	715 639	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)]] -> 715,638 vertices
R(3,4,5,7,7,8)	959 285	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 239,821 vertices; then 1 quadrupling -> 959,284 vertices
R(3,4,5,7,7,9)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(3,4,5,7,7,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,4,5,7,8,8)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(3,4,5,7,8,9)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,4,5,7,8,10)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(3,4,5,7,9,9)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(3,4,5,7,9,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,4,5,7,10,10)	3 452 321	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 2 quadruplings -> 3,452,320 vertices
R(3,4,5,8,8,8)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,4,5,8,8,9)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(3,4,5,8,8,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,4,5,8,9,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,4,5,8,9,10)	3 452 321	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 2 quadruplings -> 3,452,320 vertices
R(3,4,5,8,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(3,4,5,9,9,9)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(3,4,5,9,9,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(3,4,5,9,10,10)	5 547 153	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 2 quadruplings -> 5,547,152 vertices
R(3,4,5,10,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,4,6,6,6,6)	446 504	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices
R(3,4,6,6,6,7)	532 053	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]] -> 133,013 vertices; then 1 quadrupling -> 532,052 vertices
R(3,4,6,6,6,8)	618 977	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 2 quadruplings -> 618,976 vertices
R(3,4,6,6,6,9)	806 630	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 806,629 vertices
R(3,4,6,6,6,10)	920 493	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K3]] -> 230,123 vertices; then 1 quadrupling -> 920,492 vertices
R(3,4,6,6,7,7)	898 723	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices
R(3,4,6,6,7,8)	1 070 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 1 quadrupling -> 1,070,920 vertices
R(3,4,6,6,7,9)	1 113 404	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (7,9 ; 251)]] -> 1,113,403 vertices
R(3,4,6,6,7,10)	1 291 649	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [atom (3,6,6 ; 337)] -> 20,182 vertices; then 3 quadruplings -> 1,291,648 vertices
R(3,4,6,6,8,8)	1 246 784	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 1,246,783 vertices
R(3,4,6,6,8,9)	1 477 413	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [K3]] -> 369,353 vertices; then 1 quadrupling -> 1,477,412 vertices
R(3,4,6,6,8,10)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(3,4,6,6,9,9)	2 496 110	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices
R(3,4,6,6,9,10)	2 923 533	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 1 quadrupling -> 2,923,532 vertices
R(3,4,6,6,10,10)	3 540 920	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(3,4,6,7,7,7)	1 070 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 1 quadrupling -> 1,070,920 vertices
R(3,4,6,7,7,8)	1 428 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 2 quadruplings -> 1,428,688 vertices
R(3,4,6,7,7,9)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(3,4,6,7,7,10)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,6,7,8,8)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(3,4,6,7,8,9)	2 066 453	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 516,613 vertices; then 1 quadrupling -> 2,066,452 vertices
R(3,4,6,7,8,10)	2 581 889	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [K2]] -> 40,342 vertices; then 3 quadruplings -> 2,581,888 vertices
R(3,4,6,7,9,9)	2 923 533	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 1 quadrupling -> 2,923,532 vertices
R(3,4,6,7,9,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,4,6,7,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(3,4,6,8,8,8)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(3,4,6,8,8,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,4,6,8,8,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(3,4,6,8,9,9)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,4,6,8,9,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(3,4,6,8,10,10)	5 390 849	Published bricks + theorems	[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)] -> 21,058 vertices; then 4 quadruplings -> 5,390,848 vertices
R(3,4,6,9,9,9)	4 416 022	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)]] -> 4,416,021 vertices
R(3,4,6,9,9,10)	5 760 053	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 1,440,013 vertices; then 1 quadrupling -> 5,760,052 vertices
R(3,4,6,9,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,4,6,10,10,10)	9 342 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 4 quadruplings -> 9,342,464 vertices
R(3,4,7,7,7,7)	1 802 572	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices
R(3,4,7,7,7,8)	2 154 221	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 1 quadrupling -> 2,154,220 vertices
R(3,4,7,7,7,9)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(3,4,7,7,7,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,4,7,7,8,8)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(3,4,7,7,8,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,4,7,7,8,10)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(3,4,7,7,9,9)	5 024 212	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices
R(3,4,7,7,9,10)	5 884 541	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 1 quadrupling -> 5,884,540 vertices
R(3,4,7,7,10,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(3,4,7,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,4,7,8,8,9)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(3,4,7,8,8,10)	5 251 521	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 3 quadruplings -> 5,251,520 vertices
R(3,4,7,8,9,9)	5 884 541	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 1 quadrupling -> 5,884,540 vertices
R(3,4,7,8,9,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(3,4,7,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(3,4,7,9,9,9)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(3,4,7,9,9,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(3,4,7,9,10,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,4,7,10,10,10)	13 809 281	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 3 quadruplings -> 13,809,280 vertices
R(3,4,8,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices (by monotonicity, from (3,4,7,8,8,8))
R(3,4,8,8,8,9)	5 251 521	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 3 quadruplings -> 5,251,520 vertices
R(3,4,8,8,8,10)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(3,4,8,8,9,9)	6 923 015	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 6,923,014 vertices
R(3,4,8,8,9,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(3,4,8,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices (by monotonicity, from (3,4,7,8,10,10))
R(3,4,8,9,9,9)	8 122 689	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 3 quadruplings -> 8,122,688 vertices
R(3,4,8,9,9,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,4,8,9,10,10)	13 809 281	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 3 quadruplings -> 13,809,280 vertices
R(3,4,8,10,10,10)	17 524 481	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 4 quadruplings -> 17,524,480 vertices
R(3,4,9,9,9,9)	13 664 842	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,9,9,9,10)	16 004 501	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]] -> 4,001,125 vertices; then 1 quadrupling -> 16,004,500 vertices
R(3,4,9,9,10,10)	19 384 507	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,4,9,10,10,10)	22 188 609	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 3 quadruplings -> 22,188,608 vertices
R(3,4,10,10,10,10)	28 298 753	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 4 quadruplings -> 28,298,752 vertices
R(3,5,5,5,5,5)	141 974	Our bricks + published	[template (5,5,5,3 ; 977)] x [atom (3,5,5 ; 146) (ours)] -> 141,973 vertices
R(3,5,5,5,5,6)	200 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,5,5 ; 453)] -> 50,216 vertices; then 1 quadrupling -> 200,864 vertices
R(3,5,5,5,5,7)	267 116	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 267,115 vertices
R(3,5,5,5,5,8)	338 413	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (5,5,5 ; 453)] -> 84,603 vertices; then 1 quadrupling -> 338,412 vertices
R(3,5,5,5,5,9)	471 345	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [K2]] -> 29,459 vertices; then 2 quadruplings -> 471,344 vertices
R(3,5,5,5,5,10)	593 889	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [K2]] -> 37,118 vertices; then 2 quadruplings -> 593,888 vertices
R(3,5,5,5,6,6)	334 254	Published bricks + theorems	[template (6,6,3;235)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 334,253 vertices
R(3,5,5,5,6,7)	423 477	Published bricks + theorems	[template (6,6,3;235)] x [atom (5,5,5 ; 453)] -> 105,869 vertices; then 1 quadrupling -> 423,476 vertices
R(3,5,5,5,6,8)	597 441	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 2 quadruplings -> 597,440 vertices
R(3,5,5,5,6,9)	713 217	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 3 quadruplings -> 713,216 vertices
R(3,5,5,5,6,10)	860 929	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 4 quadruplings -> 860,928 vertices
R(3,5,5,5,7,7)	672 791	Published bricks + theorems	[template (7,7,3;472)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 672,790 vertices
R(3,5,5,5,7,8)	852 377	Published bricks + theorems	[template (7,7,3;472)] x [atom (5,5,5 ; 453)] -> 213,094 vertices; then 1 quadrupling -> 852,376 vertices
R(3,5,5,5,7,9)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,5,5,5,7,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,5,5,5,8,8)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,5,5,5,8,9)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,5,5,5,8,10)	1 721 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 4 quadruplings -> 1,721,088 vertices
R(3,5,5,5,9,9)	1 829 831	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 1,829,830 vertices
R(3,5,5,5,9,10)	2 318 297	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)] -> 579,574 vertices; then 1 quadrupling -> 2,318,296 vertices
R(3,5,5,5,10,10)	3 272 097	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 2 quadruplings -> 3,272,096 vertices
R(3,5,5,6,6,6)	423 477	Published bricks + theorems	[template (6,6,3;235)] x [atom (5,5,5 ; 453)] -> 105,869 vertices; then 1 quadrupling -> 423,476 vertices
R(3,5,5,6,6,7)	597 441	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 2 quadruplings -> 597,440 vertices
R(3,5,5,6,6,8)	779 733	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 194,933 vertices; then 1 quadrupling -> 779,732 vertices
R(3,5,5,6,6,9)	1 006 577	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,6,6 ; 337)] -> 62,911 vertices; then 2 quadruplings -> 1,006,576 vertices
R(3,5,5,6,6,10)	1 305 113	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 326,278 vertices; then 1 quadrupling -> 1,305,112 vertices
R(3,5,5,6,7,7)	852 377	Published bricks + theorems	[template (7,7,3;472)] x [atom (5,5,5 ; 453)] -> 213,094 vertices; then 1 quadrupling -> 852,376 vertices
R(3,5,5,6,7,8)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,5,5,6,7,9)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(3,5,5,6,7,10)	1 740 225	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 3 quadruplings -> 1,740,224 vertices
R(3,5,5,6,8,8)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(3,5,5,6,8,9)	1 740 225	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 3 quadruplings -> 1,740,224 vertices
R(3,5,5,6,8,10)	2 389 761	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 3 quadruplings -> 2,389,760 vertices
R(3,5,5,6,9,9)	2 389 761	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 3 quadruplings -> 2,389,760 vertices
R(3,5,5,6,9,10)	3 272 097	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 2 quadruplings -> 3,272,096 vertices
R(3,5,5,6,10,10)	3 988 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 3 quadruplings -> 3,988,160 vertices
R(3,5,5,7,7,7)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,5,5,7,7,8)	1 559 897	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 389,974 vertices; then 1 quadrupling -> 1,559,896 vertices
R(3,5,5,7,7,9)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(3,5,5,7,7,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,5,5,7,8,8)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(3,5,5,7,8,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,5,5,7,8,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,5,7,9,9)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(3,5,5,7,9,10)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(3,5,5,7,10,10)	5 722 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 4 quadruplings -> 5,722,880 vertices
R(3,5,5,8,8,8)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,5,5,8,8,9)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(3,5,5,8,8,10)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(3,5,5,8,9,9)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(3,5,5,8,9,10)	5 722 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 4 quadruplings -> 5,722,880 vertices
R(3,5,5,8,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,5,5,9,9,9)	5 722 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 4 quadruplings -> 5,722,880 vertices
R(3,5,5,9,9,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,5,5,9,10,10)	9 342 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 4 quadruplings -> 9,342,464 vertices
R(3,5,5,10,10,10)	13 088 385	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 3 quadruplings -> 13,088,384 vertices
R(3,5,6,6,6,6)	708 521	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 708,520 vertices
R(3,5,6,6,6,7)	975 717	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 1 quadrupling -> 975,716 vertices
R(3,5,6,6,6,8)	1 259 601	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 2 quadruplings -> 1,259,600 vertices
R(3,5,6,6,6,9)	1 504 065	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 376,017 – 3 = 1,504,065; with R(6,6,6,6) >= 23,502
R(3,5,6,6,6,10)	1 698 173	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 424,543 vertices; then 1 quadrupling -> 1,698,172 vertices
R(3,5,6,6,7,7)	1 426 118	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices
R(3,5,6,6,7,8)	1 963 937	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 1 quadrupling -> 1,963,936 vertices
R(3,5,6,6,7,9)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(3,5,6,6,7,10)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,5,6,6,8,8)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(3,5,6,6,8,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,5,6,6,8,10)	3 517 233	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 2 quadruplings -> 3,517,232 vertices
R(3,5,6,6,9,9)	3 941 234	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 3,941,233 vertices
R(3,5,6,6,9,10)	5 341 457	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 1 quadrupling -> 5,341,456 vertices
R(3,5,6,6,10,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(3,5,6,7,7,7)	1 963 937	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 1 quadrupling -> 1,963,936 vertices
R(3,5,6,7,7,8)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(3,5,6,7,7,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,5,6,7,7,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,5,6,7,8,8)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,5,6,7,8,9)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,5,6,7,8,10)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(3,5,6,7,9,9)	5 341 457	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 1 quadrupling -> 5,341,456 vertices
R(3,5,6,7,9,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(3,5,6,7,10,10)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(3,5,6,8,8,8)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,5,6,8,8,9)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(3,5,6,8,8,10)	5 809 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 3 quadruplings -> 5,809,664 vertices
R(3,5,6,8,9,9)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(3,5,6,8,9,10)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(3,5,6,8,10,10)	9 559 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 4 quadruplings -> 9,559,040 vertices
R(3,5,6,9,9,9)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(3,5,6,9,9,10)	9 559 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 4 quadruplings -> 9,559,040 vertices
R(3,5,6,9,10,10)	13 088 385	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 3 quadruplings -> 13,088,384 vertices
R(3,5,6,10,10,10)	15 952 641	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 4 quadruplings -> 15,952,640 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,7,7,7,7)	2 850 422	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 2,850,421 vertices
R(3,5,7,7,7,8)	3 928 949	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 1 quadrupling -> 3,928,948 vertices
R(3,5,7,7,7,9)	5 067 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 2 quadruplings -> 5,067,424 vertices
R(3,5,7,7,7,10)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,5,7,7,8,8)	5 067 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 2 quadruplings -> 5,067,424 vertices
R(3,5,7,7,8,9)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,5,7,7,8,10)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(3,5,7,7,9,9)	7 932 974	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices
R(3,5,7,7,9,10)	10 711 349	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 1 quadrupling -> 10,711,348 vertices
R(3,5,7,7,10,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,5,7,8,8,8)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,5,7,8,8,9)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(3,5,7,8,8,10)	8 453 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 3 quadruplings -> 8,453,248 vertices
R(3,5,7,8,9,9)	10 711 349	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 1 quadrupling -> 10,711,348 vertices
R(3,5,7,8,9,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,5,7,8,10,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,5,7,9,9,9)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,5,7,9,9,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,5,7,9,10,10)	19 106 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 4 quadruplings -> 19,106,816 vertices
R(3,5,7,10,10,10)	24 007 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,5,8,8,8,8)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(3,5,8,8,8,9)	8 453 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 3 quadruplings -> 8,453,248 vertices
R(3,5,8,8,8,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,5,8,8,9,9)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,5,8,8,9,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,5,8,8,10,10)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices
R(3,5,8,9,9,9)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,5,8,9,9,10)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices
R(3,5,8,9,10,10)	24 007 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,5,8,10,10,10)	30 658 305	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 4 quadruplings -> 30,658,304 vertices
R(3,5,9,9,9,9)	21 576 074	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 21,576,073 vertices
R(3,5,9,9,9,10)	29 132 189	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 1 quadrupling -> 29,132,188 vertices
R(3,5,9,9,10,10)	37 762 625	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 2 quadruplings -> 37,762,624 vertices
R(3,5,9,10,10,10)	46 028 993	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 11,507,249 – 3 = 46,028,993; with R(9,9,9,9) >= 719,204
R(3,5,10,10,10,10)	52 353 537	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 4 quadruplings -> 52,353,536 vertices
R(3,6,6,6,6,6)	916 517	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 916,516 vertices
R(3,6,6,6,6,7)	1 339 522	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,339,521 vertices
R(3,6,6,6,6,8)	1 692 027	Our bricks	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,692,026 vertices
R(3,6,6,6,6,9)	1 968 109	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 1,968,108 vertices
R(3,6,6,6,6,10)	2 350 033	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 587,508 vertices; then 1 quadrupling -> 2,350,032 vertices
R(3,6,6,6,7,7)	1 844 756	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 1,844,755 vertices
R(3,6,6,6,7,8)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(3,6,6,6,7,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,6,6,6,7,10)	3 384 065	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 211,504 vertices; then 2 quadruplings -> 3,384,064 vertices
R(3,6,6,6,8,8)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,6,6,6,8,9)	3 902 865	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 2 quadruplings -> 3,902,864 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,6,6,6,8,10)	5 038 401	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 3 quadruplings -> 5,038,400 vertices
R(3,6,6,6,9,9)	5 123 603	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 5,123,602 vertices
R(3,6,6,6,9,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(3,6,6,6,10,10)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(3,6,6,7,7,7)	2 696 179	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 2,696,178 vertices
R(3,6,6,7,7,8)	3 594 889	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 1 quadrupling -> 3,594,888 vertices
R(3,6,6,7,7,9)	4 283 681	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 2 quadruplings -> 4,283,680 vertices
R(3,6,6,7,7,10)	5 038 401	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 3 quadruplings -> 5,038,400 vertices
R(3,6,6,7,8,8)	4 283 681	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 2 quadruplings -> 4,283,680 vertices
R(3,6,6,7,8,9)	5 704 469	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 1 quadrupling -> 5,704,468 vertices
R(3,6,6,7,8,10)	6 811 409	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 425,713 vertices; then 2 quadruplings -> 6,811,408 vertices
R(3,6,6,7,9,9)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(3,6,6,7,9,10)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,6,6,7,10,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,6,6,8,8,8)	4 724 667	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 4,724,666 vertices
R(3,6,6,8,8,9)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(3,6,6,8,8,10)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,6,6,8,9,9)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,6,6,8,9,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,6,6,8,10,10)	14 068 929	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 3 quadruplings -> 14,068,928 vertices
R(3,6,6,9,9,9)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,6,6,9,9,10)	15 764 933	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 3,941,233 vertices; then 1 quadrupling -> 15,764,932 vertices
R(3,6,6,9,10,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(3,6,6,10,10,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(3,6,7,7,7,7)	3 700 025	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 3,700,024 vertices
R(3,6,7,7,7,8)	5 067 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 2 quadruplings -> 5,067,424 vertices
R(3,6,7,7,7,9)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,6,7,7,7,10)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(3,6,7,7,8,8)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 – 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,6,7,7,8,9)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(3,6,7,7,8,10)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,6,7,7,9,9)	10 312 865	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 10,312,864 vertices
R(3,6,7,7,9,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,6,7,7,10,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,6,7,8,8,8)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(3,6,7,8,8,9)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,6,7,8,8,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,6,7,8,9,9)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,6,7,8,9,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,6,7,8,10,10)	18 006 737	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(3,6,7,9,9,9)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,6,7,9,9,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(3,6,7,9,10,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(3,6,7,10,10,10)	33 632 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 4 quadruplings -> 33,632,000 vertices
R(3,6,8,8,8,8)	8 453 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 3 quadruplings -> 8,453,248 vertices
R(3,6,8,8,8,9)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,6,8,8,8,10)	14 068 929	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 3 quadruplings -> 14,068,928 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,6,8,8,9,9)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,6,8,8,9,10)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices
R(3,6,8,8,10,10)	24 007 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,6,8,9,9,9)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(3,6,8,9,9,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(3,6,8,9,10,10)	33 632 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 4 quadruplings -> 33,632,000 vertices
R(3,6,8,10,10,10)	37 369 857	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 5 quadruplings -> 37,369,856 vertices
R(3,6,9,9,9,9)	28 048 895	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 28,048,894 vertices
R(3,6,9,9,9,10)	37 762 625	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 2 quadruplings -> 37,762,624 vertices
R(3,6,9,9,10,10)	46 028 993	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 11,507,249 – 3 = 46,028,993; with R(9,9,9,9) >= 719,204
R(3,6,9,10,10,10)	52 353 537	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 4 quadruplings -> 52,353,536 vertices
R(3,6,10,10,10,10)	65 295 233	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]
R(3,7,7,7,7,7)	5 407 726	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 5,407,725 vertices
R(3,7,7,7,7,8)	7 210 285	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 1 quadrupling -> 7,210,284 vertices
R(3,7,7,7,7,9)	8 616 881	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 2 quadruplings -> 8,616,880 vertices
R(3,7,7,7,7,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,7,7,7,8,8)	8 616 881	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 2 quadruplings -> 8,616,880 vertices
R(3,7,7,7,8,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,7,7,7,8,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,7,7,7,9,9)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(3,7,7,7,9,10)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,7,7,7,10,10)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,7,7,8,8,8)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,7,7,8,8,9)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(3,7,7,8,8,10)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,7,7,8,9,9)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,7,7,8,9,10)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,7,7,8,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,7,7,9,9,9)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,7,7,9,9,10)	31 731 893	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 1 quadrupling -> 31,731,892 vertices
R(3,7,7,9,10,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,7,7,10,10,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,7,8,8,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,7,8,8,8,9)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,7,8,8,8,10)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,7,8,8,9,9)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,7,8,8,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,7,8,8,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,7,8,9,9,9)	31 731 893	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 1 quadrupling -> 31,731,892 vertices
R(3,7,8,9,9,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,7,8,9,10,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,7,8,10,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,7,9,9,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,7,9,9,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,7,9,9,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,7,9,10,10,10)	77 538 025	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,7,10,10,10,10)	96 030 209	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,8,8,8,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices (by monotonicity, from (3,7,8,8,8,8))
R(3,8,8,8,8,9)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,8,8,8,8,10)	28 318 145	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 3 quadruplings -> 28,318,144 vertices
R(3,8,8,8,9,9)	28 318 145	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 3 quadruplings -> 28,318,144 vertices
R(3,8,8,8,9,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,8,8,8,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices (by monotonicity, from (3,7,8,8,10,10))
R(3,8,8,9,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,8,8,9,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,8,8,9,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,8,8,10,10,10)	77 018 945	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 3 quadruplings -> 77,018,944 vertices
R(3,8,9,9,9,9)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,8,9,9,9,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,8,9,9,10,10)	77 018 945	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 3 quadruplings -> 77,018,944 vertices
R(3,8,9,10,10,10)	96 030 209	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,8,10,10,10,10)	113 195 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 110,542 vertices; then 5 quadruplings -> 113,195,008 vertices
R(3,9,9,9,9,9)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,9,9,9,9,10)	86 304 293	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 21,576,073 vertices; then 1 quadrupling -> 86,304,292 vertices
R(3,9,9,9,10,10)	116 528 753	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 2 quadruplings -> 116,528,752 vertices
R(3,9,9,10,10,10)	151 050 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 3 quadruplings -> 151,050,496 vertices
R(3,9,10,10,10,10)	184 115 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 46,028,993 – 3 = 184,115,969; with R(9,9,9,9) >= 719,204
R(3,10,10,10,10,10)	195 895 745	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]
R(4,4,4,4,4,4)	32 006	Published	published value
R(4,4,4,4,4,5)	37 943	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 37,942 vertices
R(4,4,4,4,4,6)	49 962	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 49,961 vertices
R(4,4,4,4,4,7)	72 727	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 72,726 vertices
R(4,4,4,4,4,8)	79 314	Published bricks + theorems	[template (8,3 ; 20)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 79,313 vertices
R(4,4,4,4,4,9)	104 359	Published bricks + theorems	[template (9,3 ; 26)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 104,358 vertices
R(4,4,4,4,4,10)	121 056	Published bricks + theorems	[template (10,3 ; 30)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 121,055 vertices
R(4,4,4,4,5,5)	68 243	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,3;10)] x [atom (4,4,4 ; 127)]] -> 68,242 vertices
R(4,4,4,4,5,6)	91 049	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 22,762 vertices; then 1 quadrupling -> 91,048 vertices
R(4,4,4,4,5,7)	130 820	Our bricks	[template (5,3;10)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 130,819 vertices
R(4,4,4,4,5,8)	174 525	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 43,631 vertices; then 1 quadrupling -> 174,524 vertices
R(4,4,4,4,5,9)	232 577	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)] -> 14,536 vertices; then 2 quadruplings -> 232,576 vertices
R(4,4,4,4,5,10)	244 753	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [atom (4,4,4 ; 127)] -> 15,297 vertices; then 2 quadruplings -> 244,752 vertices
R(4,4,4,4,6,6)	147 990	Published bricks + theorems	[template (6,6,3;235)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 147,989 vertices
R(4,4,4,4,6,7)	159 897	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,3;12)] x [atom (4,4,4 ; 127)]] -> 159,896 vertices
R(4,4,4,4,6,8)	189 742	Our bricks + published	[template (8,3 ; 20)] x [[template (4,6,3 ; 80) (ours)] x [atom (4,4,4 ; 127)]] -> 189,741 vertices
R(4,4,4,4,6,9)	272 969	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,3;10)] x [atom (4,4,4 ; 127)]] -> 68,242 vertices; then 1 quadrupling -> 272,968 vertices
R(4,4,4,4,6,10)	289 604	Our bricks + published	[template (10,3 ; 30)] x [[template (4,6,3 ; 80) (ours)] x [atom (4,4,4 ; 127)]] -> 289,603 vertices
R(4,4,4,4,7,7)	297 875	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices
R(4,4,4,4,7,8)	297 875	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices (by monotonicity, from (4,4,4,4,7,7))
R(4,4,4,4,7,9)	364 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 22,762 vertices; then 2 quadruplings -> 364,192 vertices
R(4,4,4,4,7,10)	421 525	Our bricks + published	[template (10,3 ; 30)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 421,524 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,4,8,8)	410 450	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 410,449 vertices
R(4,4,4,4,8,9)	523 277	Our bricks	[template (5,3;10)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 130,819 vertices; then 1 quadrupling -> 523,276 vertices
R(4,4,4,4,8,10)	523 277	Our bricks + published	[template (5,3;10)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 130,819 vertices; then 1 quadrupling -> 523,276 vertices (by monotonicity, from (4,4,4,4,8,9))
R(4,4,4,4,9,9)	810 155	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 810,154 vertices
R(4,4,4,4,9,10)	930 305	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)] -> 14,536 vertices; then 3 quadruplings -> 930,304 vertices
R(4,4,4,4,10,10)	1 007 574	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,5,5,5)	123 430	Published bricks + theorems	[template (5,5,5,3 ; 977)] x [atom (4,4,4 ; 127)] -> 123,429 vertices
R(4,4,4,5,5,6)	153 590	Our bricks + published	[template (5,3;10)] x [[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)]] -> 153,589 vertices
R(4,4,4,5,5,7)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(4,4,4,5,5,8)	283 809	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 70,952 vertices; then 1 quadrupling -> 283,808 vertices
R(4,4,4,5,5,9)	378 257	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)] -> 23,641 vertices; then 2 quadruplings -> 378,256 vertices
R(4,4,4,5,5,10)	475 361	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)] -> 29,710 vertices; then 2 quadruplings -> 475,360 vertices
R(4,4,4,5,6,6)	266 261	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices
R(4,4,4,5,6,7)	357 173	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 1 quadrupling -> 357,172 vertices
R(4,4,4,5,6,8)	473 361	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 2 quadruplings -> 473,360 vertices
R(4,4,4,5,6,9)	493 717	Published bricks + theorems	[template (5,5,5,3 ; 977)] x [atom (4,4,4 ; 127)] -> 123,429 vertices; then 1 quadrupling -> 493,716 vertices
R(4,4,4,5,6,10)	633 121	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 2 quadruplings -> 633,120 vertices
R(4,4,4,5,7,7)	535 928	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices
R(4,4,4,5,7,8)	714 845	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 1 quadrupling -> 714,844 vertices
R(4,4,4,5,7,9)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(4,4,4,5,7,10)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices (by monotonicity, from (4,4,4,5,7,9))
R(4,4,4,5,8,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(4,4,4,5,8,9)	985 009	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 1 quadrupling -> 985,008 vertices
R(4,4,4,5,8,10)	1 312 881	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 2 quadruplings -> 1,312,880 vertices
R(4,4,4,5,9,9)	1 457 708	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices
R(4,4,4,5,9,10)	1 944 245	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 1 quadrupling -> 1,944,244 vertices
R(4,4,4,5,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(4,4,4,6,6,6)	325 430	Published bricks + theorems	[template (6,3;12)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 325,429 vertices
R(4,4,4,6,6,7)	443 767	Published bricks + theorems	[template (7,3;16)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 443,766 vertices
R(4,4,4,6,6,8)	562 104	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,103 vertices
R(4,4,4,6,6,9)	739 609	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 739,608 vertices
R(4,4,4,6,6,10)	857 946	Published bricks + theorems	[template (10,3 ; 30)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 857,945 vertices
R(4,4,4,6,7,7)	655 023	Published bricks + theorems	[template (6,3;12)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 655,022 vertices
R(4,4,4,6,7,8)	714 845	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 1 quadrupling -> 714,844 vertices (by monotonicity, from (4,4,4,5,7,8))
R(4,4,4,6,7,9)	1 065 041	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 1 quadrupling -> 1,065,040 vertices
R(4,4,4,6,7,10)	1 091 873	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,3;10)] x [atom (4,4,4 ; 127)]] -> 68,242 vertices; then 2 quadruplings -> 1,091,872 vertices
R(4,4,4,6,8,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices (by monotonicity, from (4,4,4,5,8,8))
R(4,4,4,6,8,9)	1 428 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 2 quadruplings -> 1,428,688 vertices
R(4,4,4,6,8,10)	1 428 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 2 quadruplings -> 1,428,688 vertices (by monotonicity, from (4,4,4,6,8,9))
R(4,4,4,6,9,9)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(4,4,4,6,9,10)	1 974 865	Published bricks + theorems	[template (5,5,5,3 ; 977)] x [atom (4,4,4 ; 127)] -> 123,429 vertices; then 2 quadruplings -> 1,974,864 vertices
R(4,4,4,6,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices (by monotonicity, from (4,4,4,5,10,10))
R(4,4,4,7,7,7)	893 212	Published bricks + theorems	[template (7,3;16)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 893,211 vertices
R(4,4,4,7,7,8)	1 191 497	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 1 quadrupling -> 1,191,496 vertices
R(4,4,4,7,7,9)	1 488 684	Published bricks + theorems	[template (9,3 ; 26)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,488,683 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,7,7,10)	1 726 873	Published bricks + theorems	[template (10,3 ; 30)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,726,872 vertices
R(4,4,4,7,8,8)	1 230 817	Published bricks + theorems	[template (7,3;16)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 1,230,816 vertices
R(4,4,4,7,8,9)	2 143 709	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 1 quadrupling -> 2,143,708 vertices
R(4,4,4,7,8,10)	2 143 709	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 1 quadrupling -> 2,143,708 vertices (by monotonicity, from (4,4,4,7,8,9))
R(4,4,4,7,9,9)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(4,4,4,7,9,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(4,4,4,7,10,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices (by monotonicity, from (4,4,4,7,9,10))
R(4,4,4,8,8,8)	1 559 034	Published bricks + theorems	[template (8,3 ; 20)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 1,559,033 vertices
R(4,4,4,8,8,9)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(4,4,4,8,8,10)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices (by monotonicity, from (4,4,4,8,8,9))
R(4,4,4,8,9,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(4,4,4,8,9,10)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(4,4,4,8,10,10)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices (by monotonicity, from (4,4,4,8,9,10))
R(4,4,4,9,9,9)	4 049 184	Published bricks + theorems	[template (9,3 ; 26)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,049,183 vertices
R(4,4,4,9,9,10)	5 830 829	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 1 quadrupling -> 5,830,828 vertices
R(4,4,4,9,10,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(4,4,4,10,10,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices (by monotonicity, from (4,4,4,9,10,10))
R(4,4,5,5,5,5)	157 626	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-17 (4,4;17)]] -> 157,625 vertices
R(4,4,5,5,5,6)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(4,4,5,5,5,7)	300 902	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-17 (4,4;17)]] -> 300,901 vertices
R(4,4,5,5,5,8)	379 329	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,4 ; 54)] -> 5,927 vertices; then 3 quadruplings -> 379,328 vertices
R(4,4,5,5,5,9)	526 017	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (4,4,4 ; 127)] -> 8,219 vertices; then 3 quadruplings -> 526,016 vertices
R(4,4,5,5,5,10)	687 425	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 10,741 vertices; then 3 quadruplings -> 687,424 vertices
R(4,4,5,5,6,6)	376 754	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 376,753 vertices
R(4,4,5,5,6,7)	493 717	Published bricks + theorems	[template (5,5,5,3 ; 977)] x [atom (4,4,4 ; 127)] -> 123,429 vertices; then 1 quadrupling -> 493,716 vertices
R(4,4,5,5,6,8)	630 609	Our bricks + published	[template (6,6,3;235)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 39,413 vertices; then 2 quadruplings -> 630,608 vertices
R(4,4,5,5,6,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(4,4,5,5,6,10)	984 321	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices; then 4 quadruplings -> 984,320 vertices
R(4,4,5,5,7,7)	758 268	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 758,267 vertices
R(4,4,5,5,7,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(4,4,5,5,7,9)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(4,4,5,5,7,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(4,4,5,5,8,8)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(4,4,5,5,8,9)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(4,4,5,5,8,10)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(4,4,5,5,9,9)	2 063 628	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,063,627 vertices
R(4,4,5,5,9,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(4,4,5,5,10,10)	3 452 321	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 2 quadruplings -> 3,452,320 vertices
R(4,4,5,6,6,6)	518 997	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 518,996 vertices
R(4,4,5,6,6,7)	718 908	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 718,907 vertices
R(4,4,5,6,6,8)	903 441	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 903,440 vertices
R(4,4,5,6,6,9)	1 203 305	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,203,304 vertices
R(4,4,5,6,6,10)	1 269 623	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,10 ; 91) (ours)]] -> 1,269,622 vertices
R(4,4,5,6,7,7)	1 065 041	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 1 quadrupling -> 1,065,040 vertices
R(4,4,5,6,7,8)	1 428 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 2 quadruplings -> 1,428,688 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,5,6,7,9)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(4,4,5,6,7,10)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(4,4,5,6,8,8)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(4,4,5,6,8,9)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(4,4,5,6,8,10)	2 532 481	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 3 quadruplings -> 2,532,480 vertices
R(4,4,5,6,9,9)	2 842 752	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,842,751 vertices
R(4,4,5,6,9,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(4,4,5,6,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(4,4,5,7,7,7)	1 446 899	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 1,446,898 vertices
R(4,4,5,7,7,8)	2 143 709	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 1 quadrupling -> 2,143,708 vertices
R(4,4,5,7,7,9)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(4,4,5,7,7,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(4,4,5,7,8,8)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(4,4,5,7,8,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(4,4,5,7,8,10)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(4,4,5,7,9,9)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(4,4,5,7,9,10)	5 830 829	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 1 quadrupling -> 5,830,828 vertices
R(4,4,5,7,10,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(4,4,5,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(4,4,5,8,8,9)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(4,4,5,8,8,10)	5 251 521	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 3 quadruplings -> 5,251,520 vertices
R(4,4,5,8,9,9)	5 251 521	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 3 quadruplings -> 5,251,520 vertices
R(4,4,5,8,9,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(4,4,5,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(4,4,5,9,9,9)	6 590 974	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 6,590,973 vertices
R(4,4,5,9,9,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(4,4,5,9,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(4,4,5,10,10,10)	13 809 281	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 3 quadruplings -> 13,809,280 vertices
R(4,4,6,6,6,6)	899 598	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 899,597 vertices
R(4,4,6,6,6,7)	1 065 041	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 1 quadrupling -> 1,065,040 vertices
R(4,4,6,6,6,8)	1 065 041	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 1 quadrupling -> 1,065,040 vertices (by monotonicity, from (4,4,6,6,6,7))
R(4,4,6,6,6,9)	1 543 800	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [Paley-17 (4,4;17)]] -> 1,543,799 vertices
R(4,4,6,6,6,10)	1 671 674	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (4,10 ; 91) (ours)]] -> 1,671,673 vertices
R(4,4,6,6,7,7)	1 810 727	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices
R(4,4,6,6,7,8)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(4,4,6,6,7,9)	2 248 413	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,103 vertices; then 1 quadrupling -> 2,248,412 vertices
R(4,4,6,6,7,10)	2 958 433	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 739,608 vertices; then 1 quadrupling -> 2,958,432 vertices
R(4,4,6,6,8,8)	2 495 478	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 2,495,477 vertices
R(4,4,6,6,8,9)	2 875 629	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 718,907 vertices; then 1 quadrupling -> 2,875,628 vertices
R(4,4,6,6,8,10)	2 958 433	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 739,608 vertices; then 1 quadrupling -> 2,958,432 vertices (by monotonicity, from (4,4,6,6,7,10))
R(4,4,6,6,9,9)	4 927 440	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices
R(4,4,6,6,9,10)	4 927 440	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices (by monotonicity, from (4,4,6,6,9,9))
R(4,4,6,6,10,10)	6 150 558	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (4,4,3 ; 34)] x [atome (10,10 ; 797)]]
R(4,4,6,7,7,7)	2 143 709	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 1 quadrupling -> 2,143,708 vertices
R(4,4,6,7,7,8)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(4,4,6,7,7,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(4,4,6,7,7,10)	4 260 161	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 2 quadruplings -> 4,260,160 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,6,7,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(4,4,6,7,8,9)	4 260 161	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 2 quadruplings -> 4,260,160 vertices
R(4,4,6,7,8,10)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(4,4,6,7,9,9)	5 830 829	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 1 quadrupling -> 5,830,828 vertices
R(4,4,6,7,9,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(4,4,6,7,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(4,4,6,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices (by monotonicity, from (4,4,5,8,8,8))
R(4,4,6,8,8,9)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(4,4,6,8,8,10)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices (by monotonicity, from (4,4,6,7,8,10))
R(4,4,6,8,9,9)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(4,4,6,8,9,10)	7 923 713	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 5 quadruplings -> 7,923,712 vertices
R(4,4,6,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices (by monotonicity, from (4,4,5,8,10,10))
R(4,4,6,9,9,9)	8 451 320	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [Paley-17 (4,4;17)]] -> 8,451,319 vertices
R(4,4,6,9,9,10)	11 371 005	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,842,751 vertices; then 1 quadrupling -> 11,371,004 vertices
R(4,4,6,9,10,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(4,4,6,10,10,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices (by monotonicity, from (4,4,6,9,10,10))
R(4,4,7,7,7,7)	3 644 330	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices
R(4,4,7,7,7,8)	3 644 330	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices (by monotonicity, from (4,4,7,7,7,7))
R(4,4,7,7,7,9)	4 765 985	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 2 quadruplings -> 4,765,984 vertices
R(4,4,7,7,7,10)	5 954 733	Published bricks + theorems	[template (9,3 ; 26)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,488,683 vertices; then 1 quadrupling -> 5,954,732 vertices
R(4,4,7,7,8,8)	5 022 947	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices
R(4,4,7,7,8,9)	5 954 733	Published bricks + theorems	[template (9,3 ; 26)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,488,683 vertices; then 1 quadrupling -> 5,954,732 vertices
R(4,4,7,7,8,10)	8 574 833	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 2 quadruplings -> 8,574,832 vertices
R(4,4,7,7,9,9)	9 918 050	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices
R(4,4,7,7,9,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,4,7,7,10,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,4,7,8,8,8)	5 022 947	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices (by monotonicity, from (4,4,7,7,8,8))
R(4,4,7,8,8,9)	8 574 833	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 2 quadruplings -> 8,574,832 vertices
R(4,4,7,8,8,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,4,7,8,9,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,4,7,8,9,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,4,7,8,10,10)	15 760 129	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 3 quadruplings -> 15,760,128 vertices
R(4,4,7,9,9,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,4,7,9,9,10)	16 196 733	Published bricks + theorems	[template (9,3 ; 26)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,049,183 vertices; then 1 quadrupling -> 16,196,732 vertices
R(4,4,7,9,10,10)	23 323 313	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 2 quadruplings -> 23,323,312 vertices
R(4,4,7,10,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,4,8,8,8,8)	6 921 218	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 6,921,217 vertices
R(4,4,8,8,8,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,4,8,8,8,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices (by monotonicity, from (4,4,7,8,8,10))
R(4,4,8,8,9,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,4,8,8,9,10)	15 760 129	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 3 quadruplings -> 15,760,128 vertices
R(4,4,8,8,10,10)	17 058 598	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,8,3 ; 650)] x [[modele (4,4,3 ; 34)] x [atome (10,10 ; 797)]]
R(4,4,8,9,9,9)	15 760 129	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 3 quadruplings -> 15,760,128 vertices
R(4,4,8,9,9,10)	23 323 313	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 2 quadruplings -> 23,323,312 vertices
R(4,4,8,9,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,8,10,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices (by monotonicity, from (4,4,7,10,10,10))
R(4,4,9,9,9,9)	26 974 580	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices
R(4,4,9,9,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,4,9,9,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(4,4,9,10,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(4,4,10,10,10,10)	41 871 210	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,4,3 ; 34)] x [[modele generique 2m-1 de atome (10,10 ; 797)] x [atome (10,10 ; 797)]]
R(4,5,5,5,5,5)	244 283	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,3;10)] x [atom (5,5,5 ; 453)]] -> 244,282 vertices
R(4,5,5,5,5,6)	342 809	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 85,702 vertices; then 1 quadrupling -> 342,808 vertices
R(4,5,5,5,5,7)	493 717	Published bricks + theorems	[template (5,5,5,3 ; 977)] x [atom (4,4,4 ; 127)] -> 123,429 vertices; then 1 quadrupling -> 493,716 vertices
R(4,5,5,5,5,8)	657 065	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 164,266 vertices; then 1 quadrupling -> 657,064 vertices
R(4,5,5,5,5,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(4,5,5,5,5,10)	1 139 457	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [K2]] -> 17,804 vertices; then 3 quadruplings -> 1,139,456 vertices
R(4,5,5,5,6,6)	590 423	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 590,422 vertices
R(4,5,5,5,6,7)	747 853	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 186,963 vertices; then 1 quadrupling -> 747,852 vertices
R(4,5,5,5,6,8)	1 009 473	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [K3]] -> 63,092 vertices; then 2 quadruplings -> 1,009,472 vertices
R(4,5,5,5,6,9)	1 399 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 3 quadruplings -> 1,399,616 vertices
R(4,5,5,5,6,10)	1 703 169	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 4 quadruplings -> 1,703,168 vertices
R(4,5,5,5,7,7)	1 184 303	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,184,302 vertices
R(4,5,5,5,7,8)	1 500 101	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 375,025 vertices; then 1 quadrupling -> 1,500,100 vertices
R(4,5,5,5,7,9)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(4,5,5,5,7,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(4,5,5,5,8,8)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(4,5,5,5,8,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(4,5,5,5,8,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(4,5,5,5,9,9)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(4,5,5,5,9,10)	4 181 381	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 1,045,345 vertices; then 1 quadrupling -> 4,181,380 vertices
R(4,5,5,5,10,10)	5 547 153	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 2 quadruplings -> 5,547,152 vertices
R(4,5,5,6,6,6)	813 323	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Paley-101 (6,6;101)]] -> 813,322 vertices
R(4,5,5,6,6,7)	1 131 647	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 1,131,646 vertices
R(4,5,5,6,6,8)	1 428 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 2 quadruplings -> 1,428,688 vertices
R(4,5,5,6,6,9)	1 918 017	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 119,876 vertices; then 2 quadruplings -> 1,918,016 vertices
R(4,5,5,6,6,10)	2 475 905	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 3 quadruplings -> 2,475,904 vertices
R(4,5,5,6,7,7)	1 631 423	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,631,422 vertices
R(4,5,5,6,7,8)	2 066 453	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 516,613 vertices; then 1 quadrupling -> 2,066,452 vertices
R(4,5,5,6,7,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(4,5,5,6,7,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(4,5,5,6,8,8)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(4,5,5,6,8,9)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(4,5,5,6,8,10)	4 073 473	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 4 quadruplings -> 4,073,472 vertices
R(4,5,5,6,9,9)	4 547 423	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 4,547,422 vertices
R(4,5,5,6,9,10)	5 760 053	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 1,440,013 vertices; then 1 quadrupling -> 5,760,052 vertices
R(4,5,5,6,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(4,5,5,7,7,7)	2 269 917	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,269,916 vertices
R(4,5,5,7,7,8)	3 033 069	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 758,267 vertices; then 1 quadrupling -> 3,033,068 vertices
R(4,5,5,7,7,9)	3 837 137	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 239,821 vertices; then 2 quadruplings -> 3,837,136 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,5,7,7,10)	5 077 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 3 quadruplings -> 5,077,120 vertices
R(4,5,5,7,8,8)	3 837 137	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 239,821 vertices; then 2 quadruplings -> 3,837,136 vertices
R(4,5,5,7,8,9)	5 077 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 3 quadruplings -> 5,077,120 vertices
R(4,5,5,7,8,10)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(4,5,5,7,9,9)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(4,5,5,7,9,10)	8 254 509	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,063,627 vertices; then 1 quadrupling -> 8,254,508 vertices
R(4,5,5,7,10,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(4,5,5,8,8,8)	5 077 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 3 quadruplings -> 5,077,120 vertices
R(4,5,5,8,8,9)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(4,5,5,8,8,10)	8 122 689	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 3 quadruplings -> 8,122,688 vertices
R(4,5,5,8,9,9)	8 122 689	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 3 quadruplings -> 8,122,688 vertices
R(4,5,5,8,9,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(4,5,5,8,10,10)	13 809 281	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 3 quadruplings -> 13,809,280 vertices
R(4,5,5,9,9,9)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(4,5,5,9,9,10)	13 809 281	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 3 quadruplings -> 13,809,280 vertices
R(4,5,5,9,10,10)	17 524 481	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 4 quadruplings -> 17,524,480 vertices
R(4,5,5,10,10,10)	22 188 609	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 3 quadruplings -> 22,188,608 vertices
R(4,5,6,6,6,6)	1 410 023	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,410,022 vertices
R(4,5,6,6,6,7)	1 786 013	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 1 quadrupling -> 1,786,012 vertices
R(4,5,6,6,6,8)	2 128 209	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]] -> 133,013 vertices; then 2 quadruplings -> 2,128,208 vertices
R(4,5,6,6,6,9)	2 547 263	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 2,547,262 vertices
R(4,5,6,6,6,10)	3 226 517	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 806,629 vertices; then 1 quadrupling -> 3,226,516 vertices
R(4,5,6,6,7,7)	2 838 083	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 2,838,082 vertices
R(4,5,6,6,7,8)	3 594 889	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 1 quadrupling -> 3,594,888 vertices
R(4,5,6,6,7,9)	4 283 681	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 2 quadruplings -> 4,283,680 vertices
R(4,5,6,6,7,10)	4 813 217	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,203,304 vertices; then 1 quadrupling -> 4,813,216 vertices
R(4,5,6,6,8,8)	4 283 681	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 2 quadruplings -> 4,283,680 vertices
R(4,5,6,6,8,9)	4 987 133	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 1,246,783 vertices; then 1 quadrupling -> 4,987,132 vertices
R(4,5,6,6,8,10)	5 909 649	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [K3]] -> 369,353 vertices; then 2 quadruplings -> 5,909,648 vertices
R(4,5,6,6,9,9)	7 882 463	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices
R(4,5,6,6,9,10)	9 984 437	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 1 quadrupling -> 9,984,436 vertices
R(4,5,6,6,10,10)	11 694 129	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 2 quadruplings -> 11,694,128 vertices
R(4,5,6,7,7,7)	3 594 889	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 1 quadrupling -> 3,594,888 vertices
R(4,5,6,7,7,8)	4 283 681	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 2 quadruplings -> 4,283,680 vertices
R(4,5,6,7,7,9)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(4,5,6,7,7,10)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(4,5,6,7,8,8)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(4,5,6,7,8,9)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(4,5,6,7,8,10)	8 265 809	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 516,613 vertices; then 2 quadruplings -> 8,265,808 vertices
R(4,5,6,7,9,9)	9 984 437	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 1 quadrupling -> 9,984,436 vertices
R(4,5,6,7,9,10)	11 694 129	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 2 quadruplings -> 11,694,128 vertices
R(4,5,6,7,10,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(4,5,6,8,8,8)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(4,5,6,8,8,9)	8 265 809	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 516,613 vertices; then 2 quadruplings -> 8,265,808 vertices
R(4,5,6,8,8,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(4,5,6,8,9,9)	11 694 129	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 2 quadruplings -> 11,694,128 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,6,8,9,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(4,5,6,8,10,10)	17 524 481	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 4 quadruplings -> 17,524,480 vertices
R(4,5,6,9,9,9)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(4,5,6,9,9,10)	18 189 689	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 4,547,422 vertices; then 1 quadrupling -> 18,189,688 vertices
R(4,5,6,9,10,10)	23 040 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 1,440,013 vertices; then 2 quadruplings -> 23,040,208 vertices
R(4,5,6,10,10,10)	30 658 305	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 4 quadruplings -> 30,658,304 vertices
R(4,5,7,7,7,7)	5 692 343	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 5,692,342 vertices
R(4,5,7,7,7,8)	7 210 285	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 1 quadrupling -> 7,210,284 vertices
R(4,5,7,7,7,9)	8 616 881	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 2 quadruplings -> 8,616,880 vertices
R(4,5,7,7,7,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,5,7,7,8,8)	8 616 881	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 2 quadruplings -> 8,616,880 vertices
R(4,5,7,7,8,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,5,7,7,8,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,5,7,7,9,9)	15 865 943	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices
R(4,5,7,7,9,10)	20 096 845	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 1 quadrupling -> 20,096,844 vertices
R(4,5,7,7,10,10)	23 538 161	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 2 quadruplings -> 23,538,160 vertices
R(4,5,7,8,8,8)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,5,7,8,8,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,5,7,8,8,10)	15 760 129	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 3 quadruplings -> 15,760,128 vertices
R(4,5,7,8,9,9)	20 096 845	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 1 quadrupling -> 20,096,844 vertices
R(4,5,7,8,9,10)	23 538 161	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 2 quadruplings -> 23,538,160 vertices
R(4,5,7,8,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,5,7,9,9,9)	23 538 161	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 2 quadruplings -> 23,538,160 vertices
R(4,5,7,9,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,5,7,9,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(4,5,7,10,10,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(4,5,8,8,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,5,8,8,8,9)	15 760 129	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 3 quadruplings -> 15,760,128 vertices
R(4,5,8,8,8,10)	21 006 081	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 4 quadruplings -> 21,006,080 vertices
R(4,5,8,8,9,9)	23 538 161	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 2 quadruplings -> 23,538,160 vertices
R(4,5,8,8,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,5,8,8,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(4,5,8,9,9,9)	27 692 057	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 6,923,014 vertices; then 1 quadrupling -> 27,692,056 vertices
R(4,5,8,9,9,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(4,5,8,9,10,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(4,5,8,10,10,10)	55 237 121	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 4 quadruplings -> 55,237,120 vertices
R(4,5,9,9,9,9)	43 152 143	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 43,152,142 vertices
R(4,5,9,9,9,10)	54 659 365	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 1 quadrupling -> 54,659,364 vertices
R(4,5,9,9,10,10)	64 018 001	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]] -> 4,001,125 vertices; then 2 quadruplings -> 64,018,000 vertices
R(4,5,9,10,10,10)	77 538 025	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,5,10,10,10,10)	88 754 433	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 4 quadruplings -> 88,754,432 vertices
R(4,6,6,6,6,6)	1 856 534	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,856,533 vertices
R(4,6,6,6,6,7)	2 702 547	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,702,546 vertices
R(4,6,6,6,6,8)	2 843 552	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,843,551 vertices
R(4,6,6,6,6,9)	3 692 154	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,9 ; 68)]] -> 3,692,153 vertices
R(4,6,6,6,6,10)	4 951 542	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,10 ; 91) (ours)]] -> 4,951,541 vertices
R(4,6,6,6,7,7)	3 736 813	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 3,736,812 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,6,6,6,7,8)	4 260 161	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 2 quadruplings -> 4,260,160 vertices
R(4,6,6,6,7,9)	5 640 089	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,410,022 vertices; then 1 quadrupling -> 5,640,088 vertices
R(4,6,6,6,7,10)	6 175 197	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [Paley-17 (4,4;17)]] -> 1,543,799 vertices; then 1 quadrupling -> 6,175,196 vertices
R(4,6,6,6,8,8)	5 184 014	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 5,184,013 vertices
R(4,6,6,6,8,9)	7 144 049	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 2 quadruplings -> 7,144,048 vertices
R(4,6,6,6,8,10)	7 144 049	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 2 quadruplings -> 7,144,048 vertices (by monotonicity, from (4,6,6,6,8,9))
R(4,6,6,6,9,9)	10 378 580	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 10,378,579 vertices
R(4,6,6,6,9,10)	10 378 580	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 10,378,579 vertices (by monotonicity, from (4,6,6,6,9,9))
R(4,6,6,6,10,10)	14 722 790	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,6,3 ; 80)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(4,6,6,7,7,7)	5 439 662	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,439,661 vertices
R(4,6,6,7,7,8)	7 242 905	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 1 quadrupling -> 7,242,904 vertices
R(4,6,6,7,7,9)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(4,6,6,7,7,10)	9 966 563	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,10 ; 91) (ours)]] -> 9,966,562 vertices
R(4,6,6,7,8,8)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(4,6,6,7,8,9)	11 352 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 2,838,082 vertices; then 1 quadrupling -> 11,352,328 vertices
R(4,6,6,7,8,10)	11 833 729	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 739,608 vertices; then 2 quadruplings -> 11,833,728 vertices
R(4,6,6,7,9,9)	15 108 057	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 15,108,056 vertices
R(4,6,6,7,9,10)	19 709 757	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 1 quadrupling -> 19,709,756 vertices
R(4,6,6,7,10,10)	21 431 907	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,7,3 ; 116)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(4,6,6,8,8,8)	7 940 072	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 7,940,071 vertices
R(4,6,6,8,8,9)	14 379 553	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 2 quadruplings -> 14,379,552 vertices
R(4,6,6,8,8,10)	14 379 553	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 2 quadruplings -> 14,379,552 vertices (by monotonicity, from (4,6,6,8,8,9))
R(4,6,6,8,9,9)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(4,6,6,8,9,10)	19 948 529	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 1,246,783 vertices; then 2 quadruplings -> 19,948,528 vertices
R(4,6,6,8,10,10)	22 550 096	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,8,3 ; 122)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(4,6,6,9,9,9)	20 494 524	Our bricks + published	[modele (6,6,3;235)] x [[modele (4,9,3 ; 157)] x [atome (9,9 ; 562)]] [new template (4,9;157), v21]
R(4,6,6,9,9,10)	31 529 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices; then 1 quadrupling -> 31,529,848 vertices
R(4,6,6,9,10,10)	39 937 745	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 2 quadruplings -> 39,937,744 vertices
R(4,6,6,10,10,10)	39 937 745	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 2 quadruplings -> 39,937,744 vertices (by monotonicity, from (4,6,6,9,10,10))
R(4,6,7,7,7,7)	7 494 922	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 7,494,921 vertices
R(4,6,7,7,7,8)	8 574 833	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 2 quadruplings -> 8,574,832 vertices
R(4,6,7,7,7,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,6,7,7,7,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,6,7,7,8,8)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,6,7,7,8,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,6,7,7,8,10)	17 040 641	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 3 quadruplings -> 17,040,640 vertices
R(4,6,7,7,9,9)	20 890 162	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices
R(4,6,7,7,9,10)	23 323 313	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 2 quadruplings -> 23,323,312 vertices
R(4,6,7,7,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,6,7,8,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,6,7,8,8,9)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(4,6,7,8,8,10)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(4,6,7,8,9,9)	23 323 313	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 2 quadruplings -> 23,323,312 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,6,7,8,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,6,7,8,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(4,6,7,9,9,9)	31 529 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices; then 1 quadrupling -> 31,529,848 vertices
R(4,6,7,9,9,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(4,6,7,9,10,10)	46 776 513	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 3 quadruplings -> 46,776,512 vertices
R(4,6,7,10,10,10)	57 466 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 6 quadruplings -> 57,466,880 vertices
R(4,6,8,8,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices (by monotonicity, from (4,5,8,8,8,8))
R(4,6,8,8,8,9)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(4,6,8,8,8,10)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices (by monotonicity, from (4,6,7,8,8,10))
R(4,6,8,8,9,9)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(4,6,8,8,9,10)	33 063 233	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 516,613 vertices; then 3 quadruplings -> 33,063,232 vertices
R(4,6,8,8,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices (by monotonicity, from (4,5,8,8,10,10))
R(4,6,8,9,9,9)	39 937 745	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 2 quadruplings -> 39,937,744 vertices
R(4,6,8,9,9,10)	46 776 513	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 3 quadruplings -> 46,776,512 vertices
R(4,6,8,9,10,10)	57 466 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 6 quadruplings -> 57,466,880 vertices
R(4,6,8,10,10,10)	57 466 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 6 quadruplings -> 57,466,880 vertices (by monotonicity, from (4,6,7,10,10,10))
R(4,6,9,9,9,9)	56 816 992	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 56,816,991 vertices
R(4,6,9,9,9,10)	57 466 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 6 quadruplings -> 57,466,880 vertices
R(4,6,9,9,10,10)	80 598 757	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,6,3 ; 80)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,6,9,10,10,10)	92 160 833	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 1,440,013 vertices; then 3 quadruplings -> 92,160,832 vertices
R(4,6,10,10,10,10)	100 237 130	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,6,3 ; 80)] x [[modele generique 2m-1 de atome (10,10 ; 797)] x [atome (10,10 ; 797)]]
R(4,7,7,7,7,7)	10 910 327	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,910,326 vertices
R(4,7,7,7,7,8)	14 577 317	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 1 quadrupling -> 14,577,316 vertices
R(4,7,7,7,7,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,7,7,7,7,10)	20 060 564	Our bricks + published	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,10 ; 91) (ours)]] -> 20,060,563 vertices
R(4,7,7,7,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,7,7,7,8,9)	22 769 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 5,692,342 vertices; then 1 quadrupling -> 22,769,368 vertices
R(4,7,7,7,8,10)	23 818 929	Published bricks + theorems	[template (9,3 ; 26)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,488,683 vertices; then 2 quadruplings -> 23,818,928 vertices
R(4,7,7,7,9,9)	30 409 727	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 30,409,726 vertices
R(4,7,7,7,9,10)	39 672 197	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 1 quadrupling -> 39,672,196 vertices
R(4,7,7,7,10,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,7,7,8,8,8)	20 091 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 1 quadrupling -> 20,091,784 vertices
R(4,7,7,8,8,9)	28 841 137	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 2 quadruplings -> 28,841,136 vertices
R(4,7,7,8,8,10)	34 299 329	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 3 quadruplings -> 34,299,328 vertices
R(4,7,7,8,9,9)	39 672 197	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 1 quadrupling -> 39,672,196 vertices
R(4,7,7,8,9,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,7,7,8,10,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,7,7,9,9,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,7,7,9,9,10)	63 463 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 1 quadrupling -> 63,463,768 vertices
R(4,7,7,9,10,10)	80 387 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 2 quadruplings -> 80,387,376 vertices
R(4,7,7,10,10,10)	93 293 249	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 3 quadruplings -> 93,293,248 vertices
R(4,7,8,8,8,8)	20 930 047	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 20,930,046 vertices
R(4,7,8,8,8,9)	34 467 521	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 3 quadruplings -> 34,467,520 vertices
R(4,7,8,8,8,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,7,8,8,9,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,7,8,8,9,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,7,8,8,10,10)	63 040 513	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 4 quadruplings -> 63,040,512 vertices
R(4,7,8,9,9,9)	63 463 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 1 quadrupling -> 63,463,768 vertices
R(4,7,8,9,9,10)	80 387 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 2 quadruplings -> 80,387,376 vertices
R(4,7,8,9,10,10)	94 152 641	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 3 quadruplings -> 94,152,640 vertices
R(4,7,8,10,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(4,7,9,9,9,9)	82 708 277	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 82,708,276 vertices
R(4,7,9,9,9,10)	107 898 317	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 1 quadrupling -> 107,898,316 vertices
R(4,7,9,9,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(4,7,9,10,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(4,7,10,10,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(4,8,8,8,8,8)	22 022 052	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 22,022,051 vertices
R(4,8,8,8,8,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,8,8,8,8,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices (by monotonicity, from (4,7,8,8,8,10))
R(4,8,8,8,9,9)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,8,8,8,9,10)	63 040 513	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 4 quadruplings -> 63,040,512 vertices
R(4,8,8,8,10,10)	63 040 513	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 4 quadruplings -> 63,040,512 vertices (by monotonicity, from (4,7,8,8,10,10))
R(4,8,8,9,9,9)	80 387 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 2 quadruplings -> 80,387,376 vertices
R(4,8,8,9,9,10)	94 152 641	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 3 quadruplings -> 94,152,640 vertices
R(4,8,8,9,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(4,8,8,10,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices (by monotonicity, from (4,7,8,10,10,10))
R(4,8,9,9,9,9)	94 152 641	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 3 quadruplings -> 94,152,640 vertices
R(4,8,9,9,9,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(4,8,9,9,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(4,8,9,10,10,10)	180 244 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 6 quadruplings -> 180,244,480 vertices
R(4,8,10,10,10,10)	180 244 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 6 quadruplings -> 180,244,480 vertices (by monotonicity, from (4,8,9,10,10,10))
R(4,9,9,9,9,9)	112 195 581	Our bricks + published	[modele (4,9,3 ; 157)] x [[modele (9,9,3 ; 1282)] x [atome (9,9 ; 562)]] [new template (4,9;157), v21]
R(4,9,9,9,9,10)	172 608 569	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 43,152,142 vertices; then 1 quadrupling -> 172,608,568 vertices
R(4,9,9,9,10,10)	218 637 457	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 2 quadruplings -> 218,637,456 vertices
R(4,9,9,10,10,10)	256 072 001	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]] -> 4,001,125 vertices; then 3 quadruplings -> 256,072,000 vertices
R(4,9,10,10,10,10)	310 152 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,10,10,10,10,10)	310 152 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,5,5,5,5,5)	441 606	Published	published value
R(5,5,5,5,5,6)	559 941	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 139,985 vertices; then 1 quadrupling -> 559,940 vertices
R(5,5,5,5,5,7)	803 457	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,5,5 ; 453)] -> 50,216 vertices; then 2 quadruplings -> 803,456 vertices
R(5,5,5,5,5,8)	1 068 461	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 267,115 vertices; then 1 quadrupling -> 1,068,460 vertices
R(5,5,5,5,5,9)	1 353 649	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (5,5,5 ; 453)] -> 84,603 vertices; then 2 quadruplings -> 1,353,648 vertices
R(5,5,5,5,5,10)	1 885 377	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [K2]] -> 29,459 vertices; then 3 quadruplings -> 1,885,376 vertices
R(5,5,5,5,6,6)	964 362	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 964,361 vertices
R(5,5,5,5,6,7)	1 337 013	Published bricks + theorems	[template (6,6,3;235)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 334,253 vertices; then 1 quadrupling -> 1,337,012 vertices
R(5,5,5,5,6,8)	1 693 905	Published bricks + theorems	[template (6,6,3;235)] x [atom (5,5,5 ; 453)] -> 105,869 vertices; then 2 quadruplings -> 1,693,904 vertices
R(5,5,5,5,6,9)	2 389 761	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 3 quadruplings -> 2,389,760 vertices
R(5,5,5,5,6,10)	2 852 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 4 quadruplings -> 2,852,864 vertices
R(5,5,5,5,7,7)	1 934 366	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,934,365 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,5,5,7,8)	2 691 161	Published bricks + theorems	[template (7,7,3;472)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 672,790 vertices; then 1 quadrupling -> 2,691,160 vertices
R(5,5,5,5,7,9)	3 409 505	Published bricks + theorems	[template (7,7,3;472)] x [atom (5,5,5 ; 453)] -> 213,094 vertices; then 2 quadruplings -> 3,409,504 vertices
R(5,5,5,5,7,10)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(5,5,5,5,8,8)	3 409 505	Published bricks + theorems	[template (7,7,3;472)] x [atom (5,5,5 ; 453)] -> 213,094 vertices; then 2 quadruplings -> 3,409,504 vertices
R(5,5,5,5,8,9)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(5,5,5,5,8,10)	5 722 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 4 quadruplings -> 5,722,880 vertices
R(5,5,5,5,9,9)	5 722 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 4 quadruplings -> 5,722,880 vertices
R(5,5,5,5,9,10)	7 319 321	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 1,829,830 vertices; then 1 quadrupling -> 7,319,320 vertices
R(5,5,5,5,10,10)	9 273 185	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)] -> 579,574 vertices; then 2 quadruplings -> 9,273,184 vertices
R(5,5,5,6,6,6)	1 337 013	Published bricks + theorems	[template (6,6,3;235)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 334,253 vertices; then 1 quadrupling -> 1,337,012 vertices
R(5,5,5,6,6,7)	1 840 286	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-101 (6,6;101)]] -> 1,840,285 vertices
R(5,5,5,6,6,8)	2 389 761	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 3 quadruplings -> 2,389,760 vertices
R(5,5,5,6,6,9)	3 118 929	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 194,933 vertices; then 2 quadruplings -> 3,118,928 vertices
R(5,5,5,6,6,10)	4 026 305	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,6,6 ; 337)] -> 62,911 vertices; then 3 quadruplings -> 4,026,304 vertices
R(5,5,5,6,7,7)	2 691 161	Published bricks + theorems	[template (7,7,3;472)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 672,790 vertices; then 1 quadrupling -> 2,691,160 vertices
R(5,5,5,6,7,8)	3 409 505	Published bricks + theorems	[template (7,7,3;472)] x [atom (5,5,5 ; 453)] -> 213,094 vertices; then 2 quadruplings -> 3,409,504 vertices
R(5,5,5,6,7,9)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(5,5,5,6,7,10)	5 809 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 3 quadruplings -> 5,809,664 vertices
R(5,5,5,6,8,8)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(5,5,5,6,8,9)	5 809 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 3 quadruplings -> 5,809,664 vertices
R(5,5,5,6,8,10)	6 960 897	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)] -> 27,191 vertices; then 4 quadruplings -> 6,960,896 vertices
R(5,5,5,6,9,9)	7 427 487	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 7,427,486 vertices
R(5,5,5,6,9,10)	9 559 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 4 quadruplings -> 9,559,040 vertices
R(5,5,5,6,10,10)	13 088 385	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 3 quadruplings -> 13,088,384 vertices
R(5,5,5,7,7,7)	3 691 212	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)]] -> 3,691,211 vertices
R(5,5,5,7,7,8)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(5,5,5,7,7,9)	6 239 585	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 389,974 vertices; then 2 quadruplings -> 6,239,584 vertices
R(5,5,5,7,7,10)	8 122 689	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 3 quadruplings -> 8,122,688 vertices
R(5,5,5,7,8,8)	6 239 585	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 389,974 vertices; then 2 quadruplings -> 6,239,584 vertices
R(5,5,5,7,8,9)	8 122 689	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 3 quadruplings -> 8,122,688 vertices
R(5,5,5,7,8,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(5,5,5,7,9,9)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(5,5,5,7,9,10)	13 711 361	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 5 quadruplings -> 13,711,360 vertices
R(5,5,5,7,10,10)	19 106 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 4 quadruplings -> 19,106,816 vertices
R(5,5,5,8,8,8)	8 122 689	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 3 quadruplings -> 8,122,688 vertices
R(5,5,5,8,8,9)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(5,5,5,8,8,10)	13 711 361	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 5 quadruplings -> 13,711,360 vertices
R(5,5,5,8,9,9)	13 711 361	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 5 quadruplings -> 13,711,360 vertices
R(5,5,5,8,9,10)	19 106 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 4 quadruplings -> 19,106,816 vertices
R(5,5,5,8,10,10)	22 891 521	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 5 quadruplings -> 22,891,520 vertices
R(5,5,5,9,9,9)	19 106 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 4 quadruplings -> 19,106,816 vertices
R(5,5,5,9,9,10)	22 891 521	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 5 quadruplings -> 22,891,520 vertices
R(5,5,5,9,10,10)	30 658 305	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 4 quadruplings -> 30,658,304 vertices
R(5,5,5,10,10,10)	37 369 857	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 5 quadruplings -> 37,369,856 vertices
R(5,5,6,6,6,6)	2 303 042	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,303,041 vertices
R(5,5,6,6,6,7)	2 834 081	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 708,520 vertices; then 1 quadrupling -> 2,834,080 vertices
R(5,5,6,6,6,8)	3 902 865	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 2 quadruplings -> 3,902,864 vertices
R(5,5,6,6,6,9)	5 038 401	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 3 quadruplings -> 5,038,400 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,6,6,6,10)	6 016 257	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,504,065 – 3 = 6,016,257; with R(6,6,6,6) >= 23,502
R(5,5,6,6,7,7)	4 635 540	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 4,635,539 vertices
R(5,5,6,6,7,8)	5 704 469	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 1 quadrupling -> 5,704,468 vertices
R(5,5,6,6,7,9)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(5,5,6,6,7,10)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(5,5,6,6,8,8)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(5,5,6,6,8,9)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(5,5,6,6,8,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(5,5,6,6,9,9)	12 874 694	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 12,874,693 vertices
R(5,5,6,6,9,10)	15 764 933	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 3,941,233 vertices; then 1 quadrupling -> 15,764,932 vertices
R(5,5,6,6,10,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(5,5,6,7,7,7)	5 704 469	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 1 quadrupling -> 5,704,468 vertices
R(5,5,6,7,7,8)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(5,5,6,7,7,9)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(5,5,6,7,7,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(5,5,6,7,8,8)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(5,5,6,7,8,9)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(5,5,6,7,8,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(5,5,6,7,9,9)	15 764 933	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 3,941,233 vertices; then 1 quadrupling -> 15,764,932 vertices
R(5,5,6,7,9,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(5,5,6,7,10,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(5,5,6,8,8,8)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(5,5,6,8,8,9)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(5,5,6,8,8,10)	19 106 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 4 quadruplings -> 19,106,816 vertices
R(5,5,6,8,9,9)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(5,5,6,8,9,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(5,5,6,8,10,10)	33 632 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 4 quadruplings -> 33,632,000 vertices
R(5,5,6,9,9,9)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(5,5,6,9,9,10)	33 632 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 4 quadruplings -> 33,632,000 vertices
R(5,5,6,9,10,10)	38 236 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 5 quadruplings -> 38,236,160 vertices
R(5,5,6,10,10,10)	52 353 537	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 4 quadruplings -> 52,353,536 vertices
R(5,5,7,7,7,7)	9 297 498	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 9,297,497 vertices
R(5,5,7,7,7,8)	11 401 685	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 2,850,421 vertices; then 1 quadrupling -> 11,401,684 vertices
R(5,5,7,7,7,9)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(5,5,7,7,7,10)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(5,5,7,7,8,8)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(5,5,7,7,8,9)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(5,5,7,7,8,10)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(5,5,7,7,9,9)	25 914 378	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 25,914,377 vertices
R(5,5,7,7,9,10)	31 731 893	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 1 quadrupling -> 31,731,892 vertices
R(5,5,7,7,10,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(5,5,7,8,8,8)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(5,5,7,8,8,9)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(5,5,7,8,8,10)	28 318 145	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 3 quadruplings -> 28,318,144 vertices
R(5,5,7,8,9,9)	31 731 893	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 1 quadrupling -> 31,731,892 vertices
R(5,5,7,8,9,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(5,5,7,8,10,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,5,7,9,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,7,9,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,5,7,9,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(5,5,7,10,10,10)	76 427 265	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 5 quadruplings -> 76,427,264 vertices
R(5,5,8,8,8,8)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(5,5,8,8,8,9)	28 318 145	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 3 quadruplings -> 28,318,144 vertices
R(5,5,8,8,8,10)	33 812 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (8,8 ; 281)] -> 132,082 vertices; then 4 quadruplings -> 33,812,992 vertices
R(5,5,8,8,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(5,5,8,8,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,5,8,8,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(5,5,8,9,9,9)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,5,8,9,9,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(5,5,8,9,10,10)	77 018 945	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 3 quadruplings -> 77,018,944 vertices
R(5,5,8,10,10,10)	96 030 209	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(5,5,9,9,9,9)	70 481 838	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 70,481,837 vertices
R(5,5,9,9,9,10)	86 304 293	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 21,576,073 vertices; then 1 quadrupling -> 86,304,292 vertices
R(5,5,9,9,10,10)	116 528 753	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 2 quadruplings -> 116,528,752 vertices
R(5,5,9,10,10,10)	151 050 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 3 quadruplings -> 151,050,496 vertices
R(5,5,10,10,10,10)	184 115 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 46,028,993 – 3 = 184,115,969; with R(9,9,9,9) >= 719,204
R(5,6,6,6,6,6)	3 172 557	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 3,172,556 vertices
R(5,6,6,6,6,7)	4 394 580	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 4,394,579 vertices
R(5,6,6,6,6,8)	5 522 601	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 5,522,600 vertices
R(5,6,6,6,6,9)	7 355 633	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 7,355,632 vertices
R(5,6,6,6,6,10)	7 872 433	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 1,968,108 vertices; then 1 quadrupling -> 7,872,432 vertices
R(5,6,6,6,7,7)	6 385 692	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices
R(5,6,6,6,7,8)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(5,6,6,6,7,9)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(5,6,6,6,7,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(5,6,6,6,8,8)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(5,6,6,6,8,9)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(5,6,6,6,8,10)	15 611 457	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 3 quadruplings -> 15,611,456 vertices
R(5,6,6,6,9,9)	17 735 547	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 17,735,546 vertices
R(5,6,6,6,9,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(5,6,6,6,10,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(5,6,6,7,7,7)	8 845 367	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 8,845,366 vertices
R(5,6,6,7,7,8)	11 352 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 2,838,082 vertices; then 1 quadrupling -> 11,352,328 vertices
R(5,6,6,7,7,9)	14 805 346	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 14,805,345 vertices
R(5,6,6,7,7,10)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(5,6,6,7,8,8)	14 379 553	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 2 quadruplings -> 14,379,552 vertices
R(5,6,6,7,8,9)	18 542 157	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 4,635,539 vertices; then 1 quadrupling -> 18,542,156 vertices
R(5,6,6,7,8,10)	22 817 873	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 2 quadruplings -> 22,817,872 vertices
R(5,6,6,7,9,9)	24 567 018	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 24,567,017 vertices
R(5,6,6,7,9,10)	31 529 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices; then 1 quadrupling -> 31,529,848 vertices
R(5,6,6,7,10,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(5,6,6,8,8,8)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(5,6,6,8,8,9)	22 817 873	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 2 quadruplings -> 22,817,872 vertices
R(5,6,6,8,8,10)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(5,6,6,8,9,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,6,6,8,9,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(5,6,6,8,10,10)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices
R(5,6,6,9,9,9)	41 120 195	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 41,120,194 vertices
R(5,6,6,9,9,10)	51 498 773	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 12,874,693 vertices; then 1 quadrupling -> 51,498,772 vertices
R(5,6,6,9,10,10)	63 059 729	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 3,941,233 vertices; then 2 quadruplings -> 63,059,728 vertices
R(5,6,6,10,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(5,6,7,7,7,7)	12 807 777	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices
R(5,6,7,7,7,8)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(5,6,7,7,7,9)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(5,6,7,7,7,10)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(5,6,7,7,8,8)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(5,6,7,7,8,9)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(5,6,7,7,8,10)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(5,6,7,7,9,9)	35 698 377	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices
R(5,6,7,7,9,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(5,6,7,7,10,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,6,7,8,8,8)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(5,6,7,8,8,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(5,6,7,8,8,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(5,6,7,8,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(5,6,7,8,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,6,7,8,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(5,6,7,9,9,9)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,6,7,9,9,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(5,6,7,9,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(5,6,7,10,10,10)	110 369 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 4 quadruplings -> 110,369,024 vertices
R(5,6,8,8,8,8)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(5,6,8,8,8,9)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(5,6,8,8,8,10)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices
R(5,6,8,8,9,9)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(5,6,8,8,9,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(5,6,8,8,10,10)	77 018 945	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 3 quadruplings -> 77,018,944 vertices
R(5,6,8,9,9,9)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(5,6,8,9,9,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(5,6,8,9,10,10)	110 369 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 4 quadruplings -> 110,369,024 vertices
R(5,6,8,10,10,10)	134 528 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 5 quadruplings -> 134,528,000 vertices
R(5,6,9,9,9,9)	97 092 327	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 97,092,326 vertices
R(5,6,9,9,9,10)	116 528 753	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 2 quadruplings -> 116,528,752 vertices
R(5,6,9,9,10,10)	151 050 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 3 quadruplings -> 151,050,496 vertices
R(5,6,9,10,10,10)	184 115 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 46,028,993 – 3 = 184,115,969; with R(9,9,9,9) >= 719,204
R(5,6,10,10,10,10)	209 414 145	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 5 quadruplings -> 209,414,144 vertices
R(5,7,7,7,7,7)	17 741 144	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 17,741,143 vertices
R(5,7,7,7,7,8)	22 769 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 5,692,342 vertices; then 1 quadrupling -> 22,769,368 vertices
R(5,7,7,7,7,9)	29 695 069	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 29,695,068 vertices
R(5,7,7,7,7,10)	34 467 521	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 3 quadruplings -> 34,467,520 vertices
R(5,7,7,7,8,8)	28 841 137	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 2 quadruplings -> 28,841,136 vertices
R(5,7,7,7,8,9)	37 189 989	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 9,297,497 vertices; then 1 quadrupling -> 37,189,988 vertices
R(5,7,7,7,8,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,7,7,7,9,9)	49 448 864	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 49,448,863 vertices
R(5,7,7,7,9,10)	63 463 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 1 quadrupling -> 63,463,768 vertices
R(5,7,7,7,10,10)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(5,7,7,8,8,8)	34 467 521	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 3 quadruplings -> 34,467,520 vertices
R(5,7,7,8,8,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(5,7,7,8,8,10)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(5,7,7,8,9,9)	63 463 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 1 quadrupling -> 63,463,768 vertices
R(5,7,7,8,9,10)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(5,7,7,8,10,10)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(5,7,7,9,9,9)	82 767 349	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 82,767,348 vertices
R(5,7,7,9,9,10)	103 657 509	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 25,914,377 vertices; then 1 quadrupling -> 103,657,508 vertices
R(5,7,7,9,10,10)	126 927 569	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 2 quadruplings -> 126,927,568 vertices
R(5,7,7,10,10,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(5,7,8,8,8,8)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(5,7,8,8,8,9)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(5,7,8,8,8,10)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(5,7,8,8,9,9)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(5,7,8,8,9,10)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(5,7,8,8,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(5,7,8,9,9,9)	103 657 509	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 25,914,377 vertices; then 1 quadrupling -> 103,657,508 vertices
R(5,7,8,9,9,10)	126 927 569	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 2 quadruplings -> 126,927,568 vertices
R(5,7,8,9,10,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(5,7,8,10,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(5,7,9,9,9,9)	134 490 854	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 134,490,853 vertices
R(5,7,9,9,9,10)	172 608 569	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 43,152,142 vertices; then 1 quadrupling -> 172,608,568 vertices
R(5,7,9,9,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(5,7,9,10,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(5,7,10,10,10,10)	310 152 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,8,8,8,8,8)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(5,8,8,8,8,9)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(5,8,8,8,8,10)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(5,8,8,8,9,9)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(5,8,8,8,9,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(5,8,8,8,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(5,8,8,9,9,9)	126 927 569	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 2 quadruplings -> 126,927,568 vertices
R(5,8,8,9,9,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(5,8,8,9,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(5,8,8,10,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(5,8,9,9,9,9)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(5,8,9,9,9,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(5,8,9,9,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(5,8,9,10,10,10)	310 152 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,8,10,10,10,10)	384 120 833	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(5,9,9,9,9,9)	225 110 359	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 225,110,358 vertices
R(5,9,9,9,9,10)	281 927 349	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 70,481,837 vertices; then 1 quadrupling -> 281,927,348 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,9,9,9,10,10)	345 217 169	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 21,576,073 vertices; then 2 quadruplings -> 345,217,168 vertices
R(5,9,9,10,10,10)	466 115 009	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 3 quadruplings -> 466,115,008 vertices
R(5,9,10,10,10,10)	604 201 985	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 4 quadruplings -> 604,201,984 vertices
R(5,10,10,10,10,10)	736 463 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 184,115,969 – 3 = 736,463,873; with R(9,9,9,9) >= 719,204
R(6,6,6,6,6,6)	5 499 102	Published	published value
R(6,6,6,6,6,7)	5 663 370	Our bricks	[modele (6,6,3;235)] x [[modele (6,6,3;235)] x [atome (6,7 ; 104)]] [new atom (6,7;104), v19]
R(6,6,6,6,6,8)	5 640 089	Our bricks	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,410,022 vertices; then 1 quadrupling -> 5,640,088 vertices (by monotonicity, from (6,6,6,6,6,7))
R(6,6,6,6,6,9)	9 934 338	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 9,934,337 vertices
R(6,6,6,6,6,10)	11 084 214	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (6,10 ; 203)]] -> 11,084,213 vertices
R(6,6,6,6,7,7)	11 068 703	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices
R(6,6,6,6,7,8)	11 068 703	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices (by monotonicity, from (6,6,6,6,7,7))
R(6,6,6,6,7,9)	13 712 502	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (7,9 ; 251)]] -> 13,712,501 vertices
R(6,6,6,6,7,10)	15 902 742	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (7,10 ; 291)]] -> 15,902,741 vertices
R(6,6,6,6,8,8)	15 355 182	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 15,355,181 vertices
R(6,6,6,6,8,9)	17 578 317	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 4,394,579 vertices; then 1 quadrupling -> 17,578,316 vertices
R(6,6,6,6,8,10)	18 695 298	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (8,10 ; 342)]] -> 18,695,297 vertices
R(6,6,6,6,9,9)	30 741 618	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices
R(6,6,6,6,9,10)	30 741 618	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices (by monotonicity, from (6,6,6,6,9,9))
R(6,6,6,6,10,10)	43 609 278	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,6,6,7,7,7)	11 399 345	Our bricks	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (6,7 ; 104)]] [new atom (6,7;104), v19]
R(6,6,6,7,7,8)	14 947 249	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 3,736,812 vertices; then 1 quadrupling -> 14,947,248 vertices
R(6,6,6,7,7,9)	19 996 037	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 19,996,036 vertices
R(6,6,6,7,7,10)	22 560 353	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,410,022 vertices; then 2 quadruplings -> 22,560,352 vertices
R(6,6,6,7,8,8)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(6,6,6,7,8,9)	25 542 765	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices; then 1 quadrupling -> 25,542,764 vertices
R(6,6,6,7,8,10)	28 576 193	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 3 quadruplings -> 28,576,192 vertices
R(6,6,6,7,9,9)	31 529 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices; then 1 quadrupling -> 31,529,848 vertices
R(6,6,6,7,9,10)	41 514 317	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 10,378,579 vertices; then 1 quadrupling -> 41,514,316 vertices
R(6,6,6,7,10,10)	46 776 513	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 3 quadruplings -> 46,776,512 vertices
R(6,6,6,8,8,8)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices (by monotonicity, from (5,6,6,8,8,8))
R(6,6,6,8,8,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(6,6,6,8,8,10)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices (by monotonicity, from (5,6,6,8,8,10))
R(6,6,6,8,9,9)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(6,6,6,8,9,10)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices
R(6,6,6,8,10,10)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices (by monotonicity, from (5,6,6,8,10,10))
R(6,6,6,9,9,9)	54 386 850	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)]] -> 54,386,849 vertices
R(6,6,6,9,9,10)	70 942 185	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 17,735,546 vertices; then 1 quadrupling -> 70,942,184 vertices
R(6,6,6,9,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(6,6,6,10,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices (by monotonicity, from (5,6,6,10,10,10))
R(6,6,7,7,7,7)	22 278 974	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices
R(6,6,7,7,7,8)	22 278 974	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices (by monotonicity, from (6,6,7,7,7,7))
R(6,6,7,7,7,9)	28 971 617	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 2 quadruplings -> 28,971,616 vertices
R(6,6,7,7,7,10)	32 009 363	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (7,10 ; 291)]] -> 32,009,362 vertices
R(6,6,7,7,8,8)	30 907 223	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices
R(6,6,7,7,8,9)	35 381 465	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 8,845,366 vertices; then 1 quadrupling -> 35,381,464 vertices

Parameters	Bound (R >=)	Origin	Construction
R(6,6,7,7,8,10)	45 409 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 2,838,082 vertices; then 2 quadruplings -> 45,409,312 vertices
R(6,6,7,7,9,9)	61 877 357	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices
R(6,6,7,7,9,10)	61 877 357	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices (by monotonicity, from (6,6,7,7,9,9))
R(6,6,7,7,10,10)	87 777 647	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,6,7,8,8,8)	30 907 223	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices (by monotonicity, from (6,6,7,7,8,8))
R(6,6,7,8,8,9)	45 409 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 2,838,082 vertices; then 2 quadruplings -> 45,409,312 vertices
R(6,6,7,8,8,10)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(6,6,7,8,9,9)	61 877 357	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices (by monotonicity, from (6,6,7,7,9,9))
R(6,6,7,8,9,10)	78 839 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 2 quadruplings -> 78,839,024 vertices
R(6,6,7,8,10,10)	87 777 647	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,6,7,9,9,9)	75 069 876	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (7,9 ; 251)]] -> 75,069,875 vertices
R(6,6,7,9,9,10)	98 268 069	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 24,567,017 vertices; then 1 quadrupling -> 98,268,068 vertices
R(6,6,7,9,10,10)	126 119 393	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices; then 2 quadruplings -> 126,119,392 vertices
R(6,6,7,10,10,10)	159 750 977	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 3 quadruplings -> 159,750,976 vertices
R(6,6,8,8,8,8)	42 588 102	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 42,588,101 vertices
R(6,6,8,8,8,9)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(6,6,8,8,8,10)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices (by monotonicity, from (6,6,7,8,8,10))
R(6,6,8,8,9,9)	85 262 448	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 85,262,447 vertices
R(6,6,8,8,9,10)	91 271 489	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 3 quadruplings -> 91,271,488 vertices
R(6,6,8,8,10,10)	120 950 958	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (8,8,3 ; 650)] x [atome (10,10 ; 797)]]
R(6,6,8,9,9,9)	98 268 069	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 24,567,017 vertices; then 1 quadrupling -> 98,268,068 vertices
R(6,6,8,9,9,10)	126 119 393	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices; then 2 quadruplings -> 126,119,392 vertices
R(6,6,8,9,10,10)	162 260 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 5 quadruplings -> 162,260,992 vertices
R(6,6,8,10,10,10)	162 260 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 5 quadruplings -> 162,260,992 vertices (by monotonicity, from (6,6,8,9,10,10))
R(6,6,9,9,9,9)	168 293 370	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices
R(6,6,9,9,9,10)	168 293 370	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices (by monotonicity, from (6,6,9,9,9,9))
R(6,6,9,9,10,10)	238 735 560	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(6,6,9,10,10,10)	252 238 913	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 3,941,233 vertices; then 3 quadruplings -> 252,238,912 vertices
R(6,6,10,10,10,10)	296 904 918	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele generique 2m-1 de atome (10,10 ; 797)] x [atome (10,10 ; 797)]]
R(6,7,7,7,7,7)	22 944 497	Our bricks	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [atome (6,7 ; 104)]] [new atom (6,7;104), v19]
R(6,7,7,7,7,8)	29 979 685	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 7,494,921 vertices; then 1 quadrupling -> 29,979,684 vertices
R(6,7,7,7,7,9)	40 248 095	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (6,9 ; 182)]] -> 40,248,094 vertices
R(6,7,7,7,7,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(6,7,7,7,8,8)	34 467 521	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 3 quadruplings -> 34,467,520 vertices
R(6,7,7,7,8,9)	51 231 105	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 1 quadrupling -> 51,231,104 vertices
R(6,7,7,7,8,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(6,7,7,7,9,9)	63 463 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 1 quadrupling -> 63,463,768 vertices
R(6,7,7,7,9,10)	83 560 645	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices; then 1 quadrupling -> 83,560,644 vertices
R(6,7,7,7,10,10)	94 152 641	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 3 quadruplings -> 94,152,640 vertices
R(6,7,7,8,8,8)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(6,7,7,8,8,9)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(6,7,7,8,8,10)	68 538 881	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 4 quadruplings -> 68,538,880 vertices
R(6,7,7,8,9,9)	83 560 645	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices; then 1 quadrupling -> 83,560,644 vertices

Parameters	Bound (R >=)	Origin	Construction
R(6,7,7,8,9,10)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(6,7,7,8,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(6,7,7,9,9,9)	109 470 965	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)]] -> 109,470,964 vertices
R(6,7,7,9,9,10)	142 793 505	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 1 quadrupling -> 142,793,504 vertices
R(6,7,7,9,10,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(6,7,7,10,10,10)	187 106 049	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 4 quadruplings -> 187,106,048 vertices
R(6,7,8,8,8,8)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(6,7,8,8,8,9)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(6,7,8,8,8,10)	91 436 033	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 5 quadruplings -> 91,436,032 vertices
R(6,7,8,8,9,9)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(6,7,8,8,9,10)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(6,7,8,8,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(6,7,8,9,9,9)	142 793 505	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 1 quadrupling -> 142,793,504 vertices
R(6,7,8,9,9,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(6,7,8,9,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(6,7,8,10,10,10)	229 867 521	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 7 quadruplings -> 229,867,520 vertices
R(6,7,9,9,9,9)	172 608 569	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 43,152,142 vertices; then 1 quadrupling -> 172,608,568 vertices
R(6,7,9,9,9,10)	227 267 965	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 56,816,991 vertices; then 1 quadrupling -> 227,267,964 vertices
R(6,7,9,9,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(6,7,9,10,10,10)	341 853 185	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 4 quadruplings -> 341,853,184 vertices
R(6,7,10,10,10,10)	368 643 329	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 1,440,013 vertices; then 4 quadruplings -> 368,643,328 vertices
R(6,8,8,8,8,8)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices (by monotonicity, from (5,8,8,8,8,8))
R(6,8,8,8,8,9)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(6,8,8,8,8,10)	97 149 953	Published bricks + theorems	[template (7,7,3;472)] x [Mathon-doubling (7,7;202)] -> 94,873 vertices; then 5 quadruplings -> 97,149,952 vertices (by monotonicity, from (5,8,8,8,8,10))
R(6,8,8,8,9,9)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(6,8,8,8,9,10)	162 260 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 5 quadruplings -> 162,260,992 vertices
R(6,8,8,8,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices (by monotonicity, from (5,8,8,8,10,10))
R(6,8,8,9,9,9)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(6,8,8,9,9,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(6,8,8,9,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(6,8,8,10,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices (by monotonicity, from (5,8,8,10,10,10))
R(6,8,9,9,9,9)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(6,8,9,9,9,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(6,8,9,9,10,10)	341 853 185	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 4 quadruplings -> 341,853,184 vertices
R(6,8,9,10,10,10)	441 476 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 5 quadruplings -> 441,476,096 vertices
R(6,8,10,10,10,10)	441 476 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 5 quadruplings -> 441,476,096 vertices (by monotonicity, from (6,8,9,10,10,10))
R(6,9,9,9,9,9)	297 733 145	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)]] -> 297,733,144 vertices
R(6,9,9,9,9,10)	388 369 305	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 97,092,326 vertices; then 1 quadrupling -> 388,369,304 vertices
R(6,9,9,9,10,10)	466 115 009	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 3 quadruplings -> 466,115,008 vertices
R(6,9,9,10,10,10)	604 201 985	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 4 quadruplings -> 604,201,984 vertices
R(6,9,10,10,10,10)	736 463 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 184,115,969 – 3 = 736,463,873; with R(9,9,9,9) >= 719,204
R(6,10,10,10,10,10)	736 463 873	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)] -> 719,203 vertices; then 5 quadruplings -> 736,463,872 vertices (by monotonicity, from (5,10,10,10,10,10))
R(7,7,7,7,7,7)	44 684 915	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices
R(7,7,7,7,7,8)	44 684 915	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices (by monotonicity, from (7,7,7,7,7,7))
R(7,7,7,7,7,9)	58 309 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 2 quadruplings -> 58,309,264 vertices
R(7,7,7,7,7,10)	64 428 764	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (7,10 ; 291)]] -> 64,428,763 vertices
R(7,7,7,7,8,8)	62 210 354	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices
R(7,7,7,7,8,9)	70 964 573	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 17,741,143 vertices; then 1 quadrupling -> 70,964,572 vertices

Parameters	Bound (R >=)	Origin	Construction
R(7,7,7,7,8,10)	91 077 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 5,692,342 vertices; then 2 quadruplings -> 91,077,472 vertices
R(7,7,7,7,9,9)	124 547 675	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices
R(7,7,7,7,9,10)	124 547 675	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices (by monotonicity, from (7,7,7,7,9,9))
R(7,7,7,7,10,10)	176 680 310	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [atome (10,10 ; 797)]]
R(7,7,7,8,8,8)	62 210 354	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices (by monotonicity, from (7,7,7,7,8,8))
R(7,7,7,8,8,9)	91 077 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 5,692,342 vertices; then 2 quadruplings -> 91,077,472 vertices
R(7,7,7,8,8,10)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(7,7,7,8,9,9)	124 547 675	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices (by monotonicity, from (7,7,7,7,9,9))
R(7,7,7,8,9,10)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(7,7,7,8,10,10)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(7,7,7,9,9,9)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(7,7,7,9,9,10)	197 795 453	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 49,448,863 vertices; then 1 quadrupling -> 197,795,452 vertices
R(7,7,7,9,10,10)	253 855 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 2 quadruplings -> 253,855,072 vertices
R(7,7,7,10,10,10)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices
R(7,7,8,8,8,8)	85 722 203	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 85,722,202 vertices
R(7,7,8,8,8,9)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(7,7,8,8,8,10)	137 870 081	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 4 quadruplings -> 137,870,080 vertices
R(7,7,8,8,9,9)	171 618 002	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices
R(7,7,8,8,9,10)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(7,7,8,8,10,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(7,7,8,9,9,9)	197 795 453	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 49,448,863 vertices; then 1 quadrupling -> 197,795,452 vertices
R(7,7,8,9,9,10)	253 855 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 2 quadruplings -> 253,855,072 vertices
R(7,7,8,9,10,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(7,7,8,10,10,10)	376 610 561	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 4 quadruplings -> 376,610,560 vertices
R(7,7,9,9,9,9)	338 744 345	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices
R(7,7,9,9,9,10)	338 744 345	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices (by monotonicity, from (7,7,9,9,9,9))
R(7,7,9,9,10,10)	480 531 830	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(7,7,9,10,10,10)	507 710 273	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 3 quadruplings -> 507,710,272 vertices
R(7,7,10,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(7,8,8,8,8,8)	85 722 203	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 85,722,202 vertices (by monotonicity, from (7,7,8,8,8,8))
R(7,8,8,8,8,9)	137 870 081	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]] -> 538,555 vertices; then 4 quadruplings -> 137,870,080 vertices
R(7,8,8,8,8,10)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(7,8,8,8,9,9)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(7,8,8,8,9,10)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(7,8,8,8,10,10)	252 162 049	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 5 quadruplings -> 252,162,048 vertices
R(7,8,8,9,9,9)	253 855 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 15,865,942 vertices; then 2 quadruplings -> 253,855,072 vertices
R(7,8,8,9,9,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(7,8,8,9,10,10)	388 599 809	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 97,149,953 – 3 = 388,599,809; with R(7,7,7,7) >= 94,874
R(7,8,8,10,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(7,8,9,9,9,9)	338 744 345	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices (by monotonicity, from (7,7,9,9,9,9))
R(7,8,9,9,9,10)	431 593 265	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 2 quadruplings -> 431,593,264 vertices
R(7,8,9,9,10,10)	507 710 273	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 3 quadruplings -> 507,710,272 vertices
R(7,8,9,10,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(7,8,10,10,10,10)	720 977 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 7 quadruplings -> 720,977,920 vertices
R(7,9,9,9,9,9)	414 630 033	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 25,914,377 vertices; then 2 quadruplings -> 414,630,032 vertices

Parameters	Bound (R >=)	Origin	Construction
R(7,9,9,9,9,10)	537 963 413	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 134,490,853 vertices; then 1 quadrupling -> 537,963,412 vertices
R(7,9,9,9,10,10)	690 434 273	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 43,152,142 vertices; then 2 quadruplings -> 690,434,272 vertices
R(7,9,9,10,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(7,9,10,10,10,10)	1 083 117 569	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 6 quadruplings -> 1,083,117,568 vertices
R(7,10,10,10,10,10)	1 240 608 385	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(8,8,8,8,8,8)	118 118 282	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 118,118,281 vertices
R(8,8,8,8,8,9)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(8,8,8,8,8,10)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices (by monotonicity, from (7,8,8,8,8,10))
R(8,8,8,8,9,9)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(8,8,8,8,9,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(8,8,8,8,10,10)	335 457 998	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,8,3 ; 650)] x [[modele (8,8,3 ; 650)] x [atome (10,10 ; 797)]]
R(8,8,8,9,9,9)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(8,8,8,9,9,10)	388 599 809	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 97,149,953 – 3 = 388,599,809; with R(7,7,7,7) >= 94,874
R(8,8,8,9,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(8,8,8,10,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices (by monotonicity, from (7,8,8,10,10,10))
R(8,8,9,9,9,9)	466 762 380	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 466,762,379 vertices
R(8,8,9,9,9,10)	507 710 273	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 3 quadruplings -> 507,710,272 vertices
R(8,8,9,9,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(8,8,9,10,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(8,8,10,10,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices (by monotonicity, from (8,8,9,10,10,10))
R(8,9,9,9,9,9)	537 963 413	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 134,490,853 vertices; then 1 quadrupling -> 537,963,412 vertices
R(8,9,9,9,9,10)	690 434 273	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 43,152,142 vertices; then 2 quadruplings -> 690,434,272 vertices
R(8,9,9,9,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(8,9,9,10,10,10)	1 083 117 569	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 6 quadruplings -> 1,083,117,568 vertices
R(8,9,10,10,10,10)	1 240 608 385	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(8,10,10,10,10,10)	1 240 608 385	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(9,9,9,9,9,9)	921 298 325	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 921,298,324 vertices
R(9,9,9,9,9,10)	921 298 325	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 921,298,324 vertices (by monotonicity, from (9,9,9,9,9,9))
R(9,9,9,9,10,10)	1 306 924 160	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(9,9,9,10,10,10)	1 380 868 673	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 21,576,073 vertices; then 3 quadruplings -> 1,380,868,672 vertices
R(9,9,10,10,10,10)	1 864 460 033	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 4 quadruplings -> 1,864,460,032 vertices
R(9,10,10,10,10,10)	2 416 807 937	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 5 quadruplings -> 2,416,807,936 vertices
R(10,10,10,10,10,10)	2 416 807 937	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 5 quadruplings -> 2,416,807,936 vertices (by monotonicity, from (9,10,10,10,10,10))
R(3,3,3,3,3,3,3)	1 698	Published	published value
R(3,3,3,3,3,3,4)	3 153	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [(3,3,4;29)] -> 3,152 vertices
R(3,3,3,3,3,3,5)	5 090	Our bricks + published	[template (3,3,3,3 ; 34)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 5,089 vertices
R(3,3,3,3,3,3,6)	6 483	Our bricks	[template (3,3,3,3,3 ; 112)] x [atom (3,3,6 ; 59) (ours)] -> 6,482 vertices
R(3,3,3,3,3,3,7)	9 142	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 9,141 vertices
R(3,3,3,3,3,3,8)	11 547	Our bricks	[template (3,8,3 ; 73) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 11,546 vertices
R(3,3,3,3,3,3,9)	14 586	Our bricks + published	[template (3,3,3,3,3 ; 112)] x [atom (3,3,9 ; 132) (ours)] -> 14,585 vertices
R(3,3,3,3,3,3,10)	17 321	Our bricks	[template (3,3,3,3 ; 34)] x [atom (3,3,9 ; 132) (ours)] -> 4,330 vertices; then 1 quadrupling -> 17,320 vertices
R(3,3,3,3,3,4,4)	6 230	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Paley-17 (4,4;17)] -> 6,229 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,3,3,4,5)	10 368	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 10,367 vertices
R(3,3,3,3,3,4,6)	12 674	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 12,673 vertices
R(3,3,3,3,3,4,7)	18 447	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 18,446 vertices
R(3,3,3,3,3,4,8)	21 289	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 5,322 vertices; then 1 quadrupling -> 21,288 vertices
R(3,3,3,3,3,4,9)	26 889	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 6,722 vertices; then 1 quadrupling -> 26,888 vertices
R(3,3,3,3,3,4,10)	34 350	Our bricks + published	[template (3,3,3,3,3,3 ; 381)] x [atom (4,10 ; 91) (ours)] -> 34,349 vertices
R(3,3,3,3,3,5,5)	17 139	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 17,138 vertices
R(3,3,3,3,3,5,6)	21 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 21,656 vertices
R(3,3,3,3,3,5,7)	30 000	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 29,999 vertices
R(3,3,3,3,3,5,8)	37 770	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [atom (5,8 ; 100)] -> 37,769 vertices
R(3,3,3,3,3,5,9)	50 213	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 50,212 vertices
R(3,3,3,3,3,5,10)	58 637	Our bricks + published	[template (3,3,3,3 ; 34)] x [[template (5,9,3 ; 314) (ours)] x [K2]] -> 14,659 vertices; then 1 quadrupling -> 58,636 vertices
R(3,3,3,3,3,6,6)	38 150	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Paley-101 (6,6;101)] -> 38,149 vertices
R(3,3,3,3,3,6,7)	44 577	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 1 quadrupling -> 44,576 vertices
R(3,3,3,3,3,6,8)	53 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 2 quadruplings -> 53,808 vertices
R(3,3,3,3,3,6,9)	68 930	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [atom (6,9 ; 182)] -> 68,929 vertices
R(3,3,3,3,3,6,10)	86 273	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 21,569 – 3 = 86,273; with R(3,6,6) >= 338
R(3,3,3,3,3,7,7)	76 530	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Mathon-doubling (7,7;202)] -> 76,529 vertices
R(3,3,3,3,3,7,8)	89 421	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 1 quadrupling -> 89,420 vertices
R(3,3,3,3,3,7,9)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,3,3,3,3,7,10)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,3,3,3,3,8,8)	107 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 2 quadruplings -> 107,568 vertices
R(3,3,3,3,3,8,9)	133 505	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 3 quadruplings -> 133,504 vertices
R(3,3,3,3,3,8,10)	172 289	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 43,073 – 3 = 172,289; with R(3,7,7) >= 674
R(3,3,3,3,3,9,9)	213 330	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [atom (9,9 ; 562)] -> 213,329 vertices
R(3,3,3,3,3,9,10)	249 261	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 1 quadrupling -> 249,260 vertices
R(3,3,3,3,3,10,10)	302 630	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,3,3,3,3 ; 381)] x [atome (10,10 ; 797)]
R(3,3,3,3,4,4,4)	14 031	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [atom (4,4,4 ; 127)] -> 14,030 vertices
R(3,3,3,3,4,4,5)	18 693	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 18,692 vertices
R(3,3,3,3,4,4,6)	23 709	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,4 ; 54)] -> 5,927 vertices; then 1 quadrupling -> 23,708 vertices
R(3,3,3,3,4,4,7)	33 167	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 33,166 vertices
R(3,3,3,3,4,4,8)	42 965	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 10,741 vertices; then 1 quadrupling -> 42,964 vertices
R(3,3,3,3,4,4,9)	52 257	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [(3,3,4;29)] -> 3,266 vertices; then 2 quadruplings -> 52,256 vertices
R(3,3,3,3,4,4,10)	62 517	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,4,9 ; 241)] -> 15,629 vertices; then 1 quadrupling -> 62,516 vertices
R(3,3,3,3,4,5,5)	29 460	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K3]] -> 29,459 vertices
R(3,3,3,3,4,5,6)	41 469	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 10,367 vertices; then 1 quadrupling -> 41,468 vertices
R(3,3,3,3,4,5,7)	56 121	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,3,3,3,4,5,8)	73 785	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 18,446 vertices; then 1 quadrupling -> 73,784 vertices
R(3,3,3,3,4,5,9)	90 300	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,3,4,5,9) of 29,241 vertices] + [colouring (3,4,5,9) of 2,576 vertices] (see Chung_Robertson.cpp) -> 90,299 vertices
R(3,3,3,3,4,5,10)	116 965	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 29,241 vertices; then 1 quadrupling -> 116,964 vertices
R(3,3,3,3,4,6,6)	67 760	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 67,759 vertices
R(3,3,3,3,4,6,7)	87 477	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 1 quadrupling -> 87,476 vertices
R(3,3,3,3,4,6,8)	106 449	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 2 quadruplings -> 106,448 vertices
R(3,3,3,3,4,6,9)	121 793	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 3 quadruplings -> 121,792 vertices
R(3,3,3,3,4,6,10)	151 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,4,5 ; 86) (ours)] -> 9,479 vertices; then 2 quadruplings -> 151,664 vertices
R(3,3,3,3,4,7,7)	135 851	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,3,4,7,8)	176 021	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 1 quadrupling -> 176,020 vertices
R(3,3,3,3,4,7,9)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,3,3,3,4,7,10)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(3,3,3,3,4,8,8)	214 241	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 2 quadruplings -> 214,240 vertices
R(3,3,3,3,4,8,9)	244 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 3 quadruplings -> 244,608 vertices
R(3,3,3,3,4,8,10)	295 249	Published bricks + theorems	[template (8,8,3 ; 650)] x [(3,3,4;29)] -> 18,453 vertices; then 2 quadruplings -> 295,248 vertices
R(3,3,3,3,4,9,9)	369 940	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,3,4,9,9) of 119,759 vertices] + [colouring (3,4,9,9) of 10,662 vertices] (see Chung_Robertson.cpp) -> 369,939 vertices
R(3,3,3,3,4,9,10)	479 037	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 1 quadrupling -> 479,036 vertices
R(3,3,3,3,4,10,10)	583 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 2 quadruplings -> 583,904 vertices
R(3,3,3,3,5,5,5)	50 217	Published bricks + theorems	[template (3,3,3,3,3 ; 112)] x [atom (5,5,5 ; 453)] -> 50,216 vertices
R(3,3,3,3,5,5,6)	68 553	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 17,138 vertices; then 1 quadrupling -> 68,552 vertices
R(3,3,3,3,5,5,7)	91 731	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [(3,3;5)]] -> 91,730 vertices
R(3,3,3,3,5,5,8)	119 997	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 29,999 vertices; then 1 quadrupling -> 119,996 vertices
R(3,3,3,3,5,5,9)	153 558	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [(3,3;5)]] -> 153,557 vertices
R(3,3,3,3,5,5,10)	200 849	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 50,212 vertices; then 1 quadrupling -> 200,848 vertices
R(3,3,3,3,5,6,6)	115 707	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices
R(3,3,3,3,5,6,7)	152 597	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Paley-101 (6,6;101)] -> 38,149 vertices; then 1 quadrupling -> 152,596 vertices
R(3,3,3,3,5,6,8)	178 305	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 2 quadruplings -> 178,304 vertices
R(3,3,3,3,5,6,9)	215 233	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 3 quadruplings -> 215,232 vertices
R(3,3,3,3,5,6,10)	275 717	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [atom (6,9 ; 182)] -> 68,929 vertices; then 1 quadrupling -> 275,716 vertices
R(3,3,3,3,5,7,7)	231 480	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices
R(3,3,3,3,5,7,8)	306 117	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Mathon-doubling (7,7;202)] -> 76,529 vertices; then 1 quadrupling -> 306,116 vertices
R(3,3,3,3,5,7,9)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,3,3,5,7,10)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(3,3,3,3,5,8,8)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,3,3,5,8,9)	430 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 3 quadruplings -> 430,272 vertices
R(3,3,3,3,5,8,10)	534 017	Published bricks + theorems	[template (7,7,3;472)] x [(3,3;5)] -> 2,086 vertices; then 4 quadruplings -> 534,016 vertices
R(3,3,3,3,5,9,9)	631 080	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 631,079 vertices
R(3,3,3,3,5,9,10)	853 317	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [atom (9,9 ; 562)] -> 213,329 vertices; then 1 quadrupling -> 853,316 vertices
R(3,3,3,3,5,10,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(3,3,3,3,6,6,6)	149 361	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 1 quadrupling -> 149,360 vertices
R(3,3,3,3,6,6,7)	191 724	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (3,7,3 ; 58) (ours)] x [atom (3,6,6 ; 337)]] -> 191,723 vertices
R(3,3,3,3,6,6,8)	242 174	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (3,8,3 ; 73) (ours)] x [atom (3,6,6 ; 337)]] -> 242,173 vertices
R(3,3,3,3,6,6,9)	307 544	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,3,9 ; 132) (ours)]] -> 307,543 vertices
R(3,3,3,3,6,6,10)	387 489	Our bricks	[template (3,8,3 ; 73) (ours)] x [atom (3,6,6 ; 337)] -> 24,218 vertices; then 2 quadruplings -> 387,488 vertices
R(3,3,3,3,6,7,7)	298 545	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 1 quadrupling -> 298,544 vertices
R(3,3,3,3,6,7,8)	357 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 2 quadruplings -> 357,680 vertices
R(3,3,3,3,6,7,9)	462 825	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 1 quadrupling -> 462,824 vertices
R(3,3,3,3,6,7,10)	597 441	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 2 quadruplings -> 597,440 vertices
R(3,3,3,3,6,8,8)	440 321	Our bricks + published	[template (7,7,3;472)] x [atom (3,3,6 ; 59) (ours)] -> 27,520 vertices; then 2 quadruplings -> 440,320 vertices
R(3,3,3,3,6,8,9)	610 385	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Paley-101 (6,6;101)] -> 38,149 vertices; then 2 quadruplings -> 610,384 vertices
R(3,3,3,3,6,8,10)	713 217	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 3 quadruplings -> 713,216 vertices
R(3,3,3,3,6,9,9)	818 025	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 1 quadrupling -> 818,024 vertices
R(3,3,3,3,6,9,10)	997 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 2 quadruplings -> 997,040 vertices
R(3,3,3,3,6,10,10)	1 197 761	Our bricks + published	[template (9,9,3 ; 1282)] x [atom (3,3,6 ; 59) (ours)] -> 74,860 vertices; then 2 quadruplings -> 1,197,760 vertices
R(3,3,3,3,7,7,7)	383 244	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 383,243 vertices
R(3,3,3,3,7,7,8)	543 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 1 quadrupling -> 543,400 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,3,7,7,9)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,3,3,7,7,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,3,3,7,8,8)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,3,3,7,8,9)	925 917	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 1 quadrupling -> 925,916 vertices
R(3,3,3,3,7,8,10)	1 194 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 2 quadruplings -> 1,194,176 vertices
R(3,3,3,3,7,9,9)	1 224 465	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Mathon-doubling (7,7;202)] -> 76,529 vertices; then 2 quadruplings -> 1,224,464 vertices
R(3,3,3,3,7,9,10)	1 480 181	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,3,7,9,10) of 479,036 vertices] + [colouring (3,7,9,10) of 43,072 vertices] (see Chung_Robertson.cpp) -> 1,480,180 vertices
R(3,3,3,3,7,10,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,3,3,3,8,8,8)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,3,3,8,8,9)	1 224 465	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [Mathon-doubling (7,7;202)] -> 76,529 vertices; then 2 quadruplings -> 1,224,464 vertices
R(3,3,3,3,8,8,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,3,3,3,8,9,9)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,3,3,3,8,9,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,3,3,3,8,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,3,3,3,9,9,9)	1 721 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 4 quadruplings -> 1,721,088 vertices
R(3,3,3,3,9,9,10)	2 524 317	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 631,079 vertices; then 1 quadrupling -> 2,524,316 vertices
R(3,3,3,3,9,10,10)	3 413 265	Published bricks + theorems	[template (3,3,3,3,3,3 ; 381)] x [atom (9,9 ; 562)] -> 213,329 vertices; then 2 quadruplings -> 3,413,264 vertices
R(3,3,3,3,10,10,10)	3 988 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 3 quadruplings -> 3,988,160 vertices
R(3,3,3,4,4,4,4)	25 291	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,4,4) of 8,219 vertices] + [colouring (4,4,4,4) of 633 vertices] (see Chung_Robertson.cpp) -> 25,290 vertices
R(3,3,3,4,4,4,5)	43 229	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,4,5) of 14,030 vertices] + [colouring (4,4,4,5) of 1,138 vertices] (see Chung_Robertson.cpp) -> 43,228 vertices
R(3,3,3,4,4,4,6)	56 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,3,3,4,4,4,7)	72 024	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (3,7,3 ; 58) (ours)] x [atom (4,4,4 ; 127)]] -> 72,023 vertices
R(3,3,3,4,4,4,8)	90 974	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)]] -> 90,973 vertices
R(3,3,3,4,4,4,9)	115 249	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [atom (4,4,4 ; 127)] -> 7,203 vertices; then 2 quadruplings -> 115,248 vertices
R(3,3,3,4,4,4,10)	145 569	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 2 quadruplings -> 145,568 vertices
R(3,3,3,4,4,5,5)	56 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 1 quadrupling -> 56,120 vertices
R(3,3,3,4,4,5,6)	74 769	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 18,692 vertices; then 1 quadrupling -> 74,768 vertices
R(3,3,3,4,4,5,7)	103 705	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-17 (4,4;17)]] -> 103,704 vertices
R(3,3,3,4,4,5,8)	132 753	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,5,8) of 42,964 vertices] + [colouring (4,4,5,8) of 3,860 vertices] (see Chung_Robertson.cpp) -> 132,752 vertices
R(3,3,3,4,4,5,9)	173 501	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,5,9) of 56,120 vertices] + [colouring (4,4,5,9) of 5,140 vertices] (see Chung_Robertson.cpp) -> 173,500 vertices
R(3,3,3,4,4,5,10)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,3,3,4,4,6,6)	126 860	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 126,859 vertices
R(3,3,3,4,4,6,7)	172 913	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,6,7) of 56,120 vertices] + [colouring (4,4,6,7) of 4,552 vertices] (see Chung_Robertson.cpp) -> 172,912 vertices
R(3,3,3,4,4,6,8)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,3,3,4,4,6,9)	246 081	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices; then 3 quadruplings -> 246,080 vertices
R(3,3,3,4,4,6,10)	291 329	Published bricks + theorems	[template (5,3;10)] x [atom (4,4,4 ; 127)] -> 1,138 vertices; then 4 quadruplings -> 291,328 vertices
R(3,3,3,4,4,7,7)	255 329	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices
R(3,3,3,4,4,7,8)	317 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 1 quadrupling -> 317,320 vertices
R(3,3,3,4,4,7,9)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(3,3,3,4,4,7,10)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,3,3,4,4,8,8)	402 641	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 2 quadruplings -> 402,640 vertices
R(3,3,3,4,4,8,9)	495 233	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 3 quadruplings -> 495,232 vertices
R(3,3,3,4,4,8,10)	582 273	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [atom (4,4,4 ; 127)] -> 9,098 vertices; then 3 quadruplings -> 582,272 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,4,4,9,9)	694 889	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 694,888 vertices
R(3,3,3,4,4,9,10)	863 081	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 1 quadrupling -> 863,080 vertices
R(3,3,3,4,4,10,10)	1 095 281	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 2 quadruplings -> 1,095,280 vertices
R(3,3,3,4,5,5,5)	92 850	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 92,849 vertices
R(3,3,3,4,5,5,6)	117 837	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K3]] -> 29,459 vertices; then 1 quadrupling -> 117,836 vertices
R(3,3,3,4,5,5,7)	173 063	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,5,7) of 56,120 vertices] + [colouring (4,5,5,7) of 4,702 vertices] (see Chung_Robertson.cpp) -> 173,062 vertices
R(3,3,3,4,5,5,8)	224 481	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 2 quadruplings -> 224,480 vertices
R(3,3,3,4,5,5,9)	295 137	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 18,446 vertices; then 2 quadruplings -> 295,136 vertices
R(3,3,3,4,5,5,10)	361 197	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,5,10) of 116,964 vertices] + [colouring (4,5,5,10) of 10,304 vertices] (see Chung_Robertson.cpp) -> 361,196 vertices
R(3,3,3,4,5,6,6)	211 721	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 211,720 vertices
R(3,3,3,4,5,6,7)	271 037	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 67,759 vertices; then 1 quadrupling -> 271,036 vertices
R(3,3,3,4,5,6,8)	349 905	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 2 quadruplings -> 349,904 vertices
R(3,3,3,4,5,6,9)	425 793	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 3 quadruplings -> 425,792 vertices
R(3,3,3,4,5,6,10)	487 169	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Paley-101 (6,6;101)] -> 1,903 vertices; then 4 quadruplings -> 487,168 vertices
R(3,3,3,4,5,7,7)	424 730	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices
R(3,3,3,4,5,7,8)	543 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 1 quadrupling -> 543,400 vertices
R(3,3,3,4,5,7,9)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,3,4,5,7,10)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,3,4,5,8,8)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,3,4,5,8,9)	856 961	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 3 quadruplings -> 856,960 vertices
R(3,3,3,4,5,8,10)	978 433	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [Mathon-doubling (7,7;202)] -> 3,822 vertices; then 4 quadruplings -> 978,432 vertices
R(3,3,3,4,5,9,9)	1 183 970	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 1,183,969 vertices
R(3,3,3,4,5,9,10)	1 479 757	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,9,10) of 479,036 vertices] + [colouring (4,5,9,10) of 42,648 vertices] (see Chung_Robertson.cpp) -> 1,479,756 vertices
R(3,3,3,4,5,10,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,3,3,4,6,6,6)	265 764	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,6,3 ; 80) (ours)] x [atom (3,6,6 ; 337)]] -> 265,763 vertices
R(3,3,3,4,6,6,7)	386 854	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)]] -> 386,853 vertices
R(3,3,3,4,6,6,8)	479 505	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 119,876 vertices; then 1 quadrupling -> 479,504 vertices
R(3,3,3,4,6,6,9)	618 977	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 2 quadruplings -> 618,976 vertices
R(3,3,3,4,6,6,10)	738 945	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)] -> 11,546 vertices; then 3 quadruplings -> 738,944 vertices
R(3,3,3,4,6,7,7)	531 204	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 531,203 vertices
R(3,3,3,4,6,7,8)	704 081	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 2 quadruplings -> 704,080 vertices
R(3,3,3,4,6,7,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,3,3,4,6,7,10)	1 018 369	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [Mathon-doubling (7,7;202)] -> 15,912 vertices; then 3 quadruplings -> 1,018,368 vertices
R(3,3,3,4,6,8,8)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,3,3,4,6,8,9)	1 084 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 67,759 vertices; then 2 quadruplings -> 1,084,144 vertices
R(3,3,3,4,6,8,10)	1 399 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 3 quadruplings -> 1,399,616 vertices
R(3,3,3,4,6,9,9)	1 463 591	Our bricks + published	[template (3,3,3,3 ; 34)] x [[template (4,6,3 ; 80) (ours)] x [atom (9,9 ; 562)]] -> 1,463,590 vertices
R(3,3,3,4,6,9,10)	1 916 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 2 quadruplings -> 1,916,144 vertices
R(3,3,3,4,6,10,10)	2 335 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)] -> 36,494 vertices; then 3 quadruplings -> 2,335,616 vertices
R(3,3,3,4,7,7,7)	773 254	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 773,253 vertices
R(3,3,3,4,7,7,8)	1 021 313	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 1 quadrupling -> 1,021,312 vertices
R(3,3,3,4,7,7,9)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(3,3,3,4,7,7,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,3,3,4,7,8,8)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(3,3,3,4,7,8,9)	1 698 917	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 1 quadrupling -> 1,698,916 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,4,7,8,10)	2 030 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K3]] -> 126,917 vertices; then 2 quadruplings -> 2,030,672 vertices
R(3,3,3,4,7,9,9)	2 176 802	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,7,9,9) of 704,080 vertices] + [colouring (4,7,9,9) of 64,561 vertices] (see Chung_Robertson.cpp) -> 2,176,801 vertices
R(3,3,3,4,7,9,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,3,4,7,10,10)	3 452 321	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 2 quadruplings -> 3,452,320 vertices
R(3,3,3,4,8,8,8)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,3,3,4,8,8,9)	2 173 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 2 quadruplings -> 2,173,600 vertices
R(3,3,3,4,8,8,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,3,4,8,9,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,3,4,8,9,10)	3 452 321	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 2 quadruplings -> 3,452,320 vertices
R(3,3,3,4,8,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(3,3,3,4,9,9,9)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(3,3,3,4,9,9,10)	4 735 877	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 1,183,969 vertices; then 1 quadrupling -> 4,735,876 vertices
R(3,3,3,4,9,10,10)	5 919 025	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,9,10,10) of 1,916,144 vertices] + [colouring (4,9,10,10) of 170,592 vertices] (see Chung_Robertson.cpp) -> 5,919,024 vertices
R(3,3,3,4,10,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,3,3,5,5,5,5)	158 553	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 158,552 vertices
R(3,3,3,5,5,5,6)	200 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (5,5,5 ; 453)] -> 50,216 vertices; then 1 quadrupling -> 200,864 vertices
R(3,3,3,5,5,5,7)	274 209	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 17,138 vertices; then 2 quadruplings -> 274,208 vertices
R(3,3,3,5,5,5,8)	366 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [(3,3;5)]] -> 91,730 vertices; then 1 quadrupling -> 366,920 vertices
R(3,3,3,5,5,5,9)	479 985	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 29,999 vertices; then 2 quadruplings -> 479,984 vertices
R(3,3,3,5,5,5,10)	614 229	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [(3,3;5)]] -> 153,557 vertices; then 1 quadrupling -> 614,228 vertices
R(3,3,3,5,5,6,6)	340 304	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 340,303 vertices
R(3,3,3,5,5,6,7)	462 825	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 1 quadrupling -> 462,824 vertices
R(3,3,3,5,5,6,8)	604 177	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Paley-101 (6,6;101)] -> 37,761 vertices; then 2 quadruplings -> 604,176 vertices
R(3,3,3,5,5,6,9)	713 217	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 3 quadruplings -> 713,216 vertices
R(3,3,3,5,5,6,10)	860 929	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [atom (3,6,6 ; 337)] -> 3,363 vertices; then 4 quadruplings -> 860,928 vertices
R(3,3,3,5,5,7,7)	684 964	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 684,963 vertices
R(3,3,3,5,5,7,8)	925 917	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 1 quadrupling -> 925,916 vertices
R(3,3,3,5,5,7,9)	1 211 793	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 2 quadruplings -> 1,211,792 vertices
R(3,3,3,5,5,7,10)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,3,3,5,5,8,8)	1 211 793	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 2 quadruplings -> 1,211,792 vertices
R(3,3,3,5,5,8,9)	1 430 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 3 quadruplings -> 1,430,720 vertices
R(3,3,3,5,5,8,10)	1 721 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [K2]] -> 6,723 vertices; then 4 quadruplings -> 1,721,088 vertices
R(3,3,3,5,5,9,9)	1 869 507	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,5,3 ; 31)] x [atom (9,9 ; 562)]] -> 1,869,506 vertices
R(3,3,3,5,5,9,10)	2 524 317	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 631,079 vertices; then 1 quadrupling -> 2,524,316 vertices
R(3,3,3,5,5,10,10)	3 377 553	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [atom (9,9 ; 562)] -> 211,097 vertices; then 2 quadruplings -> 3,377,552 vertices
R(3,3,3,5,6,6,6)	462 825	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 1 quadrupling -> 462,824 vertices
R(3,3,3,5,6,6,7)	635 173	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 635,172 vertices
R(3,3,3,5,6,6,8)	802 323	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 802,322 vertices
R(3,3,3,5,6,6,9)	1 052 994	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (3,6,6 ; 337)]] -> 1,052,993 vertices
R(3,3,3,5,6,6,10)	1 305 113	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 326,278 vertices; then 1 quadrupling -> 1,305,112 vertices
R(3,3,3,5,6,7,7)	925 917	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 1 quadrupling -> 925,916 vertices
R(3,3,3,5,6,7,8)	1 211 793	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 2 quadruplings -> 1,211,792 vertices
R(3,3,3,5,6,7,9)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(3,3,3,5,6,7,10)	1 851 297	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 2 quadruplings -> 1,851,296 vertices
R(3,3,3,5,6,8,8)	1 452 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 2 quadruplings -> 1,452,416 vertices
R(3,3,3,5,6,8,9)	1 851 297	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 2 quadruplings -> 1,851,296 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,5,6,8,10)	2 416 705	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Paley-101 (6,6;101)] -> 37,761 vertices; then 3 quadruplings -> 2,416,704 vertices
R(3,3,3,5,6,9,9)	2 529 864	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,6,9,9) of 818,024 vertices] + [colouring (5,6,9,9) of 75,791 vertices] (see Chung_Robertson.cpp) -> 2,529,863 vertices
R(3,3,3,5,6,9,10)	3 377 553	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [atom (9,9 ; 562)] -> 211,097 vertices; then 2 quadruplings -> 3,377,552 vertices
R(3,3,3,5,6,10,10)	3 988 161	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 3 quadruplings -> 3,988,160 vertices
R(3,3,3,5,7,7,7)	1 274 200	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,274,199 vertices
R(3,3,3,5,7,7,8)	1 698 917	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 1 quadrupling -> 1,698,916 vertices
R(3,3,3,5,7,7,9)	2 175 286	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,7,7,9) of 704,080 vertices] + [colouring (5,7,7,9) of 63,045 vertices] (see Chung_Robertson.cpp) -> 2,175,285 vertices
R(3,3,3,5,7,7,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,3,5,7,8,8)	2 173 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 2 quadruplings -> 2,173,600 vertices
R(3,3,3,5,7,8,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,3,5,7,8,10)	3 703 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 2 quadruplings -> 3,703,664 vertices
R(3,3,3,5,7,9,9)	3 703 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 2 quadruplings -> 3,703,664 vertices
R(3,3,3,5,7,9,10)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,3,5,7,10,10)	5 920 721	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,7,10,10) of 1,916,144 vertices] + [colouring (5,7,10,10) of 172,288 vertices] (see Chung_Robertson.cpp) -> 5,920,720 vertices
R(3,3,3,5,8,8,8)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,3,5,8,8,9)	3 703 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 2 quadruplings -> 3,703,664 vertices
R(3,3,3,5,8,8,10)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,3,5,8,9,9)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,3,5,8,9,10)	5 920 721	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,8,9,10) of 1,916,144 vertices] + [colouring (5,8,9,10) of 172,288 vertices] (see Chung_Robertson.cpp) -> 5,920,720 vertices
R(3,3,3,5,8,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,3,3,5,9,9,9)	5 798 900	Our bricks + published	[template (3,3,3,3 ; 34)] x [[template (5,9,3 ; 314) (ours)] x [atom (9,9 ; 562)]] -> 5,798,899 vertices
R(3,3,3,5,9,9,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,3,3,5,9,10,10)	10 097 265	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 631,079 vertices; then 2 quadruplings -> 10,097,264 vertices
R(3,3,3,5,10,10,10)	13 510 209	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [atom (9,9 ; 562)] -> 211,097 vertices; then 3 quadruplings -> 13,510,208 vertices
R(3,3,3,6,6,6,6)	787 244	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 787,243 vertices
R(3,3,3,6,6,6,7)	975 717	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 1 quadrupling -> 975,716 vertices
R(3,3,3,6,6,6,8)	1 259 601	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 2 quadruplings -> 1,259,600 vertices
R(3,3,3,6,6,6,9)	1 504 065	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 376,017 – 3 = 1,504,065; with R(6,6,6,6) >= 23,502
R(3,3,3,6,6,6,10)	1 698 173	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 424,543 vertices; then 1 quadrupling -> 1,698,172 vertices
R(3,3,3,6,6,7,7)	1 584 574	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices
R(3,3,3,6,6,7,8)	1 963 937	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 1 quadrupling -> 1,963,936 vertices
R(3,3,3,6,6,7,9)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(3,3,3,6,6,7,10)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,3,3,6,6,8,8)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(3,3,3,6,6,8,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,3,3,6,6,8,10)	3 517 233	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,8,8 ; 940)] -> 219,827 vertices; then 2 quadruplings -> 3,517,232 vertices
R(3,3,3,6,6,9,9)	4 335 350	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices
R(3,3,3,6,6,9,10)	5 341 457	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 1 quadrupling -> 5,341,456 vertices
R(3,3,3,6,6,10,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(3,3,3,6,7,7,7)	1 963 937	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 1 quadrupling -> 1,963,936 vertices
R(3,3,3,6,7,7,8)	2 535 329	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 2 quadruplings -> 2,535,328 vertices
R(3,3,3,6,7,7,9)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices
R(3,3,3,6,7,7,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,3,3,6,7,8,8)	3 027 329	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 3 quadruplings -> 3,027,328 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,6,7,8,9)	3 703 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 2 quadruplings -> 3,703,664 vertices
R(3,3,3,6,7,8,10)	4 776 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [K2]] -> 74,636 vertices; then 3 quadruplings -> 4,776,704 vertices
R(3,3,3,6,7,9,9)	5 341 457	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 1 quadrupling -> 5,341,456 vertices
R(3,3,3,6,7,9,10)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(3,3,3,6,7,10,10)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(3,3,3,6,8,8,8)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,3,3,6,8,8,9)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,3,6,8,8,10)	5 809 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 3 quadruplings -> 5,809,664 vertices
R(3,3,3,6,8,9,9)	6 898 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 2 quadruplings -> 6,898,064 vertices
R(3,3,3,6,8,9,10)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(3,3,3,6,8,10,10)	9 559 041	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)] -> 37,340 vertices; then 4 quadruplings -> 9,559,040 vertices
R(3,3,3,6,9,9,9)	8 408 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 3 quadruplings -> 8,408,000 vertices
R(3,3,3,6,9,9,10)	10 119 453	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,6,9,9,10) of 3,272,096 vertices] + [colouring (6,9,9,10) of 303,164 vertices] (see Chung_Robertson.cpp) -> 10,119,452 vertices
R(3,3,3,6,9,10,10)	13 510 209	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [atom (9,9 ; 562)] -> 211,097 vertices; then 3 quadruplings -> 13,510,208 vertices
R(3,3,3,6,10,10,10)	15 952 641	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)] -> 62,315 vertices; then 4 quadruplings -> 15,952,640 vertices
R(3,3,3,7,7,7,7)	3 167 134	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices
R(3,3,3,7,7,7,8)	3 928 949	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 1 quadrupling -> 3,928,948 vertices
R(3,3,3,7,7,7,9)	5 067 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 2 quadruplings -> 5,067,424 vertices
R(3,3,3,7,7,7,10)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 − 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,3,3,7,7,8,8)	5 067 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 2 quadruplings -> 5,067,424 vertices
R(3,3,3,7,7,8,9)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 − 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,3,3,7,7,8,10)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(3,3,3,7,7,9,9)	8 726 264	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices
R(3,3,3,7,7,9,10)	10 711 349	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 1 quadrupling -> 10,711,348 vertices
R(3,3,3,7,7,10,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,3,3,7,8,8,8)	6 071 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,517,969 − 3 = 6,071,873; with R(7,7,7,7) >= 94,874
R(3,3,3,7,8,8,9)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(3,3,3,7,8,8,10)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,3,3,7,8,9,9)	10 711 349	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 1 quadrupling -> 10,711,348 vertices
R(3,3,3,7,8,9,10)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,3,3,7,8,10,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,3,3,7,9,9,9)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,3,3,7,9,9,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,3,3,7,9,10,10)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,3,3,7,10,10,10)	24 007 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,3,3,8,8,8,8)	7 079 537	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 2 quadruplings -> 7,079,536 vertices
R(3,3,3,8,8,8,9)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,3,3,8,8,8,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,3,3,8,8,9,9)	13 884 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 2 quadruplings -> 13,884,544 vertices
R(3,3,3,8,8,9,10)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,3,3,8,8,10,10)	19 254 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 2 quadruplings -> 19,254,736 vertices
R(3,3,3,8,9,9,9)	16 923 713	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 3 quadruplings -> 16,923,712 vertices
R(3,3,3,8,9,9,10)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,3,3,8,9,10,10)	24 007 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,3,3,8,10,10,10)	30 658 305	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 4 quadruplings -> 30,658,304 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,3,9,9,9,9)	23 733 674	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices
R(3,3,3,9,9,9,10)	29 132 189	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 1 quadrupling -> 29,132,188 vertices
R(3,3,3,9,9,10,10)	37 762 625	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 2 quadruplings -> 37,762,624 vertices
R(3,3,3,9,10,10,10)	46 028 993	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 11,507,249 – 3 = 46,028,993; with R(9,9,9,9) >= 719,204
R(3,3,3,10,10,10,10)	52 353 537	Our bricks	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 4 quadruplings -> 52,353,536 vertices
R(3,3,4,4,4,4,4)	45 528	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [atom (4,4,4 ; 127)]] -> 45,527 vertices
R(3,3,4,4,4,4,5)	75 814	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)]] -> 75,813 vertices
R(3,3,4,4,4,4,6)	99 864	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,6,3 ; 80) (ours)] x [atom (4,4,4 ; 127)]] -> 99,863 vertices
R(3,3,4,4,4,4,7)	145 354	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 145,353 vertices
R(3,3,4,4,4,4,8)	174 525	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 43,631 vertices; then 1 quadrupling -> 174,524 vertices
R(3,3,4,4,4,4,9)	232 577	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)] -> 14,536 vertices; then 2 quadruplings -> 232,576 vertices
R(3,3,4,4,4,4,10)	244 753	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [atom (4,4,4 ; 127)] -> 15,297 vertices; then 2 quadruplings -> 244,752 vertices
R(3,3,4,4,4,5,5)	126 266	Our bricks + published	[template (5,3;10)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)]] -> 126,265 vertices
R(3,3,4,4,4,5,6)	172 913	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,4,5) of 14,030 vertices] + [colouring (4,4,4,5) of 1,138 vertices] (see Chung_Robertson.cpp) -> 43,228 vertices; then 1 quadrupling -> 172,912 vertices
R(3,3,4,4,4,5,7)	236 404	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 236,403 vertices
R(3,3,4,4,4,5,8)	297 094	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)]] -> 297,093 vertices
R(3,3,4,4,4,5,9)	395 694	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)]] -> 395,693 vertices
R(3,3,4,4,4,5,10)	475 361	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)] -> 29,710 vertices; then 2 quadruplings -> 475,360 vertices
R(3,3,4,4,4,6,6)	295 844	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices
R(3,3,4,4,4,6,7)	357 173	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 1 quadrupling -> 357,172 vertices
R(3,3,4,4,4,6,8)	473 361	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 2 quadruplings -> 473,360 vertices
R(3,3,4,4,4,6,9)	507 074	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 507,073 vertices
R(3,3,4,4,4,6,10)	633 121	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 2 quadruplings -> 633,120 vertices
R(3,3,4,4,4,7,7)	595 474	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices
R(3,3,4,4,4,7,8)	714 845	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 1 quadrupling -> 714,844 vertices
R(3,3,4,4,4,7,9)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(3,3,4,4,4,7,10)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices (by monotonicity, from (3,3,4,4,4,7,9))
R(3,3,4,4,4,8,8)	952 769	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 2 quadruplings -> 952,768 vertices
R(3,3,4,4,4,8,9)	985 009	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 1 quadrupling -> 985,008 vertices
R(3,3,4,4,4,8,10)	1 312 881	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 2 quadruplings -> 1,312,880 vertices
R(3,3,4,4,4,9,9)	1 619 674	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices
R(3,3,4,4,4,9,10)	1 944 245	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 1 quadrupling -> 1,944,244 vertices
R(3,3,4,4,4,10,10)	2 591 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 2 quadruplings -> 2,591,488 vertices
R(3,3,4,4,5,5,5)	178 533	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-17 (4,4;17)]] -> 178,532 vertices
R(3,3,4,4,5,5,6)	245 910	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Paley-17 (4,4;17)]] -> 245,909 vertices
R(3,3,4,4,5,5,7)	340 815	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-17 (4,4;17)]] -> 340,814 vertices
R(3,3,4,4,5,5,8)	428 394	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [Paley-17 (4,4;17)]] -> 428,393 vertices
R(3,3,4,4,5,5,9)	570 474	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Paley-17 (4,4;17)]] -> 570,473 vertices
R(3,3,4,4,5,5,10)	694 001	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,5,9) of 56,120 vertices] + [colouring (4,4,5,9) of 5,140 vertices] (see Chung_Robertson.cpp) -> 173,500 vertices; then 1 quadrupling -> 694,000 vertices
R(3,3,4,4,5,6,6)	426 729	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 426,728 vertices
R(3,3,4,4,5,6,7)	507 437	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 126,859 vertices; then 1 quadrupling -> 507,436 vertices
R(3,3,4,4,5,6,8)	691 649	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,6,7) of 56,120 vertices] + [colouring (4,4,6,7) of 4,552 vertices] (see Chung_Robertson.cpp) -> 172,912 vertices; then 1 quadrupling -> 691,648 vertices
R(3,3,4,4,5,6,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,3,4,4,5,6,10)	984 321	Published bricks + theorems	[template (6,6,3;235)] x [Paley-17 (4,4;17)] -> 3,845 vertices; then 4 quadruplings -> 984,320 vertices
R(3,3,4,4,5,7,7)	858 852	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,4,4,5,7,8)	1 021 313	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 1 quadrupling -> 1,021,312 vertices
R(3,3,4,4,5,7,9)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(3,3,4,4,5,7,10)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,3,4,4,5,8,8)	1 269 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 2 quadruplings -> 1,269,280 vertices
R(3,3,4,4,5,8,9)	1 610 561	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 3 quadruplings -> 1,610,560 vertices
R(3,3,4,4,5,8,10)	1 980 929	Published bricks + theorems	[template (7,7,3;472)] x [Paley-17 (4,4;17)] -> 7,738 vertices; then 4 quadruplings -> 1,980,928 vertices
R(3,3,4,4,5,9,9)	2 337 372	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices
R(3,3,4,4,5,9,10)	2 779 553	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 694,888 vertices; then 1 quadrupling -> 2,779,552 vertices
R(3,3,4,4,5,10,10)	3 452 321	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 2 quadruplings -> 3,452,320 vertices
R(3,3,4,4,6,6,6)	525 542	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 525,541 vertices
R(3,3,4,4,6,6,7)	765 027	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 765,026 vertices
R(3,3,4,4,6,6,8)	900 171	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 900,170 vertices
R(3,3,4,4,6,6,9)	1 107 177	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 276,794 vertices; then 1 quadrupling -> 1,107,176 vertices
R(3,3,4,4,6,6,10)	1 375 554	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (4,10 ; 91) (ours)]] -> 1,375,553 vertices
R(3,3,4,4,6,7,7)	1 183 373	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 1 quadrupling -> 1,183,372 vertices
R(3,3,4,4,6,7,8)	1 428 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 2 quadruplings -> 1,428,688 vertices
R(3,3,4,4,6,7,9)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(3,3,4,4,6,7,10)	2 028 293	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 507,073 vertices; then 1 quadrupling -> 2,028,292 vertices
R(3,3,4,4,6,8,8)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(3,3,4,4,6,8,9)	2 029 745	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 126,859 vertices; then 2 quadruplings -> 2,029,744 vertices
R(3,3,4,4,6,8,10)	2 532 481	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)] -> 39,570 vertices; then 3 quadruplings -> 2,532,480 vertices
R(3,3,4,4,6,9,9)	2 882 981	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 2,882,980 vertices
R(3,3,4,4,6,9,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,3,4,4,6,10,10)	4 381 121	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 3 quadruplings -> 4,381,120 vertices
R(3,3,4,4,7,7,7)	1 539 782	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 1,539,781 vertices
R(3,3,4,4,7,7,8)	2 381 893	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 1 quadrupling -> 2,381,892 vertices
R(3,3,4,4,7,7,9)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(3,3,4,4,7,7,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,3,4,4,7,8,8)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(3,3,4,4,7,8,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,3,4,4,7,8,10)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(3,3,4,4,7,9,9)	4 196 742	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,196,741 vertices
R(3,3,4,4,7,9,10)	6 478 693	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 1 quadrupling -> 6,478,692 vertices
R(3,3,4,4,7,10,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(3,3,4,4,8,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,3,4,4,8,8,9)	4 085 249	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 2 quadruplings -> 4,085,248 vertices
R(3,3,4,4,8,8,10)	5 251 521	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 3 quadruplings -> 5,251,520 vertices
R(3,3,4,4,8,9,9)	5 251 521	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 3 quadruplings -> 5,251,520 vertices
R(3,3,4,4,8,9,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(3,3,4,4,8,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(3,3,4,4,9,9,9)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(3,3,4,4,9,9,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(3,3,4,4,9,10,10)	11 118 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 694,888 vertices; then 2 quadruplings -> 11,118,208 vertices
R(3,3,4,4,10,10,10)	13 809 281	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 3 quadruplings -> 13,809,280 vertices
R(3,3,4,5,5,5,5)	272 874	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [(3,3,4;29)]] -> 272,873 vertices
R(3,3,4,5,5,5,6)	375 897	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (5,5,3 ; 99) (ours)] x [(3,3,4;29)]] -> 375,896 vertices
R(3,3,4,5,5,5,7)	522 083	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (5,7 ; 79)]] -> 522,082 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,4,5,5,5,8)	692 249	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,5,7) of 56,120 vertices] + [colouring (4,5,5,7) of 4,702 vertices] (see Chung_Robertson.cpp) -> 173,062 vertices; then 1 quadrupling -> 692,248 vertices
R(3,3,4,5,5,5,9)	897 921	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 3 quadruplings -> 897,920 vertices
R(3,3,4,5,5,5,10)	1 180 545	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 18,446 vertices; then 3 quadruplings -> 1,180,544 vertices
R(3,3,4,5,5,6,6)	668 603	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 668,602 vertices
R(3,3,4,5,5,6,7)	846 881	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 211,720 vertices; then 1 quadrupling -> 846,880 vertices
R(3,3,4,5,5,6,8)	1 084 145	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 67,759 vertices; then 2 quadruplings -> 1,084,144 vertices
R(3,3,4,5,5,6,9)	1 399 617	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 3 quadruplings -> 1,399,616 vertices
R(3,3,4,5,5,6,10)	1 703 169	Published bricks + theorems	[template (6,6,3;235)] x [(3,3,4;29)] -> 6,653 vertices; then 4 quadruplings -> 1,703,168 vertices
R(3,3,4,5,5,7,7)	1 341 263	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices
R(3,3,4,5,5,7,8)	1 698 917	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 1 quadrupling -> 1,698,916 vertices
R(3,3,4,5,5,7,9)	2 173 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 2 quadruplings -> 2,173,600 vertices
R(3,3,4,5,5,7,10)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,4,5,5,8,8)	2 173 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 2 quadruplings -> 2,173,600 vertices
R(3,3,4,5,5,8,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,4,5,5,8,10)	3 427 841	Published bricks + theorems	[template (7,7,3;472)] x [(3,3,4;29)] -> 13,390 vertices; then 4 quadruplings -> 3,427,840 vertices
R(3,3,4,5,5,9,9)	3 738 863	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices
R(3,3,4,5,5,9,10)	4 735 877	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 1,183,969 vertices; then 1 quadrupling -> 4,735,876 vertices
R(3,3,4,5,5,10,10)	5 919 025	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,9,10) of 479,036 vertices] + [colouring (4,5,9,10) of 42,648 vertices] (see Chung_Robertson.cpp) -> 1,479,756 vertices; then 1 quadrupling -> 5,919,024 vertices
R(3,3,4,5,6,6,6)	898 077	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 898,076 vertices
R(3,3,4,5,6,6,7)	1 281 540	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)]] -> 1,281,539 vertices
R(3,3,4,5,6,6,8)	1 563 321	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 1,563,320 vertices
R(3,3,4,5,6,6,9)	2 082 209	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 2,082,208 vertices
R(3,3,4,5,6,6,10)	2 475 905	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (3,6,6 ; 337)] -> 38,686 vertices; then 3 quadruplings -> 2,475,904 vertices
R(3,3,4,5,6,7,7)	1 807 572	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 1,807,571 vertices
R(3,3,4,5,6,7,8)	2 173 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 2 quadruplings -> 2,173,600 vertices
R(3,3,4,5,6,7,9)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,4,5,6,7,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,3,4,5,6,8,8)	2 816 321	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 3 quadruplings -> 2,816,320 vertices
R(3,3,4,5,6,8,9)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,3,4,5,6,8,10)	4 336 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 67,759 vertices; then 3 quadruplings -> 4,336,576 vertices
R(3,3,4,5,6,9,9)	4 926 612	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,926,611 vertices
R(3,3,4,5,6,9,10)	5 919 025	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,9,10) of 479,036 vertices] + [colouring (4,5,9,10) of 42,648 vertices] (see Chung_Robertson.cpp) -> 1,479,756 vertices; then 1 quadrupling -> 5,919,024 vertices
R(3,3,4,5,6,10,10)	7 664 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 3 quadruplings -> 7,664,576 vertices
R(3,3,4,5,7,7,7)	2 570 805	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,570,804 vertices
R(3,3,4,5,7,7,8)	3 435 405	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 1 quadrupling -> 3,435,404 vertices
R(3,3,4,5,7,7,9)	4 190 890	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 4,190,889 vertices
R(3,3,4,5,7,7,10)	5 077 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 3 quadruplings -> 5,077,120 vertices
R(3,3,4,5,7,8,8)	4 085 249	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 2 quadruplings -> 4,085,248 vertices
R(3,3,4,5,7,8,9)	5 365 049	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 1 quadrupling -> 5,365,048 vertices
R(3,3,4,5,7,8,10)	6 795 665	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 2 quadruplings -> 6,795,664 vertices
R(3,3,4,5,7,9,9)	7 166 205	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)]] -> 7,166,204 vertices
R(3,3,4,5,7,9,10)	9 349 485	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 1 quadrupling -> 9,349,484 vertices
R(3,3,4,5,7,10,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,3,4,5,8,8,8)	5 077 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 3 quadruplings -> 5,077,120 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,4,5,8,8,9)	6 795 665	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 2 quadruplings -> 6,795,664 vertices
R(3,3,4,5,8,8,10)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,3,4,5,8,9,9)	8 707 205	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,7,9,9) of 704,080 vertices] + [colouring (4,7,9,9) of 64,561 vertices] (see Chung_Robertson.cpp) -> 2,176,801 vertices; then 1 quadrupling -> 8,707,204 vertices
R(3,3,4,5,8,9,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,3,4,5,8,10,10)	13 809 281	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 3 quadruplings -> 13,809,280 vertices
R(3,3,4,5,9,9,9)	11 422 442	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 11,422,441 vertices
R(3,3,4,5,9,9,10)	14 955 449	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 1 quadrupling -> 14,955,448 vertices
R(3,3,4,5,9,10,10)	18 943 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 1,183,969 vertices; then 2 quadruplings -> 18,943,504 vertices
R(3,3,4,5,10,10,10)	23 676 097	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,9,10,10) of 1,916,144 vertices] + [colouring (4,9,10,10) of 170,592 vertices] (see Chung_Robertson.cpp) -> 5,919,024 vertices; then 1 quadrupling -> 23,676,096 vertices
R(3,3,4,6,6,6,6)	1 556 670	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 1,556,669 vertices
R(3,3,4,6,6,6,7)	1 786 013	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 1 quadrupling -> 1,786,012 vertices
R(3,3,4,6,6,6,8)	2 128 209	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]] -> 133,013 vertices; then 2 quadruplings -> 2,128,208 vertices
R(3,3,4,6,6,6,9)	2 759 664	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (6,9 ; 182)]] -> 2,759,663 vertices
R(3,3,4,6,6,6,10)	3 226 517	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 806,629 vertices; then 1 quadrupling -> 3,226,516 vertices
R(3,3,4,6,6,7,7)	3 133 295	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices
R(3,3,4,6,6,7,8)	3 594 889	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 1 quadrupling -> 3,594,888 vertices
R(3,3,4,6,6,7,9)	4 283 681	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 2 quadruplings -> 4,283,680 vertices
R(3,3,4,6,6,7,10)	4 453 613	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (7,9 ; 251)]] -> 1,113,403 vertices; then 1 quadrupling -> 4,453,612 vertices
R(3,3,4,6,6,8,8)	4 317 870	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [(3,3,4;29)]] -> 4,317,869 vertices
R(3,3,4,6,6,8,9)	5 126 157	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)]] -> 1,281,539 vertices; then 1 quadrupling -> 5,126,156 vertices
R(3,3,4,6,6,8,10)	5 909 649	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [K3]] -> 369,353 vertices; then 2 quadruplings -> 5,909,648 vertices
R(3,3,4,6,6,9,9)	8 539 464	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices
R(3,3,4,6,6,9,10)	9 984 437	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 1 quadrupling -> 9,984,436 vertices
R(3,3,4,6,6,10,10)	12 113 814	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (3,3,4,3 ; 66)] x [atome (10,10 ; 797)]]
R(3,3,4,6,7,7,7)	3 594 889	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 1 quadrupling -> 3,594,888 vertices
R(3,3,4,6,7,7,8)	4 733 489	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 2 quadruplings -> 4,733,488 vertices
R(3,3,4,6,7,7,9)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(3,3,4,6,7,7,10)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(3,3,4,6,7,8,8)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(3,3,4,6,7,8,9)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(3,3,4,6,7,8,10)	8 265 809	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 516,613 vertices; then 2 quadruplings -> 8,265,808 vertices
R(3,3,4,6,7,9,9)	9 984 437	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 1 quadrupling -> 9,984,436 vertices
R(3,3,4,6,7,9,10)	11 694 129	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 2 quadruplings -> 11,694,128 vertices
R(3,3,4,6,7,10,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(3,3,4,6,8,8,8)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(3,3,4,6,8,8,9)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,3,4,6,8,8,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,3,4,6,8,9,9)	11 694 129	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 2 quadruplings -> 11,694,128 vertices
R(3,3,4,6,8,9,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(3,3,4,6,8,10,10)	17 524 481	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)] -> 68,455 vertices; then 4 quadruplings -> 17,524,480 vertices
R(3,3,4,6,9,9,9)	15 107 460	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)]] -> 15,107,459 vertices
R(3,3,4,6,9,9,10)	19 706 445	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,926,611 vertices; then 1 quadrupling -> 19,706,444 vertices
R(3,3,4,6,9,10,10)	23 676 097	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,9,10) of 479,036 vertices] + [colouring (4,5,9,10) of 42,648 vertices] (see Chung_Robertson.cpp) -> 1,479,756 vertices; then 2 quadruplings -> 23,676,096 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,4,6,10,10,10)	30 658 305	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 4 quadruplings -> 30,658,304 vertices
R(3,3,4,7,7,7,7)	6 306 422	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices
R(3,3,4,7,7,7,8)	7 210 285	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 1 quadrupling -> 7,210,284 vertices
R(3,3,4,7,7,7,9)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(3,3,4,7,7,7,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,3,4,7,7,8,8)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(3,3,4,7,7,8,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,3,4,7,7,8,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,3,4,7,7,9,9)	17 188 406	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices
R(3,3,4,7,7,9,10)	20 096 845	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 1 quadrupling -> 20,096,844 vertices
R(3,3,4,7,7,10,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices
R(3,3,4,7,8,8,8)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,3,4,7,8,8,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,3,4,7,8,8,10)	16 340 993	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 3 quadruplings -> 16,340,992 vertices
R(3,3,4,7,8,9,9)	20 096 845	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 1 quadrupling -> 20,096,844 vertices
R(3,3,4,7,8,9,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices
R(3,3,4,7,8,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,3,4,7,9,9,9)	23 538 161	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]] -> 1,471,135 vertices; then 2 quadruplings -> 23,538,160 vertices
R(3,3,4,7,9,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,3,4,7,9,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,3,4,7,10,10,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(3,3,4,8,8,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,3,4,8,8,8,9)	16 340 993	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 3 quadruplings -> 16,340,992 vertices
R(3,3,4,8,8,8,10)	21 006 081	Published bricks + theorems	[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)] -> 82,055 vertices; then 4 quadruplings -> 21,006,080 vertices
R(3,3,4,8,8,9,9)	23 684 239	Our bricks + published	[template (8,8,3 ; 650)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 23,684,238 vertices
R(3,3,4,8,8,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,3,4,8,8,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,3,4,8,9,9,9)	28 664 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)]] -> 7,166,204 vertices; then 1 quadrupling -> 28,664,816 vertices
R(3,3,4,8,9,9,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,3,4,8,9,10,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(3,3,4,8,10,10,10)	55 237 121	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 215,770 vertices; then 4 quadruplings -> 55,237,120 vertices
R(3,3,4,9,9,9,9)	46 748 160	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices
R(3,3,4,9,9,9,10)	54 659 365	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 1 quadrupling -> 54,659,364 vertices
R(3,3,4,9,9,10,10)	66 315 435	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,3,4,9,10,10,10)	77 538 025	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,3,4,10,10,10,10)	88 754 433	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K3]] -> 346,697 vertices; then 4 quadruplings -> 88,754,432 vertices
R(3,3,5,5,5,5,5)	451 940	Our bricks + published	[template (5,3;10)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (5,5,5 ; 453)]] -> 451,939 vertices
R(3,3,5,5,5,5,6)	634 209	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 158,552 vertices; then 1 quadrupling -> 634,208 vertices
R(3,3,5,5,5,5,7)	852 969	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (5,7 ; 79)]] -> 852,968 vertices
R(3,3,5,5,5,5,8)	1 096 833	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 17,138 vertices; then 3 quadruplings -> 1,096,832 vertices
R(3,3,5,5,5,5,9)	1 467 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [(3,3;5)]] -> 91,730 vertices; then 2 quadruplings -> 1,467,680 vertices
R(3,3,5,5,5,5,10)	1 919 937	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [atom (3,3,3,3,3 ; 161)] -> 29,999 vertices; then 3 quadruplings -> 1,919,936 vertices
R(3,3,5,5,5,6,6)	1 092 285	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 1,092,284 vertices
R(3,3,5,5,5,6,7)	1 361 213	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 340,303 vertices; then 1 quadrupling -> 1,361,212 vertices
R(3,3,5,5,5,6,8)	1 851 297	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 2 quadruplings -> 1,851,296 vertices
R(3,3,5,5,5,6,9)	2 416 705	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Paley-101 (6,6;101)] -> 37,761 vertices; then 3 quadruplings -> 2,416,704 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,5,5,5,6,10)	2 852 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)] -> 11,144 vertices; then 4 quadruplings -> 2,852,864 vertices
R(3,3,5,5,5,7,7)	2 190 963	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices
R(3,3,5,5,5,7,8)	2 739 853	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 684,963 vertices; then 1 quadrupling -> 2,739,852 vertices
R(3,3,5,5,5,7,9)	3 703 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 2 quadruplings -> 3,703,664 vertices
R(3,3,5,5,5,7,10)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,5,5,5,8,8)	3 703 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 2 quadruplings -> 3,703,664 vertices
R(3,3,5,5,5,8,9)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,5,5,5,8,10)	5 722 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)] -> 22,355 vertices; then 4 quadruplings -> 5,722,880 vertices
R(3,3,5,5,5,9,9)	6 107 043	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 6,107,042 vertices
R(3,3,5,5,5,9,10)	7 478 025	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,5,3 ; 31)] x [atom (9,9 ; 562)]] -> 1,869,506 vertices; then 1 quadrupling -> 7,478,024 vertices
R(3,3,5,5,5,10,10)	10 097 265	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 631,079 vertices; then 2 quadruplings -> 10,097,264 vertices
R(3,3,5,5,6,6,6)	1 504 650	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Paley-101 (6,6;101)]] -> 1,504,649 vertices
R(3,3,5,5,6,6,7)	2 084 403	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-101 (6,6;101)]] -> 2,084,402 vertices
R(3,3,5,5,6,6,8)	2 619 534	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [Paley-101 (6,6;101)]] -> 2,619,533 vertices
R(3,3,5,5,6,6,9)	3 488 886	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Paley-101 (6,6;101)]] -> 3,488,885 vertices
R(3,3,5,5,6,6,10)	4 211 973	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (3,6,6 ; 337)]] -> 1,052,993 vertices; then 1 quadrupling -> 4,211,972 vertices
R(3,3,5,5,6,7,7)	3 018 135	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 3,018,134 vertices
R(3,3,5,5,6,7,8)	3 703 665	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 2 quadruplings -> 3,703,664 vertices
R(3,3,5,5,6,7,9)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,5,5,6,7,10)	5 809 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 3 quadruplings -> 5,809,664 vertices
R(3,3,5,5,6,8,8)	4 847 169	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 3 quadruplings -> 4,847,168 vertices
R(3,3,5,5,6,8,9)	5 809 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 3 quadruplings -> 5,809,664 vertices
R(3,3,5,5,6,8,10)	7 405 185	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 3 quadruplings -> 7,405,184 vertices
R(3,3,5,5,6,9,9)	8 412 735	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 8,412,734 vertices
R(3,3,5,5,6,9,10)	10 119 453	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,6,9,9) of 818,024 vertices] + [colouring (5,6,9,9) of 75,791 vertices] (see Chung_Robertson.cpp) -> 2,529,863 vertices; then 1 quadrupling -> 10,119,452 vertices
R(3,3,5,5,6,10,10)	13 510 209	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [atom (9,9 ; 562)] -> 211,097 vertices; then 3 quadruplings -> 13,510,208 vertices
R(3,3,5,5,7,7,7)	4 180 860	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)]] -> 4,180,859 vertices
R(3,3,5,5,7,7,8)	5 365 049	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 1 quadrupling -> 5,365,048 vertices
R(3,3,5,5,7,7,9)	6 997 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)]] -> 6,997,928 vertices
R(3,3,5,5,7,7,10)	8 701 141	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,7,7,9) of 704,080 vertices] + [colouring (5,7,7,9) of 63,045 vertices] (see Chung_Robertson.cpp) -> 2,175,285 vertices; then 1 quadrupling -> 8,701,140 vertices
R(3,3,5,5,7,8,8)	6 795 665	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 2 quadruplings -> 6,795,664 vertices
R(3,3,5,5,7,8,9)	8 763 849	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 1 quadrupling -> 8,763,848 vertices
R(3,3,5,5,7,8,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,3,5,5,7,9,9)	11 653 380	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)]] -> 11,653,379 vertices
R(3,3,5,5,7,9,10)	14 955 449	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 1 quadrupling -> 14,955,448 vertices
R(3,3,5,5,7,10,10)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,3,5,5,8,8,8)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,3,5,5,8,8,9)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,3,5,5,8,8,10)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,3,5,5,8,9,9)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,3,5,5,8,9,10)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,3,5,5,8,10,10)	23 682 881	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,7,10,10) of 1,916,144 vertices] + [colouring (5,7,10,10) of 172,288 vertices] (see Chung_Robertson.cpp) -> 5,920,720 vertices; then 1 quadrupling -> 23,682,880 vertices
R(3,3,5,5,9,9,9)	19 505 409	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (9,9 ; 562)]] -> 19,505,408 vertices
R(3,3,5,5,9,9,10)	24 428 169	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 6,107,042 vertices; then 1 quadrupling -> 24,428,168 vertices
R(3,3,5,5,9,10,10)	30 658 305	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 4 quadruplings -> 30,658,304 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,5,5,10,10,10)	40 389 057	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 631,079 vertices; then 3 quadruplings -> 40,389,056 vertices
R(3,3,5,6,6,6,6)	2 608 545	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices
R(3,3,5,6,6,6,7)	3 148 973	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 787,243 vertices; then 1 quadrupling -> 3,148,972 vertices
R(3,3,5,6,6,6,8)	3 902 865	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 2 quadruplings -> 3,902,864 vertices
R(3,3,5,6,6,6,9)	5 038 401	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 3 quadruplings -> 5,038,400 vertices
R(3,3,5,6,6,6,10)	6 016 257	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 1,504,065 – 3 = 6,016,257; with R(6,6,6,6) >= 23,502
R(3,3,5,6,6,7,7)	5 250 456	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices
R(3,3,5,6,6,7,8)	6 338 293	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 1 quadrupling -> 6,338,292 vertices
R(3,3,5,6,6,7,9)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(3,3,5,6,6,7,10)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,3,5,6,6,8,8)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(3,3,5,6,6,8,9)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,3,5,6,6,8,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,3,5,6,6,9,9)	14 582 559	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices
R(3,3,5,6,6,9,10)	17 341 397	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 1 quadrupling -> 17,341,396 vertices
R(3,3,5,6,6,10,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(3,3,5,6,7,7,7)	6 338 293	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 1 quadrupling -> 6,338,292 vertices
R(3,3,5,6,7,7,8)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(3,3,5,6,7,7,9)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,3,5,6,7,7,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,3,5,6,7,8,8)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,3,5,6,7,8,9)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,3,5,6,7,8,10)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,3,5,6,7,9,9)	17 341 397	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 1 quadrupling -> 17,341,396 vertices
R(3,3,5,6,7,9,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(3,3,5,6,7,10,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(3,3,5,6,8,8,8)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,3,5,6,8,8,9)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,3,5,6,8,8,10)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,3,5,6,8,9,9)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(3,3,5,6,8,9,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(3,3,5,6,8,10,10)	33 632 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 4 quadruplings -> 33,632,000 vertices
R(3,3,5,6,9,9,9)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(3,3,5,6,9,9,10)	33 650 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 8,412,734 vertices; then 1 quadrupling -> 33,650,936 vertices
R(3,3,5,6,9,10,10)	40 477 809	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,6,9,9,10) of 3,272,096 vertices] + [colouring (6,9,9,10) of 303,164 vertices] (see Chung_Robertson.cpp) -> 10,119,452 vertices; then 1 quadrupling -> 40,477,808 vertices
R(3,3,5,6,10,10,10)	54 040 833	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [atom (9,9 ; 562)] -> 211,097 vertices; then 4 quadruplings -> 54,040,832 vertices
R(3,3,5,7,7,7,7)	10 530 837	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices
R(3,3,5,7,7,7,8)	12 668 533	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 1 quadrupling -> 12,668,532 vertices
R(3,3,5,7,7,7,9)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(3,3,5,7,7,7,10)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,3,5,7,7,8,8)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(3,3,5,7,7,8,9)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,3,5,7,7,8,10)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,3,5,7,7,9,9)	29 351 997	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices
R(3,3,5,7,7,9,10)	34 905 053	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 1 quadrupling -> 34,905,052 vertices
R(3,3,5,7,7,10,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,5,7,8,8,8)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,3,5,7,8,8,9)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,3,5,7,8,8,10)	28 318 145	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 3 quadruplings -> 28,318,144 vertices
R(3,3,5,7,8,9,9)	34 905 053	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 1 quadrupling -> 34,905,052 vertices
R(3,3,5,7,8,9,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,3,5,7,8,10,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,3,5,7,9,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,3,5,7,9,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,3,5,7,9,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,3,5,7,10,10,10)	77 554 689	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 5 quadruplings -> 77,554,688 vertices
R(3,3,5,8,8,8,8)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,3,5,8,8,8,9)	28 318 145	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 3 quadruplings -> 28,318,144 vertices
R(3,3,5,8,8,8,10)	34 777 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 4 quadruplings -> 34,777,600 vertices
R(3,3,5,8,8,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,3,5,8,8,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,3,5,8,8,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,3,5,8,9,9,9)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,3,5,8,9,9,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,3,5,8,9,10,10)	77 554 689	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 5 quadruplings -> 77,554,688 vertices
R(3,3,5,8,10,10,10)	96 030 209	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,3,5,9,9,9,9)	79 831 467	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices
R(3,3,5,9,9,9,10)	94 934 693	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 1 quadrupling -> 94,934,692 vertices
R(3,3,5,9,9,10,10)	116 528 753	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 2 quadruplings -> 116,528,752 vertices
R(3,3,5,9,10,10,10)	151 050 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 3 quadruplings -> 151,050,496 vertices
R(3,3,5,10,10,10,10)	184 115 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 46,028,993 – 3 = 184,115,969; with R(9,9,9,9) >= 719,204
R(3,3,6,6,6,6,6)	3 199 350	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,3,6 ; 59) (ours)]] -> 3,199,349 vertices
R(3,3,6,6,6,6,7)	4 487 290	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,487,289 vertices
R(3,3,6,6,6,6,8)	5 668 155	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 5,668,154 vertices
R(3,3,6,6,6,6,9)	7 196 538	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,3,9 ; 132) (ours)]] -> 7,196,537 vertices
R(3,3,6,6,6,6,10)	8 127 390	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,3,10 ; 149)]] -> 8,127,389 vertices
R(3,3,6,6,6,7,7)	6 439 715	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,3,6 ; 59) (ours)]] -> 6,439,714 vertices
R(3,3,6,6,6,7,8)	7 855 745	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 2 quadruplings -> 7,855,744 vertices
R(3,3,6,6,6,7,9)	10 434 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices; then 1 quadrupling -> 10,434,176 vertices
R(3,3,6,6,6,7,10)	12 109 313	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 4 quadruplings -> 12,109,312 vertices
R(3,3,6,6,6,8,8)	10 141 313	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 3 quadruplings -> 10,141,312 vertices
R(3,3,6,6,6,8,9)	12 595 889	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 787,243 vertices; then 2 quadruplings -> 12,595,888 vertices
R(3,3,6,6,6,8,10)	15 611 457	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 3 quadruplings -> 15,611,456 vertices
R(3,3,6,6,6,9,9)	17 517 108	Our bricks + published	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,3,6 ; 59) (ours)]] -> 17,517,107 vertices
R(3,3,6,6,6,9,10)	21 365 825	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 2 quadruplings -> 21,365,824 vertices
R(3,3,6,6,6,10,10)	27 592 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 3 quadruplings -> 27,592,256 vertices
R(3,3,6,6,7,7,7)	9 032 071	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,032,070 vertices
R(3,3,6,6,7,7,8)	12 533 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 1 quadrupling -> 12,533,176 vertices
R(3,3,6,6,7,7,9)	14 485 337	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,3,9 ; 132) (ours)]] -> 14,485,336 vertices
R(3,3,6,6,7,7,10)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(3,3,6,6,7,8,8)	14 379 553	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 2 quadruplings -> 14,379,552 vertices
R(3,3,6,6,7,8,9)	21 001 821	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 1 quadrupling -> 21,001,820 vertices
R(3,3,6,6,7,8,10)	22 817 873	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 2 quadruplings -> 22,817,872 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,6,6,7,9,9)	25 353 169	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 2 quadruplings -> 25,353,168 vertices
R(3,3,6,6,7,9,10)	34 157 853	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 1 quadrupling -> 34,157,852 vertices
R(3,3,6,6,7,10,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,3,6,6,8,8,8)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(3,3,6,6,8,8,9)	25 353 169	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 2 quadruplings -> 25,353,168 vertices
R(3,3,6,6,8,8,10)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,3,6,6,8,9,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,3,6,6,8,9,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,3,6,6,8,10,10)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices
R(3,3,6,6,9,9,9)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,3,6,6,9,9,10)	58 330 233	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 1 quadrupling -> 58,330,232 vertices
R(3,3,6,6,9,10,10)	69 365 585	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 2 quadruplings -> 69,365,584 vertices
R(3,3,6,6,10,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(3,3,6,7,7,7,7)	12 961 652	Our bricks + published	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,3,6 ; 59) (ours)]] -> 12,961,651 vertices
R(3,3,6,7,7,7,8)	15 715 793	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 2 quadruplings -> 15,715,792 vertices
R(3,3,6,7,7,7,9)	21 001 821	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 1 quadrupling -> 21,001,820 vertices
R(3,3,6,7,7,7,10)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,3,6,7,7,8,8)	20 269 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 3 quadruplings -> 20,269,696 vertices
R(3,3,6,7,7,8,9)	25 353 169	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 2 quadruplings -> 25,353,168 vertices
R(3,3,6,7,7,8,10)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,3,6,7,7,9,9)	35 258 792	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,3,6 ; 59) (ours)]] -> 35,258,791 vertices
R(3,3,6,7,7,9,10)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,3,6,7,7,10,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,3,6,7,8,8,8)	24 287 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 6,071,873 – 3 = 24,287,489; with R(7,7,7,7) >= 94,874
R(3,3,6,7,8,8,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,3,6,7,8,8,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,3,6,7,8,9,9)	42 845 393	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 2 quadruplings -> 42,845,392 vertices
R(3,3,6,7,8,9,10)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,3,6,7,8,10,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,3,6,7,9,9,9)	58 330 233	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 1 quadrupling -> 58,330,232 vertices
R(3,3,6,7,9,9,10)	69 365 585	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 2 quadruplings -> 69,365,584 vertices
R(3,3,6,7,9,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(3,3,6,7,10,10,10)	110 369 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 4 quadruplings -> 110,369,024 vertices
R(3,3,6,8,8,8,8)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(3,3,6,8,8,8,9)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,3,6,8,8,8,10)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices
R(3,3,6,8,8,9,9)	55 538 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 3 quadruplings -> 55,538,176 vertices
R(3,3,6,8,8,9,10)	67 694 849	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 4 quadruplings -> 67,694,848 vertices
R(3,3,6,8,8,10,10)	77 018 945	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)] -> 1,203,421 vertices; then 3 quadruplings -> 77,018,944 vertices
R(3,3,6,8,9,9,9)	69 365 585	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 2 quadruplings -> 69,365,584 vertices
R(3,3,6,8,9,9,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(3,3,6,8,9,10,10)	110 369 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 4 quadruplings -> 110,369,024 vertices
R(3,3,6,8,10,10,10)	134 528 001	Published bricks + theorems	[template (6,6,3;235)] x [atom (9,9 ; 562)] -> 131,375 vertices; then 5 quadruplings -> 134,528,000 vertices
R(3,3,6,9,9,9,9)	95 894 942	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,3,6 ; 59) (ours)]] -> 95,894,941 vertices
R(3,3,6,9,9,9,10)	116 528 753	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 2 quadruplings -> 116,528,752 vertices
R(3,3,6,9,9,10,10)	151 050 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 3 quadruplings -> 151,050,496 vertices
R(3,3,6,9,10,10,10)	184 115 969	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 46,028,993 – 3 = 184,115,969; with R(9,9,9,9) >= 719,204
R(3,3,6,10,10,10,10)	209 414 145	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 204,506 vertices; then 5 quadruplings -> 209,414,144 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,7,7,7,7,7)	18 052 663	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 18,052,662 vertices
R(3,3,7,7,7,7,8)	25 225 685	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 1 quadrupling -> 25,225,684 vertices
R(3,3,7,7,7,7,9)	29 156 045	Our bricks + published	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,3,9 ; 132) (ours)]] -> 29,156,044 vertices
R(3,3,7,7,7,7,10)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,3,7,7,7,8,8)	28 841 137	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 2 quadruplings -> 28,841,136 vertices
R(3,3,7,7,7,8,9)	42 123 345	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 1 quadrupling -> 42,123,344 vertices
R(3,3,7,7,7,8,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,3,7,7,7,9,9)	50 674 129	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 2 quadruplings -> 50,674,128 vertices
R(3,3,7,7,7,9,10)	68 753 621	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 1 quadrupling -> 68,753,620 vertices
R(3,3,7,7,7,10,10)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,3,7,7,8,8,8)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,3,7,7,8,8,9)	50 674 129	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 2 quadruplings -> 50,674,128 vertices
R(3,3,7,7,8,8,10)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(3,3,7,7,8,9,9)	68 753 621	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 1 quadrupling -> 68,753,620 vertices
R(3,3,7,7,8,9,10)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,3,7,7,8,10,10)	103 659 073	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,3,7,7,9,9,9)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,3,7,7,9,9,10)	117 407 985	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 1 quadrupling -> 117,407,984 vertices
R(3,3,7,7,9,10,10)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,3,7,7,10,10,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,3,7,8,8,8,8)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,3,7,8,8,8,9)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(3,3,7,8,8,8,10)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,3,7,8,8,9,9)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,3,7,8,8,9,10)	103 659 073	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,3,7,8,8,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,3,7,8,9,9,9)	117 407 985	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 1 quadrupling -> 117,407,984 vertices
R(3,3,7,8,9,9,10)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,3,7,8,9,10,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,3,7,8,10,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,3,7,9,9,9,9)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,3,7,9,9,9,10)	186 992 637	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 1 quadrupling -> 186,992,636 vertices
R(3,3,7,9,9,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,3,7,9,10,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(3,3,7,10,10,10,10)	310 152 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,3,8,8,8,8,8)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,3,8,8,8,8,9)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,3,8,8,8,8,10)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(3,3,8,8,8,9,9)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(3,3,8,8,8,9,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,3,8,8,8,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,3,8,8,9,9,9)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,3,8,8,9,9,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,3,8,8,9,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,3,8,8,10,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,3,8,9,9,9,9)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,3,8,9,9,9,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,3,8,9,9,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(3,3,8,9,10,10,10)	310 218 753	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 6 quadruplings -> 310,218,752 vertices
R(3,3,8,10,10,10,10)	384 120 833	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [atome (10,10 ; 797)]
R(3,3,9,9,9,9,9)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,3,9,9,9,9,10)	319 325 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 1 quadrupling -> 319,325,864 vertices
R(3,3,9,9,9,10,10)	379 738 769	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 2 quadruplings -> 379,738,768 vertices
R(3,3,9,9,10,10,10)	466 115 009	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 3 quadruplings -> 466,115,008 vertices
R(3,3,9,10,10,10,10)	604 201 985	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 4 quadruplings -> 604,201,984 vertices
R(3,3,10,10,10,10,10)	736 463 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 184,115,969 – 3 = 736,463,873; with R(9,9,9,9) >= 719,204
R(3,4,4,4,4,4,4)	96 015	Published bricks + theorems	[generic template 2m-1 of K2] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 96,014 vertices
R(3,4,4,4,4,4,5)	144 043	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)]] -> 144,042 vertices
R(3,4,4,4,4,4,6)	189 738	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (4,6,3 ; 80) (ours)] x [atom (4,4,4 ; 127)]] -> 189,737 vertices
R(3,4,4,4,4,4,7)	276 169	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 276,168 vertices
R(3,4,4,4,4,4,8)	300 555	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 300,554 vertices
R(3,4,4,4,4,4,9)	374 741	Our bricks + published	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (4,9,3 ; 157)] x [atome (4,4,4 ; 127)]] [new template (4,9;157), v21]
R(3,4,4,4,4,4,10)	435 028	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,10 ; 91) (ours)]] -> 435,027 vertices
R(3,4,4,4,4,5,5)	235 376	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (4,4,4 ; 127)]] -> 235,375 vertices
R(3,4,4,4,4,5,6)	324 239	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)]] -> 324,238 vertices
R(3,4,4,4,4,5,7)	449 164	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 449,163 vertices
R(3,4,4,4,4,5,8)	581 413	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 145,353 vertices; then 1 quadrupling -> 581,412 vertices
R(3,4,4,4,4,5,9)	751 815	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)]] -> 751,814 vertices
R(3,4,4,4,4,5,10)	930 305	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)] -> 14,536 vertices; then 3 quadruplings -> 930,304 vertices
R(3,4,4,4,4,6,6)	562 100	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,099 vertices
R(3,4,4,4,4,6,7)	591 957	Published bricks + theorems	[template (6,6,3;235)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 147,989 vertices; then 1 quadrupling -> 591,956 vertices
R(3,4,4,4,4,6,8)	719 019	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (4,6,3 ; 80) (ours)] x [atom (4,4,4 ; 127)]] -> 719,018 vertices
R(3,4,4,4,4,6,9)	963 437	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 963,436 vertices
R(3,4,4,4,4,6,10)	1 091 873	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,3;10)] x [atom (4,4,4 ; 127)]] -> 68,242 vertices; then 2 quadruplings -> 1,091,872 vertices
R(3,4,4,4,4,7,7)	1 131 397	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices
R(3,4,4,4,4,7,8)	1 191 497	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 1 quadrupling -> 1,191,496 vertices
R(3,4,4,4,4,7,9)	1 296 953	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)]] -> 324,238 vertices; then 1 quadrupling -> 1,296,952 vertices
R(3,4,4,4,4,7,10)	1 456 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 22,762 vertices; then 3 quadruplings -> 1,456,768 vertices
R(3,4,4,4,4,8,8)	1 559 030	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 1,559,029 vertices
R(3,4,4,4,4,8,9)	1 796 653	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 449,163 vertices; then 1 quadrupling -> 1,796,652 vertices
R(3,4,4,4,4,8,10)	2 093 105	Our bricks	[template (5,3;10)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 130,819 vertices; then 2 quadruplings -> 2,093,104 vertices
R(3,4,4,4,4,9,9)	3 077 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices
R(3,4,4,4,4,9,10)	3 240 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 810,154 vertices; then 1 quadrupling -> 3,240,616 vertices
R(3,4,4,4,4,10,10)	3 828 770	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(3,4,4,4,5,5,5)	371 658	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (3,5,3 ; 31)] x [atom (4,4,4 ; 127)]] -> 371,657 vertices
R(3,4,4,4,5,5,6)	511 977	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,5,3 ; 31)] x [atom (4,4,4 ; 127)]] -> 511,976 vertices
R(3,4,4,4,5,5,7)	709 214	Our bricks + published	[template (3,5,3 ; 31)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 709,213 vertices
R(3,4,4,4,5,5,8)	945 613	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 236,403 vertices; then 1 quadrupling -> 945,612 vertices
R(3,4,4,4,5,5,9)	1 188 373	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)]] -> 297,093 vertices; then 1 quadrupling -> 1,188,372 vertices
R(3,4,4,4,5,5,10)	1 582 773	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)]] -> 395,693 vertices; then 1 quadrupling -> 1,582,772 vertices
R(3,4,4,4,5,6,6)	887 534	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 887,533 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,4,4,5,6,7)	1 183 373	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 1 quadrupling -> 1,183,372 vertices
R(3,4,4,4,5,6,8)	1 428 689	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 2 quadruplings -> 1,428,688 vertices
R(3,4,4,4,5,6,9)	1 893 441	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 3 quadruplings -> 1,893,440 vertices
R(3,4,4,4,5,6,10)	2 028 293	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 507,073 vertices; then 1 quadrupling -> 2,028,292 vertices
R(3,4,4,4,5,7,7)	1 786 424	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices
R(3,4,4,4,5,7,8)	2 381 893	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 1 quadrupling -> 2,381,892 vertices
R(3,4,4,4,5,7,9)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(3,4,4,4,5,7,10)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,4,4,4,5,8,8)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(3,4,4,4,5,8,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,4,4,4,5,8,10)	3 940 033	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 2 quadruplings -> 3,940,032 vertices
R(3,4,4,4,5,9,9)	4 859 024	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices
R(3,4,4,4,5,9,10)	6 478 693	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 1 quadrupling -> 6,478,692 vertices
R(3,4,4,4,5,10,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(3,4,4,4,6,6,6)	1 153 793	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,153,792 vertices
R(3,4,4,4,6,6,7)	1 686 310	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,686,309 vertices
R(3,4,4,4,6,6,8)	2 130 075	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,130,074 vertices
R(3,4,4,4,6,6,9)	2 248 413	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,103 vertices; then 1 quadrupling -> 2,248,412 vertices
R(3,4,4,4,6,6,10)	2 958 433	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 739,608 vertices; then 1 quadrupling -> 2,958,432 vertices
R(3,4,4,4,6,7,7)	2 322 350	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 2,322,349 vertices
R(3,4,4,4,6,7,8)	2 859 377	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 2 quadruplings -> 2,859,376 vertices
R(3,4,4,4,6,7,9)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,4,4,4,6,7,10)	4 260 161	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 2 quadruplings -> 4,260,160 vertices
R(3,4,4,4,6,8,8)	3 811 073	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 3 quadruplings -> 3,811,072 vertices
R(3,4,4,4,6,8,9)	4 733 489	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 2 quadruplings -> 4,733,488 vertices
R(3,4,4,4,6,8,10)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(3,4,4,4,6,9,9)	6 316 730	Our bricks + published	[template (3,6,3 ; 40) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 6,316,729 vertices
R(3,4,4,4,6,9,10)	7 776 977	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 2 quadruplings -> 7,776,976 vertices
R(3,4,4,4,6,10,10)	10 365 953	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 3 quadruplings -> 10,365,952 vertices
R(3,4,4,4,7,7,7)	3 394 201	Our bricks + published	[template (3,7,3 ; 58) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,394,200 vertices
R(3,4,4,4,7,7,8)	4 525 585	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 1 quadrupling -> 4,525,584 vertices
R(3,4,4,4,7,7,9)	4 765 985	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 2 quadruplings -> 4,765,984 vertices
R(3,4,4,4,7,7,10)	5 954 733	Published bricks + theorems	[template (9,3 ; 26)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,488,683 vertices; then 1 quadrupling -> 5,954,732 vertices
R(3,4,4,4,7,8,8)	4 765 985	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 2 quadruplings -> 4,765,984 vertices
R(3,4,4,4,7,8,9)	7 145 693	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 1 quadrupling -> 7,145,692 vertices
R(3,4,4,4,7,8,10)	8 574 833	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 535,927 vertices; then 2 quadruplings -> 8,574,832 vertices
R(3,4,4,4,7,9,9)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(3,4,4,4,7,9,10)	12 309 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 1 quadrupling -> 12,309,504 vertices
R(3,4,4,4,7,10,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,4,8,8,8)	5 907 915	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 5,907,914 vertices
R(3,4,4,4,8,8,9)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(3,4,4,4,8,8,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,4,4,4,8,9,9)	11 661 651	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 11,661,650 vertices
R(3,4,4,4,8,9,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,4,8,10,10)	15 760 129	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 3 quadruplings -> 15,760,128 vertices
R(3,4,4,4,9,9,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,4,9,9,10)	19 436 093	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 1 quadrupling -> 19,436,092 vertices
R(3,4,4,4,9,10,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,4,4,10,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,4,4,5,5,5,5)	523 283	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 523,282 vertices
R(3,4,4,5,5,5,6)	714 129	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-17 (4,4;17)]] -> 178,532 vertices; then 1 quadrupling -> 714,128 vertices
R(3,4,4,5,5,5,7)	1 003 223	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 1,003,222 vertices
R(3,4,4,5,5,5,8)	1 363 257	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-17 (4,4;17)]] -> 340,814 vertices; then 1 quadrupling -> 1,363,256 vertices
R(3,4,4,5,5,5,9)	1 713 573	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [Paley-17 (4,4;17)]] -> 428,393 vertices; then 1 quadrupling -> 1,713,572 vertices
R(3,4,4,5,5,5,10)	2 281 893	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Paley-17 (4,4;17)]] -> 570,473 vertices; then 1 quadrupling -> 2,281,892 vertices
R(3,4,4,5,5,6,6)	1 225 238	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 1,225,237 vertices
R(3,4,4,5,5,6,7)	1 706 913	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 426,728 vertices; then 1 quadrupling -> 1,706,912 vertices
R(3,4,4,5,5,6,8)	2 029 745	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 126,859 vertices; then 2 quadruplings -> 2,029,744 vertices
R(3,4,4,5,5,6,9)	2 766 593	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,6,7) of 56,120 vertices] + [colouring (4,4,6,7) of 4,552 vertices] (see Chung_Robertson.cpp) -> 172,912 vertices; then 2 quadruplings -> 2,766,592 vertices
R(3,4,4,5,5,6,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,4,4,5,5,7,7)	2 466 114	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 2,466,113 vertices
R(3,4,4,5,5,7,8)	3 435 405	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 1 quadrupling -> 3,435,404 vertices
R(3,4,4,5,5,7,9)	4 085 249	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 2 quadruplings -> 4,085,248 vertices
R(3,4,4,5,5,7,10)	5 077 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 3 quadruplings -> 5,077,120 vertices
R(3,4,4,5,5,8,8)	4 085 249	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 2 quadruplings -> 4,085,248 vertices
R(3,4,4,5,5,8,9)	5 077 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 3 quadruplings -> 5,077,120 vertices
R(3,4,4,5,5,8,10)	6 442 241	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,4,4 ; 54)] -> 25,165 vertices; then 4 quadruplings -> 6,442,240 vertices
R(3,4,4,5,5,9,9)	6 708 534	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 6,708,533 vertices
R(3,4,4,5,5,9,10)	9 349 485	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 1 quadrupling -> 9,349,484 vertices
R(3,4,4,5,5,10,10)	11 118 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 694,888 vertices; then 2 quadruplings -> 11,118,208 vertices
R(3,4,4,5,6,6,6)	1 706 913	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 426,728 vertices; then 1 quadrupling -> 1,706,912 vertices
R(3,4,4,5,6,6,7)	2 337 954	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 2,337,953 vertices
R(3,4,4,5,6,6,8)	3 060 105	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 765,026 vertices; then 1 quadrupling -> 3,060,104 vertices
R(3,4,4,5,6,6,9)	3 913 259	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 3,913,258 vertices
R(3,4,4,5,6,6,10)	4 813 217	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,203,304 vertices; then 1 quadrupling -> 4,813,216 vertices
R(3,4,4,5,6,7,7)	3 550 133	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 887,533 vertices; then 1 quadrupling -> 3,550,132 vertices
R(3,4,4,5,6,7,8)	4 733 489	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 2 quadruplings -> 4,733,488 vertices
R(3,4,4,5,6,7,9)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(3,4,4,5,6,7,10)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(3,4,4,5,6,8,8)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(3,4,4,5,6,8,9)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(3,4,4,5,6,8,10)	8 118 977	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 126,859 vertices; then 3 quadruplings -> 8,118,976 vertices
R(3,4,4,5,6,9,9)	9 349 485	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 1 quadrupling -> 9,349,484 vertices
R(3,4,4,5,6,9,10)	11 531 921	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 2,882,980 vertices; then 1 quadrupling -> 11,531,920 vertices
R(3,4,4,5,6,10,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(3,4,4,5,7,7,7)	4 733 489	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 2 quadruplings -> 4,733,488 vertices
R(3,4,4,5,7,7,8)	7 145 693	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 1 quadrupling -> 7,145,692 vertices
R(3,4,4,5,7,7,9)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(3,4,4,5,7,7,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,4,4,5,7,8,8)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(3,4,4,5,7,8,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,4,4,5,7,8,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,5,7,9,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,5,7,9,10)	19 436 093	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 1 quadrupling -> 19,436,092 vertices
R(3,4,4,5,7,10,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,4,5,8,8,8)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,4,4,5,8,8,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,5,8,8,10)	16 340 993	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 3 quadruplings -> 16,340,992 vertices
R(3,4,4,5,8,9,9)	16 786 965	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,196,741 vertices; then 1 quadrupling -> 16,786,964 vertices
R(3,4,4,5,8,9,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices
R(3,4,4,5,8,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,4,4,5,9,9,9)	21 426 235	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 21,426,234 vertices
R(3,4,4,5,9,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,4,4,5,9,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,4,4,5,10,10,10)	44 472 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 694,888 vertices; then 3 quadruplings -> 44,472,832 vertices
R(3,4,4,6,6,6,6)	2 925 570	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 2,925,569 vertices
R(3,4,4,6,6,6,7)	3 598 389	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 899,597 vertices; then 1 quadrupling -> 3,598,388 vertices
R(3,4,4,6,6,6,8)	4 260 161	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 2 quadruplings -> 4,260,160 vertices
R(3,4,4,6,6,6,9)	5 015 658	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [atom (3,4,4 ; 54)]] -> 5,015,657 vertices
R(3,4,4,6,6,6,10)	6 175 197	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [Paley-17 (4,4;17)]] -> 1,543,799 vertices; then 1 quadrupling -> 6,175,196 vertices
R(3,4,4,6,6,7,7)	5 888 645	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices
R(3,4,4,6,6,7,8)	7 242 905	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 1 quadrupling -> 7,242,904 vertices
R(3,4,4,6,6,7,9)	8 520 297	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,130,074 vertices; then 1 quadrupling -> 8,520,296 vertices
R(3,4,4,6,6,7,10)	8 993 649	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,103 vertices; then 2 quadruplings -> 8,993,648 vertices
R(3,4,4,6,6,8,8)	8 114 520	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (3,4,4 ; 54)]] -> 8,114,519 vertices
R(3,4,4,6,6,8,9)	9 981 909	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 2,495,477 vertices; then 1 quadrupling -> 9,981,908 vertices
R(3,4,4,6,6,8,10)	11 833 729	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 739,608 vertices; then 2 quadruplings -> 11,833,728 vertices
R(3,4,4,6,6,9,9)	16 018 338	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices
R(3,4,4,6,6,9,10)	19 709 757	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 1 quadrupling -> 19,709,756 vertices
R(3,4,4,6,6,10,10)	21 432 864	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (3,4,4,3 ; 116)] x [atome (10,10 ; 797)]]
R(3,4,4,6,7,7,7)	7 242 905	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 1 quadrupling -> 7,242,904 vertices
R(3,4,4,6,7,7,8)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(3,4,4,6,7,7,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,4,4,6,7,7,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,6,7,8,8)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(3,4,4,6,7,8,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,6,7,8,10)	18 933 953	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 3 quadruplings -> 18,933,952 vertices
R(3,4,4,6,7,9,9)	19 709 757	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 1 quadrupling -> 19,709,756 vertices
R(3,4,4,6,7,9,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices
R(3,4,4,6,7,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,4,4,6,8,8,8)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(3,4,4,6,8,8,9)	18 933 953	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 3 quadruplings -> 18,933,952 vertices
R(3,4,4,6,8,8,10)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(3,4,4,6,8,9,9)	23 323 313	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 2 quadruplings -> 23,323,312 vertices
R(3,4,4,6,8,9,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(3,4,4,6,8,10,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,4,4,6,9,9,9)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(3,4,4,6,9,9,10)	41 463 809	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 4 quadruplings -> 41,463,808 vertices
R(3,4,4,6,9,10,10)	46 127 681	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 2,882,980 vertices; then 2 quadruplings -> 46,127,680 vertices
R(3,4,4,6,10,10,10)	57 466 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 6 quadruplings -> 57,466,880 vertices
R(3,4,4,7,7,7,7)	11 852 447	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices
R(3,4,4,7,7,7,8)	14 577 317	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 1 quadrupling -> 14,577,316 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,4,7,7,7,9)	18 102 337	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 2 quadruplings -> 18,102,336 vertices
R(3,4,4,7,7,7,10)	19 063 937	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 3 quadruplings -> 19,063,936 vertices
R(3,4,4,7,7,8,8)	18 102 337	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 2 quadruplings -> 18,102,336 vertices
R(3,4,4,7,7,8,9)	20 091 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 1 quadrupling -> 20,091,784 vertices
R(3,4,4,7,7,8,10)	28 582 769	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 2 quadruplings -> 28,582,768 vertices
R(3,4,4,7,7,9,9)	32 242 037	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices
R(3,4,4,7,7,9,10)	39 672 197	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 1 quadrupling -> 39,672,196 vertices
R(3,4,4,7,7,10,10)	49 238 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 2 quadruplings -> 49,238,016 vertices
R(3,4,4,7,8,8,8)	20 091 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 1 quadrupling -> 20,091,784 vertices
R(3,4,4,7,8,8,9)	28 582 769	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 2 quadruplings -> 28,582,768 vertices
R(3,4,4,7,8,8,10)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,4,4,7,8,9,9)	39 672 197	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 1 quadrupling -> 39,672,196 vertices
R(3,4,4,7,8,9,10)	49 238 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 2 quadruplings -> 49,238,016 vertices
R(3,4,4,7,8,10,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,4,7,9,9,9)	46 646 601	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 11,661,650 vertices; then 1 quadrupling -> 46,646,600 vertices
R(3,4,4,7,9,9,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,4,7,9,10,10)	77 744 369	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 2 quadruplings -> 77,744,368 vertices
R(3,4,4,7,10,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,4,4,8,8,8,8)	22 505 655	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [atom (3,4,4 ; 54)]] -> 22,505,654 vertices
R(3,4,4,8,8,8,9)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,4,4,8,8,8,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,4,4,8,8,9,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,4,4,8,8,9,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,4,8,8,10,10)	63 040 513	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 4 quadruplings -> 63,040,512 vertices
R(3,4,4,8,9,9,9)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,4,8,9,9,10)	77 744 369	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 2 quadruplings -> 77,744,368 vertices
R(3,4,4,8,9,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,4,4,8,10,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,4,4,9,9,9,9)	87 690 137	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices
R(3,4,4,9,9,9,10)	107 898 317	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 1 quadrupling -> 107,898,316 vertices
R(3,4,4,9,9,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,4,4,9,10,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,4,4,10,10,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,4,5,5,5,5,5)	855 023	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 855,022 vertices
R(3,4,5,5,5,5,6)	1 177 823	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 1,177,822 vertices
R(3,4,5,5,5,5,7)	1 638 797	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 1,638,796 vertices
R(3,4,5,5,5,5,8)	2 088 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (5,7 ; 79)]] -> 522,082 vertices; then 1 quadrupling -> 2,088,328 vertices
R(3,4,5,5,5,5,9)	2 768 993	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,5,7) of 56,120 vertices] + [colouring (4,5,5,7) of 4,702 vertices] (see Chung_Robertson.cpp) -> 173,062 vertices; then 2 quadruplings -> 2,768,992 vertices
R(3,4,5,5,5,5,10)	3 591 681	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 4 quadruplings -> 3,591,680 vertices
R(3,4,5,5,5,6,6)	2 041 823	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 2,041,822 vertices
R(3,4,5,5,5,6,7)	2 674 409	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 668,602 vertices; then 1 quadrupling -> 2,674,408 vertices
R(3,4,5,5,5,6,8)	3 387 521	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 211,720 vertices; then 2 quadruplings -> 3,387,520 vertices
R(3,4,5,5,5,6,9)	4 336 577	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 67,759 vertices; then 3 quadruplings -> 4,336,576 vertices
R(3,4,5,5,5,6,10)	5 598 465	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [atom (3,6,6 ; 337)] -> 21,869 vertices; then 4 quadruplings -> 5,598,464 vertices
R(3,4,5,5,5,7,7)	4 109 783	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 4,109,782 vertices
R(3,4,5,5,5,7,8)	5 365 049	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 1 quadrupling -> 5,365,048 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,5,5,5,7,9)	6 795 665	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 2 quadruplings -> 6,795,664 vertices
R(3,4,5,5,5,7,10)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,4,5,5,5,8,8)	6 795 665	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 2 quadruplings -> 6,795,664 vertices
R(3,4,5,5,5,8,9)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,4,5,5,5,8,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,4,5,5,5,9,9)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,4,5,5,5,9,10)	14 955 449	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 1 quadrupling -> 14,955,448 vertices
R(3,4,5,5,5,10,10)	18 943 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 1,183,969 vertices; then 2 quadruplings -> 18,943,504 vertices
R(3,4,5,5,6,6,6)	2 724 923	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (3,6,6 ; 337)]] -> 2,724,922 vertices
R(3,4,5,5,6,6,7)	3 913 497	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 3,913,496 vertices
R(3,4,5,5,6,6,8)	5 126 157	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)]] -> 1,281,539 vertices; then 1 quadrupling -> 5,126,156 vertices
R(3,4,5,5,6,6,9)	6 317 963	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (3,6,6 ; 337)]] -> 6,317,962 vertices
R(3,4,5,5,6,6,10)	8 328 833	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 2,082,208 vertices; then 1 quadrupling -> 8,328,832 vertices
R(3,4,5,5,6,7,7)	5 446 523	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 5,446,522 vertices
R(3,4,5,5,6,7,8)	7 230 285	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 1,807,571 vertices; then 1 quadrupling -> 7,230,284 vertices
R(3,4,5,5,6,7,9)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,4,5,5,6,7,10)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,4,5,5,6,8,8)	8 694 401	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 3 quadruplings -> 8,694,400 vertices
R(3,4,5,5,6,8,9)	11 265 281	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 4 quadruplings -> 11,265,280 vertices
R(3,4,5,5,6,8,10)	14 366 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 5 quadruplings -> 14,366,720 vertices
R(3,4,5,5,6,9,9)	14 955 449	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 1 quadrupling -> 14,955,448 vertices
R(3,4,5,5,6,9,10)	19 706 445	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,926,611 vertices; then 1 quadrupling -> 19,706,444 vertices
R(3,4,5,5,6,10,10)	23 676 097	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,9,10) of 479,036 vertices] + [colouring (4,5,9,10) of 42,648 vertices] (see Chung_Robertson.cpp) -> 1,479,756 vertices; then 2 quadruplings -> 23,676,096 vertices
R(3,4,5,5,7,7,7)	7 877 087	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 7,877,086 vertices
R(3,4,5,5,7,7,8)	10 283 217	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,570,804 vertices; then 1 quadrupling -> 10,283,216 vertices
R(3,4,5,5,7,7,9)	13 741 617	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 2 quadruplings -> 13,741,616 vertices
R(3,4,5,5,7,7,10)	16 763 557	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 4,190,889 vertices; then 1 quadrupling -> 16,763,556 vertices
R(3,4,5,5,7,8,8)	13 741 617	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 2 quadruplings -> 13,741,616 vertices
R(3,4,5,5,7,8,9)	16 763 557	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 4,190,889 vertices; then 1 quadrupling -> 16,763,556 vertices
R(3,4,5,5,7,8,10)	21 460 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 2 quadruplings -> 21,460,192 vertices
R(3,4,5,5,7,9,9)	21 460 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 2 quadruplings -> 21,460,192 vertices
R(3,4,5,5,7,9,10)	28 664 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)]] -> 7,166,204 vertices; then 1 quadrupling -> 28,664,816 vertices
R(3,4,5,5,7,10,10)	37 397 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 2 quadruplings -> 37,397,936 vertices
R(3,4,5,5,8,8,8)	16 340 993	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 3 quadruplings -> 16,340,992 vertices
R(3,4,5,5,8,8,9)	21 460 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 2 quadruplings -> 21,460,192 vertices
R(3,4,5,5,8,8,10)	27 182 657	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 3 quadruplings -> 27,182,656 vertices
R(3,4,5,5,8,9,9)	28 664 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)]] -> 7,166,204 vertices; then 1 quadrupling -> 28,664,816 vertices
R(3,4,5,5,8,9,10)	37 397 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 2 quadruplings -> 37,397,936 vertices
R(3,4,5,5,8,10,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(3,4,5,5,9,9,9)	34 828 817	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,7,9,9) of 704,080 vertices] + [colouring (4,7,9,9) of 64,561 vertices] (see Chung_Robertson.cpp) -> 2,176,801 vertices; then 2 quadruplings -> 34,828,816 vertices
R(3,4,5,5,9,9,10)	45 689 765	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 11,422,441 vertices; then 1 quadrupling -> 45,689,764 vertices
R(3,4,5,5,9,10,10)	59 821 793	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 2 quadruplings -> 59,821,792 vertices
R(3,4,5,5,10,10,10)	75 774 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 1,183,969 vertices; then 3 quadruplings -> 75,774,016 vertices
R(3,4,5,6,6,6,6)	4 723 463	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,5,6,6,6,7)	6 226 677	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 1,556,669 vertices; then 1 quadrupling -> 6,226,676 vertices
R(3,4,5,6,6,6,8)	7 144 049	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 2 quadruplings -> 7,144,048 vertices
R(3,4,5,6,6,6,9)	8 512 833	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]] -> 133,013 vertices; then 3 quadruplings -> 8,512,832 vertices
R(3,4,5,6,6,6,10)	11 038 653	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (6,9 ; 182)]] -> 2,759,663 vertices; then 1 quadrupling -> 11,038,652 vertices
R(3,4,5,6,6,7,7)	9 507 443	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices
R(3,4,5,6,6,7,8)	12 533 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 1 quadrupling -> 12,533,176 vertices
R(3,4,5,6,6,7,9)	14 379 553	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 2 quadruplings -> 14,379,552 vertices
R(3,4,5,6,6,7,10)	17 134 721	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 3 quadruplings -> 17,134,720 vertices
R(3,4,5,6,6,8,8)	14 379 553	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 2 quadruplings -> 14,379,552 vertices
R(3,4,5,6,6,8,9)	17 271 477	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [(3,3,4;29)]] -> 4,317,869 vertices; then 1 quadrupling -> 17,271,476 vertices
R(3,4,5,6,6,8,10)	20 504 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)]] -> 1,281,539 vertices; then 2 quadruplings -> 20,504,624 vertices
R(3,4,5,6,6,9,9)	25 867 703	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices
R(3,4,5,6,6,9,10)	34 157 853	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 1 quadrupling -> 34,157,852 vertices
R(3,4,5,6,6,10,10)	39 937 745	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 2 quadruplings -> 39,937,744 vertices
R(3,4,5,6,7,7,7)	12 533 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 1 quadrupling -> 12,533,176 vertices
R(3,4,5,6,7,7,8)	14 379 553	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 2 quadruplings -> 14,379,552 vertices
R(3,4,5,6,7,7,9)	18 933 953	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 3 quadruplings -> 18,933,952 vertices
R(3,4,5,6,7,7,10)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(3,4,5,6,7,8,8)	18 933 953	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 3 quadruplings -> 18,933,952 vertices
R(3,4,5,6,7,8,9)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(3,4,5,6,7,8,10)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(3,4,5,6,7,9,9)	34 157 853	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 1 quadrupling -> 34,157,852 vertices
R(3,4,5,6,7,9,10)	39 937 745	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 2 quadruplings -> 39,937,744 vertices
R(3,4,5,6,7,10,10)	48 455 253	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (3,3,4,3 ; 66)] x [atome (10,10 ; 797)]]
R(3,4,5,6,8,8,8)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(3,4,5,6,8,8,9)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(3,4,5,6,8,8,10)	34 777 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 4 quadruplings -> 34,777,600 vertices
R(3,4,5,6,8,9,9)	39 937 745	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 2 quadruplings -> 39,937,744 vertices
R(3,4,5,6,8,9,10)	46 776 513	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 3 quadruplings -> 46,776,512 vertices
R(3,4,5,6,8,10,10)	57 466 881	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 6 quadruplings -> 57,466,880 vertices
R(3,4,5,6,9,9,9)	46 776 513	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 3 quadruplings -> 46,776,512 vertices
R(3,4,5,6,9,9,10)	60 429 837	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)]] -> 15,107,459 vertices; then 1 quadrupling -> 60,429,836 vertices
R(3,4,5,6,9,10,10)	78 825 777	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,926,611 vertices; then 2 quadruplings -> 78,825,776 vertices
R(3,4,5,6,10,10,10)	94 704 385	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,9,10,10) of 1,916,144 vertices] + [colouring (4,9,10,10) of 170,592 vertices] (see Chung_Robertson.cpp) -> 5,919,024 vertices; then 2 quadruplings -> 94,704,384 vertices
R(3,4,5,7,7,7,7)	19 002 803	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices
R(3,4,5,7,7,7,8)	25 225 685	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 1 quadrupling -> 25,225,684 vertices
R(3,4,5,7,7,7,9)	28 841 137	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 2 quadruplings -> 28,841,136 vertices
R(3,4,5,7,7,7,10)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,4,5,7,7,8,8)	28 841 137	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 2 quadruplings -> 28,841,136 vertices
R(3,4,5,7,7,8,9)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,4,5,7,7,8,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,4,5,7,7,9,9)	52 067 003	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices
R(3,4,5,7,7,9,10)	68 753 621	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 1 quadrupling -> 68,753,620 vertices
R(3,4,5,7,7,10,10)	80 387 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 2 quadruplings -> 80,387,376 vertices
R(3,4,5,7,8,8,8)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,4,5,7,8,8,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,5,7,8,8,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,5,7,8,9,9)	68 753 621	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 1 quadrupling -> 68,753,620 vertices
R(3,4,5,7,8,9,10)	80 387 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 2 quadruplings -> 80,387,376 vertices
R(3,4,5,7,8,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,4,5,7,9,9,9)	80 387 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 2 quadruplings -> 80,387,376 vertices
R(3,4,5,7,9,9,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,4,5,7,9,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,4,5,7,10,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,4,5,8,8,8,8)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,4,5,8,8,8,9)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,5,8,8,8,10)	65 363 969	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 4 quadruplings -> 65,363,968 vertices
R(3,4,5,8,8,9,9)	80 387 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 2 quadruplings -> 80,387,376 vertices
R(3,4,5,8,8,9,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,4,5,8,8,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,4,5,8,9,9,9)	94 736 953	Our bricks + published	[template (8,8,3 ; 650)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 23,684,238 vertices; then 1 quadrupling -> 94,736,952 vertices
R(3,4,5,8,9,9,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,4,5,8,9,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,4,5,8,10,10,10)	180 244 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 6 quadruplings -> 180,244,480 vertices
R(3,4,5,9,9,9,9)	141 609 803	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices
R(3,4,5,9,9,9,10)	186 992 637	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 1 quadrupling -> 186,992,636 vertices
R(3,4,5,9,9,10,10)	218 637 457	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 2 quadruplings -> 218,637,456 vertices
R(3,4,5,9,10,10,10)	265 261 737	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,4,5,10,10,10,10)	310 152 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,4,6,6,6,6,6)	6 219 230	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 6,219,229 vertices
R(3,4,6,6,6,6,7)	9 053 307	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 9,053,306 vertices
R(3,4,6,6,6,6,8)	10 810 185	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,702,546 vertices; then 1 quadrupling -> 10,810,184 vertices
R(3,4,6,6,6,6,9)	13 164 942	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,4,9 ; 241)]] -> 13,164,941 vertices
R(3,4,6,6,6,6,10)	14 947 086	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [atom (4,10 ; 91) (ours)]] -> 14,947,085 vertices
R(3,4,6,6,6,7,7)	12 518 137	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 12,518,136 vertices
R(3,4,6,6,6,7,8)	14 947 249	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 3,736,812 vertices; then 1 quadrupling -> 14,947,248 vertices
R(3,4,6,6,6,7,9)	18 893 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 1 quadrupling -> 18,893,848 vertices
R(3,4,6,6,6,7,10)	22 560 353	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,410,022 vertices; then 2 quadruplings -> 22,560,352 vertices
R(3,4,6,6,6,8,8)	17 366 288	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 17,366,287 vertices
R(3,4,6,6,6,8,9)	24 906 705	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 1,556,669 vertices; then 2 quadruplings -> 24,906,704 vertices
R(3,4,6,6,6,8,10)	28 576 193	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 3 quadruplings -> 28,576,192 vertices
R(3,4,6,6,6,9,9)	34 059 146	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 34,059,145 vertices
R(3,4,6,6,6,9,10)	41 514 317	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 10,378,579 vertices; then 1 quadrupling -> 41,514,316 vertices
R(3,4,6,6,6,10,10)	46 776 513	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 3 quadruplings -> 46,776,512 vertices
R(3,4,6,6,7,7,7)	18 222 602	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 18,222,601 vertices
R(3,4,6,6,7,7,8)	23 554 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 1 quadrupling -> 23,554,576 vertices
R(3,4,6,6,7,7,9)	28 971 617	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 2 quadruplings -> 28,971,616 vertices
R(3,4,6,6,7,7,10)	34 081 185	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,130,074 vertices; then 2 quadruplings -> 34,081,184 vertices
R(3,4,6,6,7,8,8)	28 971 617	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 2 quadruplings -> 28,971,616 vertices
R(3,4,6,6,7,8,9)	38 029 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 1 quadrupling -> 38,029,768 vertices
R(3,4,6,6,7,8,10)	45 409 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 2,838,082 vertices; then 2 quadruplings -> 45,409,312 vertices
R(3,4,6,6,7,9,9)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,6,6,7,9,10)	64 073 349	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 1 quadrupling -> 64,073,348 vertices
R(3,4,6,6,7,10,10)	78 839 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 2 quadruplings -> 78,839,024 vertices
R(3,4,6,6,8,8,8)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(3,4,6,6,8,8,9)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(3,4,6,6,8,8,10)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(3,4,6,6,8,9,9)	60 432 225	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 15,108,056 vertices; then 1 quadrupling -> 60,432,224 vertices
R(3,4,6,6,8,9,10)	78 839 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 2 quadruplings -> 78,839,024 vertices
R(3,4,6,6,8,10,10)	85 727 625	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,7,3 ; 116)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(3,4,6,6,9,9,9)	72 072 336	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,9 ; 241)]] -> 72,072,335 vertices
R(3,4,6,6,9,9,10)	103 470 809	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 1 quadrupling -> 103,470,808 vertices
R(3,4,6,6,9,10,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(3,4,6,6,10,10,10)	159 750 977	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 3 quadruplings -> 159,750,976 vertices
R(3,4,6,7,7,7,7)	25 020 361	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 25,020,360 vertices
R(3,4,6,7,7,7,8)	29 979 685	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 7,494,921 vertices; then 1 quadrupling -> 29,979,684 vertices
R(3,4,6,7,7,7,9)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,4,6,7,7,7,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,4,6,7,7,8,8)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(3,4,6,7,7,8,9)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(3,4,6,7,7,8,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,6,7,7,9,9)	68 554 891	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 68,554,890 vertices
R(3,4,6,7,7,9,10)	83 560 645	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices; then 1 quadrupling -> 83,560,644 vertices
R(3,4,6,7,7,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,4,6,7,8,8,8)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(3,4,6,7,8,8,9)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,6,7,8,8,10)	75 735 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 4 quadruplings -> 75,735,808 vertices
R(3,4,6,7,8,9,9)	83 560 645	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices; then 1 quadrupling -> 83,560,644 vertices
R(3,4,6,7,8,9,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(3,4,6,7,8,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,4,6,7,9,9,9)	103 470 809	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 1 quadrupling -> 103,470,808 vertices
R(3,4,6,7,9,9,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(3,4,6,7,9,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,4,6,7,10,10,10)	187 106 049	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 4 quadruplings -> 187,106,048 vertices
R(3,4,6,8,8,8,8)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(3,4,6,8,8,8,9)	75 735 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 4 quadruplings -> 75,735,808 vertices
R(3,4,6,8,8,8,10)	91 436 033	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 5 quadruplings -> 91,436,032 vertices
R(3,4,6,8,8,9,9)	95 070 214	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 95,070,213 vertices
R(3,4,6,8,8,9,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(3,4,6,8,8,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,4,6,8,9,9,9)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(3,4,6,8,9,9,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(3,4,6,8,9,10,10)	193 821 009	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (3,3,4,3 ; 66)] x [atome (10,10 ; 797)]]
R(3,4,6,8,10,10,10)	229 867 521	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [atom (4,4,4 ; 127)] -> 14,030 vertices; then 7 quadruplings -> 229,867,520 vertices
R(3,4,6,9,9,9,9)	186 452 911	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 186,452,910 vertices
R(3,4,6,9,9,9,10)	227 267 965	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 56,816,991 vertices; then 1 quadrupling -> 227,267,964 vertices
R(3,4,6,9,9,10,10)	256 072 001	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]] -> 4,001,125 vertices; then 3 quadruplings -> 256,072,000 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,6,9,10,10,10)	322 395 025	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,6,3 ; 80)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,4,6,10,10,10,10)	368 643 329	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 1,440,013 vertices; then 4 quadruplings -> 368,643,328 vertices
R(3,4,7,7,7,7,7)	36 422 042	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 36,422,041 vertices
R(3,4,7,7,7,7,8)	47 409 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 1 quadrupling -> 47,409,784 vertices
R(3,4,7,7,7,7,9)	58 309 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 2 quadruplings -> 58,309,264 vertices
R(3,4,7,7,7,7,10)	72 409 345	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 3 quadruplings -> 72,409,344 vertices
R(3,4,7,7,7,8,8)	58 309 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 2 quadruplings -> 58,309,264 vertices
R(3,4,7,7,7,8,9)	76 011 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 1 quadrupling -> 76,011,208 vertices
R(3,4,7,7,7,8,10)	91 077 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 5,692,342 vertices; then 2 quadruplings -> 91,077,472 vertices
R(3,4,7,7,7,9,9)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(3,4,7,7,7,9,10)	128 968 145	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 1 quadrupling -> 128,968,144 vertices
R(3,4,7,7,7,10,10)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(3,4,7,7,8,8,8)	72 409 345	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 3 quadruplings -> 72,409,344 vertices
R(3,4,7,7,8,8,9)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(3,4,7,7,8,8,10)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(3,4,7,7,8,9,9)	128 968 145	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 1 quadrupling -> 128,968,144 vertices
R(3,4,7,7,8,9,10)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(3,4,7,7,8,10,10)	196 952 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 3 quadruplings -> 196,952,064 vertices
R(3,4,7,7,9,9,9)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(3,4,7,7,9,9,10)	208 268 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 1 quadrupling -> 208,268,008 vertices
R(3,4,7,7,9,10,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,4,7,7,10,10,10,10)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices
R(3,4,7,8,8,8,8)	80 367 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 2 quadruplings -> 80,367,136 vertices
R(3,4,7,8,8,8,9)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(3,4,7,8,8,8,10)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(3,4,7,8,8,9,9)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(3,4,7,8,8,9,10)	196 952 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 3 quadruplings -> 196,952,064 vertices
R(3,4,7,8,8,10,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(3,4,7,8,9,9,9)	208 268 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 1 quadrupling -> 208,268,008 vertices
R(3,4,7,8,9,9,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,4,7,8,9,10,10)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices
R(3,4,7,8,10,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(3,4,7,9,9,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,4,7,9,9,9,10)	350 760 545	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 1 quadrupling -> 350,760,544 vertices
R(3,4,7,9,9,10,10)	431 593 265	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 2 quadruplings -> 431,593,264 vertices
R(3,4,7,9,10,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(3,4,7,10,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(3,4,8,8,8,8,8)	83 720 185	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 20,930,046 vertices; then 1 quadrupling -> 83,720,184 vertices
R(3,4,8,8,8,8,9)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(3,4,8,8,8,8,10)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(3,4,8,8,8,9,9)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(3,4,8,8,8,9,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(3,4,8,8,8,10,10)	252 162 049	Published bricks + theorems	[template (8,8,3 ; 650)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 246,252 vertices; then 5 quadruplings -> 252,162,048 vertices
R(3,4,8,8,9,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,4,8,8,9,9,10)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,4,8,8,9,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(3,4,8,8,10,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(3,4,8,9,9,9,9)	330 833 105	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 82,708,276 vertices; then 1 quadrupling -> 330,833,104 vertices
R(3,4,8,9,9,9,10)	431 593 265	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 2 quadruplings -> 431,593,264 vertices
R(3,4,8,9,9,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(3,4,8,9,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(3,4,8,10,10,10,10)	720 977 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 7 quadruplings -> 720,977,920 vertices
R(3,4,9,9,9,9,9)	394 549 844	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,9 ; 241)]] -> 394,549,843 vertices
R(3,4,9,9,9,9,10)	566 439 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 1 quadrupling -> 566,439,208 vertices
R(3,4,9,9,9,10,10)	747 970 545	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 2 quadruplings -> 747,970,544 vertices
R(3,4,9,9,10,10,10)	874 549 825	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 3 quadruplings -> 874,549,824 vertices
R(3,4,9,10,10,10,10)	1 061 046 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,4,10,10,10,10,10)	1 240 608 385	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,5,5,5,5,5,5)	1 396 542	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 1,396,541 vertices
R(3,5,5,5,5,5,6)	1 923 807	Our bricks	[template (5,6,3 ; 136) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 1,923,806 vertices
R(3,5,5,5,5,5,7)	2 664 956	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 2,664,955 vertices
R(3,5,5,5,5,5,8)	3 411 873	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (5,7 ; 79)]] -> 852,968 vertices; then 1 quadrupling -> 3,411,872 vertices
R(3,5,5,5,5,5,9)	4 460 610	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (3,5,5 ; 146) (ours)]] -> 4,460,609 vertices
R(3,5,5,5,5,5,10)	5 870 721	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [(3,3;5)]] -> 91,730 vertices; then 3 quadruplings -> 5,870,720 vertices
R(3,5,5,5,5,6,6)	3 334 982	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 3,334,981 vertices
R(3,5,5,5,5,6,7)	4 369 137	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 1,092,284 vertices; then 1 quadrupling -> 4,369,136 vertices
R(3,5,5,5,5,6,8)	5 444 849	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 340,303 vertices; then 2 quadruplings -> 5,444,848 vertices
R(3,5,5,5,5,6,9)	7 405 185	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 3 quadruplings -> 7,405,184 vertices
R(3,5,5,5,5,6,10)	9 666 817	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Paley-101 (6,6;101)] -> 37,761 vertices; then 4 quadruplings -> 9,666,816 vertices
R(3,5,5,5,5,7,7)	6 712 650	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 6,712,649 vertices
R(3,5,5,5,5,7,8)	8 763 849	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 1 quadrupling -> 8,763,848 vertices
R(3,5,5,5,5,7,9)	10 959 409	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 684,963 vertices; then 2 quadruplings -> 10,959,408 vertices
R(3,5,5,5,5,7,10)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,5,5,5,5,8,8)	10 959 409	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 684,963 vertices; then 2 quadruplings -> 10,959,408 vertices
R(3,5,5,5,5,8,9)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,5,5,5,5,8,10)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,5,5,5,5,9,9)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,5,5,5,5,9,10)	24 428 169	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 6,107,042 vertices; then 1 quadrupling -> 24,428,168 vertices
R(3,5,5,5,5,10,10)	29 912 097	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,5,3 ; 31)] x [atom (9,9 ; 562)]] -> 1,869,506 vertices; then 2 quadruplings -> 29,912,096 vertices
R(3,5,5,5,6,6,6)	4 594 107	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 4,594,106 vertices
R(3,5,5,5,6,6,7)	6 363 690	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 6,363,689 vertices
R(3,5,5,5,6,6,8)	8 337 609	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-101 (6,6;101)]] -> 2,084,402 vertices; then 1 quadrupling -> 8,337,608 vertices
R(3,5,5,5,6,6,9)	10 651 523	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 10,651,522 vertices
R(3,5,5,5,6,6,10)	13 955 541	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Paley-101 (6,6;101)]] -> 3,488,885 vertices; then 1 quadrupling -> 13,955,540 vertices
R(3,5,5,5,6,7,7)	9 247 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 9,247,016 vertices
R(3,5,5,5,6,7,8)	12 072 537	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 3,018,134 vertices; then 1 quadrupling -> 12,072,536 vertices
R(3,5,5,5,6,7,9)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,5,5,5,6,7,10)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,5,5,5,6,8,8)	14 814 657	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 3 quadruplings -> 14,814,656 vertices
R(3,5,5,5,6,8,9)	19 388 673	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 4 quadruplings -> 19,388,672 vertices
R(3,5,5,5,6,8,10)	23 238 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]] -> 90,776 vertices; then 4 quadruplings -> 23,238,656 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,5,5,6,9,9)	25 151 367	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (3,5,5 ; 146) (ours)]] -> 25,151,366 vertices
R(3,5,5,5,6,9,10)	33 650 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 8,412,734 vertices; then 1 quadrupling -> 33,650,936 vertices
R(3,5,5,5,6,10,10)	40 477 809	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,5,6,9,9) of 818,024 vertices] + [colouring (5,6,9,9) of 75,791 vertices] (see Chung_Robertson.cpp) -> 2,529,863 vertices; then 2 quadruplings -> 40,477,808 vertices
R(3,5,5,5,7,7,7)	12 808 832	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 12,808,831 vertices
R(3,5,5,5,7,7,8)	16 723 437	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)]] -> 4,180,859 vertices; then 1 quadrupling -> 16,723,436 vertices
R(3,5,5,5,7,7,9)	21 460 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 2 quadruplings -> 21,460,192 vertices
R(3,5,5,5,7,7,10)	27 991 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)]] -> 6,997,928 vertices; then 1 quadrupling -> 27,991,712 vertices
R(3,5,5,5,7,8,8)	21 460 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 2 quadruplings -> 21,460,192 vertices
R(3,5,5,5,7,8,9)	27 991 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)]] -> 6,997,928 vertices; then 1 quadrupling -> 27,991,712 vertices
R(3,5,5,5,7,8,10)	35 055 393	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 2 quadruplings -> 35,055,392 vertices
R(3,5,5,5,7,9,9)	35 055 393	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 2 quadruplings -> 35,055,392 vertices
R(3,5,5,5,7,9,10)	46 613 517	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)]] -> 11,653,379 vertices; then 1 quadrupling -> 46,613,516 vertices
R(3,5,5,5,7,10,10)	59 821 793	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 2 quadruplings -> 59,821,792 vertices
R(3,5,5,5,8,8,8)	27 182 657	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 3 quadruplings -> 27,182,656 vertices
R(3,5,5,5,8,8,9)	35 055 393	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 2 quadruplings -> 35,055,392 vertices
R(3,5,5,5,8,8,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(3,5,5,5,8,9,9)	46 613 517	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)]] -> 11,653,379 vertices; then 1 quadrupling -> 46,613,516 vertices
R(3,5,5,5,8,9,10)	59 821 793	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 2 quadruplings -> 59,821,792 vertices
R(3,5,5,5,8,10,10)	77 554 689	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 5 quadruplings -> 77,554,688 vertices
R(3,5,5,5,9,9,9)	59 258 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 4 quadruplings -> 59,258,624 vertices
R(3,5,5,5,9,9,10)	78 021 633	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (9,9 ; 562)]] -> 19,505,408 vertices; then 1 quadrupling -> 78,021,632 vertices
R(3,5,5,5,9,10,10)	97 712 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 6,107,042 vertices; then 2 quadruplings -> 97,712,672 vertices
R(3,5,5,5,10,10,10)	122 633 217	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [K2]] -> 119,759 vertices; then 5 quadruplings -> 122,633,216 vertices
R(3,5,5,6,6,6,6)	7 963 122	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 7,963,121 vertices
R(3,5,5,6,6,6,7)	10 434 177	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices; then 1 quadrupling -> 10,434,176 vertices
R(3,5,5,6,6,6,8)	12 595 889	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 787,243 vertices; then 2 quadruplings -> 12,595,888 vertices
R(3,5,5,6,6,6,9)	15 611 457	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 3 quadruplings -> 15,611,456 vertices
R(3,5,5,6,6,6,10)	20 153 601	Published bricks + theorems	[template (6,6,3;235)] x [atom (3,6,6 ; 337)] -> 78,725 vertices; then 4 quadruplings -> 20,153,600 vertices
R(3,5,5,6,6,7,7)	16 028 333	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,5,5 ; 146) (ours)]] -> 16,028,332 vertices
R(3,5,5,6,6,7,8)	21 001 821	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 1 quadrupling -> 21,001,820 vertices
R(3,5,5,6,6,7,9)	25 353 169	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 2 quadruplings -> 25,353,168 vertices
R(3,5,5,6,6,7,10)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,5,5,6,6,8,8)	25 353 169	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 2 quadruplings -> 25,353,168 vertices
R(3,5,5,6,6,8,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,5,5,6,6,8,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,5,5,6,6,9,9)	43 595 706	Our bricks + published	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,5,5 ; 146) (ours)]] -> 43,595,705 vertices
R(3,5,5,6,6,9,10)	58 330 233	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 1 quadrupling -> 58,330,232 vertices
R(3,5,5,6,6,10,10)	69 365 585	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 2 quadruplings -> 69,365,584 vertices
R(3,5,5,6,7,7,7)	21 001 821	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 1 quadrupling -> 21,001,820 vertices
R(3,5,5,6,7,7,8)	25 353 169	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 2 quadruplings -> 25,353,168 vertices
R(3,5,5,6,7,7,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,5,5,6,7,7,10)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,5,5,6,7,8,8)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,5,5,6,7,8,9)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,5,5,6,7,8,10)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices
R(3,5,5,6,7,9,9)	58 330 233	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 1 quadrupling -> 58,330,232 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,5,6,7,9,10)	69 365 585	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 2 quadruplings -> 69,365,584 vertices
R(3,5,5,6,7,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(3,5,5,6,8,8,8)	40 565 249	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 4 quadruplings -> 40,565,248 vertices
R(3,5,5,6,8,8,9)	48 437 249	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 5 quadruplings -> 48,437,248 vertices
R(3,5,5,6,8,8,10)	59 258 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 4 quadruplings -> 59,258,624 vertices
R(3,5,5,6,8,9,9)	69 365 585	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 2 quadruplings -> 69,365,584 vertices
R(3,5,5,6,8,9,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(3,5,5,6,8,10,10)	110 369 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 4 quadruplings -> 110,369,024 vertices
R(3,5,5,6,9,9,9)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(3,5,5,6,9,9,10)	110 369 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 4 quadruplings -> 110,369,024 vertices
R(3,5,5,6,9,10,10)	134 603 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 8,412,734 vertices; then 2 quadruplings -> 134,603,744 vertices
R(3,5,5,6,10,10,10)	161 911 233	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,6,9,9,10) of 3,272,096 vertices] + [colouring (6,9,9,10) of 303,164 vertices] (see Chung_Robertson.cpp) -> 10,119,452 vertices; then 2 quadruplings -> 161,911,232 vertices
R(3,5,5,7,7,7,7)	32 261 819	Our bricks + published	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,5,5 ; 146) (ours)]] -> 32,261,818 vertices
R(3,5,5,7,7,7,8)	42 123 345	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 1 quadrupling -> 42,123,344 vertices
R(3,5,5,7,7,7,9)	50 674 129	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 2 quadruplings -> 50,674,128 vertices
R(3,5,5,7,7,7,10)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(3,5,5,7,7,8,8)	50 674 129	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 2 quadruplings -> 50,674,128 vertices
R(3,5,5,7,7,8,9)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(3,5,5,7,7,8,10)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,5,5,7,7,9,9)	87 750 329	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,5,5 ; 146) (ours)]] -> 87,750,328 vertices
R(3,5,5,7,7,9,10)	117 407 985	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 1 quadrupling -> 117,407,984 vertices
R(3,5,5,7,7,10,10)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,5,5,7,8,8,8)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(3,5,5,7,8,8,9)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,5,5,7,8,8,10)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 − 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(3,5,5,7,8,9,9)	117 407 985	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 1 quadrupling -> 117,407,984 vertices
R(3,5,5,7,8,9,10)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,5,5,7,8,10,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,5,7,9,9,9)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,5,5,7,9,9,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,5,7,9,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,5,5,7,10,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(3,5,5,8,8,8,8)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices
R(3,5,5,8,8,8,9)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 − 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(3,5,5,8,8,8,10)	113 272 577	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,8,8 ; 940)] -> 442,471 vertices; then 4 quadruplings -> 113,272,576 vertices
R(3,5,5,8,8,9,9)	139 620 209	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 2 quadruplings -> 139,620,208 vertices
R(3,5,5,8,8,9,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,5,8,8,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,5,5,8,9,9,9)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,5,8,9,9,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,5,5,8,9,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(3,5,5,8,10,10,10)	310 218 753	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 6 quadruplings -> 310,218,752 vertices
R(3,5,5,9,9,9,9)	238 658 549	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,5,5 ; 146) (ours)]] -> 238,658,548 vertices
R(3,5,5,9,9,9,10)	319 325 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 1 quadrupling -> 319,325,864 vertices
R(3,5,5,9,9,10,10)	379 738 769	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 2 quadruplings -> 379,738,768 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,5,9,10,10,10)	466 115 009	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 3 quadruplings -> 466,115,008 vertices
R(3,5,5,10,10,10,10)	604 201 985	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 4 quadruplings -> 604,201,984 vertices
R(3,5,6,6,6,6,6)	10 627 797	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 10,627,796 vertices
R(3,5,6,6,6,6,7)	14 721 468	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 14,721,467 vertices
R(3,5,6,6,6,6,8)	18 500 241	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 18,500,240 vertices
R(3,5,6,6,6,6,9)	24 640 745	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 24,640,744 vertices
R(3,5,6,6,6,6,10)	29 422 529	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 7,355,632 vertices; then 1 quadrupling -> 29,422,528 vertices
R(3,5,6,6,6,7,7)	21 391 752	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 21,391,751 vertices
R(3,5,6,6,6,7,8)	25 542 765	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices; then 1 quadrupling -> 25,542,764 vertices
R(3,5,6,6,6,7,9)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,5,6,6,6,7,10)	41 736 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices; then 2 quadruplings -> 41,736,704 vertices
R(3,5,6,6,6,8,8)	31 422 977	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 3 quadruplings -> 31,422,976 vertices
R(3,5,6,6,6,8,9)	41 736 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices; then 2 quadruplings -> 41,736,704 vertices
R(3,5,6,6,6,8,10)	50 383 553	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 787,243 vertices; then 3 quadruplings -> 50,383,552 vertices
R(3,5,6,6,6,9,9)	58 330 233	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 1 quadrupling -> 58,330,232 vertices
R(3,5,6,6,6,9,10)	70 942 185	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 17,735,546 vertices; then 1 quadrupling -> 70,942,184 vertices
R(3,5,6,6,6,10,10)	85 463 297	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 3 quadruplings -> 85,463,296 vertices
R(3,5,6,6,7,7,7)	29 631 539	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 29,631,538 vertices
R(3,5,6,6,7,7,8)	38 029 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 1 quadrupling -> 38,029,768 vertices
R(3,5,6,6,7,7,9)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(3,5,6,6,7,7,10)	59 221 381	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 14,805,345 vertices; then 1 quadrupling -> 59,221,380 vertices
R(3,5,6,6,7,8,8)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(3,5,6,6,7,8,9)	62 115 309	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 1 quadrupling -> 62,115,308 vertices
R(3,5,6,6,7,8,10)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(3,5,6,6,7,9,9)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(3,5,6,6,7,9,10)	103 470 809	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 1 quadrupling -> 103,470,808 vertices
R(3,5,6,6,7,10,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(3,5,6,6,8,8,8)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(3,5,6,6,8,8,9)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(3,5,6,6,8,8,10)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices
R(3,5,6,6,8,9,9)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices
R(3,5,6,6,8,9,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(3,5,6,6,8,10,10)	162 260 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 5 quadruplings -> 162,260,992 vertices
R(3,5,6,6,9,9,9)	134 943 197	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 134,943,196 vertices
R(3,5,6,6,9,9,10)	169 002 341	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 42,250,585 vertices; then 1 quadrupling -> 169,002,340 vertices
R(3,5,6,6,9,10,10)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(3,5,6,6,10,10,10)	277 462 337	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 3 quadruplings -> 277,462,336 vertices
R(3,5,6,7,7,7,7)	42 756 312	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices
R(3,5,6,7,7,7,8)	51 231 105	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 1 quadrupling -> 51,231,104 vertices
R(3,5,6,7,7,7,9)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(3,5,6,7,7,7,10)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(3,5,6,7,7,8,8)	62 863 169	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 3 quadruplings -> 62,863,168 vertices
R(3,5,6,7,7,8,9)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(3,5,6,7,7,8,10)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices
R(3,5,6,7,7,9,9)	117 407 985	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 1 quadrupling -> 117,407,984 vertices
R(3,5,6,7,7,9,10)	142 793 505	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 1 quadrupling -> 142,793,504 vertices
R(3,5,6,7,7,10,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,6,7,8,8,8)	81 078 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 4 quadruplings -> 81,078,784 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,6,7,8,8,9)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices
R(3,5,6,7,8,8,10)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(3,5,6,7,8,9,9)	142 793 505	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 1 quadrupling -> 142,793,504 vertices
R(3,5,6,7,8,9,10)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,6,7,8,10,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,5,6,7,9,9,9)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,6,7,9,9,10)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(3,5,6,7,9,10,10)	277 462 337	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 3 quadruplings -> 277,462,336 vertices
R(3,5,6,7,10,10,10)	341 853 185	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 4 quadruplings -> 341,853,184 vertices
R(3,5,6,8,8,8,8)	97 149 953	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 24,287,489 – 3 = 97,149,953; with R(7,7,7,7) >= 94,874
R(3,5,6,8,8,8,9)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(3,5,6,8,8,8,10)	162 260 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 5 quadruplings -> 162,260,992 vertices
R(3,5,6,8,8,9,9)	171 381 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 3 quadruplings -> 171,381,568 vertices
R(3,5,6,8,8,9,10)	222 152 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 4 quadruplings -> 222,152,704 vertices
R(3,5,6,8,8,10,10)	270 779 393	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 5 quadruplings -> 270,779,392 vertices
R(3,5,6,8,9,9,9)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(3,5,6,8,9,9,10)	277 462 337	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 3 quadruplings -> 277,462,336 vertices
R(3,5,6,8,9,10,10)	341 853 185	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 4 quadruplings -> 341,853,184 vertices
R(3,5,6,8,10,10,10)	441 476 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 5 quadruplings -> 441,476,096 vertices
R(3,5,6,9,9,9,9)	319 325 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 1 quadrupling -> 319,325,864 vertices
R(3,5,6,9,9,9,10)	388 369 305	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 97,092,326 vertices; then 1 quadrupling -> 388,369,304 vertices
R(3,5,6,9,9,10,10)	466 115 009	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 3 quadruplings -> 466,115,008 vertices
R(3,5,6,9,10,10,10)	604 201 985	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 4 quadruplings -> 604,201,984 vertices
R(3,5,6,10,10,10,10)	736 463 873	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 184,115,969 – 3 = 736,463,873; with R(9,9,9,9) >= 719,204
R(3,5,7,7,7,7,7)	59 225 411	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 59,225,410 vertices
R(3,5,7,7,7,7,8)	76 011 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 1 quadrupling -> 76,011,208 vertices
R(3,5,7,7,7,7,9)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(3,5,7,7,7,7,10)	118 780 273	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 29,695,068 vertices; then 1 quadrupling -> 118,780,272 vertices
R(3,5,7,7,7,8,8)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(3,5,7,7,7,8,9)	124 151 661	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 31,037,915 vertices; then 1 quadrupling -> 124,151,660 vertices
R(3,5,7,7,7,8,10)	168 493 377	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 2 quadruplings -> 168,493,376 vertices
R(3,5,7,7,7,9,9)	168 493 377	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 2 quadruplings -> 168,493,376 vertices
R(3,5,7,7,7,9,10)	208 268 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 1 quadrupling -> 208,268,008 vertices
R(3,5,7,7,7,10,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,5,7,7,8,8,8)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(3,5,7,7,8,8,9)	168 493 377	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 2 quadruplings -> 168,493,376 vertices
R(3,5,7,7,8,8,10)	202 696 513	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 3 quadruplings -> 202,696,512 vertices
R(3,5,7,7,8,9,9)	208 268 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 1 quadrupling -> 208,268,008 vertices
R(3,5,7,7,8,9,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,5,7,7,8,10,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(3,5,7,7,9,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,5,7,7,9,9,10)	340 171 101	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 1 quadrupling -> 340,171,100 vertices
R(3,5,7,7,9,10,10)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(3,5,7,7,10,10,10)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(3,5,7,8,8,8,8)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,5,7,8,8,8,9)	202 696 513	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 3 quadruplings -> 202,696,512 vertices
R(3,5,7,8,8,8,10)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(3,5,7,8,8,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,5,7,8,8,9,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(3,5,7,8,8,10,10)	414 636 289	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(3,5,7,8,9,9,9)	340 171 101	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 1 quadrupling -> 340,171,100 vertices
R(3,5,7,8,9,9,10)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(3,5,7,8,9,10,10)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(3,5,7,8,10,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(3,5,7,9,9,9,9)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(3,5,7,9,9,9,10)	566 439 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 1 quadrupling -> 566,439,208 vertices
R(3,5,7,9,9,10,10)	747 970 545	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 2 quadruplings -> 747,970,544 vertices
R(3,5,7,9,10,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(3,5,7,10,10,10,10)	1 083 117 569	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 6 quadruplings -> 1,083,117,568 vertices
R(3,5,8,8,8,8,8)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(3,5,8,8,8,8,9)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(3,5,8,8,8,8,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(3,5,8,8,8,9,9)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(3,5,8,8,8,9,10)	414 636 289	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(3,5,8,8,8,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(3,5,8,8,9,9,9)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(3,5,8,8,9,9,10)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(3,5,8,8,9,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(3,5,8,8,10,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(3,5,8,9,9,9,9)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(3,5,8,9,9,9,10)	747 970 545	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 2 quadruplings -> 747,970,544 vertices
R(3,5,8,9,9,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(3,5,8,9,10,10,10)	1 083 117 569	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 6 quadruplings -> 1,083,117,568 vertices
R(3,5,8,10,10,10,10)	1 240 875 009	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 7 quadruplings -> 1,240,875,008 vertices
R(3,5,9,9,9,9,9)	738 731 152	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 738,731,151 vertices
R(3,5,9,9,9,9,10)	925 184 061	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 231,296,015 vertices; then 1 quadrupling -> 925,184,060 vertices
R(3,5,9,9,9,10,10)	1 277 303 457	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 2 quadruplings -> 1,277,303,456 vertices
R(3,5,9,9,10,10,10)	1 518 955 073	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 3 quadruplings -> 1,518,955,072 vertices
R(3,5,9,10,10,10,10)	1 864 460 033	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 4 quadruplings -> 1,864,460,032 vertices
R(3,5,10,10,10,10,10)	2 416 807 937	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 5 quadruplings -> 2,416,807,936 vertices
R(3,6,6,6,6,6,6)	18 421 518	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 18,421,517 vertices
R(3,6,6,6,6,6,7)	21 996 405	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 5,499,102 – 3 = 21,996,405; with R(6,6,6,6,6,6) >= 5,499,102
R(3,6,6,6,6,6,8)	22 560 353	Our bricks	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 1,410,022 vertices; then 2 quadruplings -> 22,560,352 vertices
R(3,6,6,6,6,6,9)	31 891 494	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 31,891,493 vertices
R(3,6,6,6,6,6,10)	39 737 349	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 9,934,337 vertices; then 1 quadrupling -> 39,737,348 vertices
R(3,6,6,6,6,7,7)	37 079 207	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices
R(3,6,6,6,6,7,8)	44 274 809	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 1 quadrupling -> 44,274,808 vertices
R(3,6,6,6,6,7,9)	44 274 809	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 1 quadrupling -> 44,274,808 vertices (by monotonicity, from (3,6,6,6,6,7,8))
R(3,6,6,6,6,7,10)	54 850 005	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (7,9 ; 251)]] -> 13,712,501 vertices; then 1 quadrupling -> 54,850,004 vertices
R(3,6,6,6,6,8,8)	51 439 386	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices
R(3,6,6,6,6,8,9)	61 420 725	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 15,355,181 vertices; then 1 quadrupling -> 61,420,724 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,6,6,6,6,8,10)	70 313 265	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 4,394,579 vertices; then 2 quadruplings -> 70,313,264 vertices
R(3,6,6,6,6,9,9)	100 884 054	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices
R(3,6,6,6,6,9,10)	122 966 469	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 1 quadrupling -> 122,966,468 vertices
R(3,6,6,6,6,10,10)	131 581 110	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(3,6,6,6,7,7,7)	44 274 809	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 1 quadrupling -> 44,274,808 vertices
R(3,6,6,6,7,7,8)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(3,6,6,6,7,7,9)	64 191 851	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 64,191,850 vertices
R(3,6,6,6,7,7,10)	79 984 145	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (6,9 ; 182)]] -> 19,996,036 vertices; then 1 quadrupling -> 79,984,144 vertices
R(3,6,6,6,7,8,8)	61 420 725	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 15,355,181 vertices; then 1 quadrupling -> 61,420,724 vertices
R(3,6,6,6,7,8,9)	85 567 005	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 21,391,751 vertices; then 1 quadrupling -> 85,567,004 vertices
R(3,6,6,6,7,8,10)	102 171 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices; then 2 quadruplings -> 102,171,056 vertices
R(3,6,6,6,7,9,9)	122 966 469	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 1 quadrupling -> 122,966,468 vertices
R(3,6,6,6,7,9,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(3,6,6,6,7,10,10)	174 437 109	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(3,6,6,6,8,8,8)	68 538 881	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]] -> 267,730 vertices; then 4 quadruplings -> 68,538,880 vertices
R(3,6,6,6,8,8,9)	102 171 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices; then 2 quadruplings -> 102,171,056 vertices
R(3,6,6,6,8,8,10)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(3,6,6,6,8,9,9)	126 119 393	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 7,882,462 vertices; then 2 quadruplings -> 126,119,392 vertices
R(3,6,6,6,8,9,10)	166 946 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices; then 3 quadruplings -> 166,946,816 vertices
R(3,6,6,6,8,10,10)	193 748 993	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 6 quadruplings -> 193,748,992 vertices
R(3,6,6,6,9,9,9)	174 588 204	Our bricks + published	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 174,588,203 vertices
R(3,6,6,6,9,9,10)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(3,6,6,6,9,10,10)	283 768 737	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 17,735,546 vertices; then 2 quadruplings -> 283,768,736 vertices
R(3,6,6,6,10,10,10)	341 853 185	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 4 quadruplings -> 341,853,184 vertices
R(3,6,6,7,7,7,7)	74 633 450	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices
R(3,6,6,7,7,7,8)	89 115 893	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 1 quadrupling -> 89,115,892 vertices
R(3,6,6,7,7,7,9)	94 218 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 2 quadruplings -> 94,218,304 vertices
R(3,6,6,7,7,7,10)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices
R(3,6,6,7,7,8,8)	103 538 249	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices
R(3,6,6,7,7,8,9)	123 628 889	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 1 quadrupling -> 123,628,888 vertices
R(3,6,6,7,7,8,10)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(3,6,6,7,7,9,9)	203 061 491	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices
R(3,6,6,7,7,9,10)	247 509 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 1 quadrupling -> 247,509,424 vertices
R(3,6,6,7,7,10,10)	264 849 155	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(3,6,6,7,8,8,8)	123 628 889	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 1 quadrupling -> 123,628,888 vertices
R(3,6,6,7,8,8,9)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(3,6,6,7,8,8,10)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(3,6,6,7,8,9,9)	247 509 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 1 quadrupling -> 247,509,424 vertices
R(3,6,6,7,8,9,10)	256 293 393	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 2 quadruplings -> 256,293,392 vertices
R(3,6,6,7,8,10,10)	351 110 585	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(3,6,6,7,9,9,9)	248 461 233	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 2 quadruplings -> 248,461,232 vertices
R(3,6,6,7,9,9,10)	336 029 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 3 quadruplings -> 336,029,120 vertices
R(3,6,6,7,9,10,10)	413 883 233	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 2 quadruplings -> 413,883,232 vertices
R(3,6,6,7,10,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(3,6,6,8,8,8,8)	142 667 796	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (3,8,8 ; 940)]] -> 142,667,795 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,6,6,8,8,8,9)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(3,6,6,8,8,8,10)	230 072 833	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 4 quadruplings -> 230,072,832 vertices
R(3,6,6,8,8,9,9)	281 600 382	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices
R(3,6,6,8,8,9,10)	341 049 789	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 85,262,447 vertices; then 1 quadrupling -> 341,049,788 vertices
R(3,6,6,8,8,10,10)	365 085 953	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,426,117 vertices; then 4 quadruplings -> 365,085,952 vertices
R(3,6,6,8,9,9,9)	341 049 789	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 85,262,447 vertices; then 1 quadrupling -> 341,049,788 vertices
R(3,6,6,8,9,9,10)	413 883 233	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 2 quadruplings -> 413,883,232 vertices
R(3,6,6,8,9,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(3,6,6,8,10,10,10)	649 043 969	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 6 quadruplings -> 649,043,968 vertices
R(3,6,6,9,9,9,9)	552 278 244	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 552,278,243 vertices
R(3,6,6,9,9,9,10)	673 173 477	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 1 quadrupling -> 673,173,476 vertices
R(3,6,6,9,9,10,10)	720 322 235	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(3,6,6,9,10,10,10)	954 942 237	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,6,6,10,10,10,10)	1 008 955 649	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 3,941,233 vertices; then 4 quadruplings -> 1,008,955,648 vertices
R(3,6,7,7,7,7,7)	89 115 893	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 1 quadrupling -> 89,115,892 vertices
R(3,6,7,7,7,7,8)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(3,6,7,7,7,7,9)	129 206 336	Our bricks + published	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 129,206,335 vertices
R(3,6,7,7,7,7,10)	160 992 377	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (6,9 ; 182)]] -> 40,248,094 vertices; then 1 quadrupling -> 160,992,376 vertices
R(3,6,7,7,7,8,8)	123 628 889	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 1 quadrupling -> 123,628,888 vertices
R(3,6,7,7,7,8,9)	171 025 245	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices; then 1 quadrupling -> 171,025,244 vertices
R(3,6,7,7,7,8,10)	204 924 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 2 quadruplings -> 204,924,416 vertices
R(3,6,7,7,7,9,9)	247 509 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 1 quadrupling -> 247,509,424 vertices
R(3,6,7,7,7,9,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,6,7,7,7,10,10)	351 110 585	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(3,6,7,7,8,8,8)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(3,6,7,7,8,8,9)	204 924 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 2 quadruplings -> 204,924,416 vertices
R(3,6,7,7,8,8,10)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(3,6,7,7,8,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(3,6,7,7,8,9,10)	336 029 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 3 quadruplings -> 336,029,120 vertices
R(3,6,7,7,8,10,10)	414 636 289	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(3,6,7,7,9,9,9)	351 414 716	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 351,414,715 vertices
R(3,6,7,7,9,9,10)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(3,6,7,7,9,10,10)	571 174 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 2 quadruplings -> 571,174,016 vertices
R(3,6,7,7,10,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(3,6,7,8,8,8,8)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(3,6,7,8,8,8,9)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(3,6,7,8,8,8,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(3,6,7,8,8,9,9)	341 049 789	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 85,262,447 vertices; then 1 quadrupling -> 341,049,788 vertices
R(3,6,7,8,8,9,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(3,6,7,8,8,10,10)	502 767 617	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 5 quadruplings -> 502,767,616 vertices
R(3,6,7,8,9,9,9)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(3,6,7,8,9,9,10)	571 174 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 2 quadruplings -> 571,174,016 vertices
R(3,6,7,8,9,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(3,6,7,8,10,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(3,6,7,9,9,9,9)	673 173 477	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 1 quadrupling -> 673,173,476 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,6,7,9,9,9,10)	747 970 545	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 2 quadruplings -> 747,970,544 vertices
R(3,6,7,9,9,10,10)	954 942 237	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,6,7,9,10,10,10)	1 109 849 345	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 4 quadruplings -> 1,109,849,344 vertices
R(3,6,7,10,10,10,10)	1 367 412 737	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 5 quadruplings -> 1,367,412,736 vertices
R(3,6,8,8,8,8,8)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(3,6,8,8,8,8,9)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(3,6,8,8,8,8,10)	388 599 809	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 97,149,953 – 3 = 388,599,809; with R(7,7,7,7) >= 94,874
R(3,6,8,8,8,9,9)	405 650 689	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 4 quadruplings -> 405,650,688 vertices
R(3,6,8,8,8,9,10)	502 767 617	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 5 quadruplings -> 502,767,616 vertices
R(3,6,8,8,8,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(3,6,8,8,9,9,9)	571 174 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 2 quadruplings -> 571,174,016 vertices
R(3,6,8,8,9,9,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(3,6,8,8,9,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(3,6,8,8,10,10,10)	1 083 117 569	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 6 quadruplings -> 1,083,117,568 vertices
R(3,6,8,9,9,9,9)	690 434 273	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 43,152,142 vertices; then 2 quadruplings -> 690,434,272 vertices
R(3,6,8,9,9,9,10)	933 283 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 3 quadruplings -> 933,283,712 vertices
R(3,6,8,9,9,10,10)	1 109 849 345	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 4 quadruplings -> 1,109,849,344 vertices
R(3,6,8,9,10,10,10)	1 367 412 737	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 5 quadruplings -> 1,367,412,736 vertices
R(3,6,8,10,10,10,10)	1 765 904 385	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 6 quadruplings -> 1,765,904,384 vertices
R(3,6,9,9,9,9,9)	955 758 506	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 955,758,505 vertices
R(3,6,9,9,9,9,10)	1 277 303 457	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 2 quadruplings -> 1,277,303,456 vertices
R(3,6,9,9,9,10,10)	1 553 477 217	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 97,092,326 vertices; then 2 quadruplings -> 1,553,477,216 vertices
R(3,6,9,9,10,10,10)	1 864 460 033	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 4 quadruplings -> 1,864,460,032 vertices
R(3,6,9,10,10,10,10)	2 416 807 937	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 5 quadruplings -> 2,416,807,936 vertices
R(3,6,10,10,10,10,10)	2 945 855 489	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 736,463,873 – 3 = 2,945,855,489; with R(9,9,9,9) >= 719,204
R(3,7,7,7,7,7,7)	149 172 026	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 149,172,025 vertices
R(3,7,7,7,7,7,8)	178 739 657	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices; then 1 quadrupling -> 178,739,656 vertices
R(3,7,7,7,7,7,9)	189 639 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 2 quadruplings -> 189,639,136 vertices
R(3,7,7,7,7,7,10)	233 237 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 3 quadruplings -> 233,237,056 vertices
R(3,7,7,7,7,8,8)	208 403 573	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,8,8 ; 940)]] -> 208,403,572 vertices
R(3,7,7,7,7,8,9)	248 841 413	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices; then 1 quadrupling -> 248,841,412 vertices
R(3,7,7,7,7,8,10)	304 044 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 2 quadruplings -> 304,044,832 vertices
R(3,7,7,7,7,9,9)	408 725 996	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices
R(3,7,7,7,7,9,10)	498 190 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 1 quadrupling -> 498,190,696 vertices
R(3,7,7,7,7,10,10)	530 073 023	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(3,7,7,7,8,8,8)	248 841 413	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices; then 1 quadrupling -> 248,841,412 vertices
R(3,7,7,7,8,8,9)	304 044 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 2 quadruplings -> 304,044,832 vertices
R(3,7,7,7,8,8,10)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(3,7,7,7,8,9,9)	498 190 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 1 quadrupling -> 498,190,696 vertices
R(3,7,7,7,8,9,10)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(3,7,7,7,8,10,10)	706 721 237	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [atome (10,10 ; 797)]]
R(3,7,7,7,9,9,9)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(3,7,7,7,9,9,10)	673 973 505	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 3 quadruplings -> 673,973,504 vertices
R(3,7,7,7,9,10,10)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(3,7,7,7,10,10,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,7,7,8,8,8,8)	289 637 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 4 quadruplings -> 289,637,376 vertices
R(3,7,7,8,8,8,9)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(3,7,7,8,8,8,10)	461 458 177	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 4 quadruplings -> 461,458,176 vertices
R(3,7,7,8,8,9,9)	566 811 023	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices
R(3,7,7,8,8,9,10)	686 472 005	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 1 quadrupling -> 686,472,004 vertices
R(3,7,7,8,8,10,10)	787 808 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 4 quadruplings -> 787,808,256 vertices
R(3,7,7,8,9,9,9)	686 472 005	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 1 quadrupling -> 686,472,004 vertices
R(3,7,7,8,9,9,10)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(3,7,7,8,9,10,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(3,7,7,8,10,10,10)	1 297 260 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 6 quadruplings -> 1,297,260,544 vertices
R(3,7,7,9,9,9,9)	1 111 636 976	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices
R(3,7,7,9,9,9,10)	1 354 977 377	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 1 quadrupling -> 1,354,977,376 vertices
R(3,7,7,9,9,10,10)	1 441 669 973	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(3,7,7,9,10,10,10)	1 922 127 317	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,7,7,10,10,10,10)	2 030 841 089	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 7,932,973 vertices; then 4 quadruplings -> 2,030,841,088 vertices
R(3,7,8,8,8,8,8)	342 888 809	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 85,722,202 vertices; then 1 quadrupling -> 342,888,808 vertices
R(3,7,8,8,8,8,9)	461 458 177	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 4 quadruplings -> 461,458,176 vertices
R(3,7,8,8,8,8,10)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices
R(3,7,8,8,8,9,9)	686 472 005	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 1 quadrupling -> 686,472,004 vertices
R(3,7,8,8,8,9,10)	810 786 049	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 4 quadruplings -> 810,786,048 vertices
R(3,7,8,8,8,10,10)	1 005 810 689	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 5 quadruplings -> 1,005,810,688 vertices
R(3,7,8,8,9,9,9)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(3,7,8,8,9,9,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(3,7,8,8,9,10,10)	1 297 260 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 6 quadruplings -> 1,297,260,544 vertices
R(3,7,8,8,10,10,10)	1 658 545 153	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(3,7,8,9,9,9,9)	1 354 977 377	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 1 quadrupling -> 1,354,977,376 vertices
R(3,7,8,9,9,9,10)	1 403 042 177	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 2 quadruplings -> 1,403,042,176 vertices
R(3,7,8,9,9,10,10)	1 922 127 317	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,7,8,9,10,10,10)	2 233 923 329	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 4 quadruplings -> 2,233,923,328 vertices
R(3,7,8,10,10,10,10)	2 742 105 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 5 quadruplings -> 2,742,105,088 vertices
R(3,7,9,9,9,9,9)	1 360 684 401	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 2 quadruplings -> 1,360,684,400 vertices
R(3,7,9,9,9,9,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(3,7,9,9,9,10,10)	2 265 756 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 2 quadruplings -> 2,265,756,832 vertices
R(3,7,9,9,10,10,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(3,7,9,10,10,10,10)	3 554 443 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 6 quadruplings -> 3,554,443,264 vertices
R(3,7,10,10,10,10,10)	4 332 470 273	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 7 quadruplings -> 4,332,470,272 vertices
R(3,8,8,8,8,8,8)	395 689 741	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [atom (3,8,8 ; 940)]] -> 395,689,740 vertices
R(3,8,8,8,8,8,9)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices
R(3,8,8,8,8,8,10)	732 000 257	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 6 quadruplings -> 732,000,256 vertices
R(3,8,8,8,8,9,9)	810 786 049	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 4 quadruplings -> 810,786,048 vertices
R(3,8,8,8,8,9,10)	1 005 810 689	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 5 quadruplings -> 1,005,810,688 vertices
R(3,8,8,8,8,10,10)	1 297 260 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 6 quadruplings -> 1,297,260,544 vertices

Parameters	Bound (R >=)	Origin	Construction
R(3,8,8,8,9,9,9)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(3,8,8,8,9,9,10)	1 297 260 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 6 quadruplings -> 1,297,260,544 vertices
R(3,8,8,8,9,10,10)	1 658 545 153	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(3,8,8,8,10,10,10)	1 990 905 857	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 6 quadruplings -> 1,990,905,856 vertices
R(3,8,8,9,9,9,9)	1 541 581 583	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 1,541,581,582 vertices
R(3,8,8,9,9,9,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(3,8,8,9,9,10,10)	2 233 923 329	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 4 quadruplings -> 2,233,923,328 vertices
R(3,8,8,9,10,10,10)	2 742 105 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 5 quadruplings -> 2,742,105,088 vertices
R(3,8,8,10,10,10,10)	3 554 443 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 6 quadruplings -> 3,554,443,264 vertices
R(3,8,9,9,9,9,9)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(3,8,9,9,9,9,10)	2 265 756 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 2 quadruplings -> 2,265,756,832 vertices
R(3,8,9,9,9,10,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(3,8,9,9,10,10,10)	3 554 443 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 6 quadruplings -> 3,554,443,264 vertices
R(3,8,9,10,10,10,10)	4 332 470 273	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 7 quadruplings -> 4,332,470,272 vertices
R(3,8,10,10,10,10,10)	4 962 433 537	Our bricks (Paley-doubling atoms, Nov 2026)	[modele decale (3,4,3 ; largeur 19, queue 2)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,9,9,9,9,9,9)	3 023 369 366	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 3,023,369,365 vertices
R(3,9,9,9,9,9,10)	3 685 193 297	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 921,298,324 vertices; then 1 quadrupling -> 3,685,193,296 vertices
R(3,9,9,9,9,10,10)	3 920 975 033	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(3,9,9,9,10,10,10)	5 227 696 637	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(3,9,9,10,10,10,10)	6 075 820 289	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 4 quadruplings -> 6,075,820,288 vertices
R(3,9,10,10,10,10,10)	7 457 840 129	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 5 quadruplings -> 7,457,840,128 vertices
R(3,10,10,10,10,10,10)	9 667 231 745	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 6 quadruplings -> 9,667,231,744 vertices
R(4,4,4,4,4,4,4)	160 024	Published	published value
R(4,4,4,4,4,4,5)	288 041	Published bricks + theorems	[template (5,3;10)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 288,040 vertices
R(4,4,4,4,4,4,6)	352 050	Published bricks + theorems	[template (6,3;12)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 352,049 vertices
R(4,4,4,4,4,4,7)	480 067	Published bricks + theorems	[template (7,3;16)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 480,066 vertices
R(4,4,4,4,4,4,8)	608 084	Published bricks + theorems	[template (8,3 ; 20)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 608,083 vertices
R(4,4,4,4,4,4,9)	800 109	Published bricks + theorems	[template (9,3 ; 26)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 800,108 vertices
R(4,4,4,4,4,4,10)	928 126	Published bricks + theorems	[template (10,3 ; 30)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 928,125 vertices
R(4,4,4,4,4,5,5)	454 883	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)]] -> 454,882 vertices
R(4,4,4,4,4,5,6)	599 183	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,6,3 ; 80) (ours)] x [atom (4,4,4 ; 127)]] -> 599,182 vertices
R(4,4,4,4,4,5,7)	872 123	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 872,122 vertices
R(4,4,4,4,4,5,8)	1 104 673	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 276,168 vertices; then 1 quadrupling -> 1,104,672 vertices
R(4,4,4,4,4,5,9)	1 306 595	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 1,306,594 vertices
R(4,4,4,4,4,5,10)	1 454 333	Our bricks + published	[template (9,3 ; 26)] x [[template (3,4,4,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 363,583 vertices; then 1 quadrupling -> 1,454,332 vertices
R(4,4,4,4,4,6,6)	976 818	Published bricks + theorems	[template (6,6,3;235)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 976,817 vertices
R(4,4,4,4,4,6,7)	1 152 161	Published bricks + theorems	[template (5,3;10)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 288,040 vertices; then 1 quadrupling -> 1,152,160 vertices
R(4,4,4,4,4,6,8)	1 208 418	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (4,8,3 ; 122) (ours)] x [atom (4,4,4 ; 127)]] -> 1,208,417 vertices
R(4,4,4,4,4,6,9)	1 819 529	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)]] -> 454,882 vertices; then 1 quadrupling -> 1,819,528 vertices
R(4,4,4,4,4,6,10)	1 819 529	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)]] -> 454,882 vertices; then 1 quadrupling -> 1,819,528 vertices (by monotonicity, from (4,4,4,4,4,6,9))
R(4,4,4,4,4,7,7)	1 966 157	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 1,966,156 vertices
R(4,4,4,4,4,7,8)	1 966 157	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 1,966,156 vertices (by monotonicity, from (4,4,4,4,4,7,7))
R(4,4,4,4,4,7,9)	2 432 333	Published bricks + theorems	[template (8,3 ; 20)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 608,083 vertices; then 1 quadrupling -> 2,432,332 vertices
R(4,4,4,4,4,7,10)	3 200 433	Published bricks + theorems	[template (9,3 ; 26)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,4,4 ; 127)]] -> 800,108 vertices; then 1 quadrupling -> 3,200,432 vertices
R(4,4,4,4,4,8,8)	2 709 208	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 2,709,207 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,4,4,8,9)	3 488 489	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 872,122 vertices; then 1 quadrupling -> 3,488,488 vertices
R(4,4,4,4,4,8,10)	3 488 489	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (4,4,4 ; 127)]] -> 872,122 vertices; then 1 quadrupling -> 3,488,488 vertices (by monotonicity, from (4,4,4,4,4,8,9))
R(4,4,4,4,4,9,9)	5 347 457	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 5,347,456 vertices
R(4,4,4,4,4,9,10)	5 347 457	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 5,347,456 vertices (by monotonicity, from (4,4,4,4,4,9,9))
R(4,4,4,4,4,10,10)	6 649 980	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,4,3 ; 34)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,4,5,5,5)	743 303	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (4,4,4 ; 127)]] -> 743,302 vertices
R(4,4,4,4,5,5,6)	1 023 923	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (4,4,4 ; 127)]] -> 1,023,922 vertices
R(4,4,4,4,5,5,7)	1 424 667	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (4,4,4 ; 127)]] -> 1,424,666 vertices
R(4,4,4,4,5,5,8)	1 796 653	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 449,163 vertices; then 1 quadrupling -> 1,796,652 vertices
R(4,4,4,4,5,5,9)	2 374 163	Our bricks	[template (4,5,3 ; 61) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)]] -> 2,374,162 vertices
R(4,4,4,4,5,5,10)	3 007 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)]] -> 751,814 vertices; then 1 quadrupling -> 3,007,256 vertices
R(4,4,4,4,5,6,6)	1 775 063	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,775,062 vertices
R(4,4,4,4,5,6,7)	2 248 397	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,099 vertices; then 1 quadrupling -> 2,248,396 vertices
R(4,4,4,4,5,6,8)	2 367 825	Published bricks + theorems	[template (6,6,3;235)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 147,989 vertices; then 2 quadruplings -> 2,367,824 vertices
R(4,4,4,4,5,6,9)	3 125 985	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)]] -> 3,125,984 vertices
R(4,4,4,4,5,6,10)	3 853 745	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 963,436 vertices; then 1 quadrupling -> 3,853,744 vertices
R(4,4,4,4,5,7,7)	3 572 843	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices
R(4,4,4,4,5,7,8)	4 525 585	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 1 quadrupling -> 4,525,584 vertices
R(4,4,4,4,5,7,9)	4 765 985	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 2 quadruplings -> 4,765,984 vertices
R(4,4,4,4,5,7,10)	5 226 377	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 1,306,594 vertices; then 1 quadrupling -> 5,226,376 vertices
R(4,4,4,4,5,8,8)	4 923 263	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 4,923,262 vertices
R(4,4,4,4,5,8,9)	6 236 117	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 1,559,029 vertices; then 1 quadrupling -> 6,236,116 vertices
R(4,4,4,4,5,8,10)	7 186 609	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 449,163 vertices; then 2 quadruplings -> 7,186,608 vertices
R(4,4,4,4,5,9,9)	9 718 043	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices
R(4,4,4,4,5,9,10)	12 309 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 1 quadrupling -> 12,309,504 vertices
R(4,4,4,4,5,10,10)	12 962 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 810,154 vertices; then 2 quadruplings -> 12,962,464 vertices
R(4,4,4,4,6,6,6)	2 337 170	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,337,169 vertices
R(4,4,4,4,6,6,7)	3 402 207	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,402,206 vertices
R(4,4,4,4,6,6,8)	3 579 716	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,579,715 vertices
R(4,4,4,4,6,6,9)	4 615 284	Our bricks + published	[modele (6,6,3;235)] x [[modele (4,9,3 ; 157)] x [atome (4,4,4 ; 127)]] [new template (4,9;157), v21]
R(4,4,4,4,6,6,10)	5 357 766	Our bricks + published	[template (6,6,3;235)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,10 ; 91) (ours)]] -> 5,357,765 vertices
R(4,4,4,4,6,7,7)	4 704 247	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 4,704,246 vertices
R(4,4,4,4,6,7,8)	4 704 247	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 4,704,246 vertices (by monotonicity, from (4,4,4,4,6,7,7))
R(4,4,4,4,6,7,9)	7 100 249	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,775,062 vertices; then 1 quadrupling -> 7,100,248 vertices
R(4,4,4,4,6,7,10)	7 278 113	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (4,4,4 ; 127)]] -> 454,882 vertices; then 2 quadruplings -> 7,278,112 vertices
R(4,4,4,4,6,8,8)	6 482 300	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 6,482,299 vertices
R(4,4,4,4,6,8,9)	8 993 585	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,099 vertices; then 2 quadruplings -> 8,993,584 vertices
R(4,4,4,4,6,8,10)	8 993 585	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,099 vertices; then 2 quadruplings -> 8,993,584 vertices (by monotonicity, from (4,4,4,4,6,8,9))
R(4,4,4,4,6,9,9)	12 795 427	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices
R(4,4,4,4,6,9,10)	12 795 427	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices (by monotonicity, from (4,4,4,4,6,9,9))
R(4,4,4,4,6,10,10)	15 919 640	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,6,3 ; 80)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,4,7,7,7)	6 847 952	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 6,847,951 vertices
R(4,4,4,4,7,7,8)	7 864 625	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 1,966,156 vertices; then 1 quadrupling -> 7,864,624 vertices
R(4,4,4,4,7,7,9)	9 289 736	Our bricks + published	[modele (7,7,3;472)] x [[modele (4,9,3 ; 157)] x [atome (4,4,4 ; 127)]] [new template (4,9;157), v21]
R(4,4,4,4,7,7,10)	10 784 219	Our bricks + published	[template (7,7,3;472)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (4,10 ; 91) (ours)]] -> 10,784,218 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,4,7,8,8)	9 436 257	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 9,436,256 vertices
R(4,4,4,4,7,8,9)	14 291 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 1 quadrupling -> 14,291,368 vertices
R(4,4,4,4,7,8,10)	14 291 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 1 quadrupling -> 14,291,368 vertices (by monotonicity, from (4,4,4,4,7,8,9))
R(4,4,4,4,7,9,9)	18 626 252	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 18,626,251 vertices
R(4,4,4,4,7,9,10)	21 389 825	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (4,4,3 ; 34)] x [atom (4,4,4 ; 127)]] -> 5,347,456 vertices; then 1 quadrupling -> 21,389,824 vertices
R(4,4,4,4,7,10,10)	23 174 157	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,7,3 ; 116)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,4,8,8,8)	9 928 586	Our bricks + published	[template (4,8,3 ; 122) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 9,928,585 vertices
R(4,4,4,4,8,8,9)	18 102 337	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 2 quadruplings -> 18,102,336 vertices
R(4,4,4,4,8,8,10)	18 102 337	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 2 quadruplings -> 18,102,336 vertices (by monotonicity, from (4,4,4,4,8,8,9))
R(4,4,4,4,8,9,9)	19 693 049	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 4,923,262 vertices; then 1 quadrupling -> 19,693,048 vertices
R(4,4,4,4,8,9,10)	24 944 465	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 1,559,029 vertices; then 2 quadruplings -> 24,944,464 vertices
R(4,4,4,4,8,10,10)	24 944 465	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 1,559,029 vertices; then 2 quadruplings -> 24,944,464 vertices (by monotonicity, from (4,4,4,4,8,9,10))
R(4,4,4,4,9,9,9)	25 266 921	Our bricks + published	[modele (4,9,3 ; 157)] x [[modele (9,9,3 ; 1282)] x [atome (4,4,4 ; 127)]] [new template (4,9;157), v21]
R(4,4,4,4,9,9,10)	38 872 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 1 quadrupling -> 38,872,168 vertices
R(4,4,4,4,9,10,10)	49 238 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 2 quadruplings -> 49,238,016 vertices
R(4,4,4,4,10,10,10)	49 238 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 2 quadruplings -> 49,238,016 vertices (by monotonicity, from (4,4,4,4,9,10,10))
R(4,4,4,4,5,5,5,5)	1 214 066	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (4,4,4 ; 127)]] -> 1,214,065 vertices
R(4,4,4,5,5,5,6)	1 672 437	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (4,4,4 ; 127)]] -> 1,672,436 vertices
R(4,4,4,5,5,5,7)	2 316 762	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (4,4,4 ; 127)]] -> 2,316,761 vertices
R(4,4,4,5,5,5,8)	2 911 524	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)]] -> 2,911,523 vertices
R(4,4,4,5,5,5,9)	3 877 804	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (4,4,4 ; 127)]] -> 3,877,803 vertices
R(4,4,4,5,5,5,10)	4 753 489	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (5,8,3 ; 236) (ours)] x [atom (4,4,4 ; 127)]] -> 297,093 vertices; then 2 quadruplings -> 4,753,488 vertices
R(4,4,4,5,5,6,6)	2 899 274	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,899,273 vertices
R(4,4,4,5,5,6,7)	3 550 133	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 887,533 vertices; then 1 quadrupling -> 3,550,132 vertices
R(4,4,4,5,5,6,8)	4 733 489	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 2 quadruplings -> 4,733,488 vertices
R(4,4,4,5,5,6,9)	5 714 753	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 3 quadruplings -> 5,714,752 vertices
R(4,4,4,5,5,6,10)	7 573 761	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 4 quadruplings -> 7,573,760 vertices
R(4,4,4,5,5,7,7)	5 835 648	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices
R(4,4,4,5,5,7,8)	7 145 693	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 1 quadrupling -> 7,145,692 vertices
R(4,4,4,5,5,7,9)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(4,4,4,5,5,7,10)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,4,4,5,5,8,8)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices
R(4,4,4,5,5,8,9)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,4,4,5,5,8,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,4,4,5,5,9,9)	15 872 808	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 15,872,807 vertices
R(4,4,4,5,5,9,10)	19 436 093	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 1 quadrupling -> 19,436,092 vertices
R(4,4,4,5,5,10,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices
R(4,4,4,5,6,6,6)	3 993 897	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,993,896 vertices
R(4,4,4,5,6,6,7)	5 532 288	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 5,532,287 vertices
R(4,4,4,5,6,6,8)	6 952 341	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,952,340 vertices
R(4,4,4,5,6,6,9)	9 259 925	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 9,259,924 vertices
R(4,4,4,5,6,6,10)	9 259 925	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 9,259,924 vertices (by monotonicity, from (4,4,4,5,6,6,9))
R(4,4,4,5,6,7,7)	8 038 902	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices
R(4,4,4,5,6,7,8)	9 527 569	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 2 quadruplings -> 9,527,568 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,5,6,7,9)	11 597 093	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,899,273 vertices; then 1 quadrupling -> 11,597,092 vertices
R(4,4,4,5,6,7,10)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,4,4,5,6,8,8)	11 437 505	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 3 quadruplings -> 11,437,504 vertices
R(4,4,4,5,6,8,9)	15 244 289	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 4 quadruplings -> 15,244,288 vertices
R(4,4,4,5,6,8,10)	18 933 953	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 3 quadruplings -> 18,933,952 vertices
R(4,4,4,5,6,9,9)	21 865 602	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices
R(4,4,4,5,6,9,10)	25 914 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 2 quadruplings -> 25,914,768 vertices
R(4,4,4,5,6,10,10)	31 107 905	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 3 quadruplings -> 31,107,904 vertices
R(4,4,4,5,7,7,7)	11 135 369	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 11,135,368 vertices
R(4,4,4,5,7,7,8)	14 291 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 1 quadrupling -> 14,291,368 vertices
R(4,4,4,5,7,7,9)	18 638 344	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 18,638,343 vertices
R(4,4,4,5,7,7,10)	19 063 937	Published bricks + theorems	[template (7,7,3;472)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 297,874 vertices; then 3 quadruplings -> 19,063,936 vertices
R(4,4,4,5,7,8,8)	18 102 337	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 2 quadruplings -> 18,102,336 vertices
R(4,4,4,5,7,8,9)	23 342 589	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 1 quadrupling -> 23,342,588 vertices
R(4,4,4,5,7,8,10)	28 582 769	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 2 quadruplings -> 28,582,768 vertices
R(4,4,4,5,7,9,9)	30 287 909	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 30,287,908 vertices
R(4,4,4,5,7,9,10)	38 872 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 1 quadrupling -> 38,872,168 vertices
R(4,4,4,5,7,10,10)	49 238 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 2 quadruplings -> 49,238,016 vertices
R(4,4,4,5,8,8,8)	19 282 791	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,282,790 vertices
R(4,4,4,5,8,8,9)	28 582 769	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 2 quadruplings -> 28,582,768 vertices
R(4,4,4,5,8,8,10)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,4,5,8,9,9)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,4,5,8,9,10)	49 238 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 2 quadruplings -> 49,238,016 vertices
R(4,4,4,5,8,10,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,4,4,5,9,9,9)	50 695 804	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 50,695,803 vertices
R(4,4,4,5,9,9,10)	63 491 229	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 15,872,807 vertices; then 1 quadrupling -> 63,491,228 vertices
R(4,4,4,5,9,10,10)	77 744 369	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 2 quadruplings -> 77,744,368 vertices
R(4,4,4,5,10,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(4,4,4,6,6,6,6)	6 922 758	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices
R(4,4,4,6,6,6,7)	7 100 249	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,775,062 vertices; then 1 quadrupling -> 7,100,248 vertices
R(4,4,4,6,6,6,8)	7 100 249	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,775,062 vertices; then 1 quadrupling -> 7,100,248 vertices (by monotonicity, from (4,4,4,6,6,6,7))
R(4,4,4,6,6,6,9)	11 865 540	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 11,865,539 vertices
R(4,4,4,6,6,6,10)	11 988 390	Published bricks + theorems	[template (6,6,3;235)] x [[generic template 2m-1 of atom (4,4,4 ; 127)] x [atom (6,10 ; 203)]] -> 11,988,389 vertices
R(4,4,4,6,6,7,7)	13 934 267	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices
R(4,4,4,6,6,7,8)	13 934 267	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices (by monotonicity, from (4,4,4,6,6,7,7))
R(4,4,4,6,6,7,9)	15 975 585	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,993,896 vertices; then 1 quadrupling -> 15,975,584 vertices
R(4,4,4,6,6,7,10)	17 635 581	Our bricks + published	[template (6,6,3;235)] x [[template (4,9,3 ; 150) (ours)] x [atom (4,4,4 ; 127)]] -> 4,408,895 vertices; then 1 quadrupling -> 17,635,580 vertices
R(4,4,4,6,6,8,8)	19 200 738	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices
R(4,4,4,6,6,8,9)	22 129 149	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 5,532,287 vertices; then 1 quadrupling -> 22,129,148 vertices
R(4,4,4,6,6,8,10)	22 129 149	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 5,532,287 vertices; then 1 quadrupling -> 22,129,148 vertices (by monotonicity, from (4,4,4,6,6,8,9))
R(4,4,4,6,6,9,9)	37 900 380	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices
R(4,4,4,6,6,9,10)	37 900 380	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices (by monotonicity, from (4,4,4,6,6,9,9))
R(4,4,4,6,6,10,10)	47 154 378	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,6,7,7,7)	14 291 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 1 quadrupling -> 14,291,368 vertices
R(4,4,4,6,7,7,8)	18 816 985	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 4,704,246 vertices; then 1 quadrupling -> 18,816,984 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,6,7,7,9)	23 883 200	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 23,883,199 vertices
R(4,4,4,6,7,7,10)	28 400 993	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,775,062 vertices; then 2 quadruplings -> 28,400,992 vertices
R(4,4,4,6,7,8,8)	19 693 049	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 4,923,262 vertices; then 1 quadrupling -> 19,693,048 vertices
R(4,4,4,6,7,8,9)	32 155 605	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 1 quadrupling -> 32,155,604 vertices
R(4,4,4,6,7,8,10)	35 974 337	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,099 vertices; then 3 quadruplings -> 35,974,336 vertices
R(4,4,4,6,7,9,9)	38 872 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 1 quadrupling -> 38,872,168 vertices
R(4,4,4,6,7,9,10)	51 181 705	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices; then 1 quadrupling -> 51,181,704 vertices
R(4,4,4,6,7,10,10)	51 849 857	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (4,3;6)] x [atom (4,4,4 ; 127)]] -> 810,154 vertices; then 3 quadruplings -> 51,849,856 vertices
R(4,4,4,6,8,8,8)	19 693 049	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 4,923,262 vertices; then 1 quadrupling -> 19,693,048 vertices (by monotonicity, from (4,4,4,6,7,8,8))
R(4,4,4,6,8,8,9)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,4,6,8,8,10)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices (by monotonicity, from (4,4,4,5,8,8,10))
R(4,4,4,6,8,9,9)	46 388 369	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,899,273 vertices; then 2 quadruplings -> 46,388,368 vertices
R(4,4,4,6,8,9,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,4,4,6,8,10,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices (by monotonicity, from (4,4,4,5,8,10,10))
R(4,4,4,6,9,9,9)	64 956 230	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 64,956,229 vertices
R(4,4,4,6,9,9,10)	87 462 405	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 1 quadrupling -> 87,462,404 vertices
R(4,4,4,6,9,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(4,4,4,6,10,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices (by monotonicity, from (4,4,4,5,10,10,10))
R(4,4,4,7,7,7,7)	28 046 840	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices
R(4,4,4,7,7,7,8)	28 046 840	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices (by monotonicity, from (4,4,4,7,7,7,7))
R(4,4,4,7,7,7,9)	32 155 605	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 1 quadrupling -> 32,155,604 vertices
R(4,4,4,7,7,7,10)	35 974 337	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 562,099 vertices; then 3 quadruplings -> 35,974,336 vertices
R(4,4,4,7,7,8,8)	38 647 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices
R(4,4,4,7,7,8,9)	44 541 473	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 11,135,368 vertices; then 1 quadrupling -> 44,541,472 vertices
R(4,4,4,7,7,8,10)	57 165 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 2 quadruplings -> 57,165,472 vertices
R(4,4,4,7,7,9,9)	76 286 660	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices
R(4,4,4,7,7,9,10)	76 286 660	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices (by monotonicity, from (4,4,4,7,7,9,9))
R(4,4,4,7,7,10,10)	94 913 297	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,7,8,8,8)	38 647 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices (by monotonicity, from (4,4,4,7,7,8,8))
R(4,4,4,7,8,8,9)	57 165 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 2 quadruplings -> 57,165,472 vertices
R(4,4,4,7,8,8,10)	72 409 345	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 3 quadruplings -> 72,409,344 vertices
R(4,4,4,7,8,9,9)	76 286 660	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices (by monotonicity, from (4,4,4,7,7,9,9))
R(4,4,4,7,8,9,10)	93 370 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 2 quadruplings -> 93,370,352 vertices
R(4,4,4,7,8,10,10)	99 777 857	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 1,559,029 vertices; then 3 quadruplings -> 99,777,856 vertices
R(4,4,4,7,9,9,9)	93 370 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 2 quadruplings -> 93,370,352 vertices
R(4,4,4,7,9,9,10)	121 151 633	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 30,287,908 vertices; then 1 quadrupling -> 121,151,632 vertices
R(4,4,4,7,9,10,10)	155 488 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 2 quadruplings -> 155,488,672 vertices
R(4,4,4,7,10,10,10)	196 952 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 3 quadruplings -> 196,952,064 vertices
R(4,4,4,8,8,8,8)	53 253 328	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 53,253,327 vertices
R(4,4,4,8,8,8,9)	72 409 345	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 3 quadruplings -> 72,409,344 vertices
R(4,4,4,8,8,8,10)	72 409 345	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 3 quadruplings -> 72,409,344 vertices (by monotonicity, from (4,4,4,7,8,8,10))
R(4,4,4,8,8,9,9)	105 116 865	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices
R(4,4,4,8,8,9,10)	114 331 073	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 3 quadruplings -> 114,331,072 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,4,8,8,10,10)	130 782 868	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,8,3 ; 650)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,8,9,9,9)	121 151 633	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 30,287,908 vertices; then 1 quadrupling -> 121,151,632 vertices
R(4,4,4,8,9,9,10)	155 488 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 2 quadruplings -> 155,488,672 vertices
R(4,4,4,8,9,9,10,10)	196 952 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 3 quadruplings -> 196,952,064 vertices
R(4,4,4,8,10,10,10)	196 952 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 3 quadruplings -> 196,952,064 vertices (by monotonicity, from (4,4,4,7,10,10,10))
R(4,4,4,9,9,9,9)	207 480 290	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices
R(4,4,4,9,9,9,10)	207 480 290	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices (by monotonicity, from (4,4,4,9,9,9,9))
R(4,4,4,9,9,10,10)	258 139 997	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,4,9,10,10,10)	310 977 473	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 3 quadruplings -> 310,977,472 vertices
R(4,4,4,10,10,10,10)	321 012 600	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de atome (4,4,4 ; 127)] x [[modele generique 2m-1 de atome (10,10 ; 797)] x [atome (10,10 ; 797)]]
R(4,4,5,5,5,5,5)	1 628 483	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,5,3 ; 61) (ours)] x [atom (5,5,5 ; 453)]] -> 1,628,482 vertices
R(4,4,5,5,5,5,6)	2 169 237	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (5,5,5,3 ; 977)] x [Paley-17 (4,4;17)]] -> 2,169,236 vertices
R(4,4,5,5,5,5,7)	3 121 523	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (5,5,5 ; 453)]] -> 3,121,522 vertices
R(4,4,5,5,5,5,8)	3 953 913	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (5,5,5 ; 453)]] -> 988,478 vertices; then 1 quadrupling -> 3,953,912 vertices
R(4,4,5,5,5,5,9)	5 453 025	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-17 (4,4;17)]] -> 340,814 vertices; then 2 quadruplings -> 5,453,024 vertices
R(4,4,5,5,5,5,10)	6 854 289	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [Paley-17 (4,4;17)]] -> 428,393 vertices; then 2 quadruplings -> 6,854,288 vertices
R(4,4,5,5,5,6,6)	3 760 014	Published bricks + theorems	[template (6,6,3;235)] x [[template (5,5,5,3 ; 977)] x [Paley-17 (4,4;17)]] -> 3,760,013 vertices
R(4,4,5,5,5,6,7)	4 900 949	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 1,225,237 vertices; then 1 quadrupling -> 4,900,948 vertices
R(4,4,5,5,5,6,8)	6 827 649	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 426,728 vertices; then 2 quadruplings -> 6,827,648 vertices
R(4,4,5,5,5,6,9)	8 118 977	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 126,859 vertices; then 3 quadruplings -> 8,118,976 vertices
R(4,4,5,5,5,6,10)	11 066 369	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,4,6,7) of 56,120 vertices] + [colouring (4,4,6,7) of 4,552 vertices] (see Chung_Robertson.cpp) -> 172,912 vertices; then 3 quadruplings -> 11,066,368 vertices
R(4,4,5,5,5,7,7)	7 568 231	Published bricks + theorems	[template (7,7,3;472)] x [[template (5,5,5,3 ; 977)] x [Paley-17 (4,4;17)]] -> 7,568,230 vertices
R(4,4,5,5,5,7,8)	9 864 453	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 2,466,113 vertices; then 1 quadrupling -> 9,864,452 vertices
R(4,4,5,5,5,7,9)	13 741 617	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 2 quadruplings -> 13,741,616 vertices
R(4,4,5,5,5,7,10)	16 340 993	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 3 quadruplings -> 16,340,992 vertices
R(4,4,5,5,5,8,8)	13 741 617	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 2 quadruplings -> 13,741,616 vertices
R(4,4,5,5,5,8,9)	16 340 993	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 255,328 vertices; then 3 quadruplings -> 16,340,992 vertices
R(4,4,5,5,5,8,10)	20 308 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,4,4,3 ; 116) (ours)] x [K2]] -> 79,330 vertices; then 4 quadruplings -> 20,308,480 vertices
R(4,4,5,5,5,9,9)	20 583 671	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (5,5,5,3 ; 977)] x [Paley-17 (4,4;17)]] -> 20,583,670 vertices
R(4,4,5,5,5,9,10)	26 834 133	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 6,708,533 vertices; then 1 quadrupling -> 26,834,132 vertices
R(4,4,5,5,5,10,10)	37 397 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 2 quadruplings -> 37,397,936 vertices
R(4,4,5,5,6,6,6)	4 900 949	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 1,225,237 vertices; then 1 quadrupling -> 4,900,948 vertices
R(4,4,5,5,6,6,7)	7 100 249	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,775,062 vertices; then 1 quadrupling -> 7,100,248 vertices
R(4,4,5,5,6,6,8)	9 351 813	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 2,337,953 vertices; then 1 quadrupling -> 9,351,812 vertices
R(4,4,5,5,6,6,9)	12 240 417	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 765,026 vertices; then 2 quadruplings -> 12,240,416 vertices
R(4,4,5,5,6,6,10)	15 653 033	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 3,913,258 vertices; then 1 quadrupling -> 15,653,032 vertices
R(4,4,5,5,6,7,7)	11 597 093	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,899,273 vertices; then 1 quadrupling -> 11,597,092 vertices
R(4,4,5,5,6,7,8)	14 200 529	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 887,533 vertices; then 2 quadruplings -> 14,200,528 vertices
R(4,4,5,5,6,7,9)	18 933 953	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 3 quadruplings -> 18,933,952 vertices
R(4,4,5,5,6,7,10)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(4,4,5,5,6,8,8)	18 933 953	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 3 quadruplings -> 18,933,952 vertices
R(4,4,5,5,6,8,9)	22 859 009	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 4 quadruplings -> 22,859,008 vertices
R(4,4,5,5,6,8,10)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices
R(4,4,5,5,6,9,9)	30 295 041	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 5 quadruplings -> 30,295,040 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,5,5,6,9,10)	37 397 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 2 quadruplings -> 37,397,936 vertices
R(4,4,5,5,6,10,10)	46 127 681	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 2,882,980 vertices; then 2 quadruplings -> 46,127,680 vertices
R(4,4,5,5,7,7,7)	14 291 369	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 1 quadrupling -> 14,291,368 vertices
R(4,4,5,5,7,7,8)	23 342 589	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 1 quadrupling -> 23,342,588 vertices
R(4,4,5,5,7,7,9)	28 582 769	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 2 quadruplings -> 28,582,768 vertices
R(4,4,5,5,7,7,10)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,5,5,7,8,8)	28 582 769	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 2 quadruplings -> 28,582,768 vertices
R(4,4,5,5,7,8,9)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,5,5,7,8,10)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,4,5,5,7,9,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,4,5,5,7,9,10)	63 491 229	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 15,872,807 vertices; then 1 quadrupling -> 63,491,228 vertices
R(4,4,5,5,7,10,10)	77 744 369	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 2 quadruplings -> 77,744,368 vertices
R(4,4,5,5,8,8,8)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,5,5,8,8,9)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices
R(4,4,5,5,8,8,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,4,5,5,8,9,9)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,4,5,5,8,9,10)	77 744 369	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 2 quadruplings -> 77,744,368 vertices
R(4,4,5,5,8,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(4,4,5,5,9,9,9)	67 147 857	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,196,741 vertices; then 2 quadruplings -> 67,147,856 vertices
R(4,4,5,5,9,9,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(4,4,5,5,9,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(4,4,5,5,10,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(4,4,5,6,6,6,6)	8 096 369	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 8,096,368 vertices
R(4,4,5,6,6,6,7)	11 702 277	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 2,925,569 vertices; then 1 quadrupling -> 11,702,276 vertices
R(4,4,5,6,6,6,8)	14 393 553	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 899,597 vertices; then 2 quadruplings -> 14,393,552 vertices
R(4,4,5,6,6,6,9)	17 040 641	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 3 quadruplings -> 17,040,640 vertices
R(4,4,5,6,6,6,10)	20 062 629	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [atom (3,4,4 ; 54)]] -> 5,015,657 vertices; then 1 quadrupling -> 20,062,628 vertices
R(4,4,5,6,6,7,7)	16 296 530	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 16,296,529 vertices
R(4,4,5,6,6,7,8)	23 554 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 1 quadrupling -> 23,554,576 vertices
R(4,4,5,6,6,7,9)	28 971 617	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 2 quadruplings -> 28,971,616 vertices
R(4,4,5,6,6,7,10)	37 039 697	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 9,259,924 vertices; then 1 quadrupling -> 37,039,696 vertices
R(4,4,5,6,6,8,8)	28 971 617	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 2 quadruplings -> 28,971,616 vertices
R(4,4,5,6,6,8,9)	34 081 185	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,130,074 vertices; then 2 quadruplings -> 34,081,184 vertices
R(4,4,5,6,6,8,10)	39 927 633	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 2,495,477 vertices; then 2 quadruplings -> 39,927,632 vertices
R(4,4,5,6,6,9,9)	44 346 947	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]]] -> 44,346,946 vertices
R(4,4,5,6,6,9,10)	64 073 349	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 1 quadrupling -> 64,073,348 vertices
R(4,4,5,6,6,10,10)	78 839 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 2 quadruplings -> 78,839,024 vertices
R(4,4,5,6,7,7,7)	23 554 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 1 quadrupling -> 23,554,576 vertices
R(4,4,5,6,7,7,8)	32 155 605	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 1 quadrupling -> 32,155,604 vertices
R(4,4,5,6,7,7,9)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,5,6,7,7,10)	46 388 369	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,899,273 vertices; then 2 quadruplings -> 46,388,368 vertices
R(4,4,5,6,7,8,8)	38 110 273	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 3 quadruplings -> 38,110,272 vertices
R(4,4,5,6,7,8,9)	46 388 369	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,899,273 vertices; then 2 quadruplings -> 46,388,368 vertices
R(4,4,5,6,7,8,10)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,4,5,6,7,9,9)	64 073 349	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 1 quadrupling -> 64,073,348 vertices
R(4,4,5,6,7,9,10)	87 462 405	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 1 quadrupling -> 87,462,404 vertices
R(4,4,5,6,7,10,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(4,4,5,6,8,8,8)	45 750 017	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 4 quadruplings -> 45,750,016 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,5,6,8,8,9)	60 977 153	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 5 quadruplings -> 60,977,152 vertices
R(4,4,5,6,8,8,10)	75 735 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 4 quadruplings -> 75,735,808 vertices
R(4,4,5,6,8,9,9)	78 839 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 2 quadruplings -> 78,839,024 vertices
R(4,4,5,6,8,9,10)	103 659 073	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 3 quadruplings -> 103,659,072 vertices
R(4,4,5,6,8,10,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(4,4,5,6,9,9,9)	93 293 249	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,457,707 vertices; then 3 quadruplings -> 93,293,248 vertices
R(4,4,5,6,9,9,10)	124 431 617	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 4 quadruplings -> 124,431,616 vertices
R(4,4,5,6,9,10,10)	165 855 233	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 5 quadruplings -> 165,855,232 vertices
R(4,4,5,6,10,10,10)	184 510 721	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 2,882,980 vertices; then 3 quadruplings -> 184,510,720 vertices
R(4,4,5,7,7,7,7)	32 798 957	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]]] -> 32,798,956 vertices
R(4,4,5,7,7,7,8)	47 409 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 1 quadrupling -> 47,409,784 vertices
R(4,4,5,7,7,7,9)	58 309 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 2 quadruplings -> 58,309,264 vertices
R(4,4,5,7,7,7,10)	74 553 373	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 18,638,343 vertices; then 1 quadrupling -> 74,553,372 vertices
R(4,4,5,7,7,8,8)	58 309 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 2 quadruplings -> 58,309,264 vertices
R(4,4,5,7,7,8,9)	74 553 373	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 18,638,343 vertices; then 1 quadrupling -> 74,553,372 vertices
R(4,4,5,7,7,8,10)	93 370 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 2 quadruplings -> 93,370,352 vertices
R(4,4,5,7,7,9,9)	93 370 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 2 quadruplings -> 93,370,352 vertices
R(4,4,5,7,7,9,10)	128 968 145	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 1 quadrupling -> 128,968,144 vertices
R(4,4,5,7,7,10,10)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(4,4,5,7,8,8,8)	72 409 345	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 3 quadruplings -> 72,409,344 vertices
R(4,4,5,7,8,8,9)	93 370 353	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 2 quadruplings -> 93,370,352 vertices
R(4,4,5,7,8,8,10)	114 331 073	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 3 quadruplings -> 114,331,072 vertices
R(4,4,5,7,8,9,9)	128 968 145	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 1 quadrupling -> 128,968,144 vertices
R(4,4,5,7,8,9,10)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(4,4,5,7,8,10,10)	196 952 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 3 quadruplings -> 196,952,064 vertices
R(4,4,5,7,9,9,9)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(4,4,5,7,9,9,10)	202 783 213	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 50,695,803 vertices; then 1 quadrupling -> 202,783,212 vertices
R(4,4,5,7,9,10,10)	253 964 913	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 15,872,807 vertices; then 2 quadruplings -> 253,964,912 vertices
R(4,4,5,7,10,10,10)	310 977 473	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 3 quadruplings -> 310,977,472 vertices
R(4,4,5,8,8,8,8)	80 367 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 2 quadruplings -> 80,367,136 vertices
R(4,4,5,8,8,8,9)	114 331 073	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 3 quadruplings -> 114,331,072 vertices
R(4,4,5,8,8,8,10)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,4,5,8,8,9,9)	158 688 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 2 quadruplings -> 158,688,784 vertices
R(4,4,5,8,8,9,10)	196 952 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 3 quadruplings -> 196,952,064 vertices
R(4,4,5,8,8,10,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(4,4,5,8,9,9,9)	186 586 401	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 11,661,650 vertices; then 2 quadruplings -> 186,586,400 vertices
R(4,4,5,8,9,9,10)	253 964 913	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 15,872,807 vertices; then 2 quadruplings -> 253,964,912 vertices
R(4,4,5,8,9,10,10)	310 977 473	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 3 quadruplings -> 310,977,472 vertices
R(4,4,5,8,10,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,4,5,9,9,9,9)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(4,4,5,9,9,9,10)	350 760 545	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 1 quadrupling -> 350,760,544 vertices
R(4,4,5,9,9,10,10)	431 593 265	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 2 quadruplings -> 431,593,264 vertices
R(4,4,5,9,10,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(4,4,5,10,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(4,4,6,6,6,6,6)	10 507 982	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]]] -> 10,507,981 vertices
R(4,4,6,6,6,6,7)	15 975 585	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,993,896 vertices; then 1 quadrupling -> 15,975,584 vertices
R(4,4,6,6,6,6,8)	17 092 332	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 17,092,331 vertices
R(4,4,6,6,6,6,9)	22 489 909	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 22,489,908 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,6,6,6,6,10)	26 088 294	Published bricks + theorems	[template (10,3 ; 30)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 26,088,293 vertices
R(4,4,6,6,6,7,7)	27 691 029	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 1 quadrupling -> 27,691,028 vertices
R(4,4,6,6,6,7,8)	28 400 993	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 1,775,062 vertices; then 2 quadruplings -> 28,400,992 vertices
R(4,4,6,6,6,7,9)	37 039 697	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 9,259,924 vertices; then 1 quadrupling -> 37,039,696 vertices
R(4,4,6,6,6,7,10)	47 462 157	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 11,865,539 vertices; then 1 quadrupling -> 47,462,156 vertices
R(4,4,6,6,6,8,8)	29 178 842	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [K3]]] -> 29,178,841 vertices
R(4,4,6,6,6,8,9)	46 809 105	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 2,925,569 vertices; then 2 quadruplings -> 46,809,104 vertices
R(4,4,6,6,6,8,10)	47 462 157	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 11,865,539 vertices; then 1 quadrupling -> 47,462,156 vertices (by monotonicity, from (4,4,6,6,6,7,10))
R(4,4,6,6,6,9,9)	57 739 712	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 57,739,711 vertices
R(4,4,6,6,6,9,10)	68 162 561	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 266,260 vertices; then 4 quadruplings -> 68,162,560 vertices
R(4,4,6,6,6,10,10)	78 839 025	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 2 quadruplings -> 78,839,024 vertices (by monotonicity, from (4,4,5,6,6,10,10))
R(4,4,6,6,7,7,7)	32 155 605	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 1 quadrupling -> 32,155,604 vertices
R(4,4,6,6,7,7,8)	55 737 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 1 quadrupling -> 55,737,064 vertices
R(4,4,6,6,7,7,9)	55 737 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 1 quadrupling -> 55,737,064 vertices (by monotonicity, from (4,4,6,6,7,7,8))
R(4,4,6,6,7,7,10)	63 902 337	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,993,896 vertices; then 2 quadruplings -> 63,902,336 vertices
R(4,4,6,6,7,8,8)	55 737 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 1 quadrupling -> 55,737,064 vertices (by monotonicity, from (4,4,6,6,7,7,8))
R(4,4,6,6,7,8,9)	76 802 949	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 1 quadrupling -> 76,802,948 vertices
R(4,4,6,6,7,8,10)	88 516 593	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 5,532,287 vertices; then 2 quadruplings -> 88,516,592 vertices
R(4,4,6,6,7,9,9)	94 218 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 2 quadruplings -> 94,218,304 vertices
R(4,4,6,6,7,9,10)	151 601 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 1 quadrupling -> 151,601,516 vertices
R(4,4,6,6,7,10,10)	151 601 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 1 quadrupling -> 151,601,516 vertices (by monotonicity, from (4,4,6,6,7,9,10))
R(4,4,6,6,8,8,8)	55 737 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 1 quadrupling -> 55,737,064 vertices (by monotonicity, from (4,4,6,6,7,7,8))
R(4,4,6,6,8,8,9)	94 218 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 2 quadruplings -> 94,218,304 vertices
R(4,4,6,6,8,8,10)	94 218 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 2 quadruplings -> 94,218,304 vertices (by monotonicity, from (4,4,6,6,8,8,9))
R(4,4,6,6,8,9,9)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices
R(4,4,6,6,8,9,10)	151 601 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 1 quadrupling -> 151,601,516 vertices (by monotonicity, from (4,4,6,6,7,9,10))
R(4,4,6,6,8,10,10)	151 601 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 1 quadrupling -> 151,601,516 vertices (by monotonicity, from (4,4,6,6,7,9,10))
R(4,4,6,6,9,9,9)	136 324 737	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,130,074 vertices; then 3 quadruplings -> 136,324,736 vertices
R(4,4,6,6,9,9,10)	177 387 785	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]]] -> 44,346,946 vertices; then 1 quadrupling -> 177,387,784 vertices
R(4,4,6,6,9,10,10)	256 293 393	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 2 quadruplings -> 256,293,392 vertices
R(4,4,6,6,10,10,10)	256 293 393	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 2 quadruplings -> 256,293,392 vertices (by monotonicity, from (4,4,6,6,9,10,10))
R(4,4,6,7,7,7,7)	55 737 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 1 quadrupling -> 55,737,064 vertices
R(4,4,6,7,7,7,8)	57 165 473	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 2 quadruplings -> 57,165,472 vertices
R(4,4,6,7,7,7,9)	75 267 937	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 4,704,246 vertices; then 2 quadruplings -> 75,267,936 vertices
R(4,4,6,7,7,7,10)	95 532 797	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 23,883,199 vertices; then 1 quadrupling -> 95,532,796 vertices
R(4,4,6,7,7,8,8)	76 802 949	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 1 quadrupling -> 76,802,948 vertices
R(4,4,6,7,7,8,9)	95 532 797	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 23,883,199 vertices; then 1 quadrupling -> 95,532,796 vertices
R(4,4,6,7,7,8,10)	128 622 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 2 quadruplings -> 128,622,416 vertices
R(4,4,6,7,7,9,9)	151 601 517	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 1 quadrupling -> 151,601,516 vertices
R(4,4,6,7,7,9,10)	155 488 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 2 quadruplings -> 155,488,672 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,6,7,7,10,10)	204 726 817	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices; then 2 quadruplings -> 204,726,816 vertices
R(4,4,6,7,8,8,8)	78 772 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 4,923,262 vertices; then 2 quadruplings -> 78,772,192 vertices
R(4,4,6,7,8,8,9)	128 622 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 2 quadruplings -> 128,622,416 vertices
R(4,4,6,7,8,8,10)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,4,6,7,8,9,9)	155 488 673	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 2 quadruplings -> 155,488,672 vertices
R(4,4,6,7,8,9,10)	204 726 817	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices; then 2 quadruplings -> 204,726,816 vertices
R(4,4,6,7,8,10,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(4,4,6,7,9,9,9)	202 783 213	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 50,695,803 vertices; then 1 quadrupling -> 202,783,212 vertices
R(4,4,6,7,9,9,10)	259 824 917	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 64,956,229 vertices; then 1 quadrupling -> 259,824,916 vertices
R(4,4,6,7,9,10,10)	349 849 617	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 2 quadruplings -> 349,849,616 vertices
R(4,4,6,7,10,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,4,6,8,8,8,8)	80 927 792	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [K3]]] -> 80,927,791 vertices
R(4,4,6,8,8,8,9)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,4,6,8,8,8,10)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices (by monotonicity, from (4,4,5,8,8,8,10))
R(4,4,6,8,8,9,9)	185 553 473	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 2,899,273 vertices; then 3 quadruplings -> 185,553,472 vertices
R(4,4,6,8,8,9,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(4,4,6,8,8,10,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices (by monotonicity, from (4,4,5,8,8,10,10))
R(4,4,6,8,9,9,9)	256 293 393	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 2 quadruplings -> 256,293,392 vertices
R(4,4,6,8,9,9,10)	349 849 617	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 2 quadruplings -> 349,849,616 vertices
R(4,4,6,8,9,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,4,6,8,10,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices (by monotonicity, from (4,4,5,8,10,10,10))
R(4,4,6,9,9,9,9)	316 088 830	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 316,088,829 vertices
R(4,4,6,9,9,9,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,4,6,9,9,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(4,4,6,9,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(4,4,6,10,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices (by monotonicity, from (4,4,5,10,10,10,10))
R(4,4,7,7,7,7,7)	61 933 757	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]]] -> 61,933,756 vertices
R(4,4,7,7,7,7,8)	112 187 357	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 1 quadrupling -> 112,187,356 vertices
R(4,4,7,7,7,7,9)	112 187 357	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 1 quadrupling -> 112,187,356 vertices (by monotonicity, from (4,4,7,7,7,7,8))
R(4,4,7,7,7,7,10)	128 622 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 2 quadruplings -> 128,622,416 vertices
R(4,4,7,7,7,8,8)	112 187 357	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 1 quadrupling -> 112,187,356 vertices (by monotonicity, from (4,4,7,7,7,7,8))
R(4,4,7,7,7,8,9)	154 590 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 1 quadrupling -> 154,590,544 vertices
R(4,4,7,7,7,8,10)	178 165 889	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 11,135,368 vertices; then 2 quadruplings -> 178,165,888 vertices
R(4,4,7,7,7,9,9)	189 639 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 2 quadruplings -> 189,639,136 vertices
R(4,4,7,7,7,9,10)	305 146 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 1 quadrupling -> 305,146,636 vertices
R(4,4,7,7,7,10,10)	305 146 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 1 quadrupling -> 305,146,636 vertices (by monotonicity, from (4,4,7,7,7,9,10))
R(4,4,7,7,8,8,8)	154 590 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 1 quadrupling -> 154,590,544 vertices
R(4,4,7,7,8,8,9)	189 639 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 2 quadruplings -> 189,639,136 vertices
R(4,4,7,7,8,8,10)	228 661 889	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 3,572,842 vertices; then 3 quadruplings -> 228,661,888 vertices
R(4,4,7,7,8,9,9)	305 146 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 1 quadrupling -> 305,146,636 vertices
R(4,4,7,7,8,9,10)	305 146 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 1 quadrupling -> 305,146,636 vertices (by monotonicity, from (4,4,7,7,7,9,10))
R(4,4,7,7,8,10,10)	379 653 185	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]

Parameters	Bound (R >=)	Origin	Construction
R(4,4,7,7,9,9,9)	305 146 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 1 quadrupling -> 305,146,636 vertices (by monotonicity, from (4,4,7,7,8,9,9))
R(4,4,7,7,9,9,10)	373 481 409	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 3 quadruplings -> 373,481,408 vertices
R(4,4,7,7,9,10,10)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,4,7,7,10,10,10)	621 954 689	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 9,718,042 vertices; then 3 quadruplings -> 621,954,688 vertices
R(4,4,7,8,8,8,8)	154 590 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 1 quadrupling -> 154,590,544 vertices (by monotonicity, from (4,4,7,7,8,8,8))
R(4,4,7,8,8,8,9)	233 237 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 3 quadruplings -> 233,237,056 vertices
R(4,4,7,8,8,8,10)	289 637 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 4 quadruplings -> 289,637,376 vertices
R(4,4,7,8,8,9,9)	305 146 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 1 quadrupling -> 305,146,636 vertices (by monotonicity, from (4,4,7,7,8,9,9))
R(4,4,7,8,8,9,10)	420 467 457	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 1 quadrupling -> 420,467,456 vertices
R(4,4,7,8,8,10,10)	457 324 289	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 4 quadruplings -> 457,324,288 vertices
R(4,4,7,8,9,9,9)	420 467 457	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 1 quadrupling -> 420,467,456 vertices
R(4,4,7,8,9,9,10)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,4,7,8,9,10,10)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,4,7,8,10,10,10)	787 808 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 4 quadruplings -> 787,808,256 vertices
R(4,4,7,9,9,9,9)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,4,7,9,9,9,10)	829 921 157	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 1 quadrupling -> 829,921,156 vertices
R(4,4,7,9,9,10,10)	829 921 157	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 1 quadrupling -> 829,921,156 vertices (by monotonicity, from (4,4,7,9,9,9,10))
R(4,4,7,9,10,10,10)	1 032 559 985	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,4,7,10,10,10,10)	1 243 909 889	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 4 quadruplings -> 1,243,909,888 vertices
R(4,4,8,8,8,8,8)	154 590 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 1 quadrupling -> 154,590,544 vertices (by monotonicity, from (4,4,7,7,8,8,8))
R(4,4,8,8,8,8,9)	289 637 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 4 quadruplings -> 289,637,376 vertices
R(4,4,8,8,8,8,10)	289 637 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 4 quadruplings -> 289,637,376 vertices (by monotonicity, from (4,4,7,8,8,8,10))
R(4,4,8,8,8,9,9)	373 481 409	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 5,835,647 vertices; then 3 quadruplings -> 373,481,408 vertices
R(4,4,8,8,8,9,10)	457 324 289	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 4 quadruplings -> 457,324,288 vertices
R(4,4,8,8,8,10,10)	457 324 289	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 4 quadruplings -> 457,324,288 vertices (by monotonicity, from (4,4,7,8,8,10,10))
R(4,4,8,8,9,9,9)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,4,8,8,9,9,10)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,4,8,8,9,10,10)	787 808 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 4 quadruplings -> 787,808,256 vertices
R(4,4,8,8,10,10,10)	787 808 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 4 quadruplings -> 787,808,256 vertices (by monotonicity, from (4,4,7,8,10,10,10))
R(4,4,8,9,9,9,9)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,4,8,9,9,9,10)	829 921 157	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 1 quadrupling -> 829,921,156 vertices (by monotonicity, from (4,4,7,9,9,9,10))
R(4,4,8,9,9,10,10)	1 015 859 649	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 15,872,807 vertices; then 3 quadruplings -> 1,015,859,648 vertices
R(4,4,8,9,10,10,10)	1 243 909 889	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 4 quadruplings -> 1,243,909,888 vertices
R(4,4,8,10,10,10,10)	1 243 909 889	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 4 quadruplings -> 1,243,909,888 vertices (by monotonicity, from (4,4,7,10,10,10,10))
R(4,4,9,9,9,9,9)	746 345 601	Our bricks + published	[template (3,8,3 ; 73) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 11,661,650 vertices; then 3 quadruplings -> 746,345,600 vertices
R(4,4,9,9,9,9,10)	1 015 859 649	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 15,872,807 vertices; then 3 quadruplings -> 1,015,859,648 vertices
R(4,4,9,9,9,10,10)	1 403 042 177	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 2 quadruplings -> 1,403,042,176 vertices
R(4,4,9,9,10,10,10)	1 726 373 057	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 3 quadruplings -> 1,726,373,056 vertices
R(4,4,9,10,10,10,10)	1 990 905 857	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 6 quadruplings -> 1,990,905,856 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,4,10,10,10,10,10)	1 990 905 857	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 6 quadruplings -> 1,990,905,856 vertices (by monotonicity, from (4,4,9,10,10,10,10))
R(4,5,5,5,5,5,5)	2 660 183	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (5,5,5 ; 453)]] -> 2,660,182 vertices
R(4,5,5,5,5,5,6)	3 664 523	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (5,5,5 ; 453)]] -> 3,664,522 vertices
R(4,5,5,5,5,5,7)	5 098 687	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (5,5,5 ; 453)]] -> 5,098,686 vertices
R(4,5,5,5,5,5,8)	6 429 765	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (5,5,5 ; 453)]] -> 1,607,441 vertices; then 1 quadrupling -> 6,429,764 vertices
R(4,5,5,5,5,5,9)	8 496 443	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (5,5,5 ; 453)]] -> 8,496,442 vertices
R(4,5,5,5,5,5,10)	11 075 969	Our bricks + published	Chung-Robertson construction (non-linear): 3 x [colouring (3,3,4,5,5,7) of 56,120 vertices] + [colouring (4,5,5,7) of 4,702 vertices] (see Chung_Robertson.cpp) -> 173,062 vertices; then 3 quadruplings -> 11,075,968 vertices
R(4,5,5,5,5,6,6)	6 352 103	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 6,352,102 vertices
R(4,5,5,5,5,6,7)	8 045 981	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 2,011,495 vertices; then 1 quadrupling -> 8,045,980 vertices
R(4,5,5,5,5,6,8)	10 697 633	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 668,602 vertices; then 2 quadruplings -> 10,697,632 vertices
R(4,5,5,5,5,6,9)	13 550 081	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Paley-101 (6,6;101)]] -> 211,720 vertices; then 3 quadruplings -> 13,550,080 vertices
R(4,5,5,5,5,6,10)	17 346 305	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 67,759 vertices; then 4 quadruplings -> 17,346,304 vertices
R(4,5,5,5,5,7,7)	12 785 603	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 12,785,602 vertices
R(4,5,5,5,5,7,8)	16 195 081	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 4,048,770 vertices; then 1 quadrupling -> 16,195,080 vertices
R(4,5,5,5,5,7,9)	21 460 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 2 quadruplings -> 21,460,192 vertices
R(4,5,5,5,5,7,10)	27 182 657	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 3 quadruplings -> 27,182,656 vertices
R(4,5,5,5,5,8,8)	21 460 193	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 2 quadruplings -> 21,460,192 vertices
R(4,5,5,5,5,8,9)	27 182 657	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 424,729 vertices; then 3 quadruplings -> 27,182,656 vertices
R(4,5,5,5,5,8,10)	34 777 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 4 quadruplings -> 34,777,600 vertices
R(4,5,5,5,5,9,9)	34 777 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 4 quadruplings -> 34,777,600 vertices
R(4,5,5,5,5,9,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(4,5,5,5,5,10,10)	59 821 793	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 2 quadruplings -> 59,821,792 vertices
R(4,5,5,5,6,6,6)	8 363 606	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 8,363,605 vertices
R(4,5,5,5,6,6,7)	12 174 867	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 12,174,866 vertices
R(4,5,5,5,6,6,8)	15 165 465	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (3,6,6 ; 337)]] -> 3,791,366 vertices; then 1 quadrupling -> 15,165,464 vertices
R(4,5,5,5,6,6,9)	20 504 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Paley-101 (6,6;101)]] -> 1,281,539 vertices; then 2 quadruplings -> 20,504,624 vertices
R(4,5,5,5,6,6,10)	25 271 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (3,6,6 ; 337)]] -> 6,317,962 vertices; then 1 quadrupling -> 25,271,848 vertices
R(4,5,5,5,6,7,7)	16 834 381	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 16,834,380 vertices
R(4,5,5,5,6,7,8)	21 786 089	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 5,446,522 vertices; then 1 quadrupling -> 21,786,088 vertices
R(4,5,5,5,6,7,9)	28 921 137	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 1,807,571 vertices; then 2 quadruplings -> 28,921,136 vertices
R(4,5,5,5,6,7,10)	34 777 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 4 quadruplings -> 34,777,600 vertices
R(4,5,5,5,6,8,8)	28 921 137	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 1,807,571 vertices; then 2 quadruplings -> 28,921,136 vertices
R(4,5,5,5,6,8,9)	34 777 601	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [(3,3;5)]] -> 135,850 vertices; then 4 quadruplings -> 34,777,600 vertices
R(4,5,5,5,6,8,10)	45 061 121	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [K2]] -> 44,005 vertices; then 5 quadruplings -> 45,061,120 vertices
R(4,5,5,5,6,9,9)	45 786 301	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)]] -> 45,786,300 vertices
R(4,5,5,5,6,9,10)	59 821 793	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 2 quadruplings -> 59,821,792 vertices
R(4,5,5,5,6,10,10)	78 825 777	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,926,611 vertices; then 2 quadruplings -> 78,825,776 vertices
R(4,5,5,5,7,7,7)	24 505 742	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 24,505,741 vertices
R(4,5,5,5,7,7,8)	30 312 345	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 7,578,086 vertices; then 1 quadrupling -> 30,312,344 vertices
R(4,5,5,5,7,7,9)	41 132 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,570,804 vertices; then 2 quadruplings -> 41,132,864 vertices
R(4,5,5,5,7,7,10)	54 966 465	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 3 quadruplings -> 54,966,464 vertices
R(4,5,5,5,7,8,8)	41 132 865	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,570,804 vertices; then 2 quadruplings -> 41,132,864 vertices
R(4,5,5,5,7,8,9)	54 966 465	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 3 quadruplings -> 54,966,464 vertices
R(4,5,5,5,7,8,10)	67 054 225	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 4,190,889 vertices; then 2 quadruplings -> 67,054,224 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,5,5,7,9,9)	67 054 225	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 4,190,889 vertices; then 2 quadruplings -> 67,054,224 vertices
R(4,5,5,5,7,9,10)	85 840 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 3 quadruplings -> 85,840,768 vertices
R(4,5,5,5,7,10,10)	114 659 265	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)]] -> 7,166,204 vertices; then 2 quadruplings -> 114,659,264 vertices
R(4,5,5,5,8,8,8)	54 966 465	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 858,851 vertices; then 3 quadruplings -> 54,966,464 vertices
R(4,5,5,5,8,8,9)	67 054 225	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 4,190,889 vertices; then 2 quadruplings -> 67,054,224 vertices
R(4,5,5,5,8,8,10)	85 840 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 3 quadruplings -> 85,840,768 vertices
R(4,5,5,5,8,9,9)	85 840 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 3 quadruplings -> 85,840,768 vertices
R(4,5,5,5,8,9,10)	114 659 265	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)]] -> 7,166,204 vertices; then 2 quadruplings -> 114,659,264 vertices
R(4,5,5,5,8,10,10)	149 591 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 3 quadruplings -> 149,591,744 vertices
R(4,5,5,5,9,9,9)	114 659 265	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (4,7,3 ; 116) (ours)] x [atom (9,9 ; 562)]] -> 7,166,204 vertices; then 2 quadruplings -> 114,659,264 vertices
R(4,5,5,5,9,9,10)	149 591 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 2,337,371 vertices; then 3 quadruplings -> 149,591,744 vertices
R(4,5,5,5,9,10,10)	182 759 057	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 11,422,441 vertices; then 2 quadruplings -> 182,759,056 vertices
R(4,5,5,5,10,10,10)	239 287 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 3 quadruplings -> 239,287,168 vertices
R(4,5,5,6,6,6,6)	13 035 218	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]]] -> 13,035,217 vertices
R(4,5,5,6,6,6,7)	18 893 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 1 quadrupling -> 18,893,848 vertices
R(4,5,5,6,6,6,8)	24 906 705	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 1,556,669 vertices; then 2 quadruplings -> 24,906,704 vertices
R(4,5,5,6,6,6,9)	28 576 193	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 446,503 vertices; then 3 quadruplings -> 28,576,192 vertices
R(4,5,5,6,6,6,10)	34 051 329	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]] -> 133,013 vertices; then 4 quadruplings -> 34,051,328 vertices
R(4,5,5,6,6,7,7)	26 237 484	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]]] -> 26,237,483 vertices
R(4,5,5,6,6,7,8)	38 029 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 1 quadrupling -> 38,029,768 vertices
R(4,5,5,6,6,7,9)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(4,5,5,6,6,7,10)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(4,5,5,6,6,8,8)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(4,5,5,6,6,8,9)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(4,5,5,6,6,8,10)	69 085 905	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [(3,3,4;29)]] -> 4,317,869 vertices; then 2 quadruplings -> 69,085,904 vertices
R(4,5,5,6,6,9,9)	71 626 478	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 71,626,477 vertices
R(4,5,5,6,6,9,10)	103 470 809	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 1 quadrupling -> 103,470,808 vertices
R(4,5,5,6,6,10,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(4,5,5,6,7,7,7)	38 029 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 1 quadrupling -> 38,029,768 vertices
R(4,5,5,6,7,7,8)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(4,5,5,6,7,7,9)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(4,5,5,6,7,7,10)	75 735 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 4 quadruplings -> 75,735,808 vertices
R(4,5,5,6,7,8,8)	57 518 209	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 898,722 vertices; then 3 quadruplings -> 57,518,208 vertices
R(4,5,5,6,7,8,9)	75 735 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 4 quadruplings -> 75,735,808 vertices
R(4,5,5,6,7,8,10)	91 436 033	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 5 quadruplings -> 91,436,032 vertices
R(4,5,5,6,7,9,9)	103 470 809	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 1 quadrupling -> 103,470,808 vertices
R(4,5,5,6,7,9,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(4,5,5,6,7,10,10)	159 750 977	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 3 quadruplings -> 159,750,976 vertices
R(4,5,5,6,8,8,8)	75 735 809	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 4 quadruplings -> 75,735,808 vertices
R(4,5,5,6,8,8,9)	91 436 033	Our bricks + published	[template (3,6,6,3 ; 707) (ours)] x [atom (4,4,4 ; 127)] -> 89,293 vertices; then 5 quadruplings -> 91,436,032 vertices
R(4,5,5,6,8,8,10)	121 180 161	Published bricks + theorems	[template (6,6,3;235)] x [atom (4,4,4 ; 127)] -> 29,585 vertices; then 6 quadruplings -> 121,180,160 vertices
R(4,5,5,6,8,9,9)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(4,5,5,6,8,9,10)	159 750 977	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 3 quadruplings -> 159,750,976 vertices
R(4,5,5,6,8,10,10)	193 821 009	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (3,3,4,3 ; 66)] x [atome (10,10 ; 797)]]
R(4,5,5,6,9,9,9)	159 750 977	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 2,496,109 vertices; then 3 quadruplings -> 159,750,976 vertices
R(4,5,5,6,9,9,10)	187 106 049	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]] -> 730,883 vertices; then 4 quadruplings -> 187,106,048 vertices
R(4,5,5,6,9,10,10)	241 719 345	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (6,9 ; 182)]] -> 15,107,459 vertices; then 2 quadruplings -> 241,719,344 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,5,6,10,10,10)	315 303 105	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 4,926,611 vertices; then 3 quadruplings -> 315,303,104 vertices
R(4,5,5,7,7,7,7)	52 778 334	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]]] -> 52,778,333 vertices
R(4,5,5,7,7,7,8)	76 011 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 1 quadrupling -> 76,011,208 vertices
R(4,5,5,7,7,7,9)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(4,5,5,7,7,7,10)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(4,5,5,7,7,8,8)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(4,5,5,7,7,8,9)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(4,5,5,7,7,8,10)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,5,5,7,7,9,9)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,5,5,7,7,9,10)	208 268 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 1 quadrupling -> 208,268,008 vertices
R(4,5,5,7,7,10,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(4,5,5,7,8,8,8)	115 364 545	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 3 quadruplings -> 115,364,544 vertices
R(4,5,5,7,8,8,9)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,5,5,7,8,8,10)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(4,5,5,7,8,9,9)	208 268 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 1 quadrupling -> 208,268,008 vertices
R(4,5,5,7,8,9,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(4,5,5,7,8,10,10)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices
R(4,5,5,7,9,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(4,5,5,7,9,9,10)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices
R(4,5,5,7,9,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,5,5,7,10,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(4,5,5,8,8,8,8)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,5,5,8,8,8,9)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(4,5,5,8,8,8,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(4,5,5,8,8,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(4,5,5,8,8,9,10)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices
R(4,5,5,8,8,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,5,5,8,9,9,9)	321 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 3 quadruplings -> 321,549,504 vertices
R(4,5,5,8,9,9,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,5,5,8,9,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(4,5,5,8,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(4,5,5,9,9,9,9)	392 110 194	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 392,110,193 vertices
R(4,5,5,9,9,9,10)	566 439 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 1 quadrupling -> 566,439,208 vertices
R(4,5,5,9,9,10,10)	747 970 545	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 2 quadruplings -> 747,970,544 vertices
R(4,5,5,9,10,10,10)	874 549 825	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 3 quadruplings -> 874,549,824 vertices
R(4,5,5,10,10,10,10)	1 061 046 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,5,6,6,6,6,6)	18 893 849	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 1 quadrupling -> 18,893,848 vertices
R(4,5,6,6,6,6,7)	27 691 029	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 1 quadrupling -> 27,691,028 vertices
R(4,5,6,6,6,6,8)	36 213 225	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 9,053,306 vertices; then 1 quadrupling -> 36,213,224 vertices
R(4,5,6,6,6,6,9)	43 240 737	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,702,546 vertices; then 2 quadruplings -> 43,240,736 vertices
R(4,5,6,6,6,6,10)	47 403 189	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (4,9,3 ; 150) (ours)] x [K2]]] -> 11,850,797 vertices; then 1 quadrupling -> 47,403,188 vertices
R(4,5,6,6,6,7,7)	38 029 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 1 quadrupling -> 38,029,768 vertices
R(4,5,6,6,6,7,8)	50 132 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 2 quadruplings -> 50,132,704 vertices
R(4,5,6,6,6,7,9)	59 788 993	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 3,736,812 vertices; then 2 quadruplings -> 59,788,992 vertices
R(4,5,6,6,6,7,10)	75 575 393	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 2 quadruplings -> 75,575,392 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,6,6,6,8,8)	59 788 993	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 3,736,812 vertices; then 2 quadruplings -> 59,788,992 vertices
R(4,5,6,6,6,8,9)	75 575 393	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 2 quadruplings -> 75,575,392 vertices
R(4,5,6,6,6,8,10)	99 626 817	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 1,556,669 vertices; then 3 quadruplings -> 99,626,816 vertices
R(4,5,6,6,6,9,9)	103 470 809	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 1 quadrupling -> 103,470,808 vertices
R(4,5,6,6,6,9,10)	136 631 409	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 2 quadruplings -> 136,631,408 vertices
R(4,5,6,6,6,10,10)	166 057 265	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 10,378,579 vertices; then 2 quadruplings -> 166,057,264 vertices
R(4,5,6,6,7,7,7)	55 737 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 1 quadrupling -> 55,737,064 vertices
R(4,5,6,6,7,7,8)	72 890 405	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 18,222,601 vertices; then 1 quadrupling -> 72,890,404 vertices
R(4,5,6,6,7,7,9)	94 218 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 2 quadruplings -> 94,218,304 vertices
R(4,5,6,6,7,7,10)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices
R(4,5,6,6,7,8,8)	94 218 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 2 quadruplings -> 94,218,304 vertices
R(4,5,6,6,7,8,9)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices
R(4,5,6,6,7,8,10)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(4,5,6,6,7,9,9)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(4,5,6,6,7,9,10)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(4,5,6,6,7,10,10)	256 293 393	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 2 quadruplings -> 256,293,392 vertices
R(4,5,6,6,8,8,8)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices
R(4,5,6,6,8,8,9)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(4,5,6,6,8,8,10)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(4,5,6,6,8,9,9)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(4,5,6,6,8,9,10)	256 293 393	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 2 quadruplings -> 256,293,392 vertices
R(4,5,6,6,8,10,10)	315 356 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 3 quadruplings -> 315,356,096 vertices
R(4,5,6,6,9,9,9)	241 728 897	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 15,108,056 vertices; then 2 quadruplings -> 241,728,896 vertices
R(4,5,6,6,9,9,10)	315 356 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 3 quadruplings -> 315,356,096 vertices
R(4,5,6,6,9,10,10)	413 883 233	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 2 quadruplings -> 413,883,232 vertices
R(4,5,6,6,10,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(4,5,6,7,7,7,7)	76 011 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 1 quadrupling -> 76,011,208 vertices
R(4,5,6,7,7,7,8)	100 902 737	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 2 quadruplings -> 100,902,736 vertices
R(4,5,6,7,7,7,9)	128 622 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 2 quadruplings -> 128,622,416 vertices
R(4,5,6,7,7,7,10)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,5,6,7,7,8,8)	128 622 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 2 quadruplings -> 128,622,416 vertices
R(4,5,6,7,7,8,9)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,5,6,7,7,8,10)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(4,5,6,7,7,9,9)	208 268 009	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 1 quadrupling -> 208,268,008 vertices
R(4,5,6,7,7,9,10)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(4,5,6,7,7,10,10)	349 849 617	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 2 quadruplings -> 349,849,616 vertices
R(4,5,6,7,8,8,8)	152 441 089	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 4 quadruplings -> 152,441,088 vertices
R(4,5,6,7,8,8,9)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(4,5,6,7,8,8,10)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices
R(4,5,6,7,8,9,9)	275 014 481	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 2 quadruplings -> 275,014,480 vertices
R(4,5,6,7,8,9,10)	349 849 617	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 2 quadruplings -> 349,849,616 vertices
R(4,5,6,7,8,10,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,5,6,7,9,9,9)	334 242 577	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices; then 2 quadruplings -> 334,242,576 vertices
R(4,5,6,7,9,9,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,5,6,7,9,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(4,5,6,7,10,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(4,5,6,8,8,8,8)	183 000 065	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 5 quadruplings -> 183,000,064 vertices
R(4,5,6,8,8,8,9)	243 908 609	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 6 quadruplings -> 243,908,608 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,6,8,8,8,10)	302 943 233	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 295,843 vertices; then 5 quadruplings -> 302,943,232 vertices
R(4,5,6,8,8,9,9)	334 242 577	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices; then 2 quadruplings -> 334,242,576 vertices
R(4,5,6,8,8,9,10)	414 636 289	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 4 quadruplings -> 414,636,288 vertices
R(4,5,6,8,8,10,10)	497 726 465	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 5 quadruplings -> 497,726,464 vertices
R(4,5,6,8,9,9,9)	413 883 233	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 2 quadruplings -> 413,883,232 vertices
R(4,5,6,8,9,9,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(4,5,6,8,9,10,10)	663 420 929	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 6 quadruplings -> 663,420,928 vertices
R(4,5,6,8,10,10,10)	775 284 033	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (3,3,4,3 ; 66)] x [atome (10,10 ; 797)]]
R(4,5,6,9,9,9,9)	566 439 209	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 1 quadrupling -> 566,439,208 vertices
R(4,5,6,9,9,9,10)	747 970 545	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 2 quadruplings -> 747,970,544 vertices
R(4,5,6,9,9,10,10)	909 071 857	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 56,816,991 vertices; then 2 quadruplings -> 909,071,856 vertices
R(4,5,6,9,10,10,10)	1 061 046 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,5,6,10,10,10,10)	1 289 580 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,6,3 ; 80)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,5,7,7,7,7,7)	112 187 357	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 1 quadrupling -> 112,187,356 vertices
R(4,5,7,7,7,7,8)	145 688 165	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 36,422,041 vertices; then 1 quadrupling -> 145,688,164 vertices
R(4,5,7,7,7,7,9)	189 639 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 2 quadruplings -> 189,639,136 vertices
R(4,5,7,7,7,7,10)	233 237 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 3 quadruplings -> 233,237,056 vertices
R(4,5,7,7,7,8,8)	189 639 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 2 quadruplings -> 189,639,136 vertices
R(4,5,7,7,7,8,9)	233 237 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 3 quadruplings -> 233,237,056 vertices
R(4,5,7,7,7,8,10)	304 044 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 2 quadruplings -> 304,044,832 vertices
R(4,5,7,7,7,9,9)	305 146 637	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 1 quadrupling -> 305,146,636 vertices
R(4,5,7,7,7,9,10)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(4,5,7,7,7,10,10)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,5,7,7,8,8,8)	233 237 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 3 quadruplings -> 233,237,056 vertices
R(4,5,7,7,8,8,9)	304 044 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 2 quadruplings -> 304,044,832 vertices
R(4,5,7,7,8,8,10)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(4,5,7,7,8,9,9)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(4,5,7,7,8,9,10)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,5,7,7,8,10,10)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,5,7,7,9,9,9)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,5,7,7,9,9,10)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,5,7,7,9,10,10)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(4,5,7,7,10,10,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,5,7,8,8,8,8)	289 637 377	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 4 quadruplings -> 289,637,376 vertices
R(4,5,7,8,8,8,9)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(4,5,7,8,8,8,10)	461 458 177	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 4 quadruplings -> 461,458,176 vertices
R(4,5,7,8,8,9,9)	515 872 577	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 2 quadruplings -> 515,872,576 vertices
R(4,5,7,8,8,9,10)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,5,7,8,8,10,10)	787 808 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 4 quadruplings -> 787,808,256 vertices
R(4,5,7,8,9,9,9)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,5,7,8,9,9,10)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(4,5,7,8,9,10,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,5,7,8,10,10,10)	1 286 198 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 4 quadruplings -> 1,286,198,016 vertices
R(4,5,7,9,9,9,9)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(4,5,7,9,9,9,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,5,7,9,9,10,10)	1 403 042 177	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 2 quadruplings -> 1,403,042,176 vertices
R(4,5,7,9,10,10,10)	1 726 373 057	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 3 quadruplings -> 1,726,373,056 vertices
R(4,5,7,10,10,10,10)	1 990 905 857	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 6 quadruplings -> 1,990,905,856 vertices
R(4,5,8,8,8,8,8)	321 468 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 3 quadruplings -> 321,468,544 vertices
R(4,5,8,8,8,8,9)	461 458 177	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 4 quadruplings -> 461,458,176 vertices
R(4,5,8,8,8,8,10)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices
R(4,5,8,8,8,9,9)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,5,8,8,8,9,10)	787 808 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 4 quadruplings -> 787,808,256 vertices
R(4,5,8,8,8,10,10)	975 634 433	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 7 quadruplings -> 975,634,432 vertices
R(4,5,8,8,9,9,9)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(4,5,8,8,9,9,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,5,8,8,9,10,10)	1 286 198 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 4 quadruplings -> 1,286,198,016 vertices
R(4,5,8,8,10,10,10)	1 658 545 153	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(4,5,8,9,9,9,9)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,5,8,9,9,9,10)	1 403 042 177	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 2 quadruplings -> 1,403,042,176 vertices
R(4,5,8,9,9,10,10)	1 726 373 057	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 3 quadruplings -> 1,726,373,056 vertices
R(4,5,8,9,10,10,10)	1 990 905 857	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 6 quadruplings -> 1,990,905,856 vertices
R(4,5,8,10,10,10,10)	2 653 683 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 7 quadruplings -> 2,653,683,712 vertices
R(4,5,9,9,9,9,9)	1 323 332 417	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 82,708,276 vertices; then 2 quadruplings -> 1,323,332,416 vertices
R(4,5,9,9,9,9,10)	1 726 373 057	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 3 quadruplings -> 1,726,373,056 vertices
R(4,5,9,9,9,10,10)	2 265 756 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 2 quadruplings -> 2,265,756,832 vertices
R(4,5,9,9,10,10,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(4,5,9,10,10,10,10)	3 498 199 297	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 4 quadruplings -> 3,498,199,296 vertices
R(4,5,10,10,10,10,10)	4 244 187 777	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,6,6,6,6,6,6)	31 124 910	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]]] -> 31,124,909 vertices
R(4,6,6,6,6,6,7)	32 385 473	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 8,096,368 vertices; then 1 quadrupling -> 32,385,472 vertices
R(4,6,6,6,6,6,8)	36 213 225	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 9,053,306 vertices; then 1 quadrupling -> 36,213,224 vertices (by monotonicity, from (4,5,6,6,6,6,8))
R(4,6,6,6,6,6,9)	53 848 650	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K3]]] -> 53,848,649 vertices
R(4,6,6,6,6,6,10)	55 436 574	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (4,3;6)] x [atom (6,10 ; 203)]]] -> 55,436,573 vertices
R(4,6,6,6,6,7,7)	62 648 855	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]]] -> 62,648,854 vertices
R(4,6,6,6,6,7,8)	63 902 337	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,993,896 vertices; then 2 quadruplings -> 63,902,336 vertices
R(4,6,6,6,6,7,9)	75 575 393	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 2 quadruplings -> 75,575,392 vertices
R(4,6,6,6,6,7,10)	89 959 633	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 22,489,908 vertices; then 1 quadrupling -> 89,959,632 vertices
R(4,6,6,6,6,8,8)	86 428 470	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [K3]]] -> 86,428,469 vertices
R(4,6,6,6,6,8,9)	110 764 113	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 2 quadruplings -> 110,764,112 vertices
R(4,6,6,6,6,8,10)	110 764 113	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 2 quadruplings -> 110,764,112 vertices (by monotonicity, from (4,6,6,6,6,8,9))
R(4,6,6,6,6,9,9)	171 026 490	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 171,026,489 vertices
R(4,6,6,6,6,9,10)	172 962 945	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,702,546 vertices; then 3 quadruplings -> 172,962,944 vertices
R(4,6,6,6,6,10,10)	218 061 894	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (6,6,3;235)] x [[modele (4,3;6)] x [atome (10,10 ; 797)]]]
R(4,6,6,6,7,7,7)	65 186 117	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]]] -> 16,296,529 vertices; then 1 quadrupling -> 65,186,116 vertices
R(4,6,6,6,7,7,8)	110 764 113	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 2 quadruplings -> 110,764,112 vertices
R(4,6,6,6,7,7,9)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices
R(4,6,6,6,7,7,10)	148 158 785	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 9,259,924 vertices; then 2 quadruplings -> 148,158,784 vertices
R(4,6,6,6,7,8,8)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,6,6,6,7,8,9)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(4,6,6,6,7,8,10)	189 848 625	Our bricks + published	[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 11,865,539 vertices; then 2 quadruplings -> 189,848,624 vertices
R(4,6,6,6,7,9,9)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(4,6,6,6,7,9,10)	256 293 393	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 2 quadruplings -> 256,293,392 vertices
R(4,6,6,6,7,10,10)	315 356 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 3 quadruplings -> 315,356,096 vertices
R(4,6,6,6,8,8,8)	115 886 465	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 3 quadruplings -> 115,886,464 vertices (by monotonicity, from (4,5,6,6,8,8,8))
R(4,6,6,6,8,8,9)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(4,6,6,6,8,8,10)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices (by monotonicity, from (4,5,6,6,8,8,10))
R(4,6,6,6,8,9,9)	239 155 969	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 3,736,812 vertices; then 3 quadruplings -> 239,155,968 vertices
R(4,6,6,6,8,9,10)	302 301 569	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 3 quadruplings -> 302,301,568 vertices
R(4,6,6,6,8,10,10)	315 356 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 3 quadruplings -> 315,356,096 vertices (by monotonicity, from (4,5,6,6,8,10,10))
R(4,6,6,6,9,9,9)	302 301 569	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 4,723,462 vertices; then 3 quadruplings -> 302,301,568 vertices
R(4,6,6,6,9,9,10)	413 883 233	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 2 quadruplings -> 413,883,232 vertices
R(4,6,6,6,9,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(4,6,6,6,10,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices (by monotonicity, from (4,5,6,6,10,10,10))
R(4,6,6,7,7,7,7)	126 100 562	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]]] -> 126,100,561 vertices
R(4,6,6,7,7,7,8)	128 622 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 2 quadruplings -> 128,622,416 vertices
R(4,6,6,7,7,7,9)	222 948 257	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 2 quadruplings -> 222,948,256 vertices
R(4,6,6,7,7,7,10)	222 948 257	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 2 quadruplings -> 222,948,256 vertices (by monotonicity, from (4,6,6,7,7,7,9))
R(4,6,6,7,7,8,8)	222 948 257	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 2 quadruplings -> 222,948,256 vertices
R(4,6,6,7,7,8,9)	222 948 257	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 2 quadruplings -> 222,948,256 vertices
R(4,6,6,7,7,8,10)	307 211 793	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 2 quadruplings -> 307,211,792 vertices
R(4,6,6,7,7,9,9)	344 245 625	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 344,245,624 vertices
R(4,6,6,7,7,9,10)	376 873 217	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 3 quadruplings -> 376,873,216 vertices
R(4,6,6,7,7,10,10)	606 406 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 2 quadruplings -> 606,406,064 vertices
R(4,6,6,7,8,8,8)	222 948 257	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 2 quadruplings -> 222,948,256 vertices (by monotonicity, from (4,6,6,7,7,8,8))
R(4,6,6,7,8,8,9)	307 211 793	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 2 quadruplings -> 307,211,792 vertices
R(4,6,6,7,8,8,10)	376 873 217	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 3 quadruplings -> 376,873,216 vertices
R(4,6,6,7,8,9,9)	376 873 217	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 3 quadruplings -> 376,873,216 vertices
R(4,6,6,7,8,9,10)	606 406 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 2 quadruplings -> 606,406,064 vertices
R(4,6,6,7,8,10,10)	606 406 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 2 quadruplings -> 606,406,064 vertices (by monotonicity, from (4,6,6,7,7,10,10))
R(4,6,6,7,9,9,9)	463 545 857	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 4 quadruplings -> 463,545,856 vertices
R(4,6,6,7,9,9,10)	608 476 289	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 3 quadruplings -> 608,476,288 vertices
R(4,6,6,7,9,10,10)	802 123 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 4 quadruplings -> 802,123,264 vertices
R(4,6,6,7,10,10,10)	1 025 173 569	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 3 quadruplings -> 1,025,173,568 vertices
R(4,6,6,8,8,8,8)	239 710 170	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [K3]]] -> 239,710,169 vertices
R(4,6,6,8,8,8,9)	376 873 217	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 3 quadruplings -> 376,873,216 vertices
R(4,6,6,8,8,8,10)	376 873 217	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 3 quadruplings -> 376,873,216 vertices (by monotonicity, from (4,6,6,7,8,8,10))
R(4,6,6,8,8,9,9)	474 343 140	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 474,343,139 vertices
R(4,6,6,8,8,9,10)	608 476 289	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 3 quadruplings -> 608,476,288 vertices
R(4,6,6,8,8,10,10)	608 476 289	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 3 quadruplings -> 608,476,288 vertices (by monotonicity, from (4,6,6,8,8,9,10))
R(4,6,6,8,9,9,9)	608 476 289	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 3 quadruplings -> 608,476,288 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,6,6,8,9,9,10)	802 123 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 4 quadruplings -> 802,123,264 vertices
R(4,6,6,8,9,10,10)	1 025 173 569	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 3 quadruplings -> 1,025,173,568 vertices
R(4,6,6,8,10,10,10)	1 025 173 569	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 3 quadruplings -> 1,025,173,568 vertices (by monotonicity, from (4,6,6,7,10,10,10))
R(4,6,6,9,9,9,9)	936 263 118	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 936,263,117 vertices
R(4,6,6,9,9,9,10)	1 025 173 569	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 3 quadruplings -> 1,025,173,568 vertices
R(4,6,6,9,9,10,10)	1 261 424 385	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 4 quadruplings -> 1,261,424,384 vertices
R(4,6,6,9,10,10,10)	1 655 532 929	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 3 quadruplings -> 1,655,532,928 vertices
R(4,6,6,10,10,10,10)	1 655 532 929	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 3 quadruplings -> 1,655,532,928 vertices (by monotonicity, from (4,6,6,9,10,10,10))
R(4,6,7,7,7,7,7)	131 195 825	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]]] -> 32,798,956 vertices; then 1 quadrupling -> 131,195,824 vertices
R(4,6,7,7,7,7,8)	222 948 257	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 2 quadruplings -> 222,948,256 vertices
R(4,6,7,7,7,7,9)	233 237 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 3 quadruplings -> 233,237,056 vertices
R(4,6,7,7,7,7,10)	301 071 745	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 4,704,246 vertices; then 3 quadruplings -> 301,071,744 vertices
R(4,6,7,7,7,8,8)	233 237 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 3 quadruplings -> 233,237,056 vertices
R(4,6,7,7,7,8,9)	307 211 793	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 2 quadruplings -> 307,211,792 vertices
R(4,6,7,7,7,8,10)	382 131 185	Our bricks + published	[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [atom (4,4,4 ; 127)]] -> 23,883,199 vertices; then 2 quadruplings -> 382,131,184 vertices
R(4,6,7,7,7,9,9)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(4,6,7,7,7,9,10)	606 406 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 2 quadruplings -> 606,406,064 vertices
R(4,6,7,7,7,10,10)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,6,7,7,8,8,8)	307 211 793	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 2 quadruplings -> 307,211,792 vertices
R(4,6,7,7,8,8,9)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(4,6,7,7,8,8,10)	514 489 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 3 quadruplings -> 514,489,664 vertices
R(4,6,7,7,8,9,9)	606 406 065	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 2 quadruplings -> 606,406,064 vertices
R(4,6,7,7,8,9,10)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,6,7,7,8,10,10)	818 907 265	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices; then 3 quadruplings -> 818,907,264 vertices
R(4,6,7,7,9,9,9)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,6,7,7,9,9,10)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(4,6,7,7,9,10,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,6,7,7,10,10,10)	1 399 398 465	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 3 quadruplings -> 1,399,398,464 vertices
R(4,6,7,8,8,8,8)	321 468 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 3 quadruplings -> 321,468,544 vertices
R(4,6,7,8,8,8,9)	514 489 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 3 quadruplings -> 514,489,664 vertices
R(4,6,7,8,8,8,10)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices
R(4,6,7,8,8,9,9)	634 755 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 3 quadruplings -> 634,755,136 vertices
R(4,6,7,8,8,9,10)	818 907 265	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices; then 3 quadruplings -> 818,907,264 vertices
R(4,6,7,8,8,10,10)	975 634 433	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 7 quadruplings -> 975,634,432 vertices
R(4,6,7,8,9,9,9)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(4,6,7,8,9,9,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,6,7,8,9,10,10)	1 399 398 465	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 3 quadruplings -> 1,399,398,464 vertices
R(4,6,7,8,10,10,10)	1 658 545 153	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(4,6,7,9,9,9,9)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,6,7,9,9,9,10)	1 403 042 177	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 2 quadruplings -> 1,403,042,176 vertices
R(4,6,7,9,9,10,10)	1 726 373 057	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 3 quadruplings -> 1,726,373,056 vertices
R(4,6,7,9,10,10,10)	2 186 102 529	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 4 quadruplings -> 2,186,102,528 vertices
R(4,6,7,10,10,10,10)	2 653 683 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 7 quadruplings -> 2,653,683,712 vertices
R(4,6,8,8,8,8,8)	321 468 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 3 quadruplings -> 321,468,544 vertices (by monotonicity, from (4,5,8,8,8,8,8))
R(4,6,8,8,8,8,9)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,6,8,8,8,8,10)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices (by monotonicity, from (4,5,8,8,8,8,10))
R(4,6,8,8,8,9,9)	802 123 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 4 quadruplings -> 802,123,264 vertices
R(4,6,8,8,8,9,10)	975 634 433	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 7 quadruplings -> 975,634,432 vertices
R(4,6,8,8,8,10,10)	975 634 433	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 7 quadruplings -> 975,634,432 vertices (by monotonicity, from (4,5,8,8,8,10,10))
R(4,6,8,8,9,9,9)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(4,6,8,8,9,9,10)	1 399 398 465	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 3 quadruplings -> 1,399,398,464 vertices
R(4,6,8,8,9,10,10)	1 658 545 153	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(4,6,8,8,10,10,10)	1 658 545 153	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices (by monotonicity, from (4,5,8,8,10,10,10))
R(4,6,8,9,9,9,9)	1 336 970 305	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 20,890,161 vertices; then 3 quadruplings -> 1,336,970,304 vertices
R(4,6,8,9,9,9,10)	1 658 545 153	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(4,6,8,9,9,10,10)	2 186 102 529	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 4 quadruplings -> 2,186,102,528 vertices
R(4,6,8,9,10,10,10)	2 653 683 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 7 quadruplings -> 2,653,683,712 vertices
R(4,6,8,10,10,10,10)	2 653 683 713	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 7 quadruplings -> 2,653,683,712 vertices (by monotonicity, from (4,5,8,10,10,10,10))
R(4,6,9,9,9,9,9)	1 655 532 929	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 3 quadruplings -> 1,655,532,928 vertices
R(4,6,9,9,9,9,10)	2 265 756 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 2 quadruplings -> 2,265,756,832 vertices
R(4,6,9,9,9,10,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(4,6,9,9,10,10,10)	3 636 287 425	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 56,816,991 vertices; then 3 quadruplings -> 3,636,287,424 vertices
R(4,6,9,10,10,10,10)	4 244 187 777	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,6,10,10,10,10,10)	4 244 187 777	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(4,7,7,7,7,7,7)	253 659 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]]] -> 253,659,136 vertices
R(4,7,7,7,7,7,8)	253 659 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]]] -> 253,659,136 vertices (by monotonicity, from (4,7,7,7,7,7,7))
R(4,7,7,7,7,7,9)	448 749 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 2 quadruplings -> 448,749,424 vertices
R(4,7,7,7,7,7,10)	448 749 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 2 quadruplings -> 448,749,424 vertices (by monotonicity, from (4,7,7,7,7,7,9))
R(4,7,7,7,7,8,8)	448 749 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 2 quadruplings -> 448,749,424 vertices
R(4,7,7,7,7,8,9)	448 749 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 2 quadruplings -> 448,749,424 vertices
R(4,7,7,7,7,8,10)	618 362 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 2 quadruplings -> 618,362,176 vertices
R(4,7,7,7,7,9,9)	692 904 317	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 692,904,316 vertices
R(4,7,7,7,7,9,10)	758 556 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 3 quadruplings -> 758,556,544 vertices
R(4,7,7,7,7,10,10)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices
R(4,7,7,7,8,8,8)	448 749 425	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 2 quadruplings -> 448,749,424 vertices (by monotonicity, from (4,7,7,7,7,8,8))
R(4,7,7,7,8,8,9)	618 362 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 2 quadruplings -> 618,362,176 vertices
R(4,7,7,7,8,8,10)	758 556 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 3 quadruplings -> 758,556,544 vertices
R(4,7,7,7,8,9,9)	758 556 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 3 quadruplings -> 758,556,544 vertices
R(4,7,7,7,8,9,10)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices
R(4,7,7,7,8,10,10)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices (by monotonicity, from (4,7,7,7,7,10,10))
R(4,7,7,7,9,9,9)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices
R(4,7,7,7,9,9,10)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices
R(4,7,7,7,9,10,10)	1 614 443 777	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 4 quadruplings -> 1,614,443,776 vertices
R(4,7,7,7,10,10,10)	2 063 490 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 3 quadruplings -> 2,063,490,304 vertices
R(4,7,7,8,8,8,8)	618 362 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 2 quadruplings -> 618,362,176 vertices
R(4,7,7,8,8,8,9)	758 556 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 3 quadruplings -> 758,556,544 vertices
R(4,7,7,8,8,8,10)	932 948 225	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 4 quadruplings -> 932,948,224 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,7,7,8,8,9,9)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices
R(4,7,7,8,8,9,10)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices (by monotonicity, from (4,7,7,7,8,9,10))
R(4,7,7,8,8,10,10)	1 681 869 825	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 2 quadruplings -> 1,681,869,824 vertices
R(4,7,7,8,9,9,9)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices
R(4,7,7,8,9,9,10)	1 681 869 825	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 2 quadruplings -> 1,681,869,824 vertices
R(4,7,7,8,9,10,10)	2 063 490 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 3 quadruplings -> 2,063,490,304 vertices
R(4,7,7,8,10,10,10)	2 539 020 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 4 quadruplings -> 2,539,020,544 vertices
R(4,7,7,9,9,9,9)	1 884 529 607	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 1,884,529,606 vertices
R(4,7,7,9,9,9,10)	2 063 490 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 3 quadruplings -> 2,063,490,304 vertices
R(4,7,7,9,9,10,10)	3 319 684 625	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 2 quadruplings -> 3,319,684,624 vertices
R(4,7,7,9,10,10,10)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(4,7,7,10,10,10,10)	4 130 239 937	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de atome (4,4,4 ; 127)] x [atome (10,10 ; 797)]]
R(4,7,8,8,8,8,8)	618 362 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 2 quadruplings -> 618,362,176 vertices (by monotonicity, from (4,7,7,8,8,8,8))
R(4,7,8,8,8,8,9)	932 948 225	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 4 quadruplings -> 932,948,224 vertices
R(4,7,8,8,8,8,10)	1 158 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 5 quadruplings -> 1,158,549,504 vertices
R(4,7,8,8,8,9,9)	1 220 586 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 2 quadruplings -> 1,220,586,544 vertices (by monotonicity, from (4,7,7,8,8,9,9))
R(4,7,8,8,8,9,10)	1 681 869 825	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 2 quadruplings -> 1,681,869,824 vertices
R(4,7,8,8,8,10,10)	1 829 297 153	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,786,423 vertices; then 5 quadruplings -> 1,829,297,152 vertices
R(4,7,8,8,9,9,9)	1 681 869 825	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 2 quadruplings -> 1,681,869,824 vertices
R(4,7,8,8,9,9,10)	2 063 490 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 3 quadruplings -> 2,063,490,304 vertices
R(4,7,8,8,9,10,10)	2 539 020 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 4 quadruplings -> 2,539,020,544 vertices
R(4,7,8,8,10,10,10)	3 151 233 025	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 5 quadruplings -> 3,151,233,024 vertices
R(4,7,8,9,9,9,9)	2 063 490 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 3 quadruplings -> 2,063,490,304 vertices
R(4,7,8,9,9,9,10)	3 319 684 625	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 2 quadruplings -> 3,319,684,624 vertices
R(4,7,8,9,9,10,10)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(4,7,8,9,10,10,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(4,7,8,10,10,10,10)	4 975 639 553	Published bricks + theorems	[template (3,5,3 ; 31)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 4,859,023 vertices; then 5 quadruplings -> 4,975,639,552 vertices
R(4,7,9,9,9,9,9)	2 539 020 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 4 quadruplings -> 2,539,020,544 vertices
R(4,7,9,9,9,9,10)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(4,7,9,9,9,10,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(4,7,9,9,10,10,10)	5 612 168 705	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 3 quadruplings -> 5,612,168,704 vertices
R(4,7,9,10,10,10,10)	6 905 492 225	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 4 quadruplings -> 6,905,492,224 vertices
R(4,7,10,10,10,10,10)	7 963 623 425	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 7 quadruplings -> 7,963,623,424 vertices
R(4,8,8,8,8,8,8)	664 837 180	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [[template (8,8,3 ; 650)] x [K3]]] -> 664,837,179 vertices
R(4,8,8,8,8,8,9)	1 158 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 5 quadruplings -> 1,158,549,504 vertices
R(4,8,8,8,8,8,10)	1 158 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 5 quadruplings -> 1,158,549,504 vertices (by monotonicity, from (4,7,8,8,8,8,10))
R(4,8,8,8,8,9,9)	1 614 443 777	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 4 quadruplings -> 1,614,443,776 vertices
R(4,8,8,8,8,8,9,10)	1 845 832 705	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 5 quadruplings -> 1,845,832,704 vertices
R(4,8,8,8,8,8,10,10)	1 845 832 705	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 5 quadruplings -> 1,845,832,704 vertices (by monotonicity, from (4,8,8,8,8,8,9,10))
R(4,8,8,8,8,9,9,9)	2 063 490 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 3 quadruplings -> 2,063,490,304 vertices
R(4,8,8,8,8,9,9,10)	2 539 020 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 4 quadruplings -> 2,539,020,544 vertices
R(4,8,8,8,8,9,10,10)	3 151 233 025	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 5 quadruplings -> 3,151,233,024 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,8,8,8,10,10,10)	3 151 233 025	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 5 quadruplings -> 3,151,233,024 vertices (by monotonicity, from (4,7,8,8,10,10,10))
R(4,8,8,9,9,9,9)	2 596 729 758	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 2,596,729,757 vertices
R(4,8,8,9,9,9,10)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(4,8,8,9,9,10,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(4,8,8,9,10,10,10)	5 144 792 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 5 quadruplings -> 5,144,792,064 vertices
R(4,8,8,10,10,10,10)	5 144 792 065	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 5 quadruplings -> 5,144,792,064 vertices (by monotonicity, from (4,8,8,9,10,10,10))
R(4,8,9,9,9,9,9)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(4,8,9,9,9,9,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(4,8,9,9,9,10,10)	5 612 168 705	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 3 quadruplings -> 5,612,168,704 vertices
R(4,8,9,9,10,10,10)	6 905 492 225	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 4 quadruplings -> 6,905,492,224 vertices
R(4,8,9,10,10,10,10)	7 963 623 425	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 7 quadruplings -> 7,963,623,424 vertices
R(4,8,10,10,10,10,10)	7 963 623 425	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 7 quadruplings -> 7,963,623,424 vertices (by monotonicity, from (4,7,10,10,10,10,10))
R(4,9,9,9,9,9,9)	5 125 440 407	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K3]]] -> 5,125,440,406 vertices
R(4,9,9,9,9,9,10)	5 612 168 705	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 3 quadruplings -> 5,612,168,704 vertices
R(4,9,9,9,9,10,10)	6 905 492 225	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 4 quadruplings -> 6,905,492,224 vertices
R(4,9,9,9,10,10,10)	9 063 027 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 3 quadruplings -> 9,063,027,328 vertices
R(4,9,9,10,10,10,10)	11 967 528 705	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 4 quadruplings -> 11,967,528,704 vertices
R(4,9,10,10,10,10,10)	13 992 797 185	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 5 quadruplings -> 13,992,797,184 vertices
R(4,10,10,10,10,10,10)	13 992 797 185	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 5 quadruplings -> 13,992,797,184 vertices (by monotonicity, from (4,9,10,10,10,10,10))
R(5,5,5,5,5,5,5)	4 344 970	Our bricks	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (5,5,5 ; 453)]] -> 4,344,969 vertices
R(5,5,5,5,5,5,6)	5 985 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (5,5,5 ; 453)]] -> 5,985,416 vertices
R(5,5,5,5,5,5,7)	8 291 038	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (5,5,5 ; 453)]] -> 8,291,037 vertices
R(5,5,5,5,5,5,8)	10 419 304	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,8,3 ; 236) (ours)] x [atom (5,5,5 ; 453)]] -> 10,419,303 vertices
R(5,5,5,5,5,5,9)	13 877 528	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (5,5,5 ; 453)]] -> 13,877,527 vertices
R(5,5,5,5,5,5,10)	17 549 313	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [K2]] -> 17,138 vertices; then 5 quadruplings -> 17,549,312 vertices
R(5,5,5,5,5,6,6)	10 375 106	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 10,375,105 vertices
R(5,5,5,5,5,6,7)	12 923 621	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (3,6,6 ; 337)]] -> 3,230,905 vertices; then 1 quadrupling -> 12,923,620 vertices
R(5,5,5,5,5,6,8)	17 476 545	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Paley-101 (6,6;101)]] -> 1,092,284 vertices; then 2 quadruplings -> 17,476,544 vertices
R(5,5,5,5,5,6,9)	21 507 521	Our bricks + published	[template (5,3;10)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (3,6,6 ; 337)]] -> 336,055 vertices; then 3 quadruplings -> 21,507,520 vertices
R(5,5,5,5,5,6,10)	29 620 737	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 115,706 vertices; then 4 quadruplings -> 29,620,736 vertices
R(5,5,5,5,5,7,7)	20 883 156	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 20,883,155 vertices
R(5,5,5,5,5,7,8)	25 831 397	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 6,457,849 vertices; then 1 quadrupling -> 25,831,396 vertices
R(5,5,5,5,5,7,9)	35 055 393	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 2 quadruplings -> 35,055,392 vertices
R(5,5,5,5,5,7,10)	43 058 561	Published bricks + theorems	[template (7,7,3;472)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 672,790 vertices; then 3 quadruplings -> 43,058,560 vertices
R(5,5,5,5,5,8,8)	35 055 393	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 2 quadruplings -> 35,055,392 vertices
R(5,5,5,5,5,8,9)	43 058 561	Published bricks + theorems	[template (7,7,3;472)] x [[template (5,5,5,3 ; 977)] x [K2]] -> 672,790 vertices; then 3 quadruplings -> 43,058,560 vertices
R(5,5,5,5,5,8,10)	59 258 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 4 quadruplings -> 59,258,624 vertices
R(5,5,5,5,5,9,9)	59 258 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 4 quadruplings -> 59,258,624 vertices
R(5,5,5,5,5,9,10)	77 554 689	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 5 quadruplings -> 77,554,688 vertices
R(5,5,5,5,5,10,10)	97 712 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 6,107,042 vertices; then 2 quadruplings -> 97,712,672 vertices
R(5,5,5,5,6,6,6)	14 292 237	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 14,292,236 vertices
R(5,5,5,5,6,6,7)	19 797 396	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 19,797,395 vertices
R(5,5,5,5,6,6,8)	24 879 081	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 24,879,080 vertices
R(5,5,5,5,6,6,9)	33 350 433	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Paley-101 (6,6;101)]] -> 2,084,402 vertices; then 2 quadruplings -> 33,350,432 vertices
R(5,5,5,5,6,6,10)	41 912 529	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,8,3 ; 236) (ours)] x [Paley-101 (6,6;101)]] -> 2,619,533 vertices; then 2 quadruplings -> 41,912,528 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,5,5,6,7,7)	28 767 612	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 28,767,611 vertices
R(5,5,5,5,6,7,8)	35 584 065	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [K2]]] -> 8,896,016 vertices; then 1 quadrupling -> 35,584,064 vertices
R(5,5,5,5,6,7,9)	48 290 145	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 3,018,134 vertices; then 2 quadruplings -> 48,290,144 vertices
R(5,5,5,5,6,7,10)	59 258 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 4 quadruplings -> 59,258,624 vertices
R(5,5,5,5,6,8,8)	48 290 145	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [Mathon-doubling (7,7;202)]] -> 3,018,134 vertices; then 2 quadruplings -> 48,290,144 vertices
R(5,5,5,5,6,8,9)	59 258 625	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 231,479 vertices; then 4 quadruplings -> 59,258,624 vertices
R(5,5,5,5,6,8,10)	77 554 689	Published bricks + theorems	[template (3,3,3,3,3,3 ; 377)] x [Mathon-doubling (7,7;202)] -> 75,737 vertices; then 5 quadruplings -> 77,554,688 vertices
R(5,5,5,5,6,9,9)	78 242 412	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)]] -> 78,242,411 vertices
R(5,5,5,5,6,9,10)	97 712 673	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 6,107,042 vertices; then 2 quadruplings -> 97,712,672 vertices
R(5,5,5,5,6,10,10)	134 603 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 8,412,734 vertices; then 2 quadruplings -> 134,603,744 vertices
R(5,5,5,5,7,7,7)	39 848 471	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 39,848,470 vertices
R(5,5,5,5,7,7,8)	51 142 409	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 12,785,602 vertices; then 1 quadrupling -> 51,142,408 vertices
R(5,5,5,5,7,7,9)	66 893 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)]] -> 4,180,859 vertices; then 2 quadruplings -> 66,893,744 vertices
R(5,5,5,5,7,7,10)	85 840 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 3 quadruplings -> 85,840,768 vertices
R(5,5,5,5,7,8,8)	66 893 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [Mathon-doubling (7,7;202)]] -> 4,180,859 vertices; then 2 quadruplings -> 66,893,744 vertices
R(5,5,5,5,7,8,9)	85 840 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 3 quadruplings -> 85,840,768 vertices
R(5,5,5,5,7,8,10)	111 966 849	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)]] -> 6,997,928 vertices; then 2 quadruplings -> 111,966,848 vertices
R(5,5,5,5,7,9,9)	111 966 849	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)]] -> 6,997,928 vertices; then 2 quadruplings -> 111,966,848 vertices
R(5,5,5,5,7,9,10)	140 221 569	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 3 quadruplings -> 140,221,568 vertices
R(5,5,5,5,7,10,10)	186 454 065	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)]] -> 11,653,379 vertices; then 2 quadruplings -> 186,454,064 vertices
R(5,5,5,5,8,8,8)	85 840 769	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [Mathon-doubling (7,7;202)]] -> 1,341,262 vertices; then 3 quadruplings -> 85,840,768 vertices
R(5,5,5,5,8,8,9)	111 966 849	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [Mathon-doubling (7,7;202)]] -> 6,997,928 vertices; then 2 quadruplings -> 111,966,848 vertices
R(5,5,5,5,8,8,10)	140 221 569	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 3 quadruplings -> 140,221,568 vertices
R(5,5,5,5,8,9,9)	140 221 569	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [Mathon-doubling (7,7;202)]] -> 2,190,962 vertices; then 3 quadruplings -> 140,221,568 vertices
R(5,5,5,5,8,9,10)	186 454 065	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)]] -> 11,653,379 vertices; then 2 quadruplings -> 186,454,064 vertices
R(5,5,5,5,8,10,10)	239 287 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 3 quadruplings -> 239,287,168 vertices
R(5,5,5,5,9,9,9)	186 454 065	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,7,3 ; 188) (ours)] x [atom (9,9 ; 562)]] -> 11,653,379 vertices; then 2 quadruplings -> 186,454,064 vertices
R(5,5,5,5,9,9,10)	239 287 169	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (3,3,5,3 ; 112) (ours)] x [atom (9,9 ; 562)]] -> 3,738,862 vertices; then 3 quadruplings -> 239,287,168 vertices
R(5,5,5,5,9,10,10)	312 086 529	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,9,3 ; 314) (ours)] x [atom (9,9 ; 562)]] -> 19,505,408 vertices; then 2 quadruplings -> 312,086,528 vertices
R(5,5,5,5,10,10,10)	390 850 689	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,5,3 ; 99) (ours)] x [atom (9,9 ; 562)]] -> 6,107,042 vertices; then 3 quadruplings -> 390,850,688 vertices
R(5,5,5,6,6,6,6)	24 773 214	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 24,773,213 vertices
R(5,5,5,6,6,6,7)	30 859 973	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 7,714,993 vertices; then 1 quadrupling -> 30,859,972 vertices
R(5,5,5,6,6,6,8)	41 736 705	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices; then 2 quadruplings -> 41,736,704 vertices
R(5,5,5,6,6,6,9)	50 383 553	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 787,243 vertices; then 3 quadruplings -> 50,383,552 vertices
R(5,5,5,6,6,6,10)	62 445 825	Our bricks + published	[template (6,6,3;235)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 243,929 vertices; then 4 quadruplings -> 62,445,824 vertices
R(5,5,5,6,6,7,7)	49 864 031	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 49,864,030 vertices
R(5,5,5,6,6,7,8)	62 115 309	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 1 quadrupling -> 62,115,308 vertices
R(5,5,5,6,6,7,9)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(5,5,5,6,6,7,10)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices
R(5,5,5,6,6,8,8)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(5,5,5,6,6,8,9)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices
R(5,5,5,6,6,8,10)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(5,5,5,6,6,9,9)	135 620 184	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)]] -> 135,620,183 vertices
R(5,5,5,6,6,9,10)	169 002 341	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 42,250,585 vertices; then 1 quadrupling -> 169,002,340 vertices
R(5,5,5,6,6,10,10)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(5,5,5,6,7,7,7)	62 115 309	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 1 quadrupling -> 62,115,308 vertices
R(5,5,5,6,7,7,8)	84 007 281	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 2 quadruplings -> 84,007,280 vertices
R(5,5,5,6,7,7,9)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,5,6,7,7,10)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(5,5,5,6,7,8,8)	101 412 673	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 3 quadruplings -> 101,412,672 vertices
R(5,5,5,6,7,8,9)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(5,5,5,6,7,8,10)	162 260 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 5 quadruplings -> 162,260,992 vertices
R(5,5,5,6,7,9,9)	169 002 341	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 42,250,585 vertices; then 1 quadrupling -> 169,002,340 vertices
R(5,5,5,6,7,9,10)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(5,5,5,6,7,10,10)	277 462 337	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 3 quadruplings -> 277,462,336 vertices
R(5,5,5,6,8,8,8)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(5,5,5,6,8,8,9)	162 260 993	Published bricks + theorems	[template (7,7,3;472)] x [atom (3,6,6 ; 337)] -> 158,458 vertices; then 5 quadruplings -> 162,260,992 vertices
R(5,5,5,6,8,8,10)	193 748 993	Published bricks + theorems	[template (7,7,3;472)] x [Paley-101 (6,6;101)] -> 47,302 vertices; then 6 quadruplings -> 193,748,992 vertices
R(5,5,5,6,8,9,9)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(5,5,5,6,8,9,10)	277 462 337	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 3 quadruplings -> 277,462,336 vertices
R(5,5,5,6,8,10,10)	341 853 185	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 4 quadruplings -> 341,853,184 vertices
R(5,5,5,6,9,9,9)	277 462 337	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 3 quadruplings -> 277,462,336 vertices
R(5,5,5,6,9,9,10)	341 853 185	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 4 quadruplings -> 341,853,184 vertices
R(5,5,5,6,9,10,10)	441 476 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]] -> 431,129 vertices; then 5 quadruplings -> 441,476,096 vertices
R(5,5,5,6,10,10,10)	538 414 977	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (5,6,3 ; 136) (ours)] x [atom (9,9 ; 562)]] -> 8,412,734 vertices; then 3 quadruplings -> 538,414,976 vertices
R(5,5,5,7,7,7,7)	100 367 006	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (5,5,5 ; 453)]] -> 100,367,005 vertices
R(5,5,5,7,7,7,8)	124 151 661	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 31,037,915 vertices; then 1 quadrupling -> 124,151,660 vertices
R(5,5,5,7,7,7,9)	168 493 377	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 2 quadruplings -> 168,493,376 vertices
R(5,5,5,7,7,7,10)	202 696 513	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 3 quadruplings -> 202,696,512 vertices
R(5,5,5,7,7,8,8)	168 493 377	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 2 quadruplings -> 168,493,376 vertices
R(5,5,5,7,7,8,9)	202 696 513	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 3 quadruplings -> 202,696,512 vertices
R(5,5,5,7,7,8,10)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(5,5,5,7,7,9,9)	272 979 086	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)]] -> 272,979,085 vertices
R(5,5,5,7,7,9,10)	340 171 101	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 1 quadrupling -> 340,171,100 vertices
R(5,5,5,7,7,10,10)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(5,5,5,7,8,8,8)	202 696 513	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 3 quadruplings -> 202,696,512 vertices
R(5,5,5,7,8,8,9)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(5,5,5,7,8,8,10)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(5,5,5,7,8,9,9)	340 171 101	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 1 quadrupling -> 340,171,100 vertices
R(5,5,5,7,8,9,10)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(5,5,5,7,8,10,10)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(5,5,5,7,9,9,9)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(5,5,5,7,9,9,10)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(5,5,5,7,9,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(5,5,5,7,10,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(5,5,5,8,8,8,8)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(5,5,5,8,8,8,9)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(5,5,5,8,8,8,10)	388 599 809	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 97,149,953 – 3 = 388,599,809; with R(7,7,7,7) >= 94,874
R(5,5,5,8,8,9,9)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(5,5,5,8,8,9,10)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(5,5,5,8,8,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(5,5,5,8,9,9,9)	558 480 833	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 3 quadruplings -> 558,480,832 vertices
R(5,5,5,8,9,9,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,5,8,9,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(5,5,5,8,10,10,10)	1 083 117 569	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 6 quadruplings -> 1,083,117,568 vertices
R(5,5,5,9,9,9,9)	742 433 576	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)]] -> 742,433,575 vertices
R(5,5,5,9,9,9,10)	925 184 061	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 231,296,015 vertices; then 1 quadrupling -> 925,184,060 vertices
R(5,5,5,9,9,10,10)	1 277 303 457	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 2 quadruplings -> 1,277,303,456 vertices
R(5,5,5,9,10,10,10)	1 518 955 073	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 3 quadruplings -> 1,518,955,072 vertices
R(5,5,5,10,10,10,10)	1 864 460 033	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 4 quadruplings -> 1,864,460,032 vertices
R(5,5,6,6,6,6,6)	30 859 973	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 7,714,993 vertices; then 1 quadrupling -> 30,859,972 vertices
R(5,5,6,6,6,6,7)	42 511 185	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 10,627,796 vertices; then 1 quadrupling -> 42,511,184 vertices
R(5,5,6,6,6,6,8)	58 885 869	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 14,721,467 vertices; then 1 quadrupling -> 58,885,868 vertices
R(5,5,6,6,6,6,9)	74 000 961	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 18,500,240 vertices; then 1 quadrupling -> 74,000,960 vertices
R(5,5,6,6,6,6,10)	98 562 977	Our bricks + published	[template (5,9,3 ; 314) (ours)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 24,640,744 vertices; then 1 quadrupling -> 98,562,976 vertices
R(5,5,6,6,6,7,7)	62 115 309	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 1 quadrupling -> 62,115,308 vertices
R(5,5,6,6,6,7,8)	85 567 005	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 21,391,751 vertices; then 1 quadrupling -> 85,567,004 vertices
R(5,5,6,6,6,7,9)	102 171 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices; then 2 quadruplings -> 102,171,056 vertices
R(5,5,6,6,6,7,10)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(5,5,6,6,6,8,8)	102 171 057	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices; then 2 quadruplings -> 102,171,056 vertices
R(5,5,6,6,6,8,9)	125 691 905	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 4 quadruplings -> 125,691,904 vertices
R(5,5,6,6,6,8,10)	166 946 817	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 2,608,544 vertices; then 3 quadruplings -> 166,946,816 vertices
R(5,5,6,6,6,9,9)	169 002 341	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 42,250,585 vertices; then 1 quadrupling -> 169,002,340 vertices
R(5,5,6,6,6,9,10)	233 320 929	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 2 quadruplings -> 233,320,928 vertices
R(5,5,6,6,6,10,10)	283 768 737	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 17,735,546 vertices; then 2 quadruplings -> 283,768,736 vertices
R(5,5,6,6,7,7,7)	85 567 005	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 21,391,751 vertices; then 1 quadrupling -> 85,567,004 vertices
R(5,5,6,6,7,7,8)	118 526 153	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 29,631,538 vertices; then 1 quadrupling -> 118,526,152 vertices
R(5,5,6,6,7,7,9)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(5,5,6,6,7,7,10)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(5,5,6,6,7,8,8)	152 119 073	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 2 quadruplings -> 152,119,072 vertices
R(5,5,6,6,7,8,9)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(5,5,6,6,7,8,10)	248 461 233	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 2 quadruplings -> 248,461,232 vertices
R(5,5,6,6,7,9,9)	248 461 233	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 2 quadruplings -> 248,461,232 vertices
R(5,5,6,6,7,9,10)	336 029 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 3 quadruplings -> 336,029,120 vertices
R(5,5,6,6,7,10,10)	413 883 233	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 2 quadruplings -> 413,883,232 vertices
R(5,5,6,6,8,8,8)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(5,5,6,6,8,8,9)	248 461 233	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 15,528,827 vertices; then 2 quadruplings -> 248,461,232 vertices
R(5,5,6,6,8,8,10)	336 029 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 3 quadruplings -> 336,029,120 vertices
R(5,5,6,6,8,9,9)	336 029 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 3 quadruplings -> 336,029,120 vertices
R(5,5,6,6,8,9,10)	413 883 233	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 2 quadruplings -> 413,883,232 vertices
R(5,5,6,6,8,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(5,5,6,6,9,9,9)	405 650 689	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 4 quadruplings -> 405,650,688 vertices
R(5,5,6,6,9,9,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(5,5,6,6,9,10,10)	676 009 361	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 42,250,585 vertices; then 2 quadruplings -> 676,009,360 vertices
R(5,5,6,6,10,10,10)	933 283 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 3 quadruplings -> 933,283,712 vertices
R(5,5,6,7,7,7,7)	124 151 661	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 31,037,915 vertices; then 1 quadrupling -> 124,151,660 vertices
R(5,5,6,7,7,7,8)	171 025 245	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices; then 1 quadrupling -> 171,025,244 vertices
R(5,5,6,7,7,7,9)	204 924 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 2 quadruplings -> 204,924,416 vertices
R(5,5,6,7,7,7,10)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(5,5,6,7,7,8,8)	204 924 417	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 2 quadruplings -> 204,924,416 vertices
R(5,5,6,7,7,8,9)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,6,7,7,8,10)	336 029 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 3 quadruplings -> 336,029,120 vertices
R(5,5,6,7,7,9,9)	340 171 101	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 1 quadrupling -> 340,171,100 vertices
R(5,5,6,7,7,9,10)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(5,5,6,7,7,10,10)	571 174 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 2 quadruplings -> 571,174,016 vertices
R(5,5,6,7,8,8,8)	251 452 673	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 4 quadruplings -> 251,452,672 vertices
R(5,5,6,7,8,8,9)	336 029 121	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 5,250,455 vertices; then 3 quadruplings -> 336,029,120 vertices
R(5,5,6,7,8,8,10)	405 650 689	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 4 quadruplings -> 405,650,688 vertices
R(5,5,6,7,8,9,9)	469 631 937	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 2 quadruplings -> 469,631,936 vertices
R(5,5,6,7,8,9,10)	571 174 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 2 quadruplings -> 571,174,016 vertices
R(5,5,6,7,8,10,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(5,5,6,7,9,9,9)	571 174 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 2 quadruplings -> 571,174,016 vertices
R(5,5,6,7,9,9,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(5,5,6,7,9,10,10)	933 283 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 3 quadruplings -> 933,283,712 vertices
R(5,5,6,7,10,10,10)	1 109 849 345	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 4 quadruplings -> 1,109,849,344 vertices
R(5,5,6,8,8,8,8)	324 315 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 5 quadruplings -> 324,315,136 vertices
R(5,5,6,8,8,8,9)	405 650 689	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 1,584,573 vertices; then 4 quadruplings -> 405,650,688 vertices
R(5,5,6,8,8,8,10)	502 767 617	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 5 quadruplings -> 502,767,616 vertices
R(5,5,6,8,8,9,9)	571 174 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 2 quadruplings -> 571,174,016 vertices
R(5,5,6,8,8,9,10)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(5,5,6,8,8,10,10)	888 610 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 5 quadruplings -> 888,610,816 vertices
R(5,5,6,8,9,9,9)	685 526 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 4 quadruplings -> 685,526,272 vertices
R(5,5,6,8,9,9,10)	933 283 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 3 quadruplings -> 933,283,712 vertices
R(5,5,6,8,9,10,10)	1 109 849 345	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 4 quadruplings -> 1,109,849,344 vertices
R(5,5,6,8,10,10,10)	1 367 412 737	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 5 quadruplings -> 1,367,412,736 vertices
R(5,5,6,9,9,9,9)	933 283 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 3 quadruplings -> 933,283,712 vertices
R(5,5,6,9,9,9,10)	1 277 303 457	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 2 quadruplings -> 1,277,303,456 vertices
R(5,5,6,9,9,10,10)	1 553 477 217	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 97,092,326 vertices; then 2 quadruplings -> 1,553,477,216 vertices
R(5,5,6,9,10,10,10)	1 864 460 033	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 4 quadruplings -> 1,864,460,032 vertices
R(5,5,6,10,10,10,10)	2 416 807 937	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 5 quadruplings -> 2,416,807,936 vertices
R(5,5,7,7,7,7,7)	171 025 245	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices; then 1 quadrupling -> 171,025,244 vertices
R(5,5,7,7,7,7,8)	236 901 641	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 59,225,410 vertices; then 1 quadrupling -> 236,901,640 vertices
R(5,5,7,7,7,7,9)	304 044 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 2 quadruplings -> 304,044,832 vertices
R(5,5,7,7,7,7,10)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(5,5,7,7,7,8,8)	304 044 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 2 quadruplings -> 304,044,832 vertices
R(5,5,7,7,7,8,9)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(5,5,7,7,7,8,10)	496 606 641	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 31,037,915 vertices; then 2 quadruplings -> 496,606,640 vertices
R(5,5,7,7,7,9,9)	496 606 641	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 31,037,915 vertices; then 2 quadruplings -> 496,606,640 vertices
R(5,5,7,7,7,9,10)	673 973 505	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 3 quadruplings -> 673,973,504 vertices
R(5,5,7,7,7,10,10)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(5,5,7,7,8,8,8)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(5,5,7,7,8,8,9)	496 606 641	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 31,037,915 vertices; then 2 quadruplings -> 496,606,640 vertices
R(5,5,7,7,8,8,10)	673 973 505	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 3 quadruplings -> 673,973,504 vertices
R(5,5,7,7,8,9,9)	673 973 505	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 3 quadruplings -> 673,973,504 vertices
R(5,5,7,7,8,9,10)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(5,5,7,7,8,10,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,5,7,7,9,9,9)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(5,5,7,7,9,9,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,5,7,7,9,10,10)	1 360 684 401	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 2 quadruplings -> 1,360,684,400 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,5,7,7,10,10,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,5,7,8,8,8,8)	461 458 177	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 4 quadruplings -> 461,458,176 vertices
R(5,5,7,8,8,8,9)	673 973 505	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 10,530,836 vertices; then 3 quadruplings -> 673,973,504 vertices
R(5,5,7,8,8,8,10)	810 786 049	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 4 quadruplings -> 810,786,048 vertices
R(5,5,7,8,8,9,9)	833 072 033	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 2 quadruplings -> 833,072,032 vertices
R(5,5,7,8,8,9,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,5,7,8,8,10,10)	1 297 260 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 6 quadruplings -> 1,297,260,544 vertices
R(5,5,7,8,9,9,9)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,5,7,8,9,9,10)	1 360 684 401	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 2 quadruplings -> 1,360,684,400 vertices
R(5,5,7,8,9,10,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,5,7,8,10,10,10)	2 233 923 329	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 4 quadruplings -> 2,233,923,328 vertices
R(5,5,7,9,9,9,9)	1 360 684 401	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 2 quadruplings -> 1,360,684,400 vertices
R(5,5,7,9,9,9,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,5,7,9,9,10,10)	2 265 756 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 2 quadruplings -> 2,265,756,832 vertices
R(5,5,7,9,10,10,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(5,5,7,10,10,10,10)	3 554 443 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 6 quadruplings -> 3,554,443,264 vertices
R(5,5,8,8,8,8,8)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices
R(5,5,8,8,8,8,9)	810 786 049	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 4 quadruplings -> 810,786,048 vertices
R(5,5,8,8,8,8,10)	1 005 810 689	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 5 quadruplings -> 1,005,810,688 vertices
R(5,5,8,8,8,9,9)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,5,8,8,8,9,10)	1 297 260 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 6 quadruplings -> 1,297,260,544 vertices
R(5,5,8,8,8,10,10)	1 658 545 153	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(5,5,8,8,9,9,9)	1 360 684 401	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 85,042,775 vertices; then 2 quadruplings -> 1,360,684,400 vertices
R(5,5,8,8,9,9,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,5,8,8,9,10,10)	2 233 923 329	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 4 quadruplings -> 2,233,923,328 vertices
R(5,5,8,8,10,10,10)	2 742 105 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 5 quadruplings -> 2,742,105,088 vertices
R(5,5,8,9,9,9,9)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,5,8,9,9,9,10)	2 265 756 833	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 2 quadruplings -> 2,265,756,832 vertices
R(5,5,8,9,9,10,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(5,5,8,9,10,10,10)	3 554 443 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 6 quadruplings -> 3,554,443,264 vertices
R(5,5,8,10,10,10,10)	4 332 470 273	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 7 quadruplings -> 4,332,470,272 vertices
R(5,5,9,9,9,9,9)	2 233 923 329	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 4 quadruplings -> 2,233,923,328 vertices
R(5,5,9,9,9,9,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(5,5,9,9,9,10,10)	3 700 736 241	Our bricks + published	[template (5,5,3 ; 99) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 231,296,015 vertices; then 2 quadruplings -> 3,700,736,240 vertices
R(5,5,9,9,10,10,10)	5 109 213 825	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 3 quadruplings -> 5,109,213,824 vertices
R(5,5,9,10,10,10,10)	6 075 820 289	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 4 quadruplings -> 6,075,820,288 vertices
R(5,5,10,10,10,10,10)	7 457 840 129	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 5 quadruplings -> 7,457,840,128 vertices
R(5,6,6,6,6,6,6)	49 491 905	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 49,491,904 vertices
R(5,6,6,6,6,6,7)	73 686 069	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 18,421,517 vertices; then 1 quadrupling -> 73,686,068 vertices
R(5,6,6,6,6,6,8)	87 985 617	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 21,996,405 – 3 = 87,985,617; with R(6,6,6,6,6,6) >= 5,499,102
R(5,6,6,6,6,6,9)	99 092 853	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (5,5,5 ; 453)]] -> 24,773,213 vertices; then 1 quadrupling -> 99,092,852 vertices
R(5,6,6,6,6,6,10)	127 565 973	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 31,891,493 vertices; then 1 quadrupling -> 127,565,972 vertices
R(5,6,6,6,6,7,7)	99 618 314	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 99,618,313 vertices
R(5,6,6,6,6,7,8)	148 316 825	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 1 quadrupling -> 148,316,824 vertices
R(5,6,6,6,6,7,9)	177 099 233	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 2 quadruplings -> 177,099,232 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,6,6,6,6,7,10)	177 099 233	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 2 quadruplings -> 177,099,232 vertices (by monotonicity, from (5,6,6,6,6,7,9))
R(5,6,6,6,6,8,8)	177 099 233	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 2 quadruplings -> 177,099,232 vertices
R(5,6,6,6,6,8,9)	205 757 541	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices; then 1 quadrupling -> 205,757,540 vertices
R(5,6,6,6,6,8,10)	245 682 897	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 15,355,181 vertices; then 2 quadruplings -> 245,682,896 vertices
R(5,6,6,6,6,9,9)	276 674 549	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]]] -> 276,674,548 vertices
R(5,6,6,6,6,9,10)	403 536 213	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices; then 1 quadrupling -> 403,536,212 vertices
R(5,6,6,6,6,10,10)	491 865 873	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 2 quadruplings -> 491,865,872 vertices
R(5,6,6,6,7,7,7)	148 316 825	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 1 quadrupling -> 148,316,824 vertices
R(5,6,6,6,7,7,8)	177 099 233	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 2 quadruplings -> 177,099,232 vertices
R(5,6,6,6,7,7,9)	200 530 817	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 3 quadruplings -> 200,530,816 vertices
R(5,6,6,6,7,7,10)	256 767 401	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 64,191,850 vertices; then 1 quadrupling -> 256,767,400 vertices
R(5,6,6,6,7,8,8)	205 757 541	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices; then 1 quadrupling -> 205,757,540 vertices
R(5,6,6,6,7,8,9)	256 767 401	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 64,191,850 vertices; then 1 quadrupling -> 256,767,400 vertices
R(5,6,6,6,7,8,10)	342 268 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 21,391,751 vertices; then 2 quadruplings -> 342,268,016 vertices
R(5,6,6,6,7,9,9)	403 536 213	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices; then 1 quadrupling -> 403,536,212 vertices
R(5,6,6,6,7,9,10)	491 865 873	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 2 quadruplings -> 491,865,872 vertices
R(5,6,6,6,7,10,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(5,6,6,6,8,8,8)	245 682 897	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 15,355,181 vertices; then 2 quadruplings -> 245,682,896 vertices
R(5,6,6,6,8,8,9)	342 268 017	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 21,391,751 vertices; then 2 quadruplings -> 342,268,016 vertices
R(5,6,6,6,8,8,10)	408 684 225	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 6,385,691 vertices; then 3 quadruplings -> 408,684,224 vertices
R(5,6,6,6,8,9,9)	491 865 873	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 2 quadruplings -> 491,865,872 vertices
R(5,6,6,6,8,9,10)	546 525 633	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 3 quadruplings -> 546,525,632 vertices
R(5,6,6,6,8,10,10)	697 748 433	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(5,6,6,6,9,9,9)	542 480 733	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)]] -> 135,620,183 vertices; then 1 quadrupling -> 542,480,732 vertices
R(5,6,6,6,9,9,10)	698 352 813	Our bricks + published	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 174,588,203 vertices; then 1 quadrupling -> 698,352,812 vertices
R(5,6,6,6,9,10,10)	933 283 713	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 3 quadruplings -> 933,283,712 vertices
R(5,6,6,6,10,10,10)	1 135 074 945	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 17,735,546 vertices; then 3 quadruplings -> 1,135,074,944 vertices
R(5,6,6,7,7,7,7)	200 510 753	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]]] -> 200,510,752 vertices
R(5,6,6,7,7,7,8)	298 533 797	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices; then 1 quadrupling -> 298,533,796 vertices
R(5,6,6,7,7,7,9)	356 463 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 2 quadruplings -> 356,463,568 vertices
R(5,6,6,7,7,7,10)	376 873 217	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 3 quadruplings -> 376,873,216 vertices
R(5,6,6,7,7,8,8)	356 463 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 2 quadruplings -> 356,463,568 vertices
R(5,6,6,7,7,8,9)	414 152 993	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 1 quadrupling -> 414,152,992 vertices
R(5,6,6,7,7,8,10)	494 515 553	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 2 quadruplings -> 494,515,552 vertices
R(5,6,6,7,7,9,9)	556 896 200	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]]] -> 556,896,199 vertices
R(5,6,6,7,7,9,10)	812 245 961	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 1 quadrupling -> 812,245,960 vertices
R(5,6,6,7,7,10,10)	990 037 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 2 quadruplings -> 990,037,696 vertices
R(5,6,6,7,8,8,8)	414 152 993	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 1 quadrupling -> 414,152,992 vertices
R(5,6,6,7,8,8,9)	494 515 553	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 2 quadruplings -> 494,515,552 vertices
R(5,6,6,7,8,8,10)	608 476 289	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 3 quadruplings -> 608,476,288 vertices
R(5,6,6,7,8,9,9)	812 245 961	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 1 quadrupling -> 812,245,960 vertices
R(5,6,6,7,8,9,10)	990 037 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 2 quadruplings -> 990,037,696 vertices
R(5,6,6,7,8,10,10)	1 059 396 617	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(5,6,6,7,9,9,9)	990 037 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 2 quadruplings -> 990,037,696 vertices
R(5,6,6,7,9,9,10)	1 025 173 569	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 3 quadruplings -> 1,025,173,568 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,6,6,7,9,10,10)	1 404 442 337	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(5,6,6,7,10,10,10)	1 655 532 929	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 3 quadruplings -> 1,655,532,928 vertices
R(5,6,6,8,8,8,8)	494 515 553	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 2 quadruplings -> 494,515,552 vertices
R(5,6,6,8,8,8,9)	608 476 289	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 3 quadruplings -> 608,476,288 vertices
R(5,6,6,8,8,8,10)	802 123 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 4 quadruplings -> 802,123,264 vertices
R(5,6,6,8,8,9,9)	990 037 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 2 quadruplings -> 990,037,696 vertices
R(5,6,6,8,8,9,10)	1 126 401 525	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 1 quadrupling -> 1,126,401,524 vertices
R(5,6,6,8,8,10,10)	1 404 442 337	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(5,6,6,8,9,9,9)	1 126 401 525	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 1 quadrupling -> 1,126,401,524 vertices
R(5,6,6,8,9,9,10)	1 364 199 153	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 85,262,447 vertices; then 2 quadruplings -> 1,364,199,152 vertices
R(5,6,6,8,9,10,10)	1 655 532 929	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 3 quadruplings -> 1,655,532,928 vertices
R(5,6,6,8,10,10,10)	2 186 102 529	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 4 quadruplings -> 2,186,102,528 vertices
R(5,6,6,9,9,9,9)	1 514 640 317	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]]] -> 1,514,640,316 vertices
R(5,6,6,9,9,9,10)	2 209 112 973	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 552,278,243 vertices; then 1 quadrupling -> 2,209,112,972 vertices
R(5,6,6,9,9,10,10)	2 692 693 905	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 2 quadruplings -> 2,692,693,904 vertices
R(5,6,6,9,10,10,10)	2 881 288 937	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(5,6,6,10,10,10,10)	3 819 768 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,6,7,7,7,7,7)	298 533 797	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices; then 1 quadrupling -> 298,533,796 vertices
R(5,6,7,7,7,7,8)	356 463 569	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 2 quadruplings -> 356,463,568 vertices
R(5,6,7,7,7,7,9)	403 610 945	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 3 quadruplings -> 403,610,944 vertices
R(5,6,7,7,7,7,10)	516 825 341	Our bricks + published	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 129,206,335 vertices; then 1 quadrupling -> 516,825,340 vertices
R(5,6,7,7,7,8,8)	414 152 993	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 1 quadrupling -> 414,152,992 vertices
R(5,6,7,7,7,8,9)	516 825 341	Our bricks + published	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 129,206,335 vertices; then 1 quadrupling -> 516,825,340 vertices
R(5,6,7,7,7,8,10)	684 100 977	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices; then 2 quadruplings -> 684,100,976 vertices
R(5,6,7,7,7,9,9)	812 245 961	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 1 quadrupling -> 812,245,960 vertices
R(5,6,7,7,7,9,10)	990 037 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 2 quadruplings -> 990,037,696 vertices
R(5,6,7,7,7,10,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,6,7,7,8,8,8)	514 489 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 3 quadruplings -> 514,489,664 vertices
R(5,6,7,7,8,8,9)	684 100 977	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices; then 2 quadruplings -> 684,100,976 vertices
R(5,6,7,7,8,8,10)	819 697 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 3 quadruplings -> 819,697,664 vertices
R(5,6,7,7,8,9,9)	990 037 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 2 quadruplings -> 990,037,696 vertices
R(5,6,7,7,8,9,10)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,6,7,7,8,10,10)	1 404 442 337	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(5,6,7,7,9,9,9)	1 100 057 921	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 3 quadruplings -> 1,100,057,920 vertices
R(5,6,7,7,9,9,10)	1 405 658 861	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 351,414,715 vertices; then 1 quadrupling -> 1,405,658,860 vertices
R(5,6,7,7,9,10,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,6,7,7,10,10,10)	2 284 696 065	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 3 quadruplings -> 2,284,696,064 vertices
R(5,6,7,8,8,8,8)	609 764 353	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 5 quadruplings -> 609,764,352 vertices
R(5,6,7,8,8,8,9)	819 697 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 3 quadruplings -> 819,697,664 vertices
R(5,6,7,8,8,8,10)	1 005 810 689	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 5 quadruplings -> 1,005,810,688 vertices
R(5,6,7,8,8,9,9)	1 126 401 525	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 1 quadrupling -> 1,126,401,524 vertices
R(5,6,7,8,8,9,10)	1 399 398 465	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 3 quadruplings -> 1,399,398,464 vertices
R(5,6,7,8,8,10,10)	1 658 545 153	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(5,6,7,8,9,9,9)	1 405 658 861	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 351,414,715 vertices; then 1 quadrupling -> 1,405,658,860 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,6,7,8,9,9,10)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,6,7,8,9,10,10)	2 284 696 065	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 3 quadruplings -> 2,284,696,064 vertices
R(5,6,7,8,10,10,10)	2 742 105 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 5 quadruplings -> 2,742,105,088 vertices
R(5,6,7,9,9,9,9)	2 209 112 973	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 552,278,243 vertices; then 1 quadrupling -> 2,209,112,972 vertices
R(5,6,7,9,9,9,10)	2 692 693 905	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 2 quadruplings -> 2,692,693,904 vertices
R(5,6,7,9,9,10,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(5,6,7,9,10,10,10)	3 819 768 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,6,7,10,10,10,10)	4 439 397 377	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 5 quadruplings -> 4,439,397,376 vertices
R(5,6,8,8,8,8,8)	732 000 257	Published bricks + theorems	[template (7,7,3;472)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 178,711 vertices; then 6 quadruplings -> 732,000,256 vertices
R(5,6,8,8,8,8,9)	1 005 810 689	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 5 quadruplings -> 1,005,810,688 vertices
R(5,6,8,8,8,8,10)	1 297 260 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 6 quadruplings -> 1,297,260,544 vertices
R(5,6,8,8,8,9,9)	1 364 199 153	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 85,262,447 vertices; then 2 quadruplings -> 1,364,199,152 vertices
R(5,6,8,8,8,9,10)	1 658 545 153	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 5 quadruplings -> 1,658,545,152 vertices
R(5,6,8,8,8,10,10)	2 011 070 465	Our bricks + published	[template (7,7,3;472)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 490,984 vertices; then 6 quadruplings -> 2,011,070,464 vertices
R(5,6,8,8,9,9,9)	1 878 527 745	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 3 quadruplings -> 1,878,527,744 vertices
R(5,6,8,8,9,9,10)	2 284 696 065	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 3 quadruplings -> 2,284,696,064 vertices
R(5,6,8,8,9,10,10)	2 742 105 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 5 quadruplings -> 2,742,105,088 vertices
R(5,6,8,8,10,10,10)	3 554 443 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 6 quadruplings -> 3,554,443,264 vertices
R(5,6,8,9,9,9,9)	2 692 693 905	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 2 quadruplings -> 2,692,693,904 vertices
R(5,6,8,9,9,9,10)	2 991 882 177	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 3 quadruplings -> 2,991,882,176 vertices
R(5,6,8,9,9,10,10)	3 819 768 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,6,8,9,10,10,10)	4 439 397 377	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 5 quadruplings -> 4,439,397,376 vertices
R(5,6,8,10,10,10,10)	5 469 650 945	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (3,6,6,3 ; 707) (ours)] x [K2]] -> 1,335,364 vertices; then 6 quadruplings -> 5,469,650,944 vertices
R(5,6,9,9,9,9,9)	2 969 734 301	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (5,5,5 ; 453)]] -> 742,433,575 vertices; then 1 quadrupling -> 2,969,734,300 vertices
R(5,6,9,9,9,9,10)	3 823 034 021	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 955,758,505 vertices; then 1 quadrupling -> 3,823,034,020 vertices
R(5,6,9,9,9,10,10)	5 109 213 825	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 3 quadruplings -> 5,109,213,824 vertices
R(5,6,9,9,10,10,10)	6 213 908 865	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 97,092,326 vertices; then 3 quadruplings -> 6,213,908,864 vertices
R(5,6,9,10,10,10,10)	7 457 840 129	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 5 quadruplings -> 7,457,840,128 vertices
R(5,6,10,10,10,10,10)	9 667 231 745	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]] -> 2,360,164 vertices; then 6 quadruplings -> 9,667,231,744 vertices
R(5,7,7,7,7,7,7)	402 164 222	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]]] -> 402,164,221 vertices
R(5,7,7,7,7,7,8)	596 688 101	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 149,172,025 vertices; then 1 quadrupling -> 596,688,100 vertices
R(5,7,7,7,7,7,9)	714 958 625	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices; then 2 quadruplings -> 714,958,624 vertices
R(5,7,7,7,7,7,10)	758 556 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 3 quadruplings -> 758,556,544 vertices
R(5,7,7,7,7,8,8)	714 958 625	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices; then 2 quadruplings -> 714,958,624 vertices
R(5,7,7,7,7,8,9)	833 614 289	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,8,8 ; 940)]] -> 208,403,572 vertices; then 1 quadrupling -> 833,614,288 vertices
R(5,7,7,7,7,8,10)	995 365 649	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices; then 2 quadruplings -> 995,365,648 vertices
R(5,7,7,7,7,9,9)	1 120 929 062	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]]] -> 1,120,929,061 vertices
R(5,7,7,7,7,9,10)	1 634 903 981	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 1 quadrupling -> 1,634,903,980 vertices
R(5,7,7,7,7,10,10)	1 992 762 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 2 quadruplings -> 1,992,762,784 vertices
R(5,7,7,7,8,8,8)	833 614 289	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,8,8 ; 940)]] -> 208,403,572 vertices; then 1 quadrupling -> 833,614,288 vertices
R(5,7,7,7,8,8,9)	995 365 649	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices; then 2 quadruplings -> 995,365,648 vertices
R(5,7,7,7,8,8,10)	1 216 179 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 3 quadruplings -> 1,216,179,328 vertices
R(5,7,7,7,8,9,9)	1 634 903 981	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 1 quadrupling -> 1,634,903,980 vertices
R(5,7,7,7,8,9,10)	1 992 762 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 2 quadruplings -> 1,992,762,784 vertices
R(5,7,7,7,8,10,10)	2 120 292 089	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]

Parameters	Bound (R >=)	Origin	Construction
R(5,7,7,7,9,9,9)	1 992 762 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 2 quadruplings -> 1,992,762,784 vertices
R(5,7,7,7,9,9,10)	2 063 490 305	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 3 quadruplings -> 2,063,490,304 vertices
R(5,7,7,7,9,10,10)	2 826 884 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [atome (10,10 ; 797)]]
R(5,7,7,7,10,10,10)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(5,7,7,8,8,8,8)	995 365 649	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices; then 2 quadruplings -> 995,365,648 vertices
R(5,7,7,8,8,8,9)	1 216 179 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 19,002,802 vertices; then 3 quadruplings -> 1,216,179,328 vertices
R(5,7,7,8,8,8,10)	1 614 443 777	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 4 quadruplings -> 1,614,443,776 vertices
R(5,7,7,8,8,9,9)	1 992 762 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 2 quadruplings -> 1,992,762,784 vertices
R(5,7,7,8,8,9,10)	2 267 244 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 1 quadrupling -> 2,267,244,088 vertices
R(5,7,7,8,8,10,10)	2 826 884 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [atome (10,10 ; 797)]]
R(5,7,7,8,9,9,9)	2 267 244 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 1 quadrupling -> 2,267,244,088 vertices
R(5,7,7,8,9,9,10)	2 745 888 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 2 quadruplings -> 2,745,888,016 vertices
R(5,7,7,8,9,10,10)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(5,7,7,8,10,10,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(5,7,7,9,9,9,9)	3 048 699 092	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]]] -> 3,048,699,091 vertices
R(5,7,7,9,9,9,10)	4 446 547 901	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices; then 1 quadrupling -> 4,446,547,900 vertices
R(5,7,7,9,9,10,10)	5 419 909 505	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 2 quadruplings -> 5,419,909,504 vertices
R(5,7,7,9,10,10,10)	5 766 679 889	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(5,7,7,10,10,10,10)	7 688 509 265	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,7,8,8,8,8,8)	1 158 549 505	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 5 quadruplings -> 1,158,549,504 vertices
R(5,7,8,8,8,8,9)	1 614 443 777	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 4 quadruplings -> 1,614,443,776 vertices
R(5,7,8,8,8,8,10)	1 845 832 705	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 5 quadruplings -> 1,845,832,704 vertices
R(5,7,8,8,8,9,9)	2 267 244 089	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 1 quadrupling -> 2,267,244,088 vertices
R(5,7,8,8,8,9,10)	2 745 888 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 2 quadruplings -> 2,745,888,016 vertices
R(5,7,8,8,8,10,10)	3 243 144 193	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 5 quadruplings -> 3,243,144,192 vertices
R(5,7,8,8,9,9,9)	2 745 888 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 2 quadruplings -> 2,745,888,016 vertices
R(5,7,8,8,9,9,10)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(5,7,8,8,9,10,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(5,7,8,8,10,10,10)	5 189 042 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 7 quadruplings -> 5,189,042,176 vertices
R(5,7,8,9,9,9,9)	4 446 547 901	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices; then 1 quadrupling -> 4,446,547,900 vertices
R(5,7,8,9,9,9,10)	5 419 909 505	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 2 quadruplings -> 5,419,909,504 vertices
R(5,7,8,9,9,10,10)	5 766 679 889	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(5,7,8,9,10,10,10)	7 688 509 265	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,7,8,10,10,10,10)	8 935 693 313	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 5 quadruplings -> 8,935,693,312 vertices
R(5,7,9,9,9,9,9)	5 419 909 505	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 2 quadruplings -> 5,419,909,504 vertices
R(5,7,9,9,9,9,10)	5 612 168 705	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 87,690,136 vertices; then 3 quadruplings -> 5,612,168,704 vertices
R(5,7,9,9,9,10,10)	7 688 509 265	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,7,9,9,10,10,10)	9 063 027 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 3 quadruplings -> 9,063,027,328 vertices
R(5,7,9,10,10,10,10)	11 967 528 705	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 4 quadruplings -> 11,967,528,704 vertices
R(5,7,10,10,10,10,10)	14 217 773 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 7 quadruplings -> 14,217,773,056 vertices
R(5,8,8,8,8,8,8)	1 371 555 233	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 85,722,202 vertices; then 2 quadruplings -> 1,371,555,232 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,8,8,8,8,8,9)	1 845 832 705	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 5 quadruplings -> 1,845,832,704 vertices
R(5,8,8,8,8,8,10)	2 439 057 409	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 6 quadruplings -> 2,439,057,408 vertices
R(5,8,8,8,8,9,9)	2 745 888 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 2 quadruplings -> 2,745,888,016 vertices
R(5,8,8,8,8,9,10)	3 243 144 193	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 5 quadruplings -> 3,243,144,192 vertices
R(5,8,8,8,8,10,10)	4 023 242 753	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 6 quadruplings -> 4,023,242,752 vertices
R(5,8,8,8,9,9,9)	3 332 288 129	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 3 quadruplings -> 3,332,288,128 vertices
R(5,8,8,8,9,9,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(5,8,8,8,9,10,10)	5 189 042 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 7 quadruplings -> 5,189,042,176 vertices
R(5,8,8,8,10,10,10)	6 634 180 609	Our bricks	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 6 quadruplings -> 6,634,180,608 vertices
R(5,8,8,9,9,9,9)	5 419 909 505	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 2 quadruplings -> 5,419,909,504 vertices
R(5,8,8,9,9,9,10)	6 166 326 329	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 1,541,581,582 vertices; then 1 quadrupling -> 6,166,326,328 vertices
R(5,8,8,9,9,10,10)	7 688 509 265	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,8,8,9,10,10,10)	8 935 693 313	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 5 quadruplings -> 8,935,693,312 vertices
R(5,8,8,10,10,10,10)	10 968 420 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 6 quadruplings -> 10,968,420,352 vertices
R(5,8,9,9,9,9,9)	6 166 326 329	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 1,541,581,582 vertices; then 1 quadrupling -> 6,166,326,328 vertices
R(5,8,9,9,9,9,10)	7 514 110 977	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 4 quadruplings -> 7,514,110,976 vertices
R(5,8,9,9,9,10,10)	9 063 027 329	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 3 quadruplings -> 9,063,027,328 vertices
R(5,8,9,9,10,10,10)	11 967 528 705	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 4 quadruplings -> 11,967,528,704 vertices
R(5,8,9,10,10,10,10)	14 217 773 057	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 7 quadruplings -> 14,217,773,056 vertices
R(5,8,10,10,10,10,10)	17 329 881 089	Published bricks + theorems	[template (7,7,3;472)] x [atom (9,9 ; 562)] -> 264,433 vertices; then 8 quadruplings -> 17,329,881,088 vertices
R(5,9,9,9,9,9,9)	8 291 684 912	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]]] -> 8,291,684,911 vertices
R(5,9,9,9,9,9,10)	12 093 477 461	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 3,023,369,365 vertices; then 1 quadrupling -> 12,093,477,460 vertices
R(5,9,9,9,9,10,10)	14 740 773 185	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 921,298,324 vertices; then 2 quadruplings -> 14,740,773,184 vertices
R(5,9,9,9,10,10,10)	15 683 900 129	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(5,9,9,10,10,10,10)	20 910 786 545	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(5,9,10,10,10,10,10)	24 303 281 153	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 5 quadruplings -> 24,303,281,152 vertices
R(5,10,10,10,10,10,10)	29 831 360 513	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 6 quadruplings -> 29,831,360,512 vertices
R(6,6,6,6,6,6,6)	60 490 106	Published bricks + theorems	[template (6,3;12)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 60,490,105 vertices
R(6,6,6,6,6,6,7)	82 486 507	Published bricks + theorems	[template (7,3;16)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 82,486,506 vertices
R(6,6,6,6,6,6,8)	104 482 908	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 104,482,907 vertices
R(6,6,6,6,6,6,9)	137 477 509	Published bricks + theorems	[template (9,3 ; 26)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 137,477,508 vertices
R(6,6,6,6,6,6,10)	159 473 910	Published bricks + theorems	[template (10,3 ; 30)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 159,473,909 vertices
R(6,6,6,6,6,7,7)	124 499 637	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]]] -> 31,124,909 vertices; then 1 quadrupling -> 124,499,636 vertices
R(6,6,6,6,6,7,8)	148 316 825	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 1 quadrupling -> 148,316,824 vertices (by monotonicity, from (5,6,6,6,6,7,8))
R(6,6,6,6,6,7,9)	197 967 617	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 49,491,904 vertices; then 1 quadrupling -> 197,967,616 vertices
R(6,6,6,6,6,7,10)	215 394 597	Our bricks + published	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K3]]] -> 53,848,649 vertices; then 1 quadrupling -> 215,394,596 vertices
R(6,6,6,6,6,8,8)	177 099 233	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 2 quadruplings -> 177,099,232 vertices (by monotonicity, from (5,6,6,6,6,8,8))
R(6,6,6,6,6,8,9)	294 744 273	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 18,421,517 vertices; then 2 quadruplings -> 294,744,272 vertices
R(6,6,6,6,6,8,10)	294 744 273	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 18,421,517 vertices; then 2 quadruplings -> 294,744,272 vertices (by monotonicity, from (6,6,6,6,6,8,9))
R(6,6,6,6,6,9,9)	351 942 465	Published (deduced)	quadrupling (Theorem 7 [XXER]): 4 x 87,985,617 – 3 = 351,942,465; with R(6,6,6,6,6,6) >= 5,499,102
R(6,6,6,6,6,9,10)	403 536 213	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices; then 1 quadrupling -> 403,536,212 vertices (by monotonicity, from (5,6,6,6,6,9,10))

Parameters	Bound (R >=)	Origin	Construction
R(6,6,6,6,6,10,10)	491 865 873	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 2 quadruplings -> 491,865,872 vertices (by monotonicity, from (5,6,6,6,6,10,10))
R(6,6,6,6,7,7,7)	166 030 522	Published bricks + theorems	[template (7,3;16)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 166,030,521 vertices
R(6,6,6,6,7,7,8)	250 595 417	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]]] -> 62,648,854 vertices; then 1 quadrupling -> 250,595,416 vertices
R(6,6,6,6,7,7,9)	294 744 273	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 18,421,517 vertices; then 2 quadruplings -> 294,744,272 vertices
R(6,6,6,6,7,7,10)	320 992 339	Published bricks + theorems	[template (10,3 ; 30)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 320,992,338 vertices
R(6,6,6,6,7,8,8)	255 609 345	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 3,993,896 vertices; then 3 quadruplings -> 255,609,344 vertices
R(6,6,6,6,7,8,9)	398 473 253	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]]] -> 99,618,313 vertices; then 1 quadrupling -> 398,473,252 vertices
R(6,6,6,6,7,8,10)	443 056 449	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 3 quadruplings -> 443,056,448 vertices
R(6,6,6,6,7,9,9)	593 267 297	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 2 quadruplings -> 593,267,296 vertices
R(6,6,6,6,7,9,10)	708 396 929	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 3 quadruplings -> 708,396,928 vertices
R(6,6,6,6,7,10,10)	708 396 929	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 3 quadruplings -> 708,396,928 vertices (by monotonicity, from (6,6,6,6,7,9,10))
R(6,6,6,6,8,8,8)	291 748 428	Published bricks + theorems	[template (8,3 ; 20)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]]] -> 291,748,427 vertices
R(6,6,6,6,8,8,9)	593 267 297	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 2 quadruplings -> 593,267,296 vertices
R(6,6,6,6,8,8,10)	593 267 297	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 2 quadruplings -> 593,267,296 vertices (by monotonicity, from (6,6,6,6,8,8,9))
R(6,6,6,6,8,9,9)	708 396 929	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 3 quadruplings -> 708,396,928 vertices
R(6,6,6,6,8,9,10)	823 030 161	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices; then 2 quadruplings -> 823,030,160 vertices
R(6,6,6,6,8,10,10)	828 576 252	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [[modele (6,6,3;235)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]]
R(6,6,6,6,9,9,9)	823 030 161	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices; then 2 quadruplings -> 823,030,160 vertices
R(6,6,6,6,9,9,10)	1 106 698 193	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]]] -> 276,674,548 vertices; then 1 quadrupling -> 1,106,698,192 vertices
R(6,6,6,6,9,10,10)	1 614 144 849	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices; then 2 quadruplings -> 1,614,144,848 vertices
R(6,6,6,6,10,10,10)	1 614 144 849	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices; then 2 quadruplings -> 1,614,144,848 vertices (by monotonicity, from (6,6,6,6,9,10,10))
R(6,6,6,7,7,7,7)	250 595 417	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [K3]]] -> 62,648,854 vertices; then 1 quadrupling -> 250,595,416 vertices
R(6,6,6,7,7,7,8)	298 533 797	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices; then 1 quadrupling -> 298,533,796 vertices (by monotonicity, from (5,6,6,7,7,7,8))
R(6,6,6,7,7,7,9)	443 056 449	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 3 quadruplings -> 443,056,448 vertices
R(6,6,6,7,7,7,10)	463 545 857	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 4 quadruplings -> 463,545,856 vertices
R(6,6,6,7,7,8,8)	443 056 449	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 6,922,757 vertices; then 3 quadruplings -> 443,056,448 vertices
R(6,6,6,7,7,8,9)	593 267 297	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 2 quadruplings -> 593,267,296 vertices
R(6,6,6,7,7,8,10)	608 476 289	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 3 quadruplings -> 608,476,288 vertices
R(6,6,6,7,7,9,9)	708 396 929	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 3 quadruplings -> 708,396,928 vertices
R(6,6,6,7,7,9,10)	812 245 961	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 1 quadrupling -> 812,245,960 vertices (by monotonicity, from (5,6,6,7,7,9,10))
R(6,6,6,7,7,10,10)	1 025 173 569	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 3 quadruplings -> 1,025,173,568 vertices
R(6,6,6,7,8,8,8)	463 545 857	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-17 (4,4;17)]] -> 1,810,726 vertices; then 4 quadruplings -> 463,545,856 vertices
R(6,6,6,7,8,8,9)	708 396 929	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [Paley-101 (6,6;101)]] -> 11,068,702 vertices; then 3 quadruplings -> 708,396,928 vertices
R(6,6,6,7,8,8,10)	802 123 265	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [(3,3,4;29)]] -> 3,133,294 vertices; then 4 quadruplings -> 802,123,264 vertices
R(6,6,6,7,8,9,9)	823 030 161	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices; then 2 quadruplings -> 823,030,160 vertices
R(6,6,6,7,8,9,10)	1 027 069 601	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 64,191,850 vertices; then 2 quadruplings -> 1,027,069,600 vertices
R(6,6,6,7,8,10,10)	1 261 424 385	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 4,927,439 vertices; then 4 quadruplings -> 1,261,424,384 vertices
R(6,6,6,7,9,9,9)	1 106 698 193	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]]] -> 276,674,548 vertices; then 1 quadrupling -> 1,106,698,192 vertices
R(6,6,6,7,9,9,10)	1 614 144 849	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices; then 2 quadruplings -> 1,614,144,848 vertices
R(6,6,6,7,9,10,10)	1 967 463 489	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 3 quadruplings -> 1,967,463,488 vertices
R(6,6,6,7,10,10,10)	2 186 102 529	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 4 quadruplings -> 2,186,102,528 vertices
R(6,6,6,8,8,8,8)	494 515 553	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 2 quadruplings -> 494,515,552 vertices (by monotonicity, from (5,6,6,8,8,8,8))

Parameters	Bound (R >=)	Origin	Construction
R(6,6,6,8,8,8,9)	823 030 161	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices; then 2 quadruplings -> 823,030,160 vertices
R(6,6,6,8,8,8,10)	823 030 161	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 51,439,385 vertices; then 2 quadruplings -> 823,030,160 vertices (by monotonicity, from (6,6,6,8,8,8,9))
R(6,6,6,8,8,9,9)	1 027 069 601	Our bricks + published	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 64,191,850 vertices; then 2 quadruplings -> 1,027,069,600 vertices
R(6,6,6,8,8,9,10)	1 369 072 065	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 21,391,751 vertices; then 3 quadruplings -> 1,369,072,064 vertices
R(6,6,6,8,8,10,10)	1 404 442 337	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,6,6,8,9,9,9)	1 614 144 849	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 100,884,053 vertices; then 2 quadruplings -> 1,614,144,848 vertices
R(6,6,6,8,9,9,10)	1 967 463 489	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 3 quadruplings -> 1,967,463,488 vertices
R(6,6,6,8,9,10,10)	2 186 102 529	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 4 quadruplings -> 2,186,102,528 vertices
R(6,6,6,8,10,10,10)	2 186 102 529	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 4 quadruplings -> 2,186,102,528 vertices (by monotonicity, from (5,6,6,8,10,10,10))
R(6,6,6,9,9,9,9)	1 967 463 489	Published bricks + theorems	[template (6,6,3;235)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 30,741,617 vertices; then 3 quadruplings -> 1,967,463,488 vertices
R(6,6,6,9,9,9,10)	2 209 112 973	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 552,278,243 vertices; then 1 quadrupling -> 2,209,112,972 vertices (by monotonicity, from (5,6,6,9,9,9,10))
R(6,6,6,9,9,10,10)	2 793 411 249	Our bricks + published	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 174,588,203 vertices; then 2 quadruplings -> 2,793,411,248 vertices
R(6,6,6,9,10,10,10)	3 733 134 849	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 14,582,558 vertices; then 4 quadruplings -> 3,733,134,848 vertices
R(6,6,6,10,10,10,10)	3 819 768 945	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(6,6,7,7,7,7,7)	334 184 587	Published bricks + theorems	[template (7,3;16)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]]] -> 334,184,586 vertices
R(6,6,7,7,7,7,8)	504 402 245	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]]] -> 126,100,561 vertices; then 1 quadrupling -> 504,402,244 vertices
R(6,6,7,7,7,7,9)	593 267 297	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,6,6 ; 337)]] -> 37,079,206 vertices; then 2 quadruplings -> 593,267,296 vertices
R(6,6,7,7,7,7,10)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices
R(6,6,7,7,7,8,8)	514 489 665	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 3 quadruplings -> 514,489,664 vertices
R(6,6,7,7,7,8,9)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices
R(6,6,7,7,7,8,10)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices
R(6,6,7,7,7,9,9)	1 194 135 185	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices; then 2 quadruplings -> 1,194,135,184 vertices
R(6,6,7,7,7,9,10)	1 425 854 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 3 quadruplings -> 1,425,854,272 vertices
R(6,6,7,7,7,10,10)	1 507 492 865	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 4 quadruplings -> 1,507,492,864 vertices
R(6,6,7,7,8,8,8)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices
R(6,6,7,7,8,8,9)	1 194 135 185	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices; then 2 quadruplings -> 1,194,135,184 vertices
R(6,6,7,7,8,8,10)	1 228 847 169	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 3 quadruplings -> 1,228,847,168 vertices
R(6,6,7,7,8,9,9)	1 425 854 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 3 quadruplings -> 1,425,854,272 vertices
R(6,6,7,7,8,9,10)	1 656 611 969	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 2 quadruplings -> 1,656,611,968 vertices
R(6,6,7,7,8,10,10)	2 425 624 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 3 quadruplings -> 2,425,624,256 vertices
R(6,6,7,7,9,9,9)	1 656 611 969	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 2 quadruplings -> 1,656,611,968 vertices
R(6,6,7,7,9,9,10)	2 425 624 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 3 quadruplings -> 2,425,624,256 vertices
R(6,6,7,7,9,10,10)	3 248 983 841	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 2 quadruplings -> 3,248,983,840 vertices
R(6,6,7,7,10,10,10)	3 248 983 841	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 2 quadruplings -> 3,248,983,840 vertices (by monotonicity, from (6,6,7,7,9,10,10))
R(6,6,7,8,8,8,8)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices (by monotonicity, from (6,6,7,7,8,8,8))
R(6,6,7,8,8,8,9)	1 425 854 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 3 quadruplings -> 1,425,854,272 vertices
R(6,6,7,8,8,8,10)	1 507 492 865	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,4,4 ; 54)]] -> 5,888,644 vertices; then 4 quadruplings -> 1,507,492,864 vertices
R(6,6,7,8,8,9,9)	1 656 611 969	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 2 quadruplings -> 1,656,611,968 vertices
R(6,6,7,8,8,9,10)	2 425 624 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 3 quadruplings -> 2,425,624,256 vertices
R(6,6,7,8,8,10,10)	2 433 905 153	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 4 quadruplings -> 2,433,905,152 vertices
R(6,6,7,8,9,9,9)	2 227 584 797	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]]] -> 556,896,199 vertices; then 1 quadrupling -> 2,227,584,796 vertices
R(6,6,7,8,9,9,10)	3 248 983 841	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 2 quadruplings -> 3,248,983,840 vertices

Parameters	Bound (R >=)	Origin	Construction
R(6,6,7,8,9,10,10)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,6,7,8,10,10,10)	4 100 694 273	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 16,018,337 vertices; then 4 quadruplings -> 4,100,694,272 vertices
R(6,6,7,9,9,9,9)	3 248 983 841	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 2 quadruplings -> 3,248,983,840 vertices
R(6,6,7,9,9,9,10)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,6,7,9,9,10,10)	4 237 586 465	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(6,6,7,9,10,10,10)	5 617 769 345	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,6,7,10,10,10,10)	6 622 131 713	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 4 quadruplings -> 6,622,131,712 vertices
R(6,6,8,8,8,8,8)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices (by monotonicity, from (6,6,7,7,8,8,8))
R(6,6,8,8,8,8,9)	1 656 611 969	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 2 quadruplings -> 1,656,611,968 vertices
R(6,6,8,8,8,8,10)	1 656 611 969	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 2 quadruplings -> 1,656,611,968 vertices (by monotonicity, from (6,6,8,8,8,8,9))
R(6,6,8,8,8,9,9)	1 978 062 209	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (8,8 ; 281)]] -> 30,907,222 vertices; then 3 quadruplings -> 1,978,062,208 vertices
R(6,6,8,8,8,9,10)	2 433 905 153	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 4 quadruplings -> 2,433,905,152 vertices
R(6,6,8,8,8,10,10)	2 433 905 153	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 9,507,442 vertices; then 4 quadruplings -> 2,433,905,152 vertices (by monotonicity, from (6,6,7,8,8,10,10))
R(6,6,8,8,9,9,9)	3 248 983 841	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 2 quadruplings -> 3,248,983,840 vertices
R(6,6,8,8,9,9,10)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,6,8,8,9,9,10,10)	4 505 606 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 2 quadruplings -> 4,505,606,096 vertices
R(6,6,8,8,10,10,10)	4 505 606 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 2 quadruplings -> 4,505,606,096 vertices (by monotonicity, from (6,6,8,8,9,10,10))
R(6,6,8,9,9,9,9)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,6,8,9,9,9,10)	4 505 606 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 2 quadruplings -> 4,505,606,096 vertices
R(6,6,8,9,9,10,10)	5 617 769 345	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,6,8,9,10,10,10)	6 622 131 713	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 4 quadruplings -> 6,622,131,712 vertices
R(6,6,8,10,10,10,10)	6 622 131 713	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 25,867,702 vertices; then 4 quadruplings -> 6,622,131,712 vertices (by monotonicity, from (6,6,7,10,10,10,10))
R(6,6,9,9,9,9,9)	4 505 606 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 2 quadruplings -> 4,505,606,096 vertices
R(6,6,9,9,9,9,10)	6 058 561 265	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]]] -> 1,514,640,316 vertices; then 1 quadrupling -> 6,058,561,264 vertices
R(6,6,9,9,9,10,10)	8 836 451 889	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 552,278,243 vertices; then 2 quadruplings -> 8,836,451,888 vertices
R(6,6,9,9,10,10,10)	10 770 775 617	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 3 quadruplings -> 10,770,775,616 vertices
R(6,6,9,10,10,10,10)	11 525 155 745	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(6,6,10,10,10,10,10)	11 525 155 745	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (3,6,6,3 ; 707)] x [atome (10,10 ; 797)]]
R(6,7,7,7,7,7,7)	504 402 245	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (6,6,3;235)] x [K3]]] -> 126,100,561 vertices; then 1 quadrupling -> 504,402,244 vertices
R(6,7,7,7,7,7,8)	596 688 101	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 149,172,025 vertices; then 1 quadrupling -> 596,688,100 vertices (by monotonicity, from (5,7,7,7,7,7,8))
R(6,7,7,7,7,7,9)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices
R(6,7,7,7,7,7,10)	932 948 225	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 4 quadruplings -> 932,948,224 vertices
R(6,7,7,7,7,8,8)	891 793 025	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (4,4,4 ; 127)]] -> 13,934,266 vertices; then 3 quadruplings -> 891,793,024 vertices
R(6,7,7,7,7,8,9)	1 194 135 185	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices; then 2 quadruplings -> 1,194,135,184 vertices
R(6,7,7,7,7,8,10)	1 228 847 169	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 3 quadruplings -> 1,228,847,168 vertices
R(6,7,7,7,7,9,9)	1 425 854 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 3 quadruplings -> 1,425,854,272 vertices
R(6,7,7,7,7,9,10)	1 634 903 981	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 1 quadrupling -> 1,634,903,980 vertices (by monotonicity, from (5,7,7,7,7,9,10))
R(6,7,7,7,7,10,10)	2 425 624 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 3 quadruplings -> 2,425,624,256 vertices
R(6,7,7,7,8,8,8)	932 948 225	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 4 quadruplings -> 932,948,224 vertices

Parameters	Bound (R >=)	Origin	Construction
R(6,7,7,7,8,8,9)	1 425 854 273	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-101 (6,6;101)]] -> 22,278,973 vertices; then 3 quadruplings -> 1,425,854,272 vertices
R(6,7,7,7,8,8,10)	1 614 443 777	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 4 quadruplings -> 1,614,443,776 vertices
R(6,7,7,7,8,9,9)	1 656 611 969	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 2 quadruplings -> 1,656,611,968 vertices
R(6,7,7,7,8,9,10)	2 425 624 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 3 quadruplings -> 2,425,624,256 vertices
R(6,7,7,7,8,10,10)	2 539 020 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 4 quadruplings -> 2,539,020,544 vertices
R(6,7,7,7,9,9,9)	2 425 624 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 3 quadruplings -> 2,425,624,256 vertices
R(6,7,7,7,9,9,10)	3 248 983 841	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 2 quadruplings -> 3,248,983,840 vertices
R(6,7,7,7,9,10,10)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,7,7,7,10,10,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(6,7,7,8,8,8,8)	1 228 847 169	Published bricks + theorems	[template (6,6,3;235)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 19,200,737 vertices; then 3 quadruplings -> 1,228,847,168 vertices
R(6,7,7,8,8,8,9)	1 656 611 969	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (3,8,8 ; 940)]] -> 103,538,248 vertices; then 2 quadruplings -> 1,656,611,968 vertices
R(6,7,7,8,8,8,10)	2 057 958 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 4 quadruplings -> 2,057,958,656 vertices
R(6,7,7,8,8,9,9)	2 425 624 257	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 37,900,379 vertices; then 3 quadruplings -> 2,425,624,256 vertices
R(6,7,7,8,8,9,10)	2 736 403 905	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices; then 3 quadruplings -> 2,736,403,904 vertices
R(6,7,7,8,8,10,10)	3 275 629 057	Our bricks + published	[template (4,6,3 ; 80) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 12,795,426 vertices; then 4 quadruplings -> 3,275,629,056 vertices
R(6,7,7,8,9,9,9)	3 248 983 841	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 203,061,490 vertices; then 2 quadruplings -> 3,248,983,840 vertices
R(6,7,7,8,9,9,10)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,7,7,8,9,10,10)	4 400 231 681	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 4 quadruplings -> 4,400,231,680 vertices
R(6,7,7,8,10,10,10)	5 597 593 857	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 4 quadruplings -> 5,597,593,856 vertices
R(6,7,7,9,9,9,9)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,7,7,9,9,9,10)	4 446 547 901	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices; then 1 quadrupling -> 4,446,547,900 vertices (by monotonicity, from (5,7,7,9,9,9,10))
R(6,7,7,9,9,10,10)	5 622 635 441	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 351,414,715 vertices; then 2 quadruplings -> 5,622,635,440 vertices
R(6,7,7,9,10,10,10)	7 514 110 977	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 4 quadruplings -> 7,514,110,976 vertices
R(6,7,7,10,10,10,10)	8 744 410 113	Our bricks + published	[template (6,6,3;235)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 8,539,463 vertices; then 5 quadruplings -> 8,744,410,112 vertices
R(6,7,8,8,8,8,8)	1 285 874 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [Paley-17 (4,4;17)]] -> 5,022,946 vertices; then 4 quadruplings -> 1,285,874,176 vertices
R(6,7,8,8,8,8,9)	2 057 958 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 8,038,901 vertices; then 4 quadruplings -> 2,057,958,656 vertices
R(6,7,8,8,8,8,10)	2 439 057 409	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 6 quadruplings -> 2,439,057,408 vertices
R(6,7,8,8,8,9,9)	2 736 403 905	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 42,756,311 vertices; then 3 quadruplings -> 2,736,403,904 vertices
R(6,7,8,8,8,9,10)	3 278 790 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 4 quadruplings -> 3,278,790,656 vertices
R(6,7,8,8,8,10,10)	3 902 537 729	Published bricks + theorems	[template (7,7,3;472)] x [atom (4,4,4 ; 127)] -> 59,548 vertices; then 8 quadruplings -> 3,902,537,728 vertices
R(6,7,8,8,9,9,9)	3 960 150 785	Published bricks + theorems	[template (7,7,3;472)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 61,877,356 vertices; then 3 quadruplings -> 3,960,150,784 vertices
R(6,7,8,8,9,9,10)	4 505 606 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 2 quadruplings -> 4,505,606,096 vertices
R(6,7,8,8,9,10,10)	5 617 769 345	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (6,6,3;235)] x [atome (10,10 ; 797)]]
R(6,7,8,8,10,10,10)	6 634 180 609	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 6 quadruplings -> 6,634,180,608 vertices
R(6,7,8,9,9,9,9)	4 505 606 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 2 quadruplings -> 4,505,606,096 vertices
R(6,7,8,9,9,9,10)	5 622 635 441	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 351,414,715 vertices; then 2 quadruplings -> 5,622,635,440 vertices
R(6,7,8,9,9,10,10)	7 514 110 977	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 4 quadruplings -> 7,514,110,976 vertices
R(6,7,8,9,10,10,10)	9 138 784 257	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 4 quadruplings -> 9,138,784,256 vertices
R(6,7,8,10,10,10,10)	10 614 734 849	Published bricks + theorems	[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)] -> 161,968 vertices; then 8 quadruplings -> 10,614,734,848 vertices
R(6,7,9,9,9,9,9)	6 058 561 265	Published bricks + theorems	[template (5,3;10)] x [[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]]] -> 1,514,640,316 vertices; then 1 quadrupling -> 6,058,561,264 vertices
R(6,7,9,9,9,9,10)	8 836 451 889	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 552,278,243 vertices; then 2 quadruplings -> 8,836,451,888 vertices
R(6,7,9,9,9,10,10)	10 770 775 617	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 3 quadruplings -> 10,770,775,616 vertices
R(6,7,9,9,10,10,10)	11 967 528 705	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 4 quadruplings -> 11,967,528,704 vertices
R(6,7,9,10,10,10,10)	15 279 075 777	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(6,7,10,10,10,10,10)	16 976 751 105	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (3,3,4,3 ; 66)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]

Parameters	Bound (R >=)	Origin	Construction
R(6,8,8,8,8,8,8)	1 371 555 233	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (8,8 ; 281)]] -> 85,722,202 vertices; then 2 quadruplings -> 1,371,555,232 vertices (by monotonicity, from (5,8,8,8,8,8,8))
R(6,8,8,8,8,8,9)	2 439 057 409	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 6 quadruplings -> 2,439,057,408 vertices
R(6,8,8,8,8,8,10)	2 439 057 409	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 595,473 vertices; then 6 quadruplings -> 2,439,057,408 vertices (by monotonicity, from (5,8,8,8,8,8,10))
R(6,8,8,8,8,9,9)	3 278 790 657	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 12,807,776 vertices; then 4 quadruplings -> 3,278,790,656 vertices
R(6,8,8,8,8,9,10)	4 023 242 753	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 6 quadruplings -> 4,023,242,752 vertices
R(6,8,8,8,8,10,10)	4 023 242 753	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3;5)]] -> 982,237 vertices; then 6 quadruplings -> 4,023,242,752 vertices (by monotonicity, from (5,8,8,8,8,10,10))
R(6,8,8,8,9,9,9)	4 505 606 097	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 281,600,381 vertices; then 2 quadruplings -> 4,505,606,096 vertices
R(6,8,8,8,8,9,9,10)	5 597 593 857	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 21,865,601 vertices; then 4 quadruplings -> 5,597,593,856 vertices
R(6,8,8,8,8,9,10,10)	6 634 180 609	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 6 quadruplings -> 6,634,180,608 vertices
R(6,8,8,8,10,10,10,10)	6 634 180 609	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 1,619,673 vertices; then 6 quadruplings -> 6,634,180,608 vertices (by monotonicity, from (5,8,8,8,10,10,10))
R(6,8,8,9,9,9,9)	5 622 635 441	Our bricks + published	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 351,414,715 vertices; then 2 quadruplings -> 5,622,635,440 vertices
R(6,8,8,9,9,9,10)	7 514 110 977	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 29,351,996 vertices; then 4 quadruplings -> 7,514,110,976 vertices
R(6,8,8,9,9,10,10)	9 138 784 257	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 35,698,376 vertices; then 4 quadruplings -> 9,138,784,256 vertices
R(6,8,8,9,10,10,10)	10 968 420 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 6 quadruplings -> 10,968,420,352 vertices
R(6,8,8,10,10,10,10)	10 968 420 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 2,677,837 vertices; then 6 quadruplings -> 10,968,420,352 vertices (by monotonicity, from (5,8,8,10,10,10,10))
R(6,8,9,9,9,9,9)	8 836 451 889	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 552,278,243 vertices; then 2 quadruplings -> 8,836,451,888 vertices
R(6,8,9,9,9,9,10)	10 770 775 617	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 3 quadruplings -> 10,770,775,616 vertices
R(6,8,9,9,9,10,10)	11 967 528 705	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 4 quadruplings -> 11,967,528,704 vertices
R(6,8,9,9,10,10,10)	15 279 075 777	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,6,3;235)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(6,8,9,10,10,10,10)	17 757 589 505	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 6 quadruplings -> 17,757,589,504 vertices
R(6,8,10,10,10,10,10)	17 757 589 505	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (6,6,3;235)] x [atom (9,9 ; 562)]] -> 4,335,349 vertices; then 6 quadruplings -> 17,757,589,504 vertices (by monotonicity, from (6,8,9,10,10,10,10))
R(6,9,9,9,9,9,9)	10 770 775 617	Published bricks + theorems	[template (6,6,3;235)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 168,293,369 vertices; then 3 quadruplings -> 10,770,775,616 vertices
R(6,9,9,9,9,9,10)	12 093 477 461	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 3,023,369,365 vertices; then 1 quadrupling -> 12,093,477,460 vertices (by monotonicity, from (5,9,9,9,9,9,10))
R(6,9,9,9,9,10,10)	15 292 136 081	Our bricks + published	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (6,9,3 ; 402) (ours)] x [K2]]] -> 955,758,505 vertices; then 2 quadruplings -> 15,292,136,080 vertices
R(6,9,9,9,10,10,10)	20 436 855 297	Our bricks + published	[template (3,3,5,3 ; 112) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 79,831,466 vertices; then 4 quadruplings -> 20,436,855,296 vertices
R(6,9,9,10,10,10,10)	24 855 635 457	Our bricks + published	[template (5,6,3 ; 136) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 97,092,326 vertices; then 4 quadruplings -> 24,855,635,456 vertices
R(6,9,10,10,10,10,10)	29 831 360 513	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 6 quadruplings -> 29,831,360,512 vertices
R(6,10,10,10,10,10,10)	29 831 360 513	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [(3,3;5)]] -> 7,283,047 vertices; then 6 quadruplings -> 29,831,360,512 vertices (by monotonicity, from (5,10,10,10,10,10,10))
R(7,7,7,7,7,7,7)	670 273 702	Published bricks + theorems	[template (7,3;16)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]]] -> 670,273,701 vertices
R(7,7,7,7,7,7,8)	1 014 636 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]]] -> 253,659,136 vertices; then 1 quadrupling -> 1,014,636,544 vertices
R(7,7,7,7,7,7,9)	1 194 135 185	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,6,6 ; 337)]] -> 74,633,449 vertices; then 2 quadruplings -> 1,194,135,184 vertices
R(7,7,7,7,7,7,10)	1 794 997 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 3 quadruplings -> 1,794,997,696 vertices
R(7,7,7,7,7,8,8)	1 014 636 545	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K3]]] -> 253,659,136 vertices; then 1 quadrupling -> 1,014,636,544 vertices (by monotonicity, from (7,7,7,7,7,7,8))
R(7,7,7,7,7,8,9)	1 794 997 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 3 quadruplings -> 1,794,997,696 vertices
R(7,7,7,7,7,8,10)	1 794 997 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 3 quadruplings -> 1,794,997,696 vertices
R(7,7,7,7,7,9,9)	2 386 752 401	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 149,172,025 vertices; then 2 quadruplings -> 2,386,752,400 vertices
R(7,7,7,7,7,9,10)	2 859 834 497	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices; then 3 quadruplings -> 2,859,834,496 vertices
R(7,7,7,7,7,10,10)	3 034 226 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 4 quadruplings -> 3,034,226,176 vertices
R(7,7,7,7,8,8,8)	1 794 997 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 3 quadruplings -> 1,794,997,696 vertices
R(7,7,7,7,8,8,9)	2 386 752 401	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 149,172,025 vertices; then 2 quadruplings -> 2,386,752,400 vertices
R(7,7,7,7,8,8,10)	2 473 448 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 3 quadruplings -> 2,473,448,704 vertices

Parameters	Bound (R >=)	Origin	Construction
R(7,7,7,7,8,9,9)	2 859 834 497	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices; then 3 quadruplings -> 2,859,834,496 vertices
R(7,7,7,7,8,9,10)	3 334 457 153	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,8,8 ; 940)]] -> 208,403,572 vertices; then 2 quadruplings -> 3,334,457,152 vertices
R(7,7,7,7,8,10,10)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices
R(7,7,7,7,9,9,9)	3 334 457 153	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,8,8 ; 940)]] -> 208,403,572 vertices; then 2 quadruplings -> 3,334,457,152 vertices
R(7,7,7,7,9,9,10)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices
R(7,7,7,7,9,10,10)	6 539 615 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 2 quadruplings -> 6,539,615,920 vertices
R(7,7,7,7,10,10,10)	6 539 615 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 2 quadruplings -> 6,539,615,920 vertices (by monotonicity, from (7,7,7,7,9,10,10))
R(7,7,7,8,8,8,8)	1 794 997 697	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 28,046,839 vertices; then 3 quadruplings -> 1,794,997,696 vertices (by monotonicity, from (7,7,7,7,8,8,8))
R(7,7,7,8,8,8,9)	2 859 834 497	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 44,684,914 vertices; then 3 quadruplings -> 2,859,834,496 vertices
R(7,7,7,8,8,8,10)	3 034 226 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,4,4 ; 54)]] -> 11,852,446 vertices; then 4 quadruplings -> 3,034,226,176 vertices
R(7,7,7,8,8,9,9)	3 334 457 153	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,8,8 ; 940)]] -> 208,403,572 vertices; then 2 quadruplings -> 3,334,457,152 vertices
R(7,7,7,8,8,9,10)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices
R(7,7,7,8,8,10,10)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices (by monotonicity, from (7,7,7,7,8,10,10))
R(7,7,7,8,9,9,9)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices
R(7,7,7,8,9,9,10)	6 539 615 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 2 quadruplings -> 6,539,615,920 vertices
R(7,7,7,8,9,10,10)	7 971 051 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 3 quadruplings -> 7,971,051,136 vertices
R(7,7,7,8,10,10,10)	8 253 961 217	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,4,4 ; 54)]] -> 32,242,036 vertices; then 4 quadruplings -> 8,253,961,216 vertices
R(7,7,7,9,9,9,9)	6 539 615 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 2 quadruplings -> 6,539,615,920 vertices
R(7,7,7,9,9,9,10)	7 971 051 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 3 quadruplings -> 7,971,051,136 vertices
R(7,7,7,9,9,10,10)	8 481 168 353	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(7,7,7,9,10,10,10)	13 278 738 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 3 quadruplings -> 13,278,738,496 vertices
R(7,7,7,10,10,10,10)	13 329 152 513	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 4 quadruplings -> 13,329,152,512 vertices
R(7,7,8,8,8,8,8)	2 473 448 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 3 quadruplings -> 2,473,448,704 vertices
R(7,7,8,8,8,8,9)	3 334 457 153	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (3,8,8 ; 940)]] -> 208,403,572 vertices; then 2 quadruplings -> 3,334,457,152 vertices
R(7,7,8,8,8,8,10)	3 731 792 897	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [Paley-17 (4,4;17)]] -> 3,644,329 vertices; then 5 quadruplings -> 3,731,792,896 vertices
R(7,7,8,8,8,9,9)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices
R(7,7,8,8,8,9,10)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices (by monotonicity, from (7,7,7,8,8,9,10))
R(7,7,8,8,8,10,10)	6 727 479 297	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 3 quadruplings -> 6,727,479,296 vertices
R(7,7,8,8,9,9,9)	6 539 615 921	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 408,725,995 vertices; then 2 quadruplings -> 6,539,615,920 vertices
R(7,7,8,8,9,9,10)	7 971 051 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 3 quadruplings -> 7,971,051,136 vertices
R(7,7,8,8,9,10,10)	9 068 976 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 2 quadruplings -> 9,068,976,352 vertices
R(7,7,8,8,10,10,10)	10 156 082 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 9,918,049 vertices; then 5 quadruplings -> 10,156,082,176 vertices
R(7,7,8,9,9,9,9)	7 971 051 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 3 quadruplings -> 7,971,051,136 vertices
R(7,7,8,9,9,9,10)	9 068 976 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 2 quadruplings -> 9,068,976,352 vertices
R(7,7,8,9,9,10,10)	13 278 738 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 3 quadruplings -> 13,278,738,496 vertices
R(7,7,8,9,10,10,10)	13 329 152 513	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 4 quadruplings -> 13,329,152,512 vertices
R(7,7,8,10,10,10,10)	17 600 926 721	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 5 quadruplings -> 17,600,926,720 vertices
R(7,7,9,9,9,9,9)	9 068 976 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 2 quadruplings -> 9,068,976,352 vertices
R(7,7,9,9,9,9,10)	13 278 738 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 3 quadruplings -> 13,278,738,496 vertices
R(7,7,9,9,9,10,10)	17 786 191 601	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices; then 2 quadruplings -> 17,786,191,600 vertices
R(7,7,9,9,10,10,10)	21 679 638 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 3 quadruplings -> 21,679,638,016 vertices
R(7,7,9,10,10,10,10)	23 066 719 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(7,7,10,10,10,10,10)	27 621 968 897	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [Paley-17 (4,4;17)]] -> 26,974,579 vertices; then 5 quadruplings -> 27,621,968,896 vertices

Parameters	Bound (R >=)	Origin	Construction
R(7,8,8,8,8,8,8)	2 473 448 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 3 quadruplings -> 2,473,448,704 vertices (by monotonicity, from (7,7,8,8,8,8,8))
R(7,8,8,8,8,8,9)	3 981 462 593	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (8,8 ; 281)]] -> 62,210,353 vertices; then 3 quadruplings -> 3,981,462,592 vertices
R(7,8,8,8,8,8,10)	4 634 198 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 6 quadruplings -> 4,634,198,016 vertices
R(7,8,8,8,8,9,9)	4 882 346 177	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 76,286,659 vertices; then 3 quadruplings -> 4,882,346,176 vertices (by monotonicity, from (7,7,8,8,8,9,9))
R(7,8,8,8,8,9,10)	6 727 479 297	Published bricks + theorems	[template (8,8,3 ; 650)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 105,116,864 vertices; then 3 quadruplings -> 6,727,479,296 vertices
R(7,8,8,8,8,10,10)	7 383 330 817	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 6 quadruplings -> 7,383,330,816 vertices
R(7,8,8,8,9,9,9)	7 971 051 137	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 124,547,674 vertices; then 3 quadruplings -> 7,971,051,136 vertices
R(7,8,8,8,9,9,10)	9 068 976 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 2 quadruplings -> 9,068,976,352 vertices
R(7,8,8,8,9,10,10)	11 307 539 777	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (7,7,3;472)] x [atome (10,10 ; 797)]]
R(7,8,8,8,10,10,10)	12 604 932 097	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 3,077,376 vertices; then 6 quadruplings -> 12,604,932,096 vertices
R(7,8,8,9,9,9,9)	9 068 976 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 2 quadruplings -> 9,068,976,352 vertices
R(7,8,8,9,9,9,10)	13 278 738 497	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (4,4,4 ; 127)]] -> 207,480,289 vertices; then 3 quadruplings -> 13,278,738,496 vertices
R(7,8,8,9,9,10,10)	13 329 152 513	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 4 quadruplings -> 13,329,152,512 vertices
R(7,8,8,9,10,10,10)	17 600 926 721	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 5 quadruplings -> 17,600,926,720 vertices
R(7,8,8,10,10,10,10)	20 579 168 257	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 5,024,211 vertices; then 6 quadruplings -> 20,579,168,256 vertices
R(7,8,9,9,9,9,9)	12 194 796 365	Published bricks + theorems	[template (5,3;10)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]]] -> 3,048,699,091 vertices; then 1 quadrupling -> 12,194,796,364 vertices
R(7,8,9,9,9,9,10)	17 786 191 601	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices; then 2 quadruplings -> 17,786,191,600 vertices
R(7,8,9,9,9,10,10)	21 679 638 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 3 quadruplings -> 21,679,638,016 vertices
R(7,8,9,9,10,10,10)	23 066 719 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(7,8,9,10,10,10,10)	30 754 037 057	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(7,8,10,10,10,10,10)	31 854 493 697	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[generic template 2m-1 of K2] x [atom (4,4,4 ; 127)]] -> 486,061 vertices; then 8 quadruplings -> 31,854,493,696 vertices
R(7,9,9,9,9,9,9)	17 786 191 601	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices; then 2 quadruplings -> 17,786,191,600 vertices
R(7,9,9,9,9,9,10)	21 679 638 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 3 quadruplings -> 21,679,638,016 vertices
R(7,9,9,9,9,10,10)	23 066 719 553	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(7,9,9,9,10,10,10)	30 754 037 057	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(7,9,9,10,10,10,10)	36 252 109 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 4 quadruplings -> 36,252,109,312 vertices
R(7,9,10,10,10,10,10)	47 870 114 817	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 5 quadruplings -> 47,870,114,816 vertices
R(7,10,10,10,10,10,10)	55 971 188 737	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 13,664,841 vertices; then 6 quadruplings -> 55,971,188,736 vertices
R(8,8,8,8,8,8,8)	2 473 448 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (4,4,4 ; 127)]] -> 38,647,636 vertices; then 3 quadruplings -> 2,473,448,704 vertices (by monotonicity, from (7,7,8,8,8,8,8))
R(8,8,8,8,8,8,9)	4 634 198 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 6 quadruplings -> 4,634,198,016 vertices
R(8,8,8,8,8,8,10)	4 634 198 017	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [atom (4,4,4 ; 127)]] -> 1,131,396 vertices; then 6 quadruplings -> 4,634,198,016 vertices (by monotonicity, from (7,8,8,8,8,8,10))
R(8,8,8,8,8,9,9)	6 457 775 105	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [(3,3,4;29)]] -> 6,306,421 vertices; then 5 quadruplings -> 6,457,775,104 vertices
R(8,8,8,8,8,9,10)	7 383 330 817	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 6 quadruplings -> 7,383,330,816 vertices
R(8,8,8,8,8,10,10)	7 383 330 817	Our bricks + published	[shifted template (3,4,3 ; width 19, tail 2) (ours)] x [[template (7,7,3;472)] x [Mathon-doubling (7,7;202)]] -> 1,802,571 vertices; then 6 quadruplings -> 7,383,330,816 vertices (by monotonicity, from (7,8,8,8,8,10,10))
R(8,8,8,8,9,9,9)	9 068 976 353	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 566,811,022 vertices; then 2 quadruplings -> 9,068,976,352 vertices
R(8,8,8,8,9,9,10)	10 983 552 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 3 quadruplings -> 10,983,552,064 vertices
R(8,8,8,8,9,10,10)	12 972 576 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 6 quadruplings -> 12,972,576,768 vertices

Parameters	Bound (R >=)	Origin	Construction
R(8,8,8,8,10,10,10)	12 972 576 769	Our bricks + published	[shifted template (3,3,3 ; width 10, tail 2) (ours)] x [[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]]] -> 3,167,133 vertices; then 6 quadruplings -> 12,972,576,768 vertices (by monotonicity, from (8,8,8,8,9,10,10))
R(8,8,8,9,9,9,9)	10 983 552 065	Published bricks + theorems	[template (7,7,3;472)] x [[template (8,8,3 ; 650)] x [atom (9,9 ; 562)]] -> 171,618,001 vertices; then 3 quadruplings -> 10,983,552,064 vertices
R(8,8,8,9,9,9,10)	13 329 152 513	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 52,067,002 vertices; then 4 quadruplings -> 13,329,152,512 vertices
R(8,8,8,9,9,10,10)	17 600 926 721	Our bricks + published	[template (7,7,3;472)] x [[template (3,3,4,3 ; 66) (ours)] x [atom (9,9 ; 562)]] -> 17,188,405 vertices; then 5 quadruplings -> 17,600,926,720 vertices
R(8,8,8,9,10,10,10)	20 756 168 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 8 quadruplings -> 20,756,168,704 vertices
R(8,8,8,10,10,10,10)	20 756 168 705	Published bricks + theorems	[template (7,7,3;472)] x [[template (7,7,3;472)] x [K2]] -> 316,714 vertices; then 8 quadruplings -> 20,756,168,704 vertices (by monotonicity, from (8,8,8,9,10,10,10))
R(8,8,9,9,9,9,9)	17 786 191 601	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 1,111,636,975 vertices; then 2 quadruplings -> 17,786,191,600 vertices
R(8,8,9,9,9,9,10)	21 679 638 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 3 quadruplings -> 21,679,638,016 vertices
R(8,8,9,9,9,10,10)	24 665 305 313	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 1,541,581,582 vertices; then 2 quadruplings -> 24,665,305,312 vertices
R(8,8,9,9,10,10,10)	30 754 037 057	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(8,8,9,10,10,10,10)	35 742 773 249	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 6 quadruplings -> 35,742,773,248 vertices
R(8,8,10,10,10,10,10)	35 742 773 249	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (7,7,3;472)] x [atom (9,9 ; 562)]] -> 8,726,263 vertices; then 6 quadruplings -> 35,742,773,248 vertices (by monotonicity, from (8,8,9,10,10,10,10))
R(8,9,9,9,9,9,9)	21 679 638 017	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 338,744,344 vertices; then 3 quadruplings -> 21,679,638,016 vertices
R(8,9,9,9,9,9,10)	24 665 305 313	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 1,541,581,582 vertices; then 2 quadruplings -> 24,665,305,312 vertices
R(8,9,9,9,9,10,10)	30 754 037 057	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,7,3;472)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(8,9,9,9,10,10,10)	36 252 109 313	Our bricks + published	[template (4,5,3 ; 61) (ours)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 141,609,802 vertices; then 4 quadruplings -> 36,252,109,312 vertices
R(8,9,9,10,10,10,10)	47 870 114 817	Our bricks + published	[template (3,3,4,3 ; 66) (ours)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 46,748,159 vertices; then 5 quadruplings -> 47,870,114,816 vertices
R(8,9,10,10,10,10,10)	56 871 092 225	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 8 quadruplings -> 56,871,092,224 vertices
R(8,10,10,10,10,10,10)	56 871 092 225	Published bricks + theorems	[template (7,7,3;472)] x [[template (9,9,3 ; 1282)] x [K2]] -> 867,784 vertices; then 8 quadruplings -> 56,871,092,224 vertices (by monotonicity, from (8,9,10,10,10,10,10))
R(9,9,9,9,9,9,9)	24 665 305 313	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (3,8,8 ; 940)]] -> 1,541,581,582 vertices; then 2 quadruplings -> 24,665,305,312 vertices
R(9,9,9,9,9,9,10)	33 166 739 645	Published bricks + theorems	[template (5,3;10)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]]] -> 8,291,684,911 vertices; then 1 quadrupling -> 33,166,739,644 vertices
R(9,9,9,9,9,10,10)	48 373 909 841	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [K2]]] -> 3,023,369,365 vertices; then 2 quadruplings -> 48,373,909,840 vertices
R(9,9,9,9,10,10,10)	58 963 092 737	Published bricks + theorems	[template (9,9,3 ; 1282)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 921,298,324 vertices; then 3 quadruplings -> 58,963,092,736 vertices
R(9,9,9,10,10,10,10)	62 735 600 513	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (9,9,3 ; 1282)] x [[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]]]
R(9,9,10,10,10,10,10)	83 643 146 177	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,9,3 ; 1282)] x [[modele (9,9,3 ; 1282)] x [atome (10,10 ; 797)]]
R(9,10,10,10,10,10,10)	97 213 124 609	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 6 quadruplings -> 97,213,124,608 vertices
R(10,10,10,10,10,10,10)	97 213 124 609	Published bricks + theorems	[template (3,3,3,3 ; 34)] x [[template (9,9,3 ; 1282)] x [atom (9,9 ; 562)]] -> 23,733,673 vertices; then 6 quadruplings -> 97,213,124,608 vertices (by monotonicity, from (9,10,10,10,10,10,10))

Section 2 - Three colours, limits 11 to 15

Parameters	Bound (R >=)	Origin	Construction
R(3,3,11)	183	Our bricks	atom (3,3,11 ; 182) (ours) -> 182 vertices
R(3,3,12)	203	Our bricks	atom (3,3,12 ; 202) (ours) -> 202 vertices
R(3,3,13)	232	Published bricks + theorems	atom (3,3,13 ; 231) -> 231 vertices
R(3,3,14)	267	Published bricks + theorems	atom (3,3,14 ; 266) -> 266 vertices
R(3,3,15)	278	Published bricks + theorems	published non-linear colouring of 277 vertices (matrix R3315_publiee_Dauth2026_N277.txt)
R(3,4,11)	365	Our bricks + published	atom (4,10 ; 91) (ours) -> 91 vertices; then 1 quadrupling -> 364 vertices
R(3,4,12)	405	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 1 quadrupling -> 404 vertices
R(3,4,13)	405	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 1 quadrupling -> 404 vertices
R(3,4,14)	549	Published bricks + theorems	published non-linear colouring of 137 vertices (matrix R413_publiee_ExTa2015_N137.txt) -> 137 vertices; then 1 quadrupling -> 548 vertices
R(3,4,15)	585	Published bricks + theorems	published non-linear colouring of 146 vertices (matrix R414_publiee_ExTa2015_N146.txt) -> 146 vertices; then 1 quadrupling -> 584 vertices
R(3,5,11)	597	Published bricks + theorems	atom (3,3,10 ; 149) -> 149 vertices; then 1 quadrupling -> 596 vertices
R(3,5,12)	729	Published bricks + theorems	atom (5,11 ; 182) -> 182 vertices; then 1 quadrupling -> 728 vertices
R(3,5,13)	809	Published bricks + theorems	atom (5,12 ; 202) -> 202 vertices; then 1 quadrupling -> 808 vertices
R(3,5,14)	925	Published bricks + theorems	atom (3,3,13 ; 231) -> 231 vertices; then 1 quadrupling -> 924 vertices
R(3,5,15)	1 065	Published bricks + theorems	atom (3,3,14 ; 266) -> 266 vertices; then 1 quadrupling -> 1,064 vertices
R(3,6,11)	813	Published bricks + theorems	atom (6,10 ; 203) -> 203 vertices; then 1 quadrupling -> 812 vertices
R(3,6,12)	897	Published bricks + theorems	atom (5,6 ; 56) -> 56 vertices; then 2 quadruplings -> 896 vertices
R(3,6,13)	1 065	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices; then 1 quadrupling -> 1,064 vertices
R(3,6,14)	1 265	Published bricks + theorems	atom (5,7 ; 79) -> 79 vertices; then 2 quadruplings -> 1,264 vertices
R(3,6,15)	1 341	Our bricks + published	[template (5,8,3 ; 236) (ours)] x [K2] -> 335 vertices; then 1 quadrupling -> 1,340 vertices
R(3,7,11)	1 349	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 1 quadrupling -> 1,348 vertices
R(3,7,12)	1 617	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 2 quadruplings -> 1,616 vertices
R(3,7,13)	1 617	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 2 quadruplings -> 1,616 vertices
R(3,7,14)	1 825	Published bricks + theorems	published non-linear colouring of 114 vertices (matrix R67_publiee_ExTa2015_N114.txt) -> 114 vertices; then 2 quadruplings -> 1,824 vertices
R(3,7,15)	2 193	Published bricks + theorems	published non-linear colouring of 137 vertices (matrix R413_publiee_ExTa2015_N137.txt) -> 137 vertices; then 2 quadruplings -> 2,192 vertices
R(3,8,11)	1 617	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 2 quadruplings -> 1,616 vertices
R(3,8,12)	1 825	Published bricks + theorems	published non-linear colouring of 114 vertices (matrix R67_publiee_ExTa2015_N114.txt) -> 114 vertices; then 2 quadruplings -> 1,824 vertices
R(3,8,13)	2 693	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 1 quadrupling -> 2,692 vertices
R(3,8,14)	3 233	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 2 quadruplings -> 3,232 vertices
R(3,8,15)	3 233	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 2 quadruplings -> 3,232 vertices
R(3,9,11)	2 249	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 1 quadrupling -> 2,248 vertices (by monotonicity, from (3,9,10))
R(3,9,12)	2 385	Published bricks + theorems	atom (3,3,10 ; 149) -> 149 vertices; then 2 quadruplings -> 2,384 vertices
R(3,9,13)	3 233	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 2 quadruplings -> 3,232 vertices
R(3,9,14)	3 233	Published bricks + theorems	atom (5,12 ; 202) -> 202 vertices; then 2 quadruplings -> 3,232 vertices
R(3,9,15)	3 761	Published bricks + theorems	atom (3,8,8 ; 940) -> 940 vertices; then 1 quadrupling -> 3,760 vertices
R(3,10,11)	3 189	Our bricks (Paley-doubling atoms, Nov 2026)	atome (10,10 ; 797)
R(3,10,12)	3 189	Our bricks (Paley-doubling atoms, Nov 2026)	atome (10,10 ; 797)
R(3,10,13)	3 233	Published bricks + theorems	atom (5,12 ; 202) -> 202 vertices; then 2 quadruplings -> 3,232 vertices
R(3,10,14)	4 017	Published bricks + theorems	atom (7,9 ; 251) -> 251 vertices; then 2 quadruplings -> 4,016 vertices
R(3,10,15)	4 497	Published bricks + theorems	atom (8,8 ; 281) -> 281 vertices; then 2 quadruplings -> 4,496 vertices
R(3,11,11)	4 782	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(3,11,12)	6 377	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)

Parameters	Bound (R >=)	Origin	Construction
R(3,11,13)	6 377	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(3,11,14)	6 377	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(3,11,15)	6 377	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(3,12,12)	6 377	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(3,12,13)	6 377	Our bricks (Paley-doubling atoms, Nov 2026)	atome (12,12 ; 1594)
R(3,12,14)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(3,12,15)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(3,13,13)	7 662	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (13,13 ; 2554)]
R(3,13,14)	10 217	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(3,13,15)	10 217	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(3,14,14)	10 217	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(3,14,15)	11 945	Our bricks (Paley-doubling atoms, Nov 2026)	atome (14,14 ; 2986)
R(3,15,15)	16 446	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (15,15 ; 5482)]
R(4,4,11)	474	Published bricks + theorems	[template (10,3 ; 30)] x [Paley-17 (4,4;17)] -> 473 vertices (by monotonicity, from (4,4,10))
R(4,4,12)	474	Published bricks + theorems	[template (10,3 ; 30)] x [Paley-17 (4,4;17)] -> 473 vertices (by monotonicity, from (4,4,10))
R(4,4,13)	474	Published bricks + theorems	[template (10,3 ; 30)] x [Paley-17 (4,4;17)] -> 473 vertices (by monotonicity, from (4,4,10))
R(4,4,14)	549	Published bricks + theorems	published non-linear colouring of 137 vertices (matrix R413_publiee_ExTa2015_N137.txt) -> 137 vertices; then 1 quadrupling -> 548 vertices
R(4,4,15)	585	Published bricks + theorems	published non-linear colouring of 146 vertices (matrix R414_publiee_ExTa2015_N146.txt) -> 146 vertices; then 1 quadrupling -> 584 vertices
R(4,5,11)	1 089	Published bricks + theorems	[generic template 2m-1 of K2] x [atom (4,10 ; 91) (ours)] -> 272 vertices; then 1 quadrupling -> 1,088 vertices
R(4,5,12)	1 457	Our bricks + published	atom (4,10 ; 91) (ours) -> 91 vertices; then 2 quadruplings -> 1,456 vertices
R(4,5,13)	1 617	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 2 quadruplings -> 1,616 vertices
R(4,5,14)	1 617	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 2 quadruplings -> 1,616 vertices
R(4,5,15)	2 193	Published bricks + theorems	published non-linear colouring of 137 vertices (matrix R413_publiee_ExTa2015_N137.txt) -> 137 vertices; then 2 quadruplings -> 2,192 vertices
R(4,6,11)	1 377	Our bricks + published	atom (3,4,5 ; 86) (ours) -> 86 vertices; then 2 quadruplings -> 1,376 vertices
R(4,6,12)	1 457	Our bricks + published	atom (4,10 ; 91) (ours) -> 91 vertices; then 2 quadruplings -> 1,456 vertices (by monotonicity, from (4,5,12))
R(4,6,13)	2 033	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 2 quadruplings -> 2,032 vertices
R(4,6,14)	2 033	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 2 quadruplings -> 2,032 vertices (by monotonicity, from (4,6,13))
R(4,6,15)	2 577	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 2 quadruplings -> 2,576 vertices
R(4,7,11)	2 277	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 1 quadrupling -> 2,276 vertices
R(4,7,12)	2 277	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 1 quadrupling -> 2,276 vertices (by monotonicity, from (4,7,11))
R(4,7,13)	2 369	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 2 quadruplings -> 2,368 vertices
R(4,7,14)	2 369	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 2 quadruplings -> 2,368 vertices (by monotonicity, from (4,7,13))
R(4,7,15)	2 577	Our bricks + published	[template (4,7,3 ; 116) (ours)] x [K2] -> 161 vertices; then 2 quadruplings -> 2,576 vertices (by monotonicity, from (4,6,15))
R(4,8,11)	2 369	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 2 quadruplings -> 2,368 vertices
R(4,8,12)	2 369	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 2 quadruplings -> 2,368 vertices (by monotonicity, from (4,8,11))
R(4,8,13)	4 577	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 1 quadrupling -> 4,576 vertices
R(4,8,14)	4 577	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 1 quadrupling -> 4,576 vertices (by monotonicity, from (4,8,13))
R(4,8,15)	4 577	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 1 quadrupling -> 4,576 vertices (by monotonicity, from (4,8,13))
R(4,9,11)	3 857	Published bricks + theorems	atom (3,4,9 ; 241) -> 241 vertices; then 2 quadruplings -> 3,856 vertices
R(4,9,12)	4 353	Published bricks + theorems	[generic template 2m-1 of K2] x [atom (4,10 ; 91) (ours)] -> 272 vertices; then 2 quadruplings -> 4,352 vertices

Parameters	Bound (R >=)	Origin	Construction
R(4,9,13)	5 825	Our bricks + published	atom (4,10 ; 91) (ours) -> 91 vertices; then 3 quadruplings -> 5,824 vertices
R(4,9,14)	6 465	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(4,9,15)	6 465	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(4,10,11)	3 984	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (10,10 ; 797)]
R(4,10,12)	4 353	Published bricks + theorems	[generic template 2m-1 of K2] x [atom (4,10 ; 91) (ours)] -> 272 vertices; then 2 quadruplings -> 4,352 vertices (by monotonicity, from (4,9,12))
R(4,10,13)	5 825	Our bricks + published	atom (4,10 ; 91) (ours) -> 91 vertices; then 3 quadruplings -> 5,824 vertices (by monotonicity, from (4,9,13))
R(4,10,14)	6 465	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(4,10,15)	8 129	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 3 quadruplings -> 8,128 vertices
R(4,11,11)	7 969	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(4,11,12)	7 969	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(4,11,13)	7 969	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(4,11,14)	8 129	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 3 quadruplings -> 8,128 vertices
R(4,11,15)	8 129	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 3 quadruplings -> 8,128 vertices (by monotonicity, from (4,10,15))
R(4,12,12)	7 969	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (12,12 ; 1594)]
R(4,12,13)	9 105	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 2 quadruplings -> 9,104 vertices
R(4,12,14)	9 105	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 2 quadruplings -> 9,104 vertices (by monotonicity, from (4,12,13))
R(4,12,15)	9 473	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 3 quadruplings -> 9,472 vertices
R(4,13,13)	12 769	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(4,13,14)	12 769	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(4,13,15)	12 769	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(4,14,14)	14 929	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (14,14 ; 2986)]
R(4,14,15)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(4,15,15)	27 409	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (15,15 ; 5482)]
R(5,5,11)	2 113	Published bricks + theorems	atom (5,9 ; 132) -> 132 vertices; then 2 quadruplings -> 2,112 vertices
R(5,5,12)	2 385	Published bricks + theorems	atom (3,3,10 ; 149) -> 149 vertices; then 2 quadruplings -> 2,384 vertices
R(5,5,13)	2 913	Published bricks + theorems	atom (5,11 ; 182) -> 182 vertices; then 2 quadruplings -> 2,912 vertices
R(5,5,14)	3 233	Published bricks + theorems	atom (5,12 ; 202) -> 202 vertices; then 2 quadruplings -> 3,232 vertices
R(5,5,15)	3 697	Published bricks + theorems	atom (3,3,13 ; 231) -> 231 vertices; then 2 quadruplings -> 3,696 vertices
R(5,6,11)	2 913	Published bricks + theorems	atom (6,9 ; 182) -> 182 vertices; then 2 quadruplings -> 2,912 vertices
R(5,6,12)	3 249	Published bricks + theorems	atom (6,10 ; 203) -> 203 vertices; then 2 quadruplings -> 3,248 vertices
R(5,6,13)	3 585	Published bricks + theorems	atom (5,6 ; 56) -> 56 vertices; then 3 quadruplings -> 3,584 vertices
R(5,6,14)	4 257	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices; then 2 quadruplings -> 4,256 vertices
R(5,6,15)	5 057	Published bricks + theorems	atom (5,7 ; 79) -> 79 vertices; then 3 quadruplings -> 5,056 vertices
R(5,7,11)	4 017	Published bricks + theorems	atom (7,9 ; 251) -> 251 vertices; then 2 quadruplings -> 4,016 vertices
R(5,7,12)	5 393	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 2 quadruplings -> 5,392 vertices
R(5,7,13)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(5,7,14)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(5,7,15)	7 297	Published bricks + theorems	published non-linear colouring of 114 vertices (matrix R67_publiee_ExTa2015_N114.txt) -> 114 vertices; then 3 quadruplings -> 7,296 vertices
R(5,8,11)	5 393	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 2 quadruplings -> 5,392 vertices
R(5,8,12)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices

Parameters	Bound (R >=)	Origin	Construction
R(5,8,13)	7 297	Published bricks + theorems	published non-linear colouring of 114 vertices (matrix R67_publiee_ExTa2015_N114.txt) -> 114 vertices; then 3 quadruplings -> 7,296 vertices
R(5,8,14)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices
R(5,8,15)	12 929	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices
R(5,9,11)	8 993	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 2 quadruplings -> 8,992 vertices
R(5,9,12)	8 993	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 2 quadruplings -> 8,992 vertices (by monotonicity, from (5,9,11))
R(5,9,13)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices
R(5,9,14)	12 929	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices
R(5,9,15)	12 929	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices
R(5,10,11)	9 561	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]
R(5,10,12)	12 753	Our bricks (Paley-doubling atoms, Nov 2026)	atome (10,10 ; 797)
R(5,10,13)	12 929	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices
R(5,10,14)	12 929	Published bricks + theorems	atom (5,12 ; 202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices
R(5,10,15)	16 065	Published bricks + theorems	atom (7,9 ; 251) -> 251 vertices; then 3 quadruplings -> 16,064 vertices
R(5,11,11)	14 342	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (5,3;10)] x [atome (11,11 ; 1594)]
R(5,11,12)	19 125	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(5,11,13)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(5,11,14)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(5,11,15)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(5,12,12)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(5,12,13)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(5,12,14)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (12,12 ; 1594)
R(5,12,15)	25 857	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 4 quadruplings -> 25,856 vertices
R(5,13,13)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (12,12 ; 1594)
R(5,13,14)	30 645	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (13,13 ; 2554)]
R(5,13,15)	40 865	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(5,14,14)	40 865	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(5,14,15)	40 865	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(5,15,15)	49 334	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (5,3;10)] x [atome (15,15 ; 5482)]
R(6,6,11)	3 617	Published bricks + theorems	[template (5,3;10)] x [Paley-101 (6,6;101)] -> 904 vertices; then 1 quadrupling -> 3,616 vertices
R(6,6,12)	3 617	Published bricks + theorems	[template (5,3;10)] x [Paley-101 (6,6;101)] -> 904 vertices; then 1 quadrupling -> 3,616 vertices (by monotonicity, from (6,6,11))
R(6,6,13)	5 393	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 2 quadruplings -> 5,392 vertices
R(6,6,14)	5 393	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 2 quadruplings -> 5,392 vertices (by monotonicity, from (6,6,13))
R(6,6,15)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(6,7,11)	5 393	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 2 quadruplings -> 5,392 vertices
R(6,7,12)	5 825	Our bricks + published	atom (4,10 ; 91) (ours) -> 91 vertices; then 3 quadruplings -> 5,824 vertices
R(6,7,13)	7 253	Published bricks + theorems	[template (5,3;10)] x [Mathon-doubling (7,7;202)] -> 1,813 vertices; then 1 quadrupling -> 7,252 vertices

Parameters	Bound (R >=)	Origin	Construction
R(6,7,14)	8 129	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 3 quadruplings -> 8,128 vertices
R(6,7,15)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices
R(6,8,11)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices
R(6,8,12)	6 465	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 3 quadruplings -> 6,464 vertices (by monotonicity, from (5,8,12))
R(6,8,13)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices
R(6,8,14)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices (by monotonicity, from (5,8,14))
R(6,8,15)	12 929	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices
R(6,9,11)	9 329	Our bricks + published	[template (6,9,3 ; 402) (ours)] x [K2] -> 583 vertices; then 2 quadruplings -> 9,328 vertices
R(6,9,12)	11 649	Published bricks + theorems	atom (6,9 ; 182) -> 182 vertices; then 3 quadruplings -> 11,648 vertices
R(6,9,13)	12 993	Published bricks + theorems	atom (6,10 ; 203) -> 203 vertices; then 3 quadruplings -> 12,992 vertices
R(6,9,14)	14 337	Published bricks + theorems	atom (5,6 ; 56) -> 56 vertices; then 4 quadruplings -> 14,336 vertices
R(6,9,15)	17 025	Our bricks + published	[template (5,7,3 ; 188) (ours)] x [K2] -> 266 vertices; then 3 quadruplings -> 17,024 vertices
R(6,10,11)	10 033	Published bricks + theorems	[template (9,3 ; 26)] x [Paley-101 (6,6;101)] -> 2,508 vertices; then 1 quadrupling -> 10,032 vertices
R(6,10,12)	12 753	Our bricks (Paley-doubling atoms, Nov 2026)	atome (10,10 ; 797)
R(6,10,13)	14 465	Published bricks + theorems	[template (5,3;10)] x [Paley-101 (6,6;101)] -> 904 vertices; then 2 quadruplings -> 14,464 vertices
R(6,10,14)	14 465	Published bricks + theorems	[template (5,3;10)] x [Paley-101 (6,6;101)] -> 904 vertices; then 2 quadruplings -> 14,464 vertices (by monotonicity, from (6,10,13))
R(6,10,15)	21 569	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 3 quadruplings -> 21,568 vertices
R(6,11,11)	17 529	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,3;12)] x [atome (11,11 ; 1594)]
R(6,11,12)	19 125	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(6,11,13)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(6,11,14)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(6,11,15)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(6,12,12)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(6,12,13)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(6,12,14)	25 505	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(6,12,15)	25 857	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 4 quadruplings -> 25,856 vertices
R(6,13,13)	28 089	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,3;12)] x [atome (13,13 ; 2554)]
R(6,13,14)	30 645	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (13,13 ; 2554)]
R(6,13,15)	40 865	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(6,14,14)	40 865	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(6,14,15)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices
R(6,15,15)	60 297	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,3;12)] x [atome (15,15 ; 5482)]
R(7,7,11)	6 529	Published bricks + theorems	[template (9,3 ; 26)] x [Paley-17 (4,4;17)] -> 408 vertices; then 2 quadruplings -> 6,528 vertices
R(7,7,12)	9 105	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 2 quadruplings -> 9,104 vertices
R(7,7,13)	9 105	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 2 quadruplings -> 9,104 vertices
R(7,7,14)	9 473	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 3 quadruplings -> 9,472 vertices
R(7,7,15)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices (by monotonicity, from (6,7,15))

Parameters	Bound (R >=)	Origin	Construction
R(7,8,11)	9 105	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 2 quadruplings -> 9,104 vertices
R(7,8,12)	9 473	Published bricks + theorems	[template (5,3;10)] x [Paley-17 (4,4;17)] -> 148 vertices; then 3 quadruplings -> 9,472 vertices
R(7,8,13)	12 085	Published bricks + theorems	[template (7,3;16)] x [Mathon-doubling (7,7;202)] -> 3,021 vertices; then 1 quadrupling -> 12,084 vertices
R(7,8,14)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(7,8,15)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(7,9,11)	12 929	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 3 quadruplings -> 12,928 vertices (by monotonicity, from (7,9,10))
R(7,9,12)	16 065	Published bricks + theorems	atom (7,9 ; 251) -> 251 vertices; then 3 quadruplings -> 16,064 vertices
R(7,9,13)	21 569	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 3 quadruplings -> 21,568 vertices
R(7,9,14)	25 857	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 4 quadruplings -> 25,856 vertices
R(7,9,15)	25 857	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 4 quadruplings -> 25,856 vertices
R(7,10,11)	15 933	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (10,10 ; 797)]
R(7,10,12)	17 409	Published bricks + theorems	[generic template 2m-1 of K2] x [atom (4,10 ; 91) (ours)] -> 272 vertices; then 3 quadruplings -> 17,408 vertices
R(7,10,13)	23 297	Our bricks + published	atom (4,10 ; 91) (ours) -> 91 vertices; then 4 quadruplings -> 23,296 vertices
R(7,10,14)	25 857	Published bricks + theorems	published non-linear colouring of 101 vertices (matrix R411_publiee_ExTa2015_N101.txt) -> 101 vertices; then 4 quadruplings -> 25,856 vertices
R(7,10,15)	29 009	Published bricks + theorems	[template (5,3;10)] x [Mathon-doubling (7,7;202)] -> 1,813 vertices; then 2 quadruplings -> 29,008 vertices
R(7,11,11)	23 902	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,3;16)] x [atome (11,11 ; 1594)]
R(7,11,12)	31 873	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(7,11,13)	31 873	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(7,11,14)	31 873	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(7,11,15)	32 513	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 4 quadruplings -> 32,512 vertices
R(7,12,12)	31 873	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(7,12,13)	31 873	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (12,12 ; 1594)]
R(7,12,14)	36 417	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 3 quadruplings -> 36,416 vertices
R(7,12,15)	36 417	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 3 quadruplings -> 36,416 vertices
R(7,13,13)	38 302	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,3;16)] x [atome (13,13 ; 2554)]
R(7,13,14)	51 073	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(7,13,15)	51 073	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(7,14,14)	51 073	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(7,14,15)	59 713	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (14,14 ; 2986)]
R(7,15,15)	82 222	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,3;16)] x [atome (15,15 ; 5482)]
R(8,8,11)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices (by monotonicity, from (8,8,9))
R(8,8,12)	10 769	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 2 quadruplings -> 10,768 vertices (by monotonicity, from (8,8,9))
R(8,8,13)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices
R(8,8,14)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices (by monotonicity, from (7,8,14))
R(8,8,15)	18 305	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 2 quadruplings -> 18,304 vertices (by monotonicity, from (7,8,14))
R(8,9,11)	17 985	Published bricks + theorems	atom (8,8 ; 281) -> 281 vertices; then 3 quadruplings -> 17,984 vertices
R(8,9,12)	21 569	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 3 quadruplings -> 21,568 vertices
R(8,9,13)	25 857	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 4 quadruplings -> 25,856 vertices

Parameters	Bound (R >=)	Origin	Construction
R(8,9,14)	29 185	Published bricks + theorems	published non-linear colouring of 114 vertices (matrix R67_publiee_ExTa2015_N114.txt) -> 114 vertices; then 4 quadruplings -> 29,184 vertices
R(8,9,15)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices
R(8,10,11)	21 569	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 3 quadruplings -> 21,568 vertices
R(8,10,12)	21 569	Published bricks + theorems	atom (3,6,6 ; 337) -> 337 vertices; then 3 quadruplings -> 21,568 vertices (by monotonicity, from (8,9,12))
R(8,10,13)	29 009	Published bricks + theorems	[template (5,3;10)] x [Mathon-doubling (7,7;202)] -> 1,813 vertices; then 2 quadruplings -> 29,008 vertices
R(8,10,14)	29 185	Published bricks + theorems	published non-linear colouring of 114 vertices (matrix R67_publiee_ExTa2015_N114.txt) -> 114 vertices; then 4 quadruplings -> 29,184 vertices
R(8,10,15)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices
R(8,11,11)	30 275	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [atome (11,11 ; 1594)]
R(8,11,12)	31 873	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(8,11,13)	32 513	Published bricks + theorems	atom (4,4,4 ; 127) -> 127 vertices; then 4 quadruplings -> 32,512 vertices
R(8,11,14)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices
R(8,11,15)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices
R(8,12,12)	31 873	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(8,12,13)	36 417	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 3 quadruplings -> 36,416 vertices
R(8,12,14)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices (by monotonicity, from (8,11,14))
R(8,12,15)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices (by monotonicity, from (8,9,15))
R(8,13,13)	48 515	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [atome (13,13 ; 2554)]
R(8,13,14)	51 073	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(8,13,15)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(8,14,14)	56 723	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [atome (14,14 ; 2986)]
R(8,14,15)	73 217	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 3 quadruplings -> 73,216 vertices
R(8,15,15)	104 147	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [atome (15,15 ; 5482)]
R(9,9,11)	29 489	Published bricks + theorems	[template (9,9,3 ; 1282)] x [K2] -> 1,843 vertices; then 2 quadruplings -> 29,488 vertices
R(9,9,12)	35 969	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 3 quadruplings -> 35,968 vertices
R(9,9,13)	35 969	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 3 quadruplings -> 35,968 vertices (by monotonicity, from (9,9,12))
R(9,9,14)	43 073	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 3 quadruplings -> 43,072 vertices
R(9,9,15)	51 713	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 4 quadruplings -> 51,712 vertices
R(9,10,11)	35 969	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 3 quadruplings -> 35,968 vertices
R(9,10,12)	38 241	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]
R(9,10,13)	51 009	Our bricks (Paley-doubling atoms, Nov 2026)	atome (10,10 ; 797)
R(9,10,14)	51 713	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 4 quadruplings -> 51,712 vertices
R(9,10,15)	51 713	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 4 quadruplings -> 51,712 vertices
R(9,11,11)	39 834	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (9,3 ; 26)] x [atome (11,11 ; 1594)]
R(9,11,12)	57 365	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (5,3;10)] x [atome (11,11 ; 1594)]
R(9,11,13)	76 497	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(9,11,14)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(9,11,15)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)

Parameters	Bound (R >=)	Origin	Construction
R(9,12,12)	76 497	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(9,12,13)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(9,12,14)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(9,12,15)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (12,12 ; 1594)
R(9,13,13)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(9,13,14)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (12,12 ; 1594)
R(9,13,15)	122 577	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (13,13 ; 2554)]
R(9,14,14)	122 577	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (13,13 ; 2554)]
R(9,14,15)	163 457	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(9,15,15)	163 457	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(10,10,11)	35 969	Published bricks + theorems	atom (9,9 ; 562) -> 562 vertices; then 3 quadruplings -> 35,968 vertices (by monotonicity, from (9,10,11))
R(10,10,12)	38 241	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (10,10 ; 797)]
R(10,10,13)	51 713	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 4 quadruplings -> 51,712 vertices
R(10,10,14)	51 713	Published bricks + theorems	Mathon-doubling (7,7;202) -> 202 vertices; then 4 quadruplings -> 51,712 vertices (by monotonicity, from (9,10,14))
R(10,10,15)	60 161	Published bricks + theorems	atom (3,8,8 ; 940) -> 940 vertices; then 3 quadruplings -> 60,160 vertices
R(10,11,11)	46 207	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (10,3 ; 30)] x [atome (11,11 ; 1594)]
R(10,11,12)	57 365	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (5,3;10)] x [atome (11,11 ; 1594)]
R(10,11,13)	76 497	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(10,11,14)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(10,11,15)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(10,12,12)	76 497	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(10,12,13)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(10,12,14)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(10,12,15)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(10,13,13)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(10,13,14)	103 425	Published bricks + theorems	Paley-101 (6,6;101) -> 101 vertices; then 5 quadruplings -> 103,424 vertices
R(10,13,15)	122 577	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (13,13 ; 2554)]
R(10,14,14)	122 577	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (13,13 ; 2554)]
R(10,14,15)	163 457	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(10,15,15)	172 289	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 4 quadruplings -> 172,288 vertices

Parameters	Bound (R >=)	Origin	Construction
R(11,11,11)	46 207	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (10,3 ; 30)] x [atome (11,11 ; 1594)]
R(11,11,12)	70 113	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (6,3;12)] x [atome (11,11 ; 1594)]
R(11,11,13)	76 497	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(11,11,14)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(11,11,15)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(11,12,12)	76 497	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(11,12,13)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(11,12,14)	102 017	Our bricks (Paley-doubling atoms, Nov 2026)	atome (11,11 ; 1594)
R(11,12,15)	121 097	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (8,3 ; 20)] x [atome (11,11 ; 1594)]
R(11,13,13)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(11,13,14)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(11,13,15)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(11,14,14)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(11,14,15)	163 457	Our bricks (Paley-doubling atoms, Nov 2026)	atome (13,13 ; 2554)
R(11,15,15)	172 289	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 4 quadruplings -> 172,288 vertices
R(12,12,12)	76 497	Our bricks (Paley-doubling atoms, Nov 2026)	[modele generique 2m-1 de K2] x [atome (11,11 ; 1594)]
R(12,12,13)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(12,12,14)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(12,12,15)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(12,13,13)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(12,13,14)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (12,12 ; 1594)]
R(12,13,15)	145 665	Published bricks + theorems	[template (6,6,3;235)] x [K3] -> 569 vertices; then 4 quadruplings -> 145,664 vertices
R(12,14,14)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (11,11 ; 1594)]
R(12,14,15)	172 289	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 4 quadruplings -> 172,288 vertices
R(12,15,15)	172 289	Published bricks + theorems	[template (7,7,3;472)] x [K2] -> 673 vertices; then 4 quadruplings -> 172,288 vertices
R(13,13,13)	127 489	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (12,12 ; 1594)]
R(13,13,14)	153 205	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (7,3;16)] x [atome (13,13 ; 2554)]
R(13,13,15)	204 289	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(13,14,14)	204 289	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]

Parameters	Bound (R >=)	Origin	Construction
R(13,14,15)	204 289	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(13,15,15)	238 849	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (14,14 ; 2986)]
R(14,14,14)	204 289	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(14,14,15)	204 289	Our bricks (Paley-doubling atoms, Nov 2026)	[modele (4,3;6)] x [atome (13,13 ; 2554)]
R(14,15,15)	292 865	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 4 quadruplings -> 292,864 vertices
R(15,15,15)	292 865	Published bricks + theorems	[template (7,7,3;472)] x [K3] -> 1,144 vertices; then 4 quadruplings -> 292,864 vertices